Evaluating users Quality of Experience of a Collaborative Virtual Reality Application

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DECLARATION OF AUTHORSHIP

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ABSTRACT

The rapid advancement of immersive technologies has transformed multimedia experiences, moving beyond traditional two-dimensional displays to Extended Reality (XR) platforms such as Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). These platforms, powered by Head-Mounted Displays (HMDs), enable interactive and collaborative experiences in shared virtual environments. This research investigates physiological synchrony and its relationship with the user Quality of Experience (QoE) during collaborative VR tasks, integrating explicit user feedback and implicit physiological metrics.

Two experimental studies were conducted to explore the relationship between physiological signals, synchrony, and collaborative effectiveness in VR. In the first study, 22 participants worked in pairs (designated as Leader and Follower) to complete a pick-and-place puzzle task. Participants were represented as minimalistic avatars—featuring cubical heads and virtual hands—and placed in separate virtual rooms connected by an opening. Through this opening, they could see each other and pass objects to complete the task. Initial evidence of group physiological synchrony was observed in electrodermal activity (EDA), pupil dilation (PD), and heart rate (HR). Leaders consistently showed higher stress levels, indicated by elevated EDA and PD, compared to Followers. Vocal sentiment analysis revealed that positive tones were associated with shorter task durations and smoother collaboration, while negative tones were linked to frustration and increased task complexity.

These findings informed the design of a second study, which aimed to validate and extend the initial results. In this experiment, 68 participants engaged in a collaborative VR task with two conditions: a closed-room (no visual contact) and an open-room (full visual contact) collaborative environment. Avatars were enhanced with humanoid faces and virtual hands for richer interaction. While the Leader and Follower roles remained, results showed that physiological synchrony—particularly in PD and HR—increased over time and was further enhanced by visual interaction.

This research is the first to empirically examine physiological synchrony across three signals (EDA, PD, HR) in collaborative VR settings. It systematically analyses role-based differences, sentiment dynamics, and pairwise synchrony, demonstrating how these physiological signals reflect collaboration quality and impact task performance. These findings lay a foundation for designing more adaptive, immersive, and engaging collaborative VR systems. Importantly, the results converge into a triangular model linking synchrony, cognitive load, and task performance—revealing that lower synchrony predicts higher load, which in turn leads to reduced task efficiency.

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"*A journey is best measured in friends, not miles.*" *– Tim Cahill.*

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LIST OF ABBREVIATIONS

2D:	**Two-Dimensional**
3D:	**Three-Dimensional**
6DoF:	**Six Degrees of Freedom**
ACC:	**Acceleration**
AR:	**Augmented Reality**
AV:	**Augmented Virtuality**
BVP:	**Blood Volume Pulse**
CAD:	**Computer Assisted Design**
CIF:	**Context Influencing Factor**
CVR:	**Collaborative Virtual Reality**
DTW:	**Dynamic Time Warping**
ECG:	**Electrocardiogram**
EDA:	**Electrodermal Activity**
FOV:	**Field Of View**
FPS:	**Frames Per Second**
HCI:	**Human-Computer Interaction**
HRI:	**Human-Robotic Interaction**
HIF:	**Human Influencing Factor**
HR:	**Heart Rate**
HRV:	**Heart Rate Variability**
HMD:	**Head Mounted Display**
IBI:	**Interbeat Interval**
ITU-T:	**International Telecoms Union**
MOS:	**Mean Opinion Score**
MR:	**Mixed Reality**
NASA-TLX:	**NASA Task Load Index**
PD:	**Pupil Dilation**
PCC:	**Pearson Correlation Coefficient**
PUN:	**Photon Unity Networking**
PPG:	**Photoplethysmography**
QoE:	**Quality of Experience**

QoS:	**Quality of Service**
RQ:	**Research Question**
SRQ:	**Sub-Research Questions**
SD:	**Standard Deviation**
SDK:	**Software Development Kit**
SIF:	**System Influencing Factor**
SIM-TLX:	**Simulation Task Load Index**
SM:	**Sensory memory**
SPSS:	**Statistical Package for Social Sciences**
ST:	**Skin Temperature**
VM:	**Virtual Manufacturing**
VR:	**Virtual Reality**
VRNQ:	**Virtual Reality Neuroscience Questionnaire**
VRISE:	**Virtual Reality-Induced Symptoms and Effects**
Wi-Fi:	**Wireless Fidelity**
XR:	**Extended Reality**

Chapter 1: INTRODUCTION

In the rapidly evolving era of digitalisation, Virtual Reality (VR) technology is becoming a key tool for improving collaboration among teams spread across different geographical locations [1]. VR allows multiple users to interact in immersive, multimodal environments, helping them communicate more effectively and understand complex information more easily. The growing accessibility of affordable immersive hardware, such as head-mounted displays (HMDs) [2], has made virtual co-presence a practical and compelling alternative to traditional digital communication. Compared to conventional methods, VR offers significant advantages in terms of user immersion, engagement, and the ability to visualise complex products and environments in 3D [3], [4].

VR supports various collaborative activities, especially within Virtual Manufacturing (VM). It enables teams to explore, evaluate, and refine concepts together in immersive 3D environments, enhancing creative collaboration [5]. Rapid prototyping and design review of a product benefits from VR by allowing real-time feedback and iterative improvements in shared virtual spaces. A significant strength of VR in this area is its capacity to support effective teamwork by providing rich visual detail and encouraging interactive design workflows, going far beyond what's possible with paper-based or flat, two-dimensional methods [6]. As collaboration becomes more immersive, understanding the human connections formed within these virtual spaces becomes increasingly essential.

A critical aspect often associated with successful VR collaborations is interpersonal synchrony—the alignment of actions, emotions, and cognitive states between collaborators [7], [8]. In many collaborative contexts, synchrony has been shown to enhance teamwork by improving communication, decision-making, and group cohesion throughout shared tasks [9]. It is commonly categorised into three types: physiological, behavioural, and emotional [10]. For instance, synchronised changes in pupil size between collaborators can indicate shared attention or cognitive load, while aligned heart rate (HR) and electrodermal activity (EDA) patterns may reflect mutual arousal and engagement, potentially facilitating better coordination [11]. Behavioural synchrony, such as mirrored movements or synchronised eye gaze, can help establish rapport and trust, facilitating smoother interactions[12]. Emotional synchrony referring to the alignment of affective states like excitement, stress, or frustration, often expressed through tone of voice, facial expressions, or language, may foster empathy and emotional connection within teams. While synchrony has been demonstrated to benefit collaborative outcomes in various real-world settings [9], [10], these effects may depend on task type and team dynamics. Research on synchrony in immersive VR environments is still emerging and often limited to basic movement mirroring scenarios or seated meditation tasks where participants passively visualise their collaborator's physiological signal [13], [14], [15].

The evaluation of Quality of Experience (QoE) is essential in ensuring that collaborative VR systems meet user expectations and support effective task performance. QoE in VR collaboration is shaped by a range of factors, including system performance [16], network capabilities [17], [18] between remote users and user interaction elements, such as usability [19], [20], immersion[21], and emotional response [22]. Among these, interpersonal synchrony has emerged as a promising metric in some collaborative contexts, offering potential insights into how participants are aligned in their interactions and emotional states. For instance, physiological synchrony, as indicated by heart rates or electrodermal activity alignment, may reflect shared engagement and focus among team members [23], which can correlate with a positive user experience in certain scenarios. This research incorporates synchrony alongside traditional performance metrics and subjective assessments to explore how it might inform QoE evaluations in VR collaborations. Rather than assuming a universal benefit, the goal is to develop a more nuanced understanding of synchrony's relevance across different collaborative settings. This holistic approach aims to identify factors that most meaningfully contribute to effective VR-based collaboration, particularly in human-centric applications such as Smart Manufacturing [24].

Evaluating QoE in collaborative VR remains vital for ensuring user satisfaction, optimised performance, and overall system usability. Through comprehensive assessment, developers can better understand the effects of performance bottlenecks such as latency[25] and rendering quality as well as refine emotional impact and perceived fairness—improving accessibility and enjoyment for diverse users [19], [20]. High QoE is especially critical in business and industrial applications, where it can support improved learning outcomes and more efficient remote work environments. Moreover, fostering a positive user experience plays a key role in enabling the broader adoption of VR technologies, building the trust and confidence necessary for their integration into both professional and personal contexts. QoE in VR typically encompasses several dimensions, including usability, immersion, presence, and emotional response [26]. Usability relates to the ease and intuitiveness of user interactions; immersion describes how absorbed users feel in the environment; presence refers to the sense of "being there"[21]; and emotional response captures users' attitudes and affective reactions [27]. To evaluate these dimensions, this research applies both objective and subjective measures, including task performance metrics (e.g., completion time, hand controller usage), workload questionnaires NASA Task Load Index (NASA-TLX), Simulation Task Load Index (SIM-TLX), Virtual Reality Neuroscience Questionnaire (VRNQ), and user self-reports. Additionally, emotional tone and engagement are assessed through voice sentiment analysis, while interpersonal synchrony is evaluated as one of several indicators contributing to a holistic understanding of QoE [28]. By triangulating these metrics, this study seeks to identify the most influential elements affecting the quality of collaborative experiences in VR design tasks [29].

Specifically, this research focuses on the QoE evaluation of collaborative VR-based pick-and-place Soma's cube solving task [30], selected as a proof of concept for Smart Manufacturing collaborative design. Key factors influencing QoE in VR collaborative tasks include system performance and network capabilities, along with user interaction elements such as usability, immersion, and—in relevant contexts—interpersonal synchrony. While synchrony has been associated with facilitating communication, decision-making, and cohesion in some collaborative scenarios [31], its role in VR remains context-dependent and not uniformly essential. Prior studies have shown that synchrony can correlate with increased team performance and engagement [2], [31], [32], [33], [34], in real-world settings. As such, this research examines synchrony as one of several metrics that may contribute to understanding QoE in VR collaboration, without presuming its universal applicability. Although prior research has investigated synchrony in immersive VR, much of it has been constrained to relatively static contexts—such as seated meditation or basic movement mirroring [13], [35], [36], [37], [38]. This study extends existing work by exploring synchrony within a dynamic and interactive task environment: a collaborative pick-and-place activity where participants manipulate virtual objects via hand controllers, navigate using teleportation, and interact as avatars. These interactions involve complex, full-body movements such as bending, rotating, and turning while wearing HMDs, thereby offering a more ecologically valid assessment of synchrony in active VR collaboration. This research does not treat synchrony as inherently beneficial but rather explores its potential contributions to collaborative experience and QoE through physiological measures.

To investigate these dynamics, two QoE-focused evaluations were conducted. Study 1 examined the impact of Leader and Follower roles on physiological synchrony (EDA, HR, and pupil dilation) and vocal sentiment during the collaborative Soma cube task. Study 2 assessed QoE under different visibility conditions—comparing scenarios in which participants could or could not see one another during collaboration. These studies were designed around research questions aimed at understanding how synchrony, in conjunction with other metrics, may influence the perceived and measured quality of collaborative VR experiences. The specific research questions are detailed in the following section.

1.1 Research Questions

This research investigates the potential of virtual reality (VR) as a platform for enhancing collaboration during design-oriented tasks. It focuses on how task roles, physiological synchrony, vocal sentiment, and avatar visibility influence collaborative outcomes. These factors are examined to understand their effect on task efficiency, user workload, emotional engagement, and overall Quality of Experience (QoE).

In this study, QoE is defined as a multi-dimensional concept that includes task efficiency, cognitive and physical workload, emotional tone, sense of presence, and the overall quality of collaboration**.** It is assessed using a combination of objective data—such as physiological signals, task completion time, and controller interactions—and subjective responses from questionnaires and voice sentiment analysis.

Team performance in this thesis refers specifically to task completion time and the number of hand controller clicks. These metrics provide quantifiable indicators of collaboration efficiency in the VR environment.

The following primary research question was developed to guide the investigation:

"How do task role and collaboration factors (specifically avatar visibility) influence physiological synchrony in collaborative VR, and how does this synchrony relate to team performance and overall QoE?"

To break down this comprehensive investigation, the research is organised into three sub-Research Questions (SRQs), each focusing on distinct aspects of collaboration and their impact on team performance and QoE.

SRQ1. How do task roles in collaborative VR influence physiological responses and user workload and how do these affect QoE?

This question explores how different collaboration roles, such as Leader and Follower, impact users' physiological signals and their reported mental and physical workload. The analysis combines physiological measurements with subjective assessments (NASA-TLX, SIM-TLX, VRNQ, and a collaboration questionnaire) to examine how task role shapes users' experience of collaboration.

Findings are detailed in Study 1 (Chapter 5, pages 50-57) and Study 2 (Chapter 6, pages 85-98).

SRQ2. How do collaboration factors, specifically vocal sentiment and avatar visibility, influence physiological synchrony, team performance and QoE in collaborative VR?

This question examines how emotional tone in speech and visual access to a collaborator affect collaborative dynamics. It explores how vocal sentiment influences task efficiency and user workload. Avatar visibility, examined only in Study 2, is assessed for its effect on coordination and the overall collaborative experience.

Vocal sentiment: Study 1 (Chapter 5, pp. 54–65) and Study 2 (Chapter 6, pp. 94–99); Avatar visibility: Study 2 (Chapter 6, pp. 94–99)

SRQ3. How does physiological synchrony relate to collaboration quality and QoE in VR?

This question evaluates whether stronger physiological synchrony—measured via HR, EDA, and pupil dilation—is associated with better collaboration outcomes. It investigates the relationship between synchrony and key indicators such as reduced workload, shorter task completion time, fewer controller interactions, and more positive user feedback on collaboration and immersion.

Discussed in Study 1 (Chapter 5, pages 54-65) and Study 2 (Chapter 6, pages 94-99).

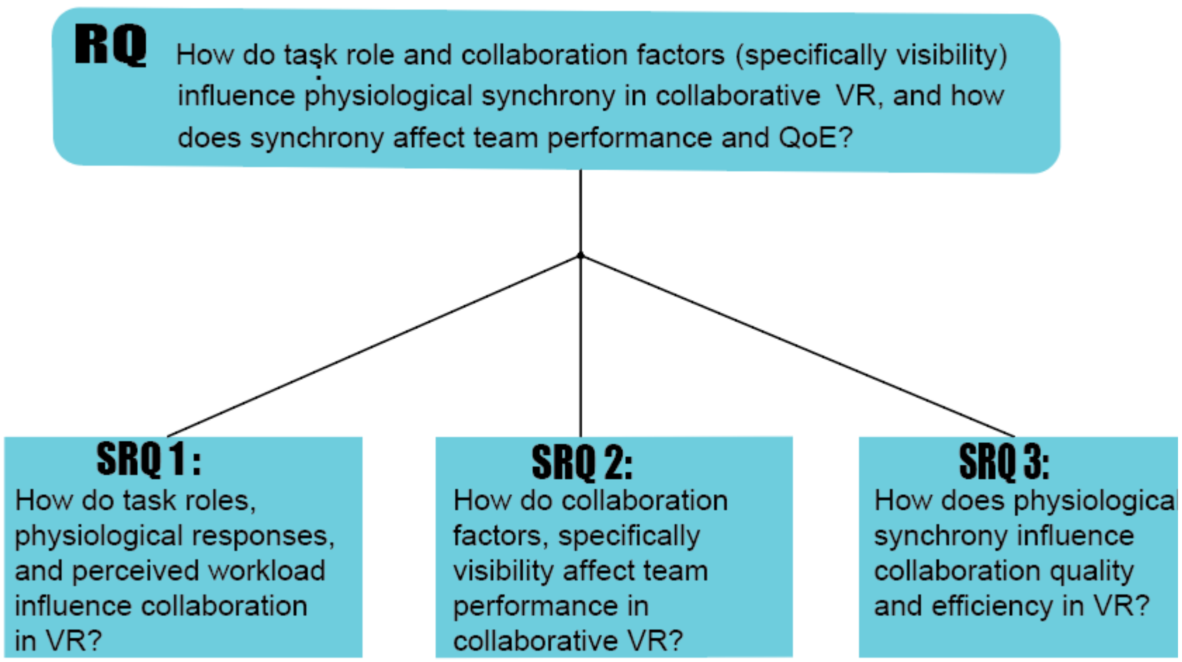


***Figure 1-1**: Three sub-research questions of the Thesis*

Together, these research questions support a structured investigation into how interpersonal synchrony and collaborative conditions influence both team performance and QoE in immersive VR environments. The following section outlines the specific research objectives based on these questions (see Figure 1-1).

1.2 Research Objectives

This research adopts a multi-stage approach to investigate the QoE in collaborative VR environments, focusing on immersive media and physiological synchrony. The methodology is structured as follows:

1. Conduct a comprehensive review and critique of existing literature on immersive media and collaboration, focusing on VR and physiological metrics, to identify the most effective approaches for capturing QoE during collaborative tasks. As part of this phase, several commercially available multiplayer platforms were tested; however, none were found suitable for physiological data capture and immersive collaboration, highlighting the need for a custom solution.
2. Design and develop a software integration platform incorporating the technologies and methods identified in the literature. The platform is developed to capture comprehensive user metrics for evaluating collaborative interactions in immersive VR environments. These include task completion time, hand and head positions, virtual block-building timings, and integrated eye-tracking metrics. The system synchronises these data streams to align gaze behaviour with interaction timings, enabling detailed user attention and coordination analysis. In addition, the platform supports capturing physiological signals and facilitates the assessment of collaboration quality.
3. Create a methodology to evaluate user QoE by integrating objective and subjective indicators. Objective metrics include task completion time and controller clicks. Physiological synchrony is computed using individual physiological signals from each collaborator, including EDA, pupil dilation, and HR. These signals are also used to evaluate physical load, mental load, and correlations between participants' physiological responses, offering more profound insight into shared cognitive and emotional states. Additionally, the system captures voice conversations during the collaborative VR task, from which sentiments are later extracted using automated sentiment analysis. Customised QoE questionnaires are also used to collect subjective user feedback.

4. Experimentally assess the effectiveness of the developed platform and evaluation methodology in providing an immersive collaborative experience. The analysis focuses on how task roles, physiological synchrony, vocal sentiment, visual access, and time contribute to collaboration quality and overall QoE. (Figure 1-2).

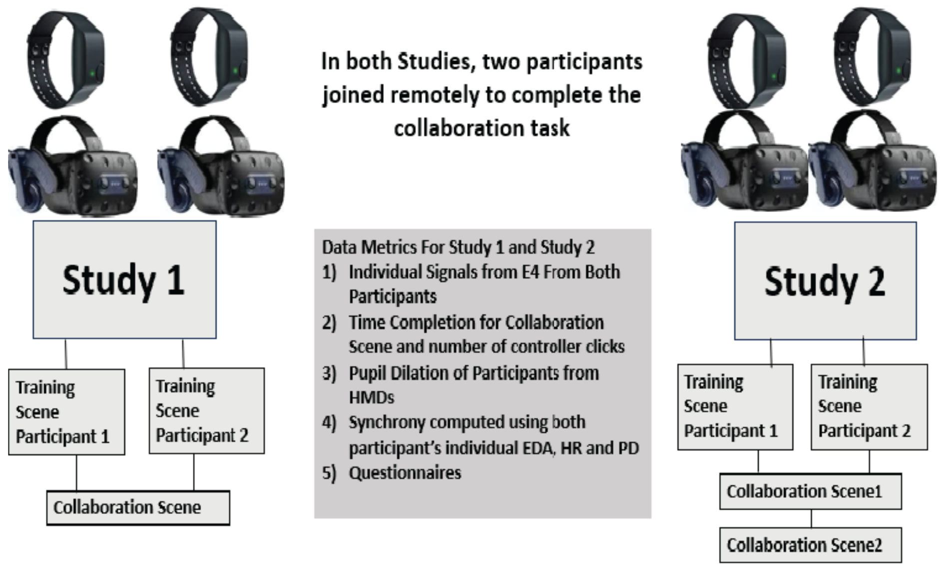


Figure 1-2: *Research Objective for Study 1 and Study 2. Study1 had one collaboration scene, and Study 2 had two collaboration scenes developed from the findings from Study1*

1.2.1 *Research Design and Objectives of Study 1*

Study 1's primary focus was to address research questions related to evaluating the QoE in collaborative tasks within VR environments. A specialised evaluation framework was meticulously designed and implemented to address the lack of a dedicated QoE measurement framework and the absence of established metrics tailored to collaborative VR tasks. This framework aimed to systematically measure users' QoE, considering the unique dynamics, challenges, and opportunities inherent in collaborative design scenarios within virtual environments.

Various physiological measures, such as EDA, HR, Interbeat Interval (IBI), Skin Temperature (ST), Blood Volume Pulse (BVP) and Eye Tracking, were integrated into the QoE assessment process, dividing participants into two groups based on their task roles (Leaders vs. Followers). This approach provided a holistic understanding of user experiences during collaborative VR design tasks. Study 1 utilised the NASA-TLX questionnaire, which includes six parameters like mental load, physical load, temporal load, performance, effort, and frustration, to evaluate cognitive and physical demands during the tasks [39]. Additionally, a customised questionnaire was employed to capture user-specific experiences and perceptions.

To explore the relationship between these subjective measures and objective performance metrics such as task completion time and hand controller clicks, correlations were analysed. Voice sentiment analysis was also conducted, as audio recordings collected using Audacity enabled an examination of how conversational dynamics during multimodal synchronous collaboration in VR influenced QoE during interactions between two users.

Towards the end of Study 1, inspired by a study utilising EDA synchrony in VR collaborative tasks [15], the focus shifted to include additional forms of synchrony, such as pupil dilation (PD) and HR synchrony, within the existing small sample size. This approach allowed for a more comprehensive assessment of the impact of collaborative physiological and behavioural synchrony on QoE. To address the limitations of study one, i.e., small sample size and use of a customised questionnaire limited the generalizability of the findings, study 2 was designed and executed.

1.2.2 *From Study 1 to Study 2*

Based on the insights and limitations observed in Study 1, Study 2 expanded to incorporate a larger sample size and more diverse collaborative conditions. To refine the evaluation, the NASA-TLX questionnaire used in Study 1 was replaced by the Simulation TLX (SIM-TLX), which includes four additional parameters beyond the original six, providing a more detailed assessment of task load [40]. The customised questionnaire was replaced with the standardised Virtual Reality Neuroscience Questionnaire (VRNQ) to capture a broader range of VR-specific experiences [41]. Additionally, an INTEREST questionnaire was introduced in Study 2 to investigate common interest factors between participants, as these may influence the level of synchrony observed during collaborative tasks. Five specific questions on collaboration were also added to assess further how participants perceived the quality and effectiveness of their collaborative interactions. Finally, drawing inspiration from a real-world collaboration study [12] which found that synchrony tends to increase over time and is influenced by visual access to collaborators. Study 2 was designed to investigate how synchrony develops during extended collaboration and improved visibility. This exploration contributes to a deeper understanding of synchrony's role in shaping collaborative experiences within VR environments.

1.2.3 *Research Design and Objectives of Study 2*

Study 2 was designed as an expanded version of Study 1 to investigate further inter-physiological synchrony between participants engaged in a collaborative VR task. This synchrony was identified as a key metric for understanding QoE in immersive environments. The study specifically aimed to explore how different visual conditions influence synchrony and user experience, building on the methodological foundation established in Study 1. In addition to physiological and behavioural synchrony, voice sentiment analysis was included to examine the role of conversational dynamics in collaborative interactions.

In Study 1, participants were represented using minimal avatars that consisted of a cube-shaped head and hands and collaborated on a puzzle-solving task. Objects were passed through a window-like opening, and coordination relied heavily on verbal communication and hand gestures. This setup provided initial insights into synchrony. It involved a collaborative task and utilised highly abstract representations of participants.

Study 2 introduced two controlled scenario variations and upgraded avatar representation to address these limitations. The same collaborative puzzle task was retained to maintain task consistency, but the simplified avatar was replaced with a humanoid avatar to enhance embodiment and ecological validity. This choice was motivated by existing research showing that more realistic avatar representations can improve user presence, emotional connection, and coordination in VR collaboration. The two scenarios were designed to manipulate visual access between collaborators systematically:

- In the first scenario, participants had no visual access to each other and relied solely on auditory communication to pass objects through a simulated escalator (or lift) system. This configuration isolated the influence of auditory-only collaboration on physiological synchrony.
- In the second scenario, participants had full visual access to each other's actions through their humanoid avatars. A grid-like boundary separated their workspaces, allowing visual and verbal interaction while preserving spatial independence. This condition enabled the exploration of how enhanced perceptual access affects synchrony and collaborative dynamics.

To control for order effects and minimise bias, the sequence of scenarios was counterbalanced between participant pairs: some completed the visual access condition first, while others began with the non-visual condition. The design of Study 2 was informed by both the findings and limitations of Study 1, as well as literature indicating the importance of visual grounding in collaboration [42]. Study 2 aimed to offer a more refined understanding of how embodiment, communication channels, and sensory access shape synchrony and QoE in collaborative VR environments by introducing controlled visibility access and increasing avatar realism.

1.3 Contributions

This research makes several original contributions to the evaluation of QoE in collaborative VR environments. The study advances methodological approaches and empirical understanding of immersive collaboration by integrating physiological synchrony, individual physiological signals, task performance metrics, and affective sentiment analysis. The key contributions and findings are as follows:

1. **Discovery and Validation of Physiological Synchrony in Collaborative VR**

This study is among the first to empirically demonstrate that physiological synchrony—specifically EDA, PD, and HR—can emerge during collaborative VR tasks. It introduces a novel multi-modal analytical framework for capturing the quality and depth of interpersonal coordination in immersive environments. This method integrates time-synchronised physiological signal processing with task role segmentation (Leader/Follower), behavioural logging (e.g., task completion time and hand controller clicks), and affective sentiment analysis of voice recordings. The novelty lies in combining these streams in real time and aligning them temporally to detect synchrony patterns dynamically during naturalistic, avatar-based collaboration. It positions physiological synchrony as a reliable, implicit metric for evaluating QoE and informs future system design in collaborative VR.

2. **Development of a Custom VR-Based Evaluation Methodology**

This research presents a multi-metric evaluation framework for collaborative VR, designed to assess QoE through synchronised physiological, behavioural, and emotional data. A custom-built multiplayer VR application (developed in Unity and Photon) captured task interactions, while physiological signals were recorded via Empatica E4 wristbands and pupil dilation from HMD eye tracking. Voice recordings, used for sentiment analysis, were collected using standard audio tools. Synchrony metrics were derived from physiological data using Dynamic Time Warping (DTW).

The framework is robust, capturing multiple dimensions of user experience, and replicable, combining commercial biosensors, validated self-report instruments, and an adaptable, researcher-developed VR platform. This contribution supports future immersive collaboration research through a transparent and modular design.

3. **Role-Based Insights from Physiological and Subjective Measures**

The study provides clear evidence of differentiated workload across collaborative roles. Collaborators assigned the Leader role experienced higher mental load, while those in the Follower role exhibited more significant physical effort. These role-based insights were derived from analysing individual physiological signals in conjunction with subjective questionnaire data, highlighting the value of combining implicit and explicit measures to understand user experience in structured VR tasks.

4. **Minor Contributions**

This study offers several minor but valuable contributions to the understanding of collaboration in VR environments. It highlights interpersonal factors between the Leader and Follower pair, such as familiarity, skill alignment, and prior VR experience, significantly enhance physiological synchrony, which in turn increases task performance and perceived QoE. Physiological synchrony between collaborators correlated with subjective experience, with synchronised pupil dilation linked to reduced mental load and higher EDA synchrony associated with faster task completion and lower frustration. Positive voice sentiment also contributed to better outcomes, emphasising the role of emotional tone. Additionally, synchrony increased over time and with visual access between collaborators, underscoring the importance of perceptual cues like avatar embodiment. These findings led to practical design recommendations for VR systems, including avatars, support for multimodal feedback, role-sensitive design, and affective cue integration to boost engagement and collaboration quality.

The findings of this research have been disseminated through peer-reviewed publications, contributing to the ongoing discourse in collaborative VR design and user experience evaluation.

The following peer-reviewed publications have resulted from this work:

- B. Moharana, C. Keighrey, and N. Murray, "*Subjective evaluation of group user QoE in a collaborative virtual environment (CVE)*" in the Proceedings of the 14th International Workshop on Immersive Mixed and Virtual Environment Systems (MMVE), June 2022, pp. 23–29. (Full international conference paper) https://dl.acm.org/doi/10.1145/3534086.3534333

- B. Moharana, C. Keighrey, and N. Murray, "*Understanding Group Quality of Experience of Immersive Virtual Reality Collaborative Design Applications*" in 37th International Manufacturing Conference (IMC-37), 2021. (Demo paper)
- B. Moharana, C. Keighrey, and N. Murray, "*Physiological Synchrony in a Collaborative Virtual Reality Task*" in IEEE 12th International Conference on Quality of Multimedia Experience (QoMEX), 2023 (Full international conference paper) https://dl.acm.org/doi/10.1145/3534086.3534333
- B. Moharana, C. Keighrey, and N. Murray, "*Voices Unveiled: Quality of Experience in Collaborative VR via AssemblyAI and NASA-TLX*" in IEEE 15th International Conference on Quality of Multimedia Experience (QoMEX), 2024. (Full international conference paper) https://ieeexplore.ieee.org/abstract/document/10598261
- B. Moharana, C. Keighrey, and N. Murray, "*Physiological Synchrony: A Novel Approach to Evaluating User Quality of Experience in Collaborative Distributed Virtual Reality Environments*" in IMVIP 25th Irish Machine Vision and Image Processing Conference, Aug 2023. (National conference paper) https://zenodo.org/records/8224881
- B. Moharana, C. Keighrey, and N. Murray, "*Role-Specific Physiological Responses and Quality of Experience in Collaborative Virtual Reality Tasks: A Comparative Study of Leaders and Followers*" in ICVR 10th International Conference on Virtual Reality (ICVR), Aug 2024. (Full International conference paper)
- B.Moharana, C.Keighrey, and N.Murray, "*Exploring the Role of Pupil Dilation and Blink Rate in Perceived Workload in Collaborative Virtual Reality*" in IEEE VR 2025.
- B.Moharana, C.Keighrey, E.Hynes, and N.Murray *"Exploring Links Between Physiological Synchrony and Task Performance Through Mesh Wall Partitions in Collaborative Virtual Reality"* -under review in Virtual Reality (Springer Nature)

1.4 Structure of the thesis

Chapter 1: Introduction—This chapter provides an overview of the research, highlighting the research gaps, questions, objectives, and contributions.

Chapter 2: Literature Review— Reviews relevant literature on collaborative design tasks, the potential of VR, QoE evaluation, and the challenges and opportunities in VR QoE evaluation and addresses the application development research questions Introduces the QoE field, including evaluation instruments (post-experience questionnaires, behavioural signals and physiological ratings), and discusses its role in assessing the influence of instruction formats and emotional and cognitive components.

Chapter 3: Collaborative Vr—This chapter introduces the system architecture developed during the research, detailing the tools, equipment, and methodologies employed. It also addresses the system architecture and tools research questions.

Chapter 4: Research Methodology—Describes the methodology employed across both studies, including research design, data collection techniques, and analytical approaches, and addresses the methodological approach to evaluate QoE and continuous improvement strategies.

Chapter 5: Experimental Study 1—This chapter starts with the first experiment– Examines the integration of EDA, HR, and Eye Tracking into QoE assessments for collaborative VR tasks, discussing how these measures contribute to a holistic understanding of user experiences.

Chapter 6: Experimental Study 2—This study applies the developed framework to assess the impact of different visibility modes on QoE, explores the link between user presence, search task performance, and overall QoE in VR environments, and addresses the multimodal synchronous collaboration challenges and their impact on QoE.

Chapter 7: Conclusion – Summarises the findings, discusses limitations, and suggests future research directions.

Chapter 2: LITERATURE REVIEW

This chapter critiques the state-of-the-art literature on collaborative Extended Reality (XR) research and QoE evaluation. It explores how these technologies are increasingly applied within smart manufacturing contexts and identifies key gaps in current research. In particular, the chapter highlights the lack of unified frameworks for assessing human-centred collaboration in immersive environments. The review sets the foundation for the research by contextualising the study's focus on physiological synchrony, immersive VR-based collaboration, and QoE evaluation.

2.1 Collaborative Design

Collaborative design is a 'human-centred' interaction that enables multiple participants to work jointly on shared projects across different times, locations, disciplines, and cultures[43]. The core value of collaborative design lies in combining diverse perspectives, which enhances creativity, problem-solving, and innovation. This process is essential in industries such as engineering, architecture, product development, and software design—where complex tasks demand coordination across teams[44].

Effective collaborative design involves more than the tools used; it requires structured communication, role distribution, mutual understanding, and conflict resolution. As such, the technology supporting collaboration plays a critical role in shaping the quality of teamwork and design outcomes.

2.1.1 *Evolution of Collaborative Design*

Historically, collaborative design relied heavily on face-to-face meetings and paper-based drawings, where team members would review and annotate physical plans[45]. As technology progressed, Computer-Aided Design (CAD) systems enabled teams to work with detailed 2D and 3D models, sharing files via email or network drives[46]. While digital tools improved accessibility, they introduced version control issues and lacked real-time interactivity.

The shift to videoconferencing and cloud-based platforms (e.g., Skype, Zoom, Google Drive) allowed remote teams to connect more fluidly. Yet, these systems still fall short of replicating the richness of in-person interaction, particularly when spatial understanding, physical co-creation, or shared immersion is needed [47].

Despite these advancements, the literature identifies a continuing gap in platforms that can fully support seamless, real-time collaboration across diverse contexts and devices, especially in the immersive environments enabled by XR.

2.1.2 *Relevance of Immersive VR to Collaborative Design*

Immersive Virtual Reality (VR) is increasingly positioned as a powerful tool for advancing collaborative design. Unlike screen-based systems, VR enables participants to interact in shared 3D spaces, manipulate virtual objects, express themselves through avatars, and communicate using gestures, voice, and gaze—all in real time. These affordances make VR particularly valuable for design scenarios requiring spatial awareness, co-presence, and embodied collaboration.

This is especially relevant in sectors like smart manufacturing, engineering, and architectural design, where teams must evaluate and iterate on complex models and environments. At the same time, immersive VR introduces new research challenges—such as maintaining user engagement, ensuring coordination across roles, and assessing Quality of Experience (QoE) through both objective and subjective metrics.

Understanding the types and structures of collaborative design in VR is essential to framing these challenges. The next section provides a categorisation of collaborative design types, drawing from the literature to define how interactions vary across modality, immersion, space-time coordination, and role structure. This typology supports the justification and scope of the current research.

2.2 Types of Collaborative Design

Collaborative design environments can be categorised along several dimensions, reflecting variations in technology use, interaction styles, immersion levels, user roles, and spatial-temporal coordination. This section synthesises typologies found in the literature [48–55] and presents six key categories relevant to immersive, XR-enabled collaboration:

Interaction Task: Distinguishes between Human-Computer Interaction (HCI)—where users collaborate via digital systems—and Human-Robot Interaction (HRI), which involves cooperation with robotic agents [48], [49].

Immersive Technology: Based on Milgram's Reality–Virtuality continuum [35], collaboration can occur in real, augmented (AR), mixed (MR), or virtual reality (VR) environments. XR serves as an umbrella term encompassing all three immersive modes.

Immersive Level: Refers to the degree of sensory immersion: fully immersive (e.g., VR headsets), semi-immersive (e.g., power walls or projection systems), and non-immersive (e.g., standard monitors) [51].

Time and Space: Collaboration can be synchronous (real-time) or asynchronous, and remote, co-located, or mixed presence, where some users are in-person and others join virtually [52].

Application-Oriented Design: Collaborative systems may target specific tasks like 3D sketching, CAD/BIM modelling, remote expert systems, or design reviews [53],[54].

Participation Model: Defines how roles are distributed—either through a master-slave model (one user directs the other) or a partnership model, where participants share control equally [55].

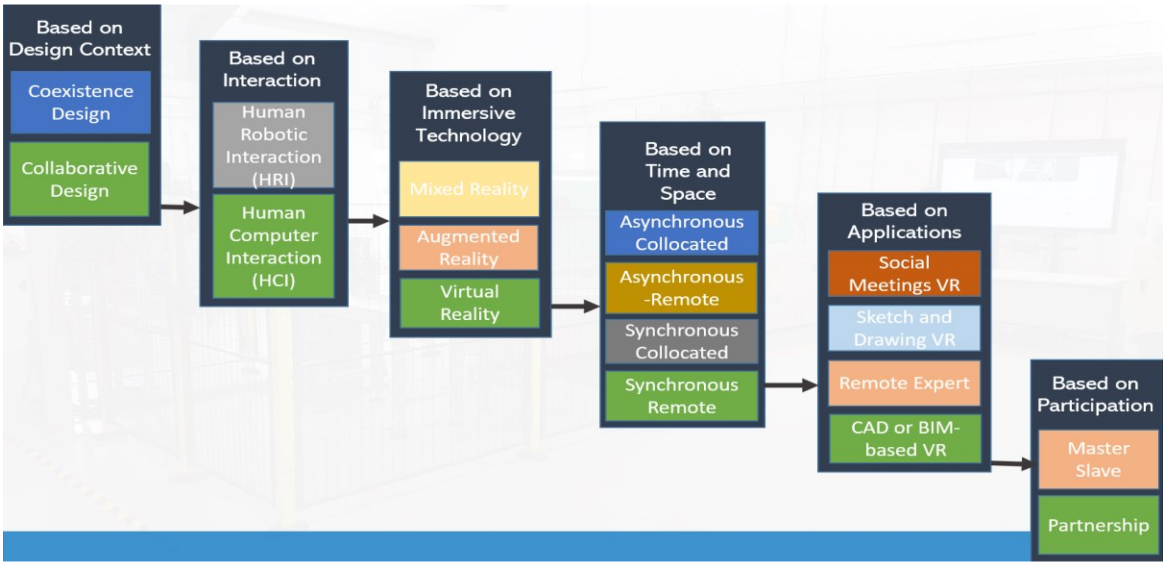


***Figure 2-1**: Collaboration types based on different classification categories.*

These categories demonstrate the diversity of collaborative design contexts and help frame the specific scenario explored in this thesis:

Collaborative Design -> Human–Computer Interaction ->Virtual Reality->Fully immersive -> Synchronous and remote -> Equal participation (partnership model).

The literature, however, lacks an integrated framework that addresses real-time collaboration in immersive VR environments while also capturing physiological, behavioural, and emotional coordination. (Figure 2-1).

2.3 XR and VR in Collaborative Design Applications

Integrating Extended Reality (XR) into collaborative design processes can potentially revolutionise how teams interact, visualise, and iterate on design projects. The state of the art in this field is defined by several cutting-edge technologies and frameworks that enhance the efficiency and creativity of collaborative design, enabling geographically dispersed teams to work together in immersive virtual environments.

2.3.1 *Extended Reality (XR) in Collaborative Design*

XR, which encompasses VR, Augmented Reality (AR), and Mixed Reality (MR), extends the possibilities of collaborative design by blending real and virtual environments. XR frameworks have been developed to support real-time, cross-reality collaboration, where designers using different devices can interact within the same virtual space [56], [57], [58]. These frameworks allow for the manipulation of both virtual and physical objects, facilitating a more integrated design process. For instance, an XR collaboration framework enables users in VR to manipulate virtual representations of physical objects [56], while users in AR can observe and interact with these objects in real-time, bridging the gap between digital and physical design elements [59]. While AR overlays digital information onto the natural world and MR seamlessly integrates virtual and physical environments, VR immerses users entirely in a computer-generated reality [60]. One notable advantage of VR is its unparalleled capacity to transport users to new environments, fostering an unmatched sense of presence and immersion. Unlike AR and MR, which enhance the real world with digital elements, VR offers a complete escape into simulated realities, making it particularly potent in gaming, industrial simulations, education, and healthcare applications. As XR advances, each component —VR, AR, and MR — uniquely shapes the future of immersive experiences, catering to diverse needs and preferences across various industries and sectors.

2.3.2 *Virtual Reality in Collaborative Design*

VR allows users to experience and interact with a 3D world that is different from the real world. This is typically achieved using a VR HMD, which covers the user's eyes and often includes sensors to track head movements, providing an immersive experience. The headset displays a 3D visual environment that changes in real-time as the user moves, giving the illusion of being present in a different location or scenario. In addition to visual immersion, VR systems often include other sensory feedback, such as sound through headphones and haptic feedback through controllers, to enhance the experience.

VR is used in various fields, including gaming, training, education, healthcare, and virtual tourism, to create experiences that would be difficult, dangerous, or impossible in the real world [61], [62], [63], [64]. VR holds significant potential to revolutionize various aspects of smart manufacturing, enhancing efficiency, reducing costs, and improving safety and product quality[65]. As a cutting-edge technology, VR creates a simulated environment where users can interact with 3D models and complex systems in real-time, which is instrumental in several key areas of manufacturing:

- **Design and Prototyping**: VR allows designers and engineers to create, manipulate, and test 3D models of products in a virtual space. This facilitates rapid prototyping, enabling quicker iteration cycles and reducing the time and resources required for developing new products [6].
- **Training and Skills Development**: Training in VR provides a risk-free environment for workers to learn and practice the operation of heavy machinery, perform maintenance tasks, and adhere to safety protocols without the physical risks associated with traditional training methods. This not only enhances learning outcomes but also significantly reduces training costs and operational downtime [65].
- **Production Planning and Simulation**: VR can be used to simulate production processes, allowing managers and engineers to optimize workflows, layout, and resource allocation before actual implementation. This predictive capability helps in identifying potential problems and inefficiencies, thereby minimizing costly disruptions once production is underway [66].
- **Maintenance, Repair, and Operations (MRO):** VR supports MRO activities by providing virtual manuals and step-by-step guides for equipment repair. Technicians can visualize and practice maintenance procedures in VR before executing them, which improves precision and reduces the time required for actual repairs. This is particularly valuable for complex or hazardous maintenance activities [67].
- **Quality Control**: VR can simulate the assembly line processes, enabling quality control engineers to inspect and assess each step virtually. This helps in detecting errors and assessing the ergonomic setup of workstations, ensuring that the final product meets quality standards while maintaining worker safety [68].
- **Remote Collaboration and Assistance**: VR enables real-time collaboration among teams located in different geographical locations. Specialists can remotely guide on-site workers through complex processes or problems using VR, which is invaluable for global operations requiring quick expert input without the delays and costs of travel [69].

2.3.3 *Applications of VR in Smart Manufacturing*

Virtual Reality (VR) has emerged as a transformative force across multiple industries, reshaping how tasks are designed, delivered, and experienced. In healthcare, VR is used for surgical training, patient rehabilitation, and phobia treatment through immersive simulations. In education, it enables experiential learning environments that enhance student engagement and knowledge retention. The retail sector leverages VR for virtual showrooms, product demos, and interactive customer experiences, while real estate professionals use it to conduct remote property tours and 3D walkthroughs. The automotive industry benefits from VR in vehicle prototyping, ergonomic testing, and immersive driver training.

In smart manufacturing, VR facilitates the evaluation and redesign of production processes with a strong emphasis on safety and ergonomics. By simulating physical tasks in immersive environments, potential risks can be identified and mitigated early, improving worker safety and operational efficiency [70]. It also enables collaborative product design and real-time workflow optimisation among distributed teams.

As applications of XR and VR continue to demonstrate improvements in design efficiency, training, and process optimisation, industries are shifting from exploratory use to practical integration. To support this, organisations increasingly rely on commercial VR frameworks that offer flexible, scalable, and customisable platforms. These tools allow companies to embed immersive technologies into existing workflows, ensuring streamlined deployment across sectors—from education labs to factory floors, retail spaces to operating rooms (Figure 2-2).

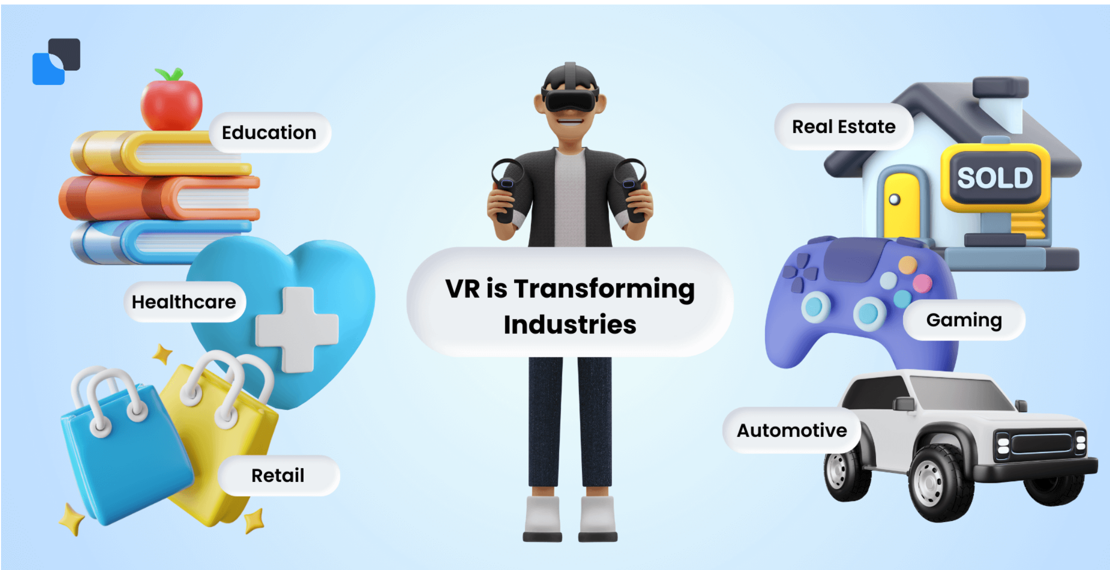


Figure 2-2*: Scope of Virtual Reality Collaborative Design*

While VR presents powerful capabilities for collaborative design, real-world implementation depends on the availability of tools that can meet domain-specific requirements. The following section examines commercially available VR platforms, assessing their strengths and limitations in supporting complex, multi-user collaborative design environments.

2.4 Commercially Available VR Tools for Collaborative Design

One of the most significant advancements in VR collaboration is the development of tools that support real-time interaction across different environments. These frameworks enable seamless collaboration between users in VR, AR, and MR, regardless of their physical location or the devices they use. Several commercially available VR tools have been designed to facilitate collaborative design processes, providing immersive environments where multiple users can work together in real time. These tools are increasingly used in various industries, including architecture, engineering, and design, to enhance teamwork and improve the visualisation and iteration of design concepts.

Given VR's significant potential in collaborative design, particularly in smart manufacturing, this research has developed and explored a variety of commercially available VR tools. However, evaluating these tools reveals essential limitations, particularly in their ability to meet the specific needs of complex, multi-user environments (Figure 2-3).

2.4.1 *Review of Existing Tools for Collaborative Design*

This research undertook an extensive survey of commercially available VR platforms designed to support multi-user, immersive collaboration. The tools explored are Improov 3, Rumii, WorldViz Vizible, Meeting VR, Facebook Horizon, Altspace VR, Insite VR, Iris Prospect, NVIDIA Holodeck, VR Sketch, Mindesk VR, Symmetry VR, Shape XR, VRChat, PixoVR, Glue, and StageVR. Each platform offered distinct features, functionalities, and collaborative tools, contributing to various options for multi-user interactions in virtual environments. A thorough examination of these frameworks aimed to identify the most suitable solutions for addressing the challenges inherent in collaborative design tasks, ensuring an informed selection process for the optimal VR platform for our research endeavours are given in Table I.

1. Improov 3: Improov 3 provides a collaborative VR platform with real-time 3D model visualization, offering features like cross-platform compatibility and live annotations for effective design collaboration.

2. Rumii: Rumii is a virtual space designed for team collaboration, integrating features such as persistent virtual rooms, spatial audio, and customizable avatars to enhance communication and teamwork in a VR environment.
3. WorldViz Vizible: WorldViz Vizible focuses on immersive presentations and collaborative meetings in VR, providing tools for interactive content creation and seamless sharing of 3D designs and experiences.
4. Meeting VR: Meeting VR offers a collaborative VR solution for meetings and presentations, enabling users to share content, interact with 3D models, and enhance communication in a virtual meeting space.
5. Facebook Horizon: Facebook Horizon is a social VR platform that offers users a shared virtual space to create, explore, and connect, emphasizing user-generated content and social interactions in a vibrant VR community.

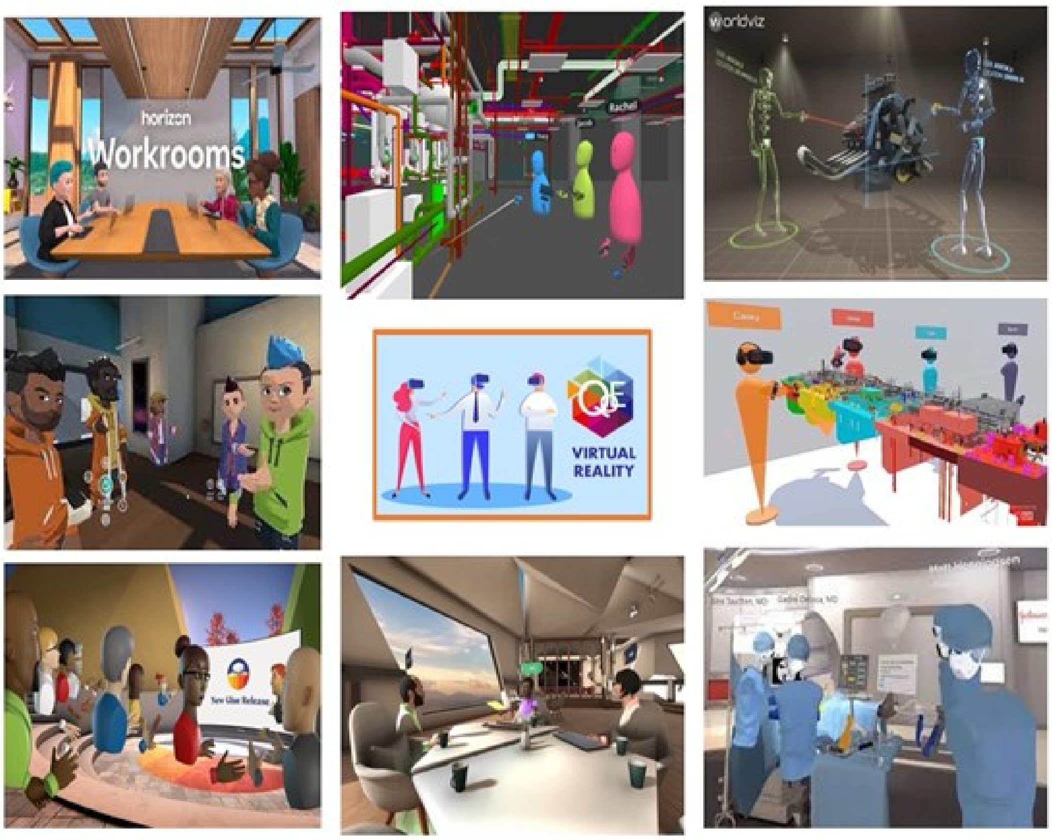


***Figure 2-3**: Commercially available VR Tools for Collaborative Design*

6. Altspace VR: Altspace VR is a social VR platform hosting events and gatherings and facilitating virtual meetups, conferences, and performances with features such as customizable environments and interactive activities.

7. Insite VR: Insite VR specializes in architectural and construction collaboration, providing tools for real-time design reviews, model exploration, and coordination among stakeholders in the AEC industry.
8. Iris Prospect: Iris Prospect focuses on architectural design collaboration, offering immersive VR experiences for design presentations, feedback gathering, and decision-making in the architectural workflow.
9. NVIDIA Holodeck: NVIDIA Holodeck is a collaborative VR platform designed for professionals, enabling realistic 3D model interactions, simulations, and virtual prototyping with advanced graphics and physics simulations.
10. VR Sketch: VR Sketch integrates with popular 3D modelling software, allowing designers to immerse themselves in their creations, make real-time adjustments, and collaborate on intricate 3D designs within a virtual space.
11. Mindesk VR: Mindesk VR provides a platform for CAD modelling in VR, allowing designers to sculpt and manipulate 3D models collaboratively, enhancing the iterative design process.
12. Symmetry VR: Symmetry VR focuses on immersive architectural design collaboration, providing tools for real-time collaboration, model examination, and decision making within virtual environments.
13. Shape XR: Shape XR offers a collaborative VR design platform with features such as 3D sketching, real-time collaboration, and spatial design tools, fostering creativity and teamwork in a virtual space.
14. VRChat: VRChat is a social VR platform known for user-generated content and interactive social experiences, allowing users to explore virtual worlds, socialize, and participate in events.
15. PixoVR: PixoVR specialises in VR training and collaboration, providing a platform for immersive learning, training simulations, and collaborative experiences in a virtual environment.
16. Glue: Glue is a VR collaboration platform offering customisable virtual spaces, real-time collaboration tools, and integrations with popular business applications to enhance remote teamwork and communication.
17. StageVR: StageVR is a collaborative VR platform focusing on virtual events, presentations, and performances, allowing users to create and share immersive experiences with interactive content and live interactions.

2.4.2 *Comparative Feature Evaluation*

The platforms were compared across several criteria—device support, interaction capabilities, collaborative features, customisation, and analytics—summarised in Table I. In evaluating commercially available VR tools, each presented unique features and capabilities for collaborative virtual experiences. Glue is a versatile VR collaboration platform offering customisable virtual spaces and real-time collaboration tools. However, our specific use case required integration with advanced technologies such as eye tracking, head tracking, and hand tracking, which proved to be cost-prohibitive within Glue. Consequently, we opted for Unity as our preferred development environment. Unity's flexibility allowed us to tailor the collaborative design tasks according to our specific needs and integrate various sensors to capture implicit and behavioural user metrics effectively. While Glue offered significant advantages, the customisation requirements for our use case led us to choose Unity, providing a comprehensive solution for developing a multiuser multimodal VR system tailored to our research objectives

In summary, the state of the art in XR collaboration is characterised by the development of advanced frameworks that enable seamless, real-time collaboration across a wide range of devices and environments. These innovations are setting the stage for more integrated and effective collaborative experiences in virtual and mixed-reality spaces, with significant implications for fields such as remote work, education, healthcare, and beyond.

This comprehensive review highlights the ongoing efforts to overcome challenges related to interoperability, platform dependency, and accessibility, paving the way for broader adoption and more sophisticated applications of XR technologies in collaborative settings.

While these tools provide value for collaborative design, they often need more customisation for specific industry applications, particularly in smart manufacturing. The literature needs to adequately address their capabilities in terms of user experience, integration with other tools, and support for complex design tasks. Furthermore, many platforms offer immersive environments and basic collaboration tools; the analysis revealed key limitations that impact their suitability for complex research-driven applications.

Table I: *Various Commercial VR Frameworks Evaluated*

Characteristics	Improov	Rumii	Vizible	Meeting	Horizon	Altspace	Insite	Iris Prospect	Holodeck	Sketch	Mindesk	Symmetry	Shape	VR Chat	Pixo	Glue	Stage
Large visualisation	✓	✓	✓	✓	✓	X	X	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Cave support	✓	?	X	✓	?	X	X	X	?	X	?	X	✓	?	✓	✓	✓
HMD support	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Desktop support	✓	✓	✓	✓	✓	✓	X	✓	✓	?	✓	X	?	✓	✓	✓	✓
Mobile support	?	✓	X	✓	✓	✓	✓	?	?	?	X	X	✓	?	?	?	?
Audio chat	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	?	✓	✓	✓	✓
Video chat	✓	X	X	✓	?	?	X	X	?	?	?	X	✓	X	?	?	?
3D video avatars	✓	✓	✓	✓	✓	X	X	✓	✓	✓	X	?	X	✓	✓	✓	✓
Non - Verbal behaviour	✓	✓	X	✓	✓	X	X	X	X	X	X	X	✓	✓	X	X	X
Avatar representation	✓	✓	✓	✓	✓	✓	X	X	✓	✓	✓	X	✓	✓	✓	✓	✓
Collaborative Interaction support	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Load arbitrary files	✓	✓	✓	?	?	X	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓
Annotations /measures on data	✓	?	✓	✓	?	X	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓
Private/public workspaces	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	*Can be*	✓	✓	✓
Shared whiteboard/draw in 3d space	✓	✓	✓	✓	?	✓	✓	✓	✓	✓	✓	✓	✓	X	✓	✓	✓
Data Analytics Feature	X	X	✓	X	?	X	X	X	✓	X	✓	X		X	✓	✓	✓
Customisation	X	X	✓	?	X	✓	X	X	X	Yes	✓	X	X	✓		X	X
Freely available	X	✓	?	X	✓	✓	X	X	X	X	X	X	X	✓	X	X	X

2.4.2.1 Limitation of Existing VR Tools for User Experience Evaluation

While commercially available VR tools offer robust platforms for virtual collaboration and interaction, they also come with certain limitations that can make them less ideal for rigorous user experience (UX) evaluation in academic or highly specialised contexts. One primary drawback is the lack of customisation and flexibility in these platforms. Commercial VR tools are typically designed with a broad user base in mind, which means they may not offer the granular control needed to tailor environments or interactions specifically for UX research purposes. This can limit the ability to conduct precise experiments or gather detailed, context-specific user data.

Moreover, these platforms often prioritise user-friendly interfaces and broad accessibility over detailed analytics and data collection features, which are critical for thorough UX evaluation [71], [72], [73], [74]. The built-in analytics provided by these tools may not be sufficient to capture the nuanced behaviours and reactions of users, making it challenging to conduct in-depth analyses. Furthermore, commercial VR platforms are frequently updated to cater to general consumer demands, which can lead to inconsistencies in research if the tools change mid-study. Another issue is the potential for data privacy and security concerns, as these platforms are generally not designed with the stringent data governance required for academic research, which could compromise the confidentiality of user data or the integrity of the research.

Lastly, licensing and using these platforms can be prohibitive, particularly for long-term studies or those requiring extensive customisation and support, making them less accessible for academic institutions or smaller research teams.

Need for a Custom Solution: These limitations highlighted the need for a custom-built VR application that could address the specific requirements of collaborative design tasks in smart manufacturing, including the integration of advanced QoE evaluation metrics. A recent study highlights the use of bibliometric methods to analyse trends and key research contributions in this area [75]. The study reveals that VR tools like Unity and Unreal Engine are frequently used, as they provide robust frameworks for developing and evaluating VR experiences. These platforms are often used with analytics tools and methodologies that assess user interaction, usability, and overall satisfaction within virtual environments. Unity is widely recognised for its versatility in VR development, offering an extensive range of features for creating interactive and immersive virtual environments. Its real-time 3D engine supports high quality, making it a popular choice for collaborative design and smart manufacturing applications.

2.5 Quality of Experience Evaluation in VR for Collaborative Design

With any new and emerging technologies, the user perceptual experience is a crucial factor towards system acceptability. Methodologies, models and frameworks are required to help us understand the utility and usability of a particular product, service, or application [76] . One such model is that of QoE.

QoE is a multidimensional construct encompassing both hedonic, utilitarian, objective (e.g., performance-related) and subjective (e.g., user-related) aspects [77]. QoE has similar motivations to User Experience (UX) evaluations [78] but is considered more from a data-driven and technical perspective. In the context of collaborative tasks through VR, we highlight how QoE can be used to inform the design of collaborative VR applications, allowing us to continuously understand the user state, engagement, etc., as they are involved in collaborative design tasks.

Over the last few years, QoE has become an important consideration for the streaming and video industry and an important research topic in the academic world. Nowadays, the users' expectations, satisfaction and (perceived) QoE are being recognized as crucial determinants for the success of the technology, even more important than technological performance and excellence, defined as Quality of Service (QoS) [79]. QoE spreads out in different disciplines or research fields, such as telecommunications, Human-Computer Interaction and user driven innovation.

2.5.1 *QoE definitions*

We present some definitions to show how this notion "QoE" is actually seen by different communities.

i. **Extension of QoS concept:** "QoE has been defined as an extension of the traditional QoS in the sense that QoE provides information regarding the delivered services from an end-user point of view" [80].

ii. **Usability as QoE:** "QoE is how a user perceives the usability of a service when in use – how satisfied he/she is with a service in terms of, e.g., usability, accessibility, retainability and integrity" [81].

iii. Business dependent QoE: "The quality of a customer's experience with business is dependent on thoughtful design of web sites, streamlined business processes that are designed to make the customer's job easier, carefully respected policies, good customer service, and excellent operational execution" [82].

iv. Degree of Delight as QoE: "QoE describes the degree of delight of the user of a service, influenced by content, network, device, application, user expectations and goals, and context of use" [83]

v. Subjective human experience as QoE: "The overall acceptability of an application or service, as perceived subjectively by the end-user" [84].

vi. Subjective human experience and Objective (Cognitive) Human factors as QoE: "QoE is a set of human-centric factors based on human subjective and objective cognitive aspects arising from the interaction of a person with technology and with business entities in a particular context" [85], [86]

The first three definitions of QoE are more inclined towards specific domains, linking QoE with QoS, HCI, and business metrics respectively. QoE is a multidisciplinary field, and different stakeholders define it according to their own needs and understanding. There was a need for a general, discipline-agnostic definition that includes human psychological aspects. Definitions IV and V fulfil this purpose well, as they encompass all the necessary aspects that impact human subjectivity. However, over time, it was also recognized that in addition to human subjective factors, human objective factors (e.g., physiological and cognitive aspects) also impact QoE. To make QoE more comprehensive and potentially more valid, it is necessary to include objective human factors along with subjective psychological measures. Thus, Definition VI is an emerging definition of QoE, incorporating both human subjective and objective aspects. As QoE matures over time, more comprehensive and exhaustive definitions are emerging to better understand its concept. This trend will continue in the future until QoE reaches full maturity. The following sections discuss how the notion of QoE is understood and treated in various disciplines.

2.5.1.1 Influencing Factors of QoE

Based on the latest definition [76] of QoE, there are three QoE influencing factors (IFs) categories: human, system, and context [87]. They are all independent variables and strongly correlated factors, resulting in a collective impact on QoE; they are illustrated in (**Figure** 2-4), along with their variations.

- Human IFs are described as the intrinsic properties or characteristics of the user (life goals, emotions, and socioeconomic position).
- System IFs are related to the quality of the technology of an application or system.
- Context IFs are defined as situational conditions experienced by the user describing a specific environment.

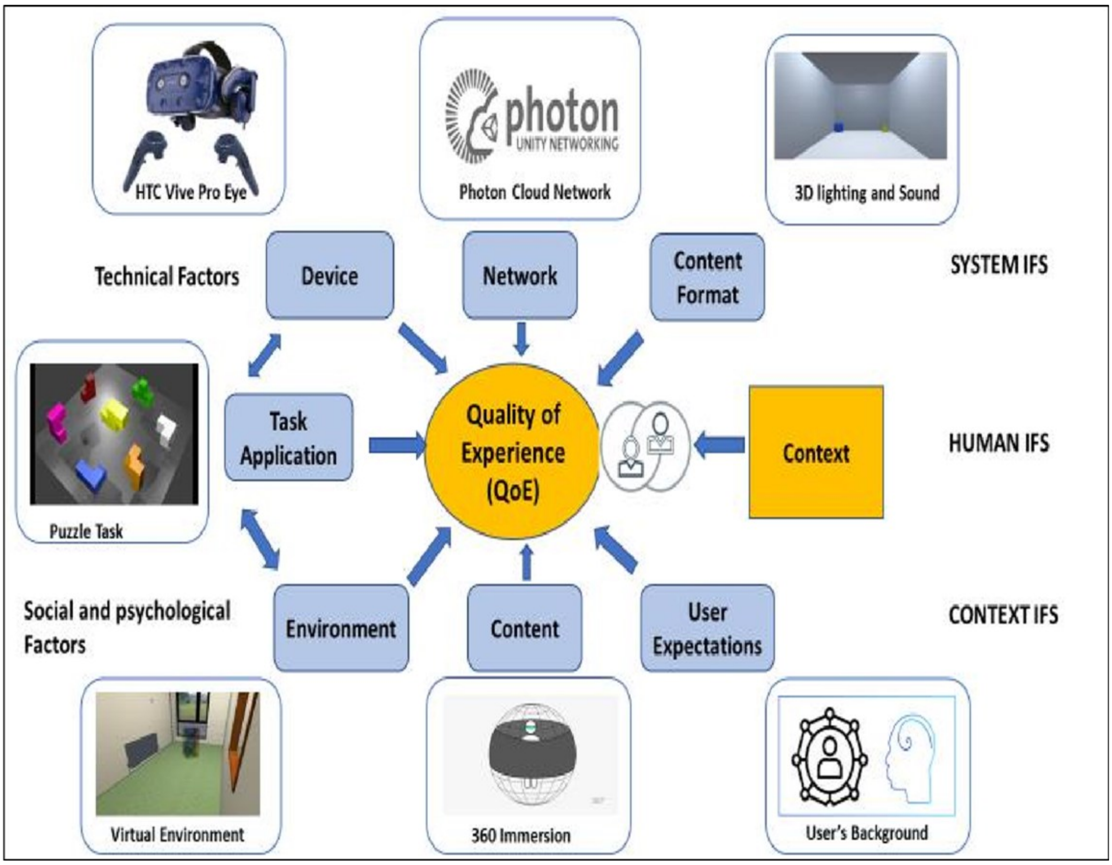


***Figure 2-4**: QoE Influencing Factors*

Review of Current Methods of QoE Evaluation

Traditional methods used for evaluating user QoE mostly relied on user surveys and scores from the user, which are too subjective and need much processing time and cost [88]. Traditionally, QoE is measured through subjective methods such as post-test questionnaires. The user's assessment in this context was via self-reporting after the experience. Such evaluation methods generally employed the mean opinion score (MOS) [89]. However, these types of evaluations are not without their flaws. They are often limited to specific application scenarios; they are often considered expensive, time-consuming, and inflexible, and sometimes the number of subjects is insufficient to represent the study's aim population [90]. As a result, the QoE community has begun to explore alternative evaluation methods [91], more specifically in terms of implicit metrics, monitoring and evaluating physiological signals as parameters to evaluate user behaviour [92]. This is now possible due to the availability of a new generation of wearable devices. Smartwatches and activity trackers observe heart rate (HR), GPS position, and step count to provide feedback on the wearer's fitness levels.

State of art research on QoE evaluations can now be categorised as shown in (Figure 2-5):

Subjective Metrics: These methods rely on users' self-reported experiences, often using standardised questionnaires like the Presence Questionnaire (PQ) [93] or the Simulator Sickness Questionnaire (SSQ) [94]. While valuable for understanding user perceptions, these methods are limited by their reliance on users' memory and the potential for biased responses.

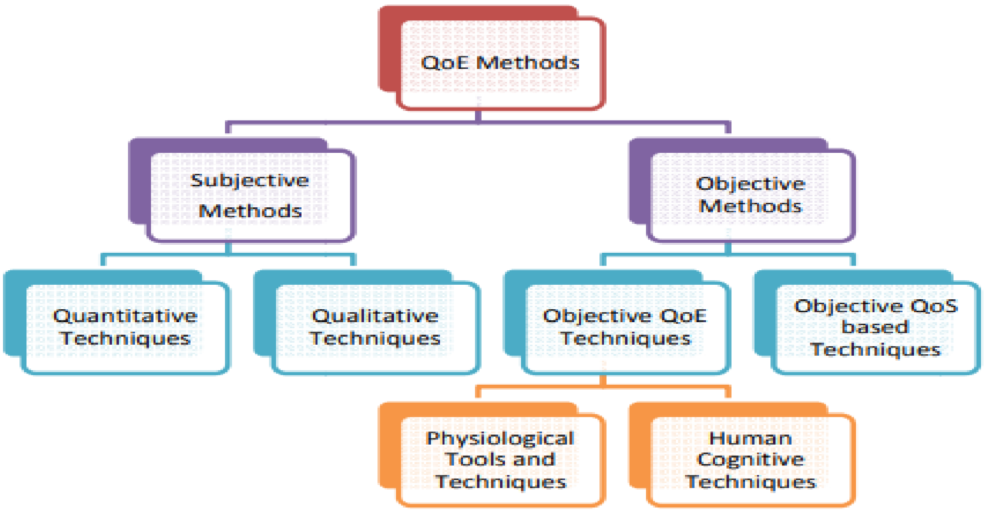


***Figure 2-5**: QoE methods of Evaluation*

Performance Metrics: Performance metrics provide insights into how intuitive and responsive the VR system is, directly impacting user satisfaction and immersion. Interaction metrics captured in the environment like head position and movement, task completion time, number of hand controller clicks, the number of errors, and teleport points [95].

Objective Metrics: Physiological metrics, such as heart rate variability (HRV), electrocardiographic signal (ECG), BVP, galvanic skin response (GSR, also known as EDA), body surface temperature, accelerometer, respiration rate, encephalography signals (EEG) and other physiological signals [96], provide real-time data on users' emotional and cognitive states during VR interactions.

Recently, physiological signals like heart rate, eye gaze, and EDA have provided an unbiased result on the user experience [88]. As shown in [84, 87], a scope of improvement is to track facial muscles, olfaction, and haptics to improve users' evaluation experiences. Another factor that raises questions about the usability of VR is simulator sickness. It causes dizziness when wearing the Head Mounted Displays (HMDs). Studies [90-91] have considered simulator sickness as a factor that affects the QoE. There are two predefined sets of questionnaire standards, the Simulator Sickness Questionnaire (SSQ) and the Virtual Reality Sickness Questionnaire (VRSQ), for the subjective evaluation of Simulator sickness. Riches et al. [85] evaluate social virtual environments with subjective evaluation based on themes. User experience evaluation rarely uses implicit and behaviour metrics in a collaborative VR application context. This opens up a new research direction for researchers to study user evaluation.

2.6 Research Gap in Current QoE Evaluation for VR Collaboration

Despite growing interest in collaborative VR, significant research gaps remain—particularly in how physiological synchrony can serve as an objective, implicit indicator of collaborative QoE. Physiological synchrony refers to the temporal alignment of physiological responses such as EDA, HR, and PD between collaborators and has shown promise in reflecting mutual engagement and shared cognitive effort in co-located interactions [97]. However, its application in remote, immersive VR collaboration remains underexplored.

Most existing work in VR collaboration focuses on individual user experience, typically using isolated physiological signals or self-report questionnaires. While self-reports are valuable, they are subject to bias, social desirability, and limitations in introspective accuracy, especially when transitioning between immersive and real-world contexts [98], [99]. Subjective reports alone may fail to capture real-time fluctuations in cognitive and emotional states, reinforcing the need for objective physiological metrics.

In this study, collaborators were assigned Leader and Follower roles in a joint VR design task. This role-based structure mirrors real-world teamwork scenarios, yet role-specific physiological analysis is rarely addressed in VR collaboration studies. For instance, while Pouliquen-Lardy et al. (2016) explored cognitive load differences between guide and manipulator roles in asymmetrical VR tasks, they did not examine physiological synchrony between participants [100]. Thus, how collaboration roles like Leader and Follower influence interpersonal alignment, and QoE remains an open question.

Voice communication is another dimension that is underutilised in synchrony-based VR research. In remote or visually constrained environments, spoken interaction becomes essential for coordination and mutual understanding. While a few studies analyse communication patterns, speech's emotional tone or sentiment and its interaction with physiological synchrony are poorly understood.

Although individual physiological signals such as EDA and HR have been used to track user state in VR [101], there is limited research on their contribution to interpersonal synchrony, especially when considering pupil dilation, a metric associated with arousal and attention. Pupil dilation synchrony, while potentially valuable for understanding shared cognitive load, has yet to be robustly integrated into collaborative VR studies.

Finally, a critical technological barrier that remains is that most commercial VR platforms cannot support the synchronised collection and analysis of multi-modal physiological data. These systems often lack customizability, preventing researchers from tracking physiological alignment, emotional tone, and task performance in real-time. This severely limits the development of holistic QoE models that can account for the complexity of collaborative VR experiences.

2.7 Developing a Custom Collaborative VR Application for QoE Evaluation

The limitations of commercially available VR frameworks and the identified gaps in QoE evaluation methods underscore the necessity of developing a custom Collaborative VR Application. This custom solution will enable the integration of advanced metrics such as eye tracking, head tracking, and hand tracking, providing a comprehensive approach to evaluating QoE in collaborative design tasks.

Our customised VR application aims to bridge these gaps by providing a platform that integrates various advanced features to offer a more comprehensive assessment of QoE. For instance, by incorporating tools to monitor physiological synchrony among users, our application will allow researchers to gain real-time insights into the emotional and cognitive states of participants, thereby offering a more objective and dynamic measure of group cohesion and user immersion. Additionally, the application’s capability to analyse voice interactions and sentiment will provide a deeper understanding of the impact of communication on collaborative tasks, while the integration of eye-tracking technology will offer detailed data on user behaviour and interaction within the virtual environment.

In summary, our research addresses the critical need for a more integrated and multifaceted approach to QoE evaluation in collaborative VR. By developing a customized application that incorporates the latest in physiological monitoring, voice and sentiment analysis, and eye-tracking technology, we aim to provide a tool that not only fills the current gaps in the literature but also sets the stage for future innovations in VR research.

2.8 Summary

This chapter has reviewed the literature on collaborative design, the use of XR and VR in smart manufacturing, and current approaches to QoE evaluation in immersive environments. It has identified key limitations in existing research, including a predominant focus on individual user metrics and self-reports, limited exploration of physiological synchrony between collaborators, insufficient analysis of distinct roles (e.g., Leader and Follower), and a lack of integration between physiological signals and voice sentiment. Furthermore, the absence of customisable VR platforms capable of synchronously capturing and analysing multi-modal data presents a practical barrier to advancing QoE research. These gaps collectively highlight the need for a more holistic, synchrony-aware framework for evaluating collaborative experiences in VR. In response, the next chapter presents the design and development of a custom-built Collaborative VR Application created to address these challenges and support detailed, role-sensitive, and physiologically informed QoE evaluation.

Chapter 3: Collaborative Vr System

This chapter presents the design and development of the Collaborative VR system that forms the foundation of this research. The system was purpose-built to evaluate QoE in immersive, multi-user environments by integrating advanced hardware, software, and data capture tools. It combines eye-tracking headsets, physiological monitoring devices, and real-time networking frameworks with a customised Unity-based application, enabling detailed monitoring of user interactions, synchrony, and communication dynamics.

3.1 Development of Collaborative VR System

The collaborative VR system developed in this research was a key vehicle used to support the evaluation of QoE in immersive environments. This chapter introduces the customised collaborative VR application designed and developed during this PhD work. The system is tailored to provide a comprehensive platform for evaluating user experiences in collaborative settings, incorporating advanced features such as physiological synchrony monitoring, voice interaction analysis, sentiment analysis, and eye-tracking capabilities.

In this chapter, the architecture and design principles of the collaborative VR system are presented, detailing how each component is integrated to create a seamless and immersive experience for participants. The chapter will also discuss the rationale behind it, including specific features, such as real-time data collection and analysis tools, that are essential for capturing the multifaceted nature of QoE in collaborative VR environments.

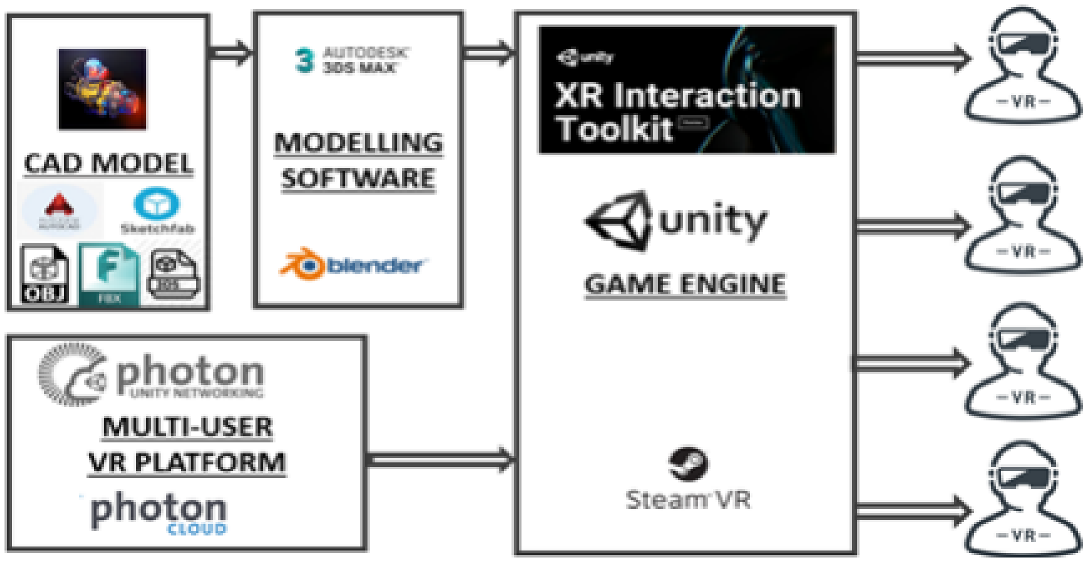


***Figure 3-1**: High Level Block Diagram of the VR Architecture*

Furthermore, we will examine the application's adaptability to various collaborative tasks and scenarios, demonstrating its potential for use in research and practical applications.

3.2 Tools for VR Environment

In VR environments, tools encompass three essential components: software, hardware, and modelling software. The software is the backbone, providing interactive and immersive experiences within the virtual space. It includes applications, platforms, and frameworks that enable users to engage with the virtual environment. On the other hand, hardware (wearable devices) constitutes the tangible equipment and devices necessary for users to access and navigate the VR world. This encompasses devices like head-mounted displays (HMDs), motion controllers, sensors, and other peripherals facilitating user interaction. Finally, modelling software is pivotal in shaping the virtual landscape, allowing designers and developers to create and manipulate 3D models and environments. Together, these components form the integral toolkit for constructing, experiencing, and interacting within the dynamic realm of VR. Figure 3-1 illustrates which components are used to build the application.

3.2.1 *Wearable Hardware Devices with Inbuilt Sensors*

A robust combination of hardware and wearable devices is essential for creating and running immersive VR applications. The following components were used in this research to ensure high-performance VR experiences and effective physiological monitoring during VR interactions:

3.2.1.1 Desktop PC with 2080 GPU Card and 32 GB RAM

A high-powered desktop PC equipped with an Nvidia GeForce RTX 2080 GPU and 32GB of RAM was used to run the VR application developed in Unity 2019.3.2f [102]. The RTX 2080 GPU provides the processing power to handle complex graphics rendering, ensuring smooth and immersive experiences in virtual environments. The 32GB of RAM allows the system to efficiently manage large datasets, multiple processes, and high-resolution textures, essential for maintaining performance in detailed VR environments. This desktop setup supports the execution of VR applications with high fidelity, enabling advanced features such as real-time lighting, particle effects, and complex 3D modelling software (see Figure 3-2).

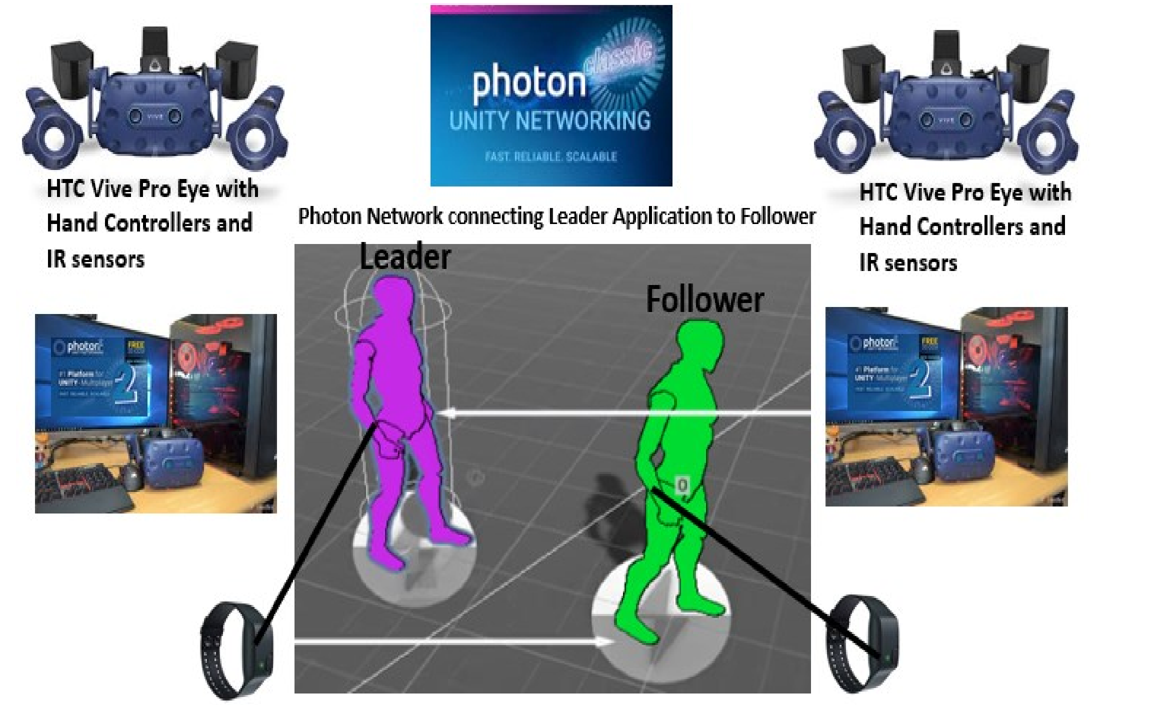


Figure 3-2*: Hardware used for Collaborative VR application Setup for measuring user interactions and task metrics*

3.2.1.2 HTC Vive Pro Eye Tethered Head-Mounted Display (HMD)

The HTC Vive Pro Eye, which is a tethered HMD, was used mainly due to its advanced eye-tracking technology that offers precise and reliable data for comprehensive analysis [103]. This headset allows us to closely monitor and understand user behaviour and cognitive processes within virtual environments, providing valuable insights into visual attention, user engagement, and psychological responses. Widely utilised for behavioural experiments in VR, the Vive Pro Eye's integration (110-degree Field of View) enhances the realism of research scenarios, making it a versatile tool for diverse studies ranging from user experience research to psychological and behavioural investigations. The headset boasts high-resolution displays at 1440x1600 per eye and a refresh rate of 90Hz. The device includes built-in sensors such as:

- **Eye-Tracking Sensors**: Track gaze origin, gaze direction, pupil position, pupil dilation, and eye openness.
- **Gyroscope**: Measures the orientation and angular velocity of the head.
- **Proximity Sensors**: Detect when the device is being worn, which helps manage power and screen activation.
- **Integrated Microphones**: Capture audio input, which can be used for voice commands, communication in multiplayer environments, and audio recording for research purposes.

- **Integrated Hi-Res Earphones**: Deliver high-quality spatial audio that enhances the immersion of the VR experience. These earphones are designed to provide clear and accurate sound, contributing to a more realistic and engaging virtual environment.

These features make the Vive Pro Eye an excellent choice for research applications, particularly in studies involving user behaviour, cognitive processes, and visual attention within VR environments.

3.2.1.3 HTC Vive Pro Hand Controllers

The HTC Vive hand controllers are integral to the VR setup, allowing users to interact with the virtual environment in a natural and intuitive way. The controllers include:

- **Accelerometers**: Measure the rate of change in velocity, which helps track movement and orientation.
- **Gyroscopes**: Provide precise rotational tracking.
- **Touch Sensors**: Detect finger placement and pressure, enabling nuanced interactions within the VR environment.

These sensors allow for precise tracking and manipulation of objects within the VR space, enhancing the user's interaction and overall immersion.

3.2.1.4 HTC Vive Base Stations (Remote Location Sensors):

The HTC Vive Base Stations, also known as Lighthouse tracking systems, are essential for accurate and precise tracking within the VR environment. These base stations include:

Infrared Emitters: Send out signals that are detected by the sensors on the HTC Vive Pro Eye HMD and controllers. These signals help determine the exact position and orientation of the user and the controllers in real-time. This setup allows for high-precision spatial tracking, which is vital for applications requiring detailed spatial interactions.

3.2.1.5 Empatica E4 Wristband

The Empatica E4 wristband is a versatile wearable device used for continuous physiological monitoring during VR interactions [104]. The wristband is equipped with several sensors, including:

- **Photoplethysmography (PPG) Sensor**: Measures BVP to derive HR and HRV, which can indicate stress levels and autonomic nervous system activity.
- **EDA Sensor**: Measures skin conductance to assess emotional arousal and stress.

- **3-Axis Accelerometer**: Tracks movement and acceleration to monitor physical activity, posture, and movement patterns.
- **Infrared Thermopile:** Measures skin temperature to monitor thermal comfort and physiological responses to environmental changes.
- **Event Marking Button**: Allows the user to mark specific events during data collection for precise analysis.

3.2.2 *Software for Building 3D Content*

3.2.2.1 3D Engine

Unity 3D offers a versatile platform for various applications beyond game creation, including video production, educational material development, and neuroscientific applications [105]. The platform's capability to create realistic 3D environments swiftly, incorporating terrains and natural/artificial objects from modelling software and integrating different SDKs to support eye tracking and hand interactions has been highlighted in Figure 3-3 , emphasising its potential for immersive virtual environments [13]. Furthermore, Unity's coordination between artists and coders within a unified environment has been recognised as a significant advantage, making it a valuable tool for various industries, including artificial intelligence environments and smart city planning [14][15]. Additionally, Unity's support for creating VR programming environments, quantum computer programming platforms, and virtual workplaces further demonstrates its adaptability to diverse applications [16][17][18].

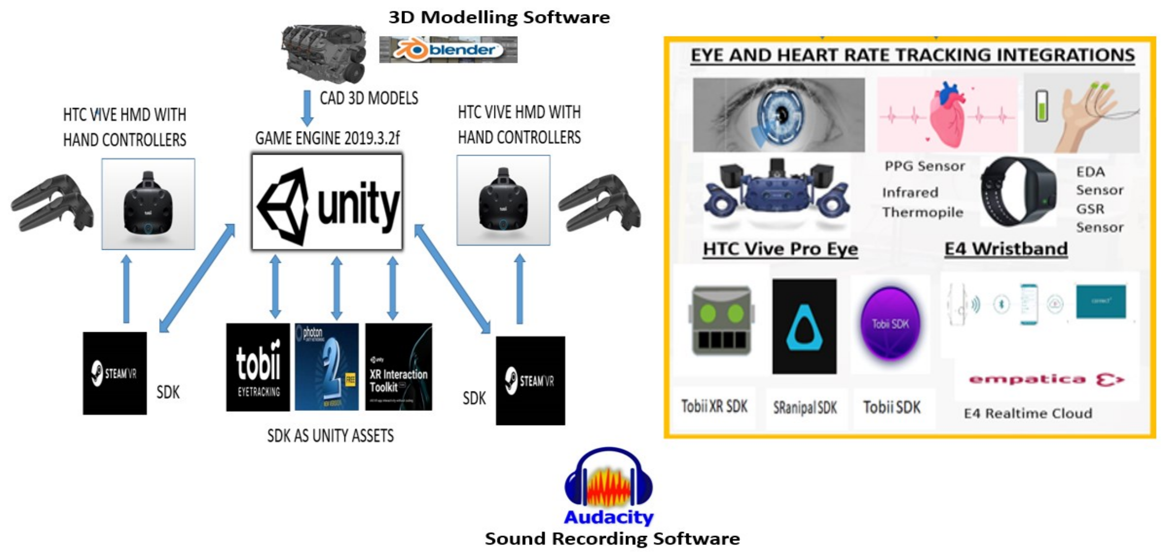


Figure 3-3*: Software SDK integrations with Unity to support eye-tracking and Hand Interactions*

Moreover, Unity's integration with XR Interaction Toolkit (XRTK) [106] and its support for OpenXR [107] emphasise its role in enabling interactions within VR and AR spaces, further solidifying its significance in facilitating immersive XR environments. The platform's ability to support the creation of virtual tours, low-cost interactive real estate applications, and 3D VR experiments showcases its potential in diverse fields, including geoinformatics, real estate, and experimental psychology [19][20][21]. Additionally, Unity's utilisation in creating multiplayer experiences through Photon Unity Networking (PUN) highlights its capability to support shared experiences and communication of game objects between players [22][23].

3.2.2.2 Multiplayer Software Development Kit (Photon Unity Networking)

Photon Unity Networking (PUN) was selected over alternatives like UNet and Bolt due to its ease of use, extensive support, and widespread community adoption [108]. UNet, Unity's now-deprecated networking solution, lacked ongoing support, making PUN a more viable option [109]. While Bolt offers a simplified API and features such as remote procedure calls (RPCs) and scene management, PUN's well-documented resources, comprehensive tutorials, and robust community support made it the preferred choice for integrating multiplayer functionality within Unity projects [110].

Developed by Exit Games, PUN simplifies the complexities of network communication, allowing developers to focus on game design rather than technical intricacies. It supports a client-server architecture, ensuring secure and cheat-resistant gameplay for real-time multiplayer games. PUN excels at synchronising game objects across the network, managing room-based multiplayer games, and integrating seamlessly with Unity's physics engine to ensure consistent interactions among players. Additionally, PUN offers cloud-based services through Photon Cloud, allowing developers to host multiplayer games without managing their own servers. Its simplicity, versatility, and efficiency make Photon Unity Networking a popular choice among Unity developers for creating engaging and immersive multiplayer experiences. In this project, Photon Voice 2 and Photon Unity Networking was used to enable voice communication and network connectivity between users.

3.2.2.3 XR Interaction Toolkit (XRTK)

The XR Interaction Toolkit (XRTK) is a framework for developing interactive extended reality (XR) experiences, particularly within virtual and augmented reality environments. The toolkit's ability to dynamically alter virtual objects and environments makes it essential for fostering rich interactions. With Unity's support for OpenXR, XRTK plays a significant role in creating immersive XR experiences[106].

The toolkit is especially valuable in educational applications, where it supports the creation of interactive XR lessons. XRTK's features enable a wide range of sensory stimulation, perceptual manipulation, and cognitive interaction within XR environments, making it a powerful tool for developing complex and engaging experiences. Compared to the Virtual Reality Toolkit (VRTK), XRTK offers stronger cross-platform compatibility, facilitating integration across a variety of XR devices [111]. Supported by a large community and regular updates, XRTK provides developers with a dynamic and evolving toolkit for crafting immersive VR experiences. Its user-friendly pre-built interaction components, like grabbing and throwing, make development more straightforward and efficient than with VRTK, which was deprecated in early 2019.

3.2.2.4 Steam VR

Integrating the SteamVR SDK with Unity is a crucial step for developers aiming to create immersive VR experiences for Steam's platform [112]. The process begins with installing the latest version of Unity, which supports OpenVR, the API used by SteamVR. Developers then download and install the SteamVR plugin from the Unity Asset Store, which provides the necessary libraries and scripts for VR development [113]. After setting up Steam and SteamVR on the system, the VR headset is connected and configured through Unity's SteamVR Input menu, allowing for the setup of controls and environment testing.

Within the Unity project, developers can add a Camera Rig prefab from SteamVR, including the VR camera and controllers, to efficiently manage input and player movement [114]. The SteamVR input system allows for customisable controller bindings, which can be configured directly within Unity using the SteamVR overlay. Developers regularly test and optimise their projects using tools like the Unity Profiler to ensure smooth performance.

Once the VR application is complete, it is built and prepared for release, following Steam's guidelines for publishing VR content. To maintain optimal performance and compatibility, developers keep their applications updated with the latest versions of Unity and the SteamVR SDK [115], [116]. They also monitor user feedback and updates from Steam to enhance the VR experience further. Advanced SteamVR features, such as tracking areas and customisable environments, are explored to enrich the VR experiences offered to users on the Steam platform.

3.2.2.5 Tobi XR SDK

The Tobii XR SDK is an essential addition to the toolkit for developing advanced VR environments, particularly when implementing eye-tracking capabilities [117]. Tobii XR SDK integrates seamlessly with existing VR frameworks like Unity and SteamVR, enabling developers to create more immersive and intuitive experiences through eye-tracking technology. Eye-tracking enhances user interaction by allowing the system to detect where the user is looking, enabling features like foveated rendering, which optimizes performance by reducing the rendering workload in peripheral vision areas [118]. This SDK is highly beneficial in applications requiring detailed user attention analysis. Moreover, Tobii XR SDK supports various VR headsets, offering developers flexibility and compatibility across different hardware platforms. Incorporating Tobii XR SDK into a VR project can significantly improve user engagement by allowing more natural and responsive interactions. It is a powerful tool for any VR development that seeks to leverage the next level of human-computer interaction [119].

3.2.2.6 SRanipal SDK

The SRanipal SDK, developed by HTC, is a crucial tool for enhancing VR environments by enabling advanced facial and eye-tracking capabilities [120]. Primarily designed to work with HTC Vive Pro Eye and Vive Facial Tracker, the SRanipal SDK allows developers to capture and utilize detailed facial expressions and eye movements, significantly enhancing the realism and interactivity of virtual experiences [121].

By integrating SRanipal SDK into VR applications, developers can create avatars that mimic the user’s real-time facial expressions and gaze direction, adding a new layer of immersion, particularly in social VR environments and applications that require nuanced user interactions. This SDK supports features such as pupil detection, gaze direction tracking, and real-time facial expression mapping, which are vital for applications in fields like virtual meetings, therapy, and educational simulations where non-verbal communication plays a critical role.

Moreover, SRanipal SDK works well with Unity and Unreal Engine, offering robust support and easy integration for VR developers. It also facilitates advanced features like foveated rendering, which optimizes the graphical performance by focusing high-resolution rendering only on the part of the scene where the user is looking, thereby reducing the load on the system and enhancing overall performance. By incorporating SRanipal SDK, developers can push the boundaries of VR experiences, making them more engaging, interactive, and lifelike.

3.2.3 *3D Modelling Software*

Blender provides Unity with a seamless pipeline for creating and importing high-quality 3D assets, making it an invaluable tool for VR development [122]. Through its robust modelling, texturing, and animation capabilities, Blender allows developers to design intricate 3D models and animations, which can be easily exported to Unity in compatible formats like FBX or glTF [123]. This interoperability ensures that the detailed work done in Blender translates accurately within Unity's environment, preserving textures, materials, and animations [124]. Additionally, Blender's ability to generate optimized assets helps maintain performance in Unity, which is crucial for creating smooth and immersive VR experiences. The open-source nature of Blender, combined with its frequent updates and strong community support, further enhances its utility as a complementary tool to Unity, enabling developers to push the boundaries of VR content creation.

3.2.4 *Sound Conversation Recording Software*

Audacity can be used alongside a VR Unity application to record high-quality audio within the virtual environment, such as voiceovers, environmental sounds, and user interactions [125], [126]. Its multi-track recording capability allows for capturing different audio elements separately, which can then be synchronised with the visual components in Unity [126]. Audacity's robust editing tools ensure the recorded audio is polished and seamlessly integrated into the VR application.

3.3 System Architecture

The system architecture for a collaborative VR application integrates these hardware and software components into a cohesive and responsive system. At its core is a high-powered desktop PC that processes data from various sensors in real-time, ensuring smooth and immersive user interactions.

This segment outlines the software tools and platforms for creating and managing the VR environment, as shown in the figure below. It includes:

- Base Station Sensors (1)
- CAD Model for 3D models as VR content (2)
- HTC Vive Pro with in-built eye-tracking sensors. HTC Vive Pro Eye is a VR headset with built-in eye-tracking capabilities. Tobii SDK and SRanipal SDK are software development kits for eye-tracking technologies (3)

- The Unity game engine for tracking time, position of users and voice data (4).
- The wristband captures various signals. The E4 Wristband is connected to the Empatica Cloud, which stores each participant's file with timestamp details in Excel format (5).
- Photon, a multi-user VR platform.

Figure 3-4 shows the required sensors and software for the development of a Collaborative VR System. The Collaborative System achieves three main functionalities:

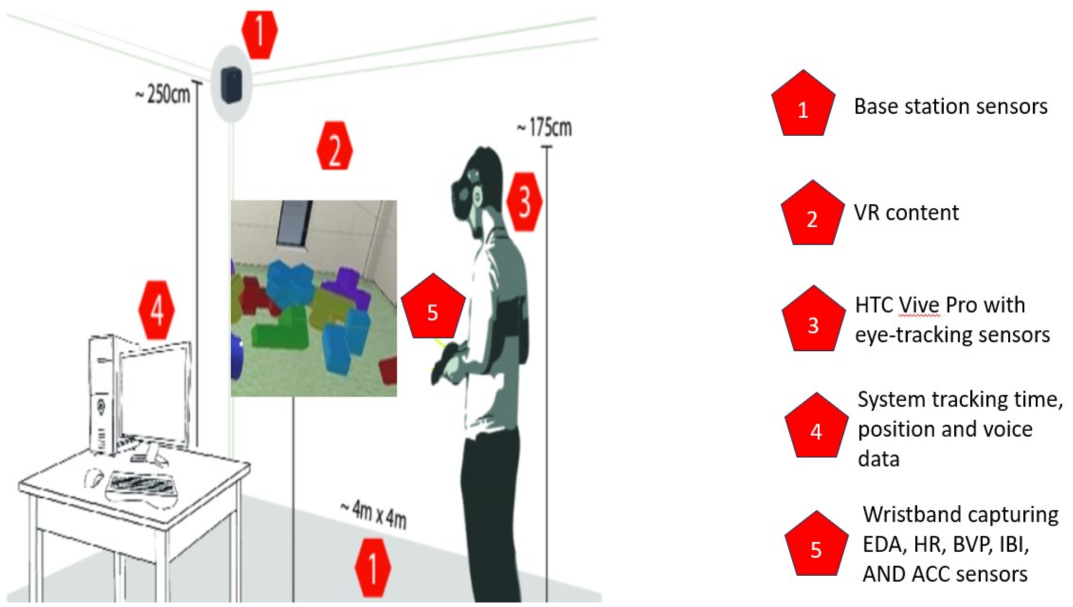


***Figure 3-4**: The flow of the Collaborative VR system with the sensors in position*

- Real-Time Data Handling: HMD, hand controller, and wristband data is continuously streamed to the desktop PC, where Unity processes it. The system handles this data in real-time, ensuring that user interactions with the virtual environment are responsive and intuitive. This real-time processing is essential for maintaining immersion, as any delays or inaccuracies in feedback could disrupt the user's experience.
- Networking and Synchronization: Photon Unity Networking ensures that all users in a collaborative session are synchronized. It manages the transmission of data, such as user movements, voice communication, and interaction with virtual objects, across the network. This synchronization is critical for collaborative tasks, where consistency in the virtual environment is necessary for effective teamwork.
- System Interaction: Users interact with the VR system through the HMD, hand controllers, and voice commands. The system captures and processes these inputs, providing immediate feedback through visual and auditory cues. The feedback loop

ensures that users are continuously aware of the impact of their actions within the virtual space, enhancing both engagement and the overall QoE.

The system architecture of the collaborative VR application is designed to leverage a powerful combination of hardware and software tools, ensuring an immersive, interactive, and responsive virtual experience. By integrating advanced devices like the HTC Vive Pro Eye and Empatica E4 Wristband with robust software platforms such as Unity 3D, Photon Unity Networking, and various SDKs, the system is capable of real-time data processing, precise user tracking, and comprehensive QoE evaluation. This architecture not only supports high-performance VR applications but also provides a flexible and scalable platform for research and professional use cases in collaborative virtual environments.

3.4 Comprehensive Data Capture in the Collaborative VR System

The collaborative VR application is meticulously designed to capture detailed data from two users simultaneously, ensuring a rich and immersive experience that adapts to their interactions and physiological responses in real-time. The system is structured to monitor and analyse every interaction with virtual objects, complete with timestamps, to provide precise insights into user behaviour and collaboration dynamics. This approach allows for a thorough understanding of how users engage with the virtual environment and each other, making the experience effective and responsive.

Various factors come into play when assessing the QoE in a Collaborative VR setting, especially when multiple users are involved. The conventional QoE evaluation concentrates on three essential factors: System, Context, and Human. We have further enriched our research by introducing a new metric, Collaboration Metrics, which comprises Synchrony, Interaction, and Voice Metrics.

User Movement and Interaction Data: The system tracks a wide array of user movements, including head and hand positions, rotations, and physical interactions with virtual objects. It also monitors the completion time for specific interactions and tasks, as well as hand controller clicks. Each interaction with virtual objects is precisely timestamped, allowing for detailed analysis of task performance and user efficiency within the virtual environment. Head movement and hand movement data, although collected, were not used during the time of writing the thesis.

Table II: Types of Data Collected from the Collaborative VR System

Data Type	Information	Purpose	Synchrony	Sentiment
User Movement and Interaction Data	Head movement, direction, rotation, hand position, rotation, physical movements, interaction with virtual objects, controller clicks, task completion	To create a responsive and immersive environment where user actions have direct impacts, to measure efficiency and ease of use, to track specific inputs, movements, and controls, and to measure task performance.	Movement synchrony is used for mirroring activities but is not used in this research context	N/A
Eye-Tracking Data	Gaze direction, focus duration, attention shifts, pupil dilation, object-looking information	To understand user attention, cognitive load, and emotional state, identify objects of interest, and optimise the virtual environment accordingly.	Pupil dilation synchrony: Measures the synchrony of pupil dilation between users to assess shared attention or emotional states.	N/A
Physiological Data	HR, EDA, BVP, Skin Temperature, Accelerometer values	To assess user stress, arousal, and engagement levels, monitor physiological responses to the VR environment, and enable real-time adjustments for comfort, immersion, and safety.	Physiological signal synchrony: Analyses the synchrony of physiological signals (e.g., heart rate, EDA) between users to evaluate shared emotional or physical states.	N/A
Voice and Communication Data	User voice communication, tone, and interaction patterns	To analyse collaboration effectiveness and ensure smooth communication between users.	Voice Synchrony is used for similar conversations but not used in this research context	Voice sentiment analysis: emotional state.

Eye-Tracking Data: The application captures detailed eye-tracking information, including gaze direction, focus duration, attention shifts, and pupil dilation. It also records which objects users are looking at and for how long. This data helps understand user attention and cognitive load, enabling the system to optimise the layout and design of the virtual environment. Additionally, pupil dilation synchrony between users is analysed to assess shared attention or emotional states during collaboration. Table II summarises the various data types, the specific information recorded, the purpose of capturing this data, the role of synchrony calculations, and how sentiment analysis is integrated into the system.

Physiological Data: The system collects various physiological signals, including HR, EDA, BVP, ST, and ACC data. These signals provide insights into users' stress levels, arousal, and overall engagement with the VR environment. Synchrony in these physiological signals between users is analysed to evaluate shared emotional or physical states, which is critical for understanding group dynamics and enhancing collaboration.

Voice and Communication Data: User voice communication, including tone and interaction patterns, is captured and analysed to ensure effective collaboration and smooth communication within the VR environment. The system also performs voice sentiment analysis to assess the emotional tone of conversations, providing insights into users' mood, satisfaction, and overall emotional state during interactions.

Environmental Interaction Data: The system monitors how users interact with virtual objects and the environment, using this data to evaluate user engagement and adjust the environment for better usability.

The collaborative VR application is designed to capture an extensive range of data from two users simultaneously, with a particular focus on interactions with virtual objects, all of which are timestamped for precise analysis. The integration of synchrony calculations and sentiment analysis further enhances the system's ability to support effective collaboration, ensuring that the virtual environment is not only immersive but also responsive to the emotional and physiological needs of its users. This comprehensive data capture and analysis framework provides a robust foundation for understanding and improving user interaction and collaboration within the VR environment.

3.5 Summary

In conclusion, the Collaborative VR system developed in this research provides a comprehensive, purpose-built platform for evaluating QoE in immersive, multi-user environments. Integrating a wide array of software and hardware components enables the synchronised capture and analysis of rich user interaction data, including physiological, behavioural, and task-based metrics, within a collaborative VR context.

The system showcases the effective fusion of advanced modelling tools, a real-time game engine, and a multi-user VR framework, facilitating both the construction of complex 3D environments and seamless participant interaction. The deliberate selection of hardware such as eye-tracking VR headsets and physiological monitoring wristbands underscores the system's commitment to capturing fine-grained user responses, offering a holistic view of user engagement through objective physiological and behavioural data.

The integration of multiple Software Development Kits (SDKs) and real-time data pipelines ensures high data fidelity, allowing for responsive analysis and adaptive system behaviour during collaborative tasks. This real-time capability is central to accurately assessing user experience as it unfolds.

Importantly, the system has been designed with scalability and flexibility in mind. Its modular architecture supports the integration of additional sensors, interaction tools, and users, making it well-suited to accommodate evolving research needs and technological advancements. This adaptability ensures that the platform remains relevant in a rapidly changing immersive technology landscape.

Overall, the Collaborative VR system reflects a multidisciplinary design philosophy, combining technical innovation with user-centred evaluation principles. It not only advances the methodological tools available for QoE research in VR but also lays the foundation for future work in immersive collaboration, human-computer interaction, and physiological computing within virtual environments.

Chapter 4: RESEARCH METHODOLOGY

This section presents how to test and validate such systems through structured research methodologies rigorously. It outlines the research methodologies used for Experiment 1 and Experiment 2 by providing an overview of the experiments, data collection, and analysis techniques to assess the system's effectiveness, ease of use, and overall impact. The chapter also covers ethical considerations, participant details, and the statistical methods to ensure reliable and consistent results. By separating the general research methods from the specific details of each study, this chapter offers a straightforward guide to understanding and evaluating the research findings presented later in this thesis.

4.1 Ethical Considerations for Human Research in VR

Conducting user experience research within VR environments introduces ethical considerations. Ethical guidelines protect participants from physical discomfort, psychological stress, and privacy breaches, particularly pertinent in studies that deeply immerse individuals in digital realms and track their physiological responses. By adhering to ethical standards, researchers safeguard participants' well-being, respect their autonomy through informed consent, and uphold the integrity of the research by ensuring data is collected, stored, and used responsibly. This section outlines the ethical framework and protocols employed to ensure participants' responsible and respectful treatment in such experiments.

All experimental procedures were submitted for review and received approval from the appropriate University Ethics Committee. The application detailed the use of VR technology, including the HMD and wristband devices, and provided a comprehensive risk assessment specific to VR environments. The review process ensured adherence to ethical guidelines pertinent to VR research, such as those related to potential simulator sickness, privacy concerns, and the intensity of immersive experiences.

Informed consent was obtained from each participant after they were given a detailed explanation of the VR experiment, the nature of the interactions within the VR environment, and the type of data to be collected via the HMD and wristband. Special attention was paid to informing participants about VR's immersive nature and the potential for psychologically impactful experiences. Participants were assured of their right to withdraw from the study at any point without any consequences.

Participant anonymity was guaranteed by assigning unique identifiers to replace any personal information. Data collected through HMDs and wristbands, potentially including biometric data, were treated with heightened security measures. Access to this sensitive information was strictly controlled, and storage was encrypted to prevent unauthorised access.

To minimise risks associated with VR, such as motion sickness or disorientation, participants were screened for pre-existing conditions that may exacerbate such effects. Clear instructions were provided on how to use the HMD and wristband safely. The physical space was secured to prevent accidents, and participants were continuously monitored for signs of discomfort, with immediate support available if needed.

The data collection process through the HMD and wristband was designed to be non-invasive and respect personal boundaries. Data use was limited strictly to the research aims, and any potential for broader data application was communicated to the participants.

Following participation, individuals were debriefed to discuss their experience and to provide additional support if the VR environment had an unexpected emotional impact. This process also gathered feedback on the ethical aspects of the VR experience, allowing for continuous improvement of the ethical protocols.

Given the evolving nature of VR technology and its impacts, the research team committed to an ongoing review of ethical practices throughout the study. This approach ensured that emerging ethical issues could be promptly identified and addressed, maintaining the welfare of participants as the primary concern.

Through this stringent ethical framework, the research aimed to respect and safeguard the well-being of all participants while exploring the frontiers of human experience within virtual realms. The ethical protocols were tailored not only to protect against known risks but also to be adaptable to the unique and dynamic challenges posed by VR technologies.

Strict ethical protocols were followed in compliance with the guidelines set by the Ethics Committee of the Athlone Institute of Technology, subsequently known as the Technological Institute of the Shannon: Midlands and Midwest, during the inception of Study 2. Participants' written consents were carefully stored and maintained securely. Privacy was paramount, with all data gathered being anonymised and securely stored to maintain participant confidentiality. Participants were provided with all relevant information before the commencement of the experiment. Supporting documents include the Participant Information Sheet (**Appendix A**), Consent Form (**Appendix B**), and the Information Protocol (**Appendix C**).

4.2 Experimental Methodology

The research employed a mixed-methods approach, combining between-subjects study designs with various quantitative and qualitative research techniques. We focused on a structured experimental approach to capture data across three dimensions: explicit feedback through questionnaires, implicit responses via physiological measures, facial expressions, and head movements, and objective performance metrics derived from task performance and completion. Experimental procedures on ITU-T recommendations P.913 and P.919 and current scholarly discourse on QoE within immersive environments were followed [127]. These standards emphasised the importance of self-directed, between-group study formations, the recruitment of gender-diverse participants, and the need to manage studies in a controlled setting, reinforced by duly processed informed consent.

Both studies utilised VR HMDs and Empatica wristbands, facilitating an immersive VR experience. The experimental task involved a collaborative VR activity designed to simulate a 'Pick and Place' puzzle challenge. This task required participants to interact with different digital shape 3D models in a shared virtual space, emulating a co-design process. The immersive nature of VR was not merely for engagement but was instrumental in evaluating QoE in a collaborative setting. The subsequent chapters provide detailed descriptions of the VR hardware employed and the 3D models used, spotlighting the bespoke configurations and modifications tailored to each study to enrich our understanding of QoE in collaborative VR design tasks.

4.2.1 *Phases of the Experiment*

The seven protocol phases reflect an overarching view of the experimental steps common to Study 1 and Study 2. The structure of these phases was informed by ITU-T P.913 [92], P.919 [93] and research in QoE of immersive experiences [2],[122]. These phases are the:

a) Information sharing and sampling phase,
b) Screening phase,
c) Video Instruction phase,
d) Baseline phase,
e) Practice phase,
f) Testing phase and the
g) Questionnaire phase.

The following subsections provide an overview of each phase at a high level.

4.2.1.1 *Information sharing and sampling phase.*

Participants were recruited through convenience sampling, primarily from the postgraduate student body of the Technological University of the Shannon: Midlands and Midwest. This was achieved through scheduling, social media contact and approaching volunteers in person. Each participant was greeted and thanked for their participation in the testing room. They were then given a full test information sheet explaining the evaluation. After reading this, giving written consent, and filling out an interest questionnaire, the participant proceeded to the screening phase. Figure 4-1 shows all the phases of the experiment in the order of data collection.

4.2.1.2 *Screening phase*

The same screening protocol was applied during each study. Each participant was first screened for visual acuity using a standard Snellen eye test [138], as seen in Fig 3.2 (a). Then, they were screened for colour perception using a digital version [139] of the standard Ishihara colour blind plates [140]. No participants were excluded from testing during the screening phase as recommended in ITU-T P.913.

Although this study involved voice-based communication during collaborative VR tasks, no formal audiometric screening was conducted. All participants self-reported normal hearing, and no difficulties were observed during interaction. However, future research may benefit from including basic hearing assessments to help ensure consistency in communication-related aspects across participants.

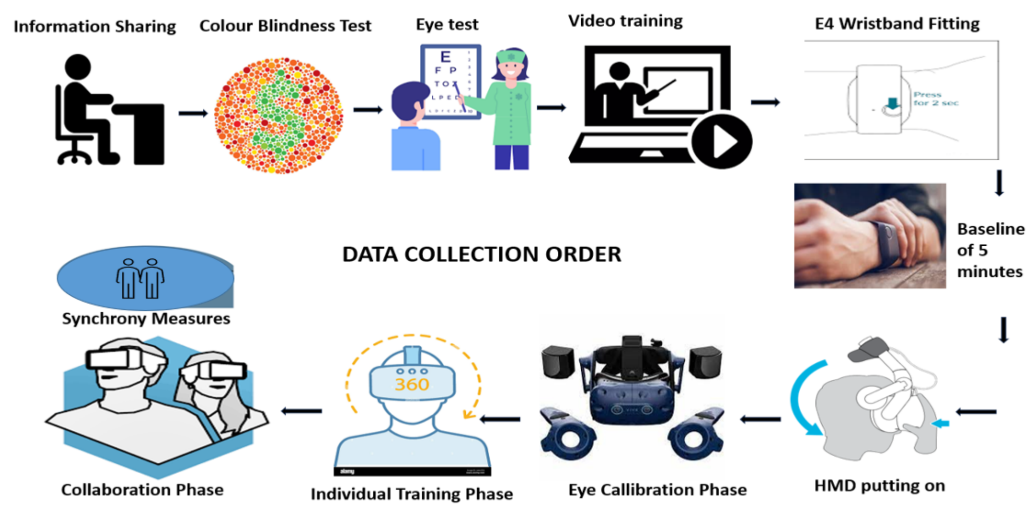


Figure 4-1*: Phases of the Experiment were designed as per ITU standards.*

4.2.1.3 Video training phase

The purpose of the video training phase was to inform the participants how to perform the required task during the testing phase. Each participant was given vocal instructions describing what they would have to do for the given testing environment they were assigned to. They were shown HTC Vive HMD and hand controllers. Each participant was allowed to ask questions before continuing to the next phase.

4.2.1.4 Baseline phase

Then, participants were fitted with E4 wristbands in their non-dominant hand and allowed to sit resting for five minutes. Each study included a five-minute baseline phase. This phase was intended to establish each participant's state before applying the technological stimulus under evaluation. Common to both studies, the data captured during the baseline phases were used to extract minimum, mean and maximum physiological ratings of each individual. The standard deviation was then calculated from the baseline mean.

4.2.1.5 Individual Training Phase

The purpose of the practice phase was to allow the participants to demonstrate that they understood the video instructions during the previous video phase. They were first fitted with the HMD. Then, once it was fitted correctly, the experimenter ran an Eye Calibration application, which allowed them to adjust the eyes correctly. Then, each participant is trained in a separate virtual room to grab and throw virtual 3D objects and move around inside the virtual environment with hand controller keys. Both studies used hand controllers to throw a ball and move around the virtual environment in this training phase.

4.2.1.6 Collaboration phase

In our exploration of QoE in collaborative virtual environments, Study 1 laid the foundational framework by focusing on individual physiological signals as participants engaged in a Soma cube puzzle task within a VR setting. Here, pairs worked together, each partner represented by a virtual avatar with a distinctive blue cube head, and hands visible through a window that facilitated the passing of objects. This initial study revealed a compelling correlation: higher

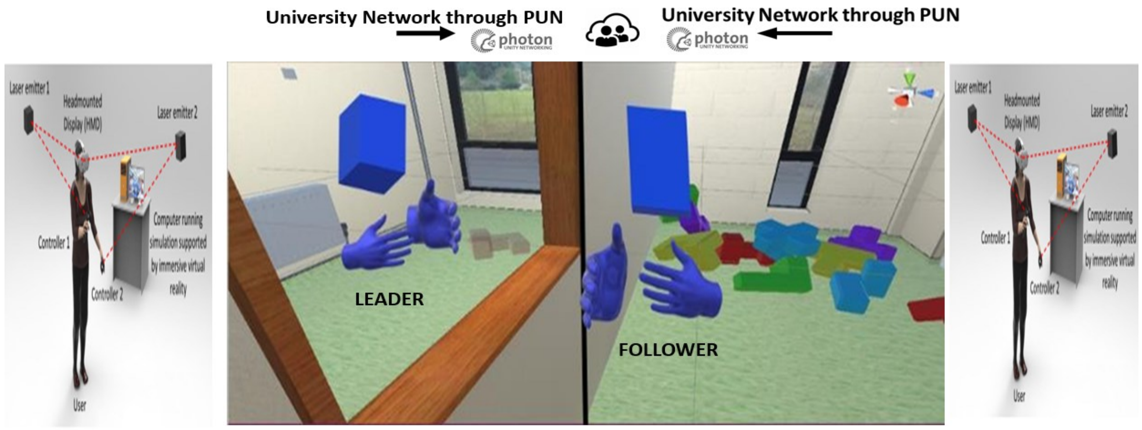


Figure 4-2: *Collaboration Phase in Study 1*

physiological synchrony between partners correlated with reduced task completion times, suggesting that deeper, unspoken connections might enhance collaborative efficiency (Figure 4-2).

Building on these findings, Study 2 was designed to delve deeper into the dynamics of physiological synchrony by examining how visual cues and interaction timing influence collaborative outcomes. To this end, we introduced significant modifications to enhance the realism and immersive quality of the participant avatars, transitioning from the abstract blue cube head to a more nuanced, realistic, unisex head. This second study was structured around two contrasting scenarios (VR Closed Phase and VR Open Phase) to dissect the effects of visibility and communication on collaboration. The first scenario restricted participants to

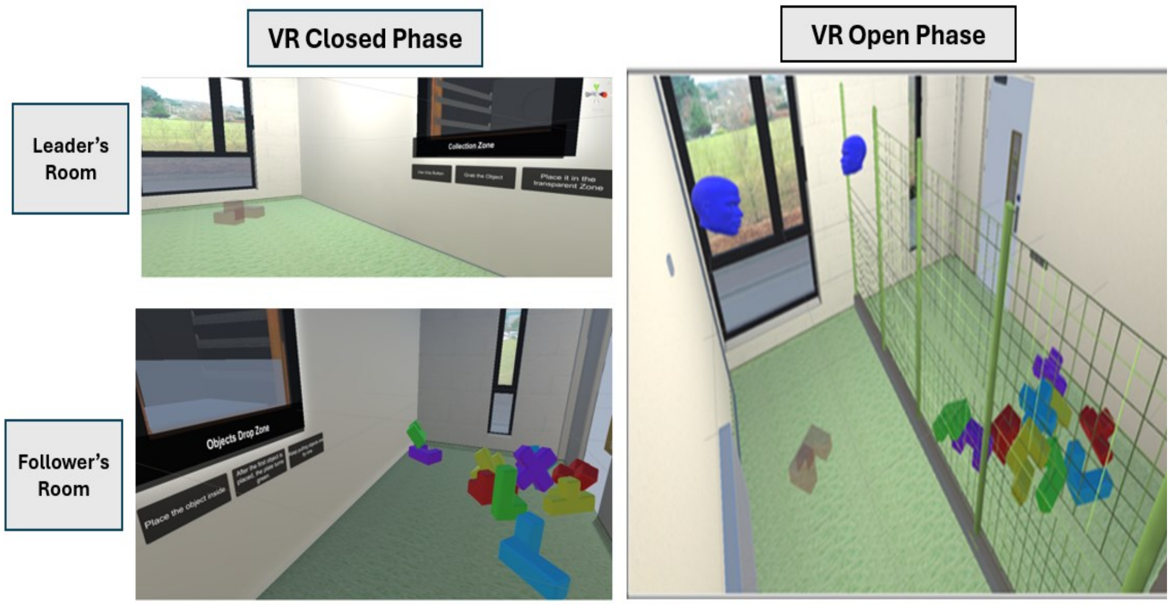


Figure 4-3: *Collaboration Phase in Study 2 consisting of two sub-phases.*

solely auditory communication, relying on an elevator system for puzzle piece exchanges, limiting visual interaction. The second scenario of Study 2 was an open workspace, where complete visibility, aural communication, and observation of each other's movements and object handling were possible (Figure 4-3).

The sequencing of these scenarios varied among participant pairs to further investigate how the order of visual and auditory experiences influenced the physiological synchrony and, by extension, the collaborative efficiency. Performance metrics, including click rates and task completion times, were meticulously recorded to assess the impact of these experimental manipulations.

The transition from Study 1 to Study 2 represented a logical progression in the research, moving from an initial understanding of the role of physiological signals in collaboration to a nuanced analysis of how these signals, and thus collaborative efficacy, are modulated by the visibility of partners and the timing of their interactions. The detailed outcomes of these studies, as discussed in Chapters 5 and 6, not only illuminate the critical role of physiological synchrony in virtual collaboration but also underscore the significance of visual presence in enhancing this synchrony and, consequently, the QoE

4.2.1.7 Questionnaire phase

Common to both studies, the participants completed a Likert scale questionnaire. For the 1st study, NASA-TLX and a customised Likert-style questionnaire were used to evaluate task performance. The Likert scale questionnaires were designed to allow the participants to report their quality judgments on the interaction, efficiency, usability, aesthetics, utility and acceptability aspects of the collaboration task under consideration for the given study. This was achieved by employing the adjectives linked to this QoE aspect in [75], which include usefulness, interest, frustration, distraction, intuitiveness, naturalness, learnability, confidence, stress, joy-of-use, ease-of-use and comfort. In addition, the recommendations of ITU-T P. 851 [145] provided guidelines for designing the questionnaire statements on interaction quality. Usability, utility and joy-of-use statements were inspired by IBM’s Post System Usability Questionnaire (PSUQ) and Computer System Usability Questionnaire (CSUQ) [146]. The Questionnaire for User Interface Satisfaction [147] helped to inform statements on ease of use. Finally, the Technology Acceptance Model [148], [149] inspired the statement about user acceptability. Common to both studies, the participants finally completed the NASA-TLX questionnaire in study 1 and SIM-TLX in Study 2 to report their subjective task load. Additionally, a state-of-the-art VRNQ questionnaire was used for Study 2.

4.3 Sample Size and Demographics

In Study 1, a sample size of 22 participants was used. These were assigned to one of two independent groups of 11 participants as calculated in ITU-T P. 913 recommendations based on gender [128]. In Study 2, a larger sample size of 68 participants was used. These were divided into two independent groups of 34 participants. This resulted in 2 participants per group as per the minimum required sample size calculated in ITU-T P. 919 [127]. Gender-balanced samples are recommended in ITU-T P. 913. To fulfil this requirement, participants were assigned to the independent test groups of Study 1. Study 1 consisted of two independent groups, with 11 males and 11 females in each group.

Study 2 consisted of two independent groups, with 38 males and 30 females. The participants in Study 1 ranged in age from 20 to 64, with a mean age of 32. In Study 2, participants ranged in age from 19 to 62, with a mean age of 32. Sixteen nationalities were represented in the Study 1 sample, including Poland, Ireland, Brazil, India, Egypt, Lithuania, the Philippines, Malaysia, Pakistan, China, Germany, Nigeria, South Africa, Portugal, Latvia and Canada. Twelve nationalities were represented in the Study 2 sample, including France, Venezuela, and Mexico, and many of them were the same nationalities as Study 1. Nationality was not controlled, and participants of different nationalities were randomly assigned to the test groups.

4.4 Standardisation of Individualistic Signals

Standardisation of signals is crucial before conducting statistical analysis, aiming to ensure the reliability, accuracy, and comparability of the data analysis. Signals, such as HR, Skin Temperature, Pupil Dilation, etc., can vary significantly between individuals, recording sessions, and even within a single session. Standardisation was performed due to several factors stated in Table III.

By standardising bio signals before statistical analysis, the aim was to improve the accuracy and reliability of their findings, make meaningful comparisons across different datasets or conditions, and derive insights based on the true underlying physiological phenomena rather than artifacts, noise, or other confounding factors [129] .

Standardising bio signals is vital for ensuring the accuracy and reliability of statistical analyses in biomedical research and other applications involving physiological data. Two

Table III: Standardisation Process Approaches and Explanation

Standardisation Benefit	Explanation
User Physiological Signals	Signals such as HR, EDA, Pupil dilation, Skin Temperature, BVP, IBI, ACC differs from person to person.
Comparability Across Subjects	Individuals can have significant physiological differences, affecting bio signal measurements. Standardisation helps adjust these signals to a standard scale, allowing for meaningful comparisons across subjects.
Baseline Removal	Bio signals often include baseline noise or variations unrelated to the specific physiological events of interest. Standardisation provides for processes to remove or reduce these baselines, focusing the analysis on the changes or patterns that matter.
Amplitude Variation	The amplitude of bio signals can vary due to several factors, including sensor placement, individual differences in anatomy or physiology, and the subject's state (e.g., resting vs. active). Standardisation normalises these amplitude differences to allow for consistent analysis and interpretation.
Noise Reduction	Bio signals are prone to various types of noise, including electronic noise from the recording equipment, interference from other electronic devices, and artefacts from subject movement. Part of the standardisation process involves filtering and cleaning the data to reduce the impact of noise.
Feature Extraction	Statistical analysis often relies on extracting features from the bio signals. Standardisation ensures that these features are extracted consistently, improving the reliability of the analysis.
Temporal Alignment	For event-related analysis, bio signal events must be temporally aligned across trials or subjects. Standardisation includes methods to align these events, facilitating accurate comparisons and statistical analysis of event-related responses.
Handling Missing or Incomplete Data	Data segments can be missing or unreliable in bio signal recording. Standardisation processes often include techniques for dealing with these gaps, such as interpolation or using models to estimate missing values.
Normalisation for Statistical Methods	Many statistical methods assume data that are normally distributed or at least scaled similarly. Standardisation helps in rescaling the data to meet these assumptions, enhancing the applicability and accuracy of statistical tests and models.

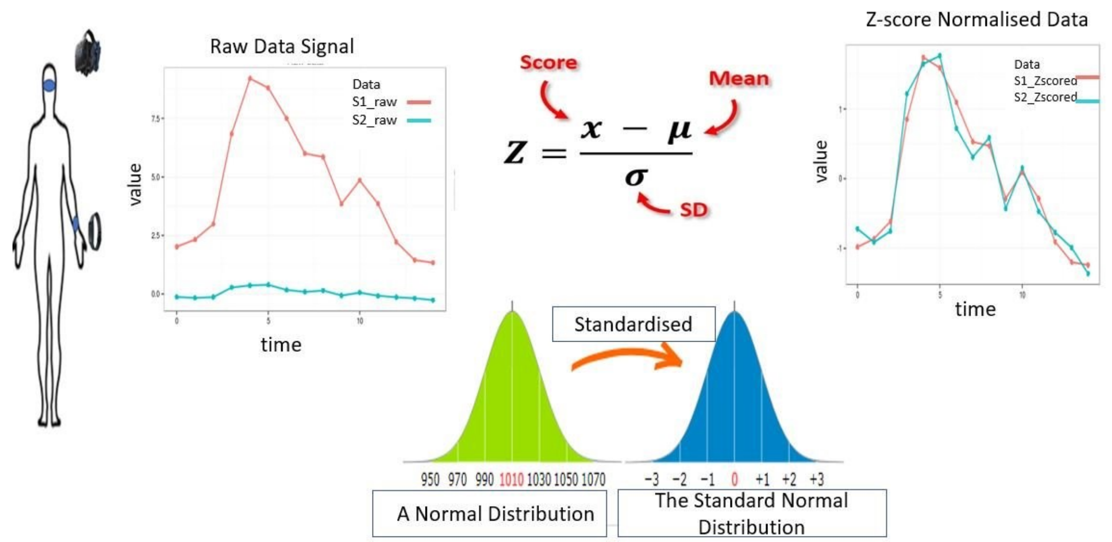


Figure 4-4: *Standardisation of Raw signals of each individual so that it gets ready for comparative analysis.*

[131]. These methods rescale the data, but they do so in slightly different ways, each with its unique advantages depending on the application [132], [133]

Zero-score normalisation and min-max scaling are both statistical techniques used to standardise the range of independent variables or features of data in data processing. They are commonly applied in the pre-processing phase in machine learning and data mining (Figure 4-4).

4.4.1 *Zero-Score Normalisation:*

Also known as Z-score normalisation or standardization, this technique involves rescaling the data so that it has a mean (μ) of 0 and a standard deviation (σ) of 1. The process of zero-score normalisation is performed according to the following formula for each feature X:

$$Z = (X - \mu) / \sigma \qquad \text{(eq. 1)}$$

Where X is the original value of the feature, μ is the mean of the feature, and σ is the standard deviation of the feature.

The result of this transformation is that the features will be centred around zero and will typically follow a standard normal distribution, which is a bell-shaped curve (Figure 4-4). Zero-score normalisation is especially useful when features have different units or very different variances, as it normalizes the feature scales without assuming any specific distribution [131], [132].

4.4.2 *Min-Max Scaling:*

Min-max scaling (or MinMax normalisation) is another technique that rescales the data to a fixed range, usually 0 to 1, or it could be -1 to 1 depending on the requirement [134], [135]. The transformation is given by:

$$X' = \frac{(X\max - X\min)}{(X - X\min)} \qquad \text{(eq. 2)}$$

X is the original value, X' is the normalised value, X_min is the minimum value for the feature X, and X_max is the maximum value for the feature X.

The min-max scaling approach is beneficial when the algorithm assumes the data is bound within a certain range, such as neural networks often assume. However, it is sensitive to outliers because the minimum and maximum values are used for scaling [135].

Standardising biosignals is a crucial step for evaluating and comparing users' data [136]. Two popular methods for this are zero-score normalisation and min-max scaling. Zero-score normalization calculates the mean and standard deviation to scale the data, making it less susceptible to outliers. Meanwhile, min-max scaling maintains the relationships among the original data values while normalising the range. The optimal method depends on the dataset's specific requirements and the machine learning algorithm. However, after evaluating both methods thoroughly, we confidently chose z-score normalisation for our QoE evaluation as it is more robust to individual bio signal variations, which can lead to outliers [134].

4.5 Statistical Analysis Methods

The Statistical Analysis section of this thesis details the comprehensive methods employed to evaluate the data collected from our studies, aimed at understanding the nuances of QoE in collaborative VR environments. Given the multifaceted nature of this research, various statistical tools and techniques were utilised to ensure a robust analysis of both quantitative and qualitative data, providing insights into participant behaviour, performance, and physiological synchrony.

- **Step 1: Data Preparation**

  The dataset was subjected to a thorough cleaning process, which involved resolving any instances of missing data, addressing outliers, and performing necessary scaling and transformations. This set a solid foundation for subsequent statistical testing.

- **Step 2: Homogeneity of Variances Assessed**

  In the context of statistical analysis, especially within the domain of bio signal processing where variability is high and normal distribution cannot be taken for granted, it is crucial to verify the uniformity and Gaussian nature of the dataset. These assessments inform the selection of statistical methodologies, as many standard parametric evaluations are predicated on normal distribution. Data homogeneity implies that the variances across different samples or groups are consistent. This uniformity is a prerequisite for the validity of certain statistical comparisons, such as those employed in ANOVA, which compare means across multiple groups to determine if at least one group mean is statistically different from the others. We conducted tests for homogeneity of variances. Levene's Test was applied due to its robustness in handling non-normal data distributions. For datasets where a normal distribution was assumed, Bartlett's Test was utilized. The results confirmed that the assumption of equal variances across groups was satisfied, validating the prerequisites for an ANOVA.

- **Step 3: Normality of Distributions Verified**

  Normality tests were performed on each subgroup and the dataset as a whole. For normality testing, the following procedures are applied:

  - **Shapiro-Wilk Test:** This test is particularly effective for smaller sample sizes. It assesses whether a sample is drawn from a normally distributed population by testing the null hypothesis that there is no deviation from normal distribution.
  - **Kolmogorov-Smirnov Test**: Appropriate for samples of any size, this test compares the empirical distribution of a sample with a specified probability distribution, such as the normal distribution, to evaluate the goodness of fit.

  These tests of normality are critical checkpoints that validate the use of parametric methods for further statistical analysis. If the assumption of normality is violated, non-parametric alternatives may be considered.

- **Step 4: Interpretation of Results**

  The tests revealed that the data largely met the requirements for homogeneity and normality. Where discrepancies were found, non-parametric tests were considered to ensure the robustness of statistical analysis. In cases where normality was not achieved, data transformation techniques were successfully employed to correct skewness and kurtosis towards normality.

- **Step 5: Statistical Analysis Undertaken**
  Appropriate statistical tests were selected and applied with the data now verified for homogeneity and normality. The analyses were congruent with the nature of the data, with parametric tests used where the assumptions of normality and homogeneity were met and non-parametric tests adopted in their absence.

4.5.1 *Descriptive Statistics*

Descriptive statistics play a vital role in data analysis by providing a detailed summary and interpretation of datasets. They present quantitative summaries and visualisations, such as mean, median, mode, standard deviation, and range, which illustrate central tendencies, variations, and frequency distributions without concluding the data itself.

In both Study 1 and Study 2 of the VR experiments, descriptive statistics were used to gain a comprehensive overview of the collected data. This approach facilitated the identification of patterns, trends, and potential outliers, which are essential for making informed decisions. Using precise measures and visual representations ensured the data was presented clearly and effectively, making it more accessible and understandable for stakeholders.

Physiological signals were analysed separately for each task role group, categorised as Leader and Follower, providing insights into the physiological differences and interactions between these roles. Additionally, the Mean Opinion Scores (MOS) from the questionnaires were calculated to summarise participants' subjective experiences. This combined analysis offered a deeper understanding of both physiological responses and participant perceptions, highlighting the dynamics between different roles and overall experiences in the VR environment.

4.5.2 *Inferential Statistics*

Inferential statistics were utilised to draw conclusions about a larger population based on data collected from physiological signals and questionnaires. In Study 1 and Study 2, individual physiological signals were analysed to understand patterns and trends, while questionnaires were used to evaluate participants' experiences and perceptions. This approach is critical for testing hypotheses, estimating population characteristics, and predicting outcomes, helping to determine whether observed differences are meaningful or due to random variation.

4.5.2.1 Hypothesis Testing, Confidence Intervals and p-value:

Hypothesis Testing: This method was used to test whether specific effects or differences exist. It starts with a null hypothesis (H0) that assumes no effect and an alternative hypothesis (H1) that suggests an effect. A test statistic was calculated from the physiological data and questionnaire responses to determine support for H0 or H1. If the p-value is less than 0.05, H0 is rejected, indicating a significant effect.

- p-Value: This indicates the probability of obtaining the observed result by chance. A p-value ≤ 0.05 suggests strong evidence against H0, leading to its rejection.
- Confidence Intervals (CIs): These intervals provide a range in which the true population parameter is likely to fall, typically at a 95% confidence level. They offer insights into the precision and reliability of the estimates derived from both physiological signals and questionnaire data.

Combining these methods allowed us to determine whether the observed effects in the physiological signals and questionnaire responses were statistically significant and where the true effect size likely lies.

4.5.2.2 Parametric and Non-Parametric Tests:

- Parametric Tests: Analysed physiological data and questionnaire responses when assumptions such as normal distribution and adequate sample size were met. Examples include t-tests (for comparing two means) and ANOVA (for comparing means across multiple groups).
- Non-Parametric Tests: Applied when the data did not meet the assumptions required for parametric tests, such as when physiological signals were not normally distributed or when the sample sizes were small. Examples include the Mann-Whitney U Test (for comparing two independent groups) and the Wilcoxon Signed-Rank Test (for comparing two related samples).

These statistical methods were selected based on the nature and distribution of the data collected from physiological signals and questionnaire responses.

4.5.3 *Correlation and Regression Analysis*

- Pearson and Spearman correlations were used to examine the relationships between continuous physiological variables, such as heart rate and task completion time, and questionnaire responses. This analysis helped identify patterns and trends in both physiological and subjective data.

- Linear regression was employed to evaluate how performance was influenced by multiple factors, such as physiological measures (e.g., EDA) and questionnaire responses (e.g., perceived task difficulty). This analysis provided valuable insights into the factors significantly impacting collaborative efficiency in VR environments.

4.5.3.1 Qualitative Analysis

Open-ended questionnaire responses were thematically analysed to understand participants' experiences and perceptions during collaborative VR tasks. This qualitative approach complemented the quantitative analysis of physiological signals, providing a comprehensive view of how participants interacted and performed in the VR environments. Common themes and patterns were identified, offering a nuanced understanding of the participant's experiences.

4.6 Summary

This chapter provides a comprehensive examination of the methodological and ethical framework essential for conducting two pivotal studies focused on understanding the QoE in collaborative VR environments. Through a structured and rigorous approach, including detailed experimental design, participant selection, and data collection, the integrity and reproducibility of the research findings are maintained.

The development and implementation of a robust ethical framework ensures participant well-being and data integrity throughout the research process, addressing the unique challenges posed by VR's immersive nature. This framework includes obtaining ethical approval, securing informed consent, ensuring confidentiality, mitigating potential risks, and facilitating participant debriefing.

The chapter adopts a mixed-methods approach, integrating quantitative and qualitative analyses to capture the multifaceted nature of QoE in collaborative VR. This approach is crucial for evaluating explicit feedback, implicit physiological responses, and objective performance metrics, providing a nuanced understanding of collaborative interactions in VR.

Statistical analyses, including descriptive and inferential techniques, are employed to extract meaningful insights from the data. Preliminary tests for homogeneity and normality, followed by appropriate parametric or non-parametric tests and advanced correlation and regression analyses, allow for a thorough exploration of the data. This analysis framework enhances understanding of the underlying dynamics of QoE in VR environments.

The structured experimental phases ensure consistency and reliability in data collection, enabling nuanced comparisons between the two studies. The comprehensive methodological and ethical framework articulated in this chapter not only strengthens the credibility of the research findings but also serves as a solid foundation for future studies in VR research.

Subsequent chapters will explore the specific methodologies and findings of each study in detail, building on the foundational principles established here. This chapter underscores the importance of methodological rigor and ethical vigilance in advancing the understanding of collaborative VR and sets the stage for deeper exploration of the insights and implications derived from the research.

Chapter 5: EXPERIMENTAL STUDY 1

This chapter evaluates the user's QoE and task performance during a collaborative task conducted in VR. This QoE evaluation is hereafter referred to as Experimental Study 1. The study investigates the impact of distinct roles Leader and Follower, on collaboration and task outcomes. Physiological synchrony between pairs was computed using three key signals: EDA, PD, and HR, exploring their relationship with subjective and objective metrics. The chapter begins with the research motivation, methodology, and experimental design, concluding with key findings that provide insights for optimising VR in collaborative tasks and enhancing user experience.

5.1 Scope and Purpose of Experimental Study 1

The integration of VR technology into smart manufacturing industries has the potential to revolutionise remote collaboration by providing immersive environments that facilitate complex design and production processes [137]. Traditional communication tools, such as videoconferencing and shared digital platforms, often fail to meet the intricate demands of collaborative design tasks due to their limitations in spatial interaction and real-time manipulation of design elements [138], [139], [140]. This inadequacy poses a significant challenge for geographically dispersed teams that require seamless coordination to achieve high efficiency and precision in manufacturing projects.

This experimental study addresses the critical need to enhance remote collaboration by evaluating the QoE and user performance in a collaborative VR setting. It investigates how distinct roles ─Leader and Follower─ impact cognitive workload, emotional engagement, and task execution, focusing on how physiological signals vary with the roles assigned. Physiological synchrony is computed using EDA, PD and HR signals between pairs of participants and how it influences subjective and objective metrics. Additionally, the study investigates the role of verbal communication patterns in shaping collaborative efficiency and user experience.

Despite the growing use of VR in training and simulation, its application in supporting multi-user collaboration remains very limited. Previous research has primarily focused on individual interactions with VR systems, leaving a gap in understanding how VR can effectively support complex, collaborative tasks that mirror real-world manufacturing challenges [6], [141]. Furthermore, while some studies have highlighted the benefits of physiological synchrony in enhancing team performance, the extent to which these findings apply in VR environments, particularly for manufacturing tasks, is not well understood [142], [143], [144].

To address these gaps, this study investigates how collaboration roles, communication patterns, and physiological synchrony influence user performance and QoE in collaborative VR settings. Precisely, the research questions explored in this study are aligned with the overarching sub-research questions (SRQs) defined in Chapter 1 and are organised as follows:

Related to SRQ1: How do task roles, physiological responses, and perceived workload influence collaboration in VR?

- How do different task roles (Leader vs. Follower) affect subjective workload and user satisfaction during collaborative VR tasks?
- What is the relationship between physiological responses (e.g., EDA, HR) and task performance across different roles?

Related to SRQ2: How do collaboration factors affect team performance and physiological synchrony in collaborative VR?

- How does verbal communication sentiment (positive or negative tone) correlate with task efficiency and user experience?

Related to SRQ3: How does physiological synchrony influence collaboration quality and efficiency in VR?

- How does physiological synchrony (EDA, HR, and pupil dilation) between participants influence perceived workload and task performance?

By addressing these targeted research questions, this study aims to deepen our understanding of how immersive VR environments can support complex, real-world collaborative manufacturing scenarios. The findings are expected to inform the design of VR-based collaboration tools that improve productivity, engagement, and coordination for distributed teams in industrial settings.

5.2 Research Methodology

This section outlines the research methodology employed in Experimental Study 1. It details the experimental design, participant selection, task procedures, and data collection methods to address the research questions. A multimodal dataset comprising subjective assessments, physiological measurements, vocal sentiments, and task performance metrics was collected. All physiological signals were analysed to identify differences between the Leader and Follower groups, while physiological synchrony was computed using EDA, PD, and HR. This multidimensional approach comprehensively evaluates cognitive, emotional, and collaborative experiences in VR settings.

5.2.1 *Experimental Design*

The study adopted a controlled experimental design within a VR environment, where participants were assigned distinct roles—Leader and Follower—to complete a collaborative task. The task involved a 'pick and place' activity using a Soma cube puzzle [145], requiring participants to work together to construct a prototype by picking up, moving, and arranging objects. This design simulated the coordination and precision necessary in real-world rooms, providing a realistic context for evaluating collaborative dynamics as shown in Figure 5-1.

5.2.2 *Participants*

22 participants were recruited for this study, with an average age of 35.86 years (SD = 12.56). The group consisted of an equal number of male and female participants, ensuring balanced representation. Most participants (68%) had prior experience with VR platforms, and nearly all (95%) were familiar with video conferencing and team-based activities, ensuring they were comfortable navigating the collaborative VR environment. All participants were screened for colour vision deficiencies using the Ishihara Test, and none reported any form of colour blindness, which was essential for accurately performing the colour-dependent task [146]. However, physiological data for three participants were not recorded due to a malfunction with one of the wristbands used for data collection. Given the paired nature of our study, this

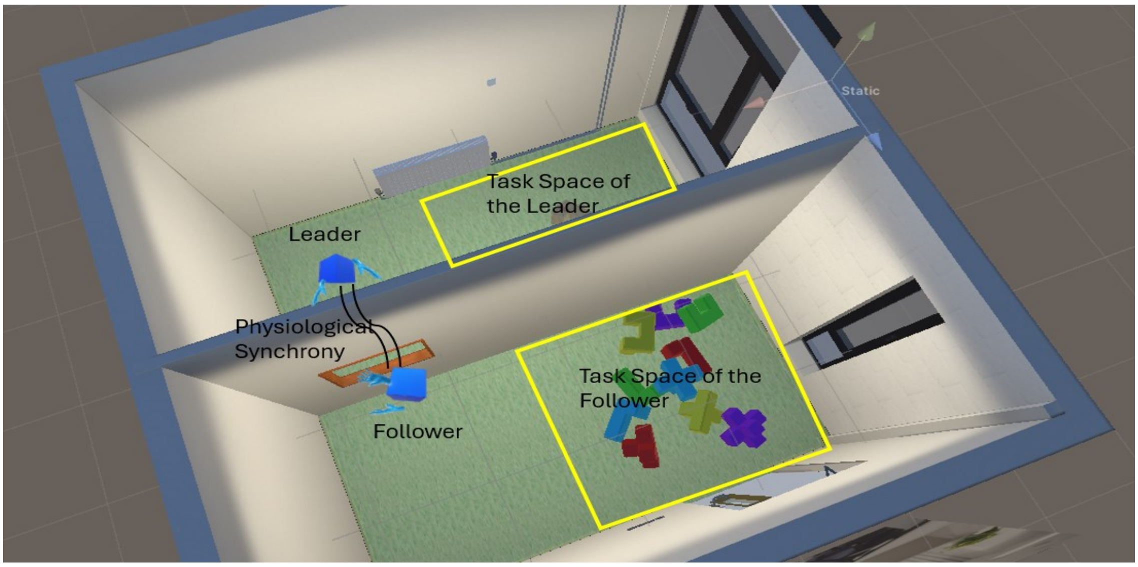


Figure 5-1*: Figure showing the views of the Leader, Follower, with the Puzzle pick and place task, and Interaction mechanism through the virtual window and Virtual avatar representation of users*

necessitated the exclusion of their corresponding partners, reducing our dataset to 8 complete pairs (16 participants) for analysis.

5.2.3 *Task Description*

The collaborative VR task involved participants working together to assemble a Soma cube puzzle using various coloured and shaped objects. The task required the construction of two Soma cubes stacked on top of each other to form a single cuboid structure as depicted in Figure 5-2. The Leader provided detailed instructions to the Follower, who then collected the specific pieces in a particular order as instructed. Each block was uniquely identified by colour and 3D shape, and the Leader had to accurately describe each piece to ensure the Follower selected the correct item. The task sequence involved picking up and placing ten blocks in a specific order,

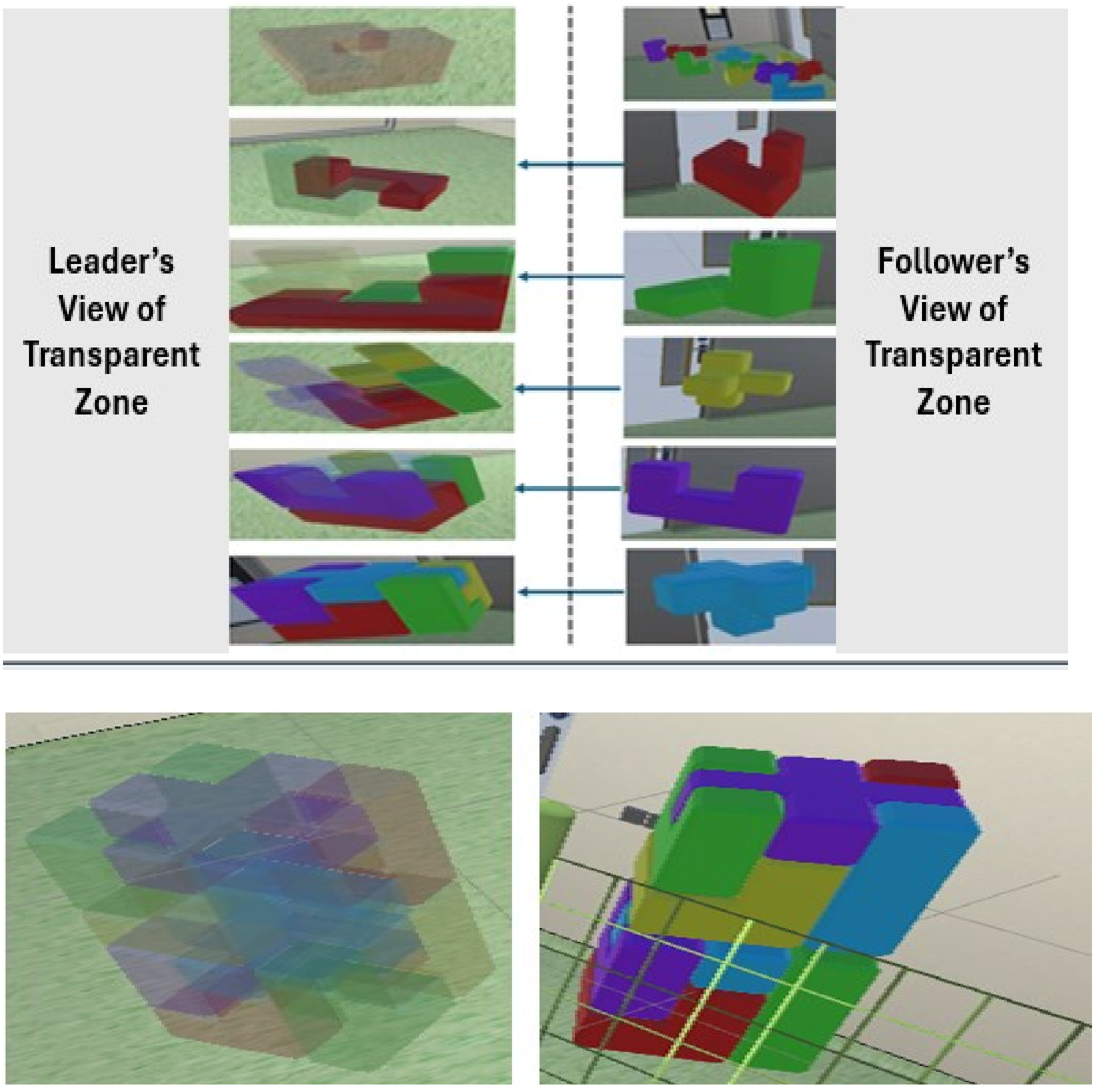


Figure 5-2*: Collaborative Task description with Leader and Follower's View in their respective Virtual Room*

with the Follower passing the selected blocks through a virtual window to the Leader's designated assembly space. This setup emphasises the need for effective communication and coordination between the two roles to achieve communication and coordination between the two roles to achieve successful task completion.

Participants were confined to separate virtual rooms, with lighting conditions kept constant to ensure uniform visibility and minimize visual distractions. Movement within the VR environment was facilitated by a teleportation feature, allowing participants to navigate within their virtual spaces seamlessly without physical movement as shown in Figure 5-3. Interaction with the virtual objects was enabled using hand controllers. The hand controllers allowed participants to grab, move, and place the virtual objects using designated buttons and controls, ensuring precise and accurate manipulation throughout the task. The users were represented with blue cube heads and virtual hands as shown in the figure.

The VR application leverages Photon Unity Networking (PUN) and Photon Voice to facilitate robust sound communication between participants, ensuring minimal network latency and maximised real-time interactivity using the University network [108], [110]. This structured and immersive setup provided a robust platform to simulate the intricate collaboration and coordination required in real-world manufacturing processes, making the task highly relevant for studying the dynamics of remote teamwork in smart manufacturing contexts.

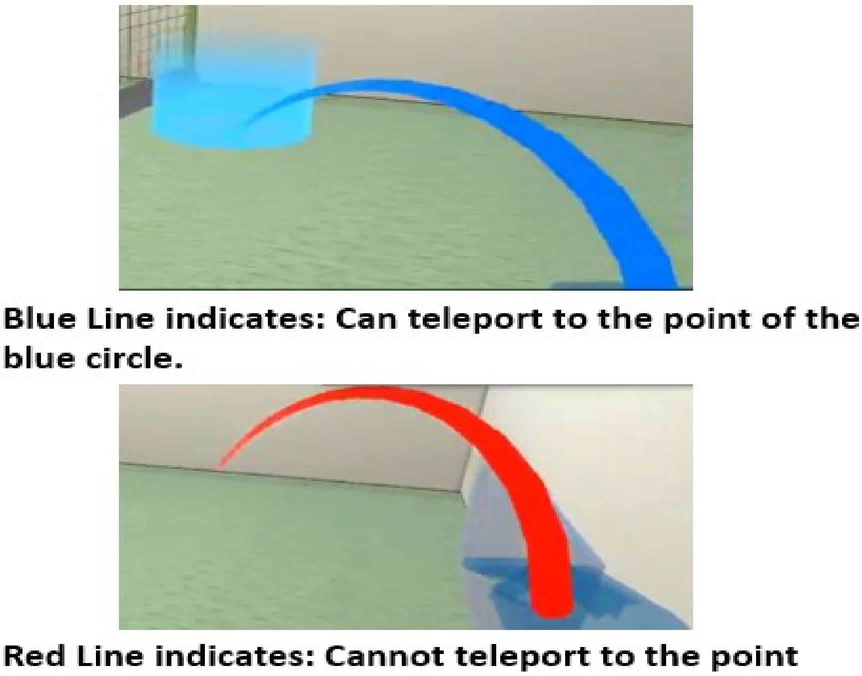


Figure 5-3: *Teleportation using hand controller click in VR*

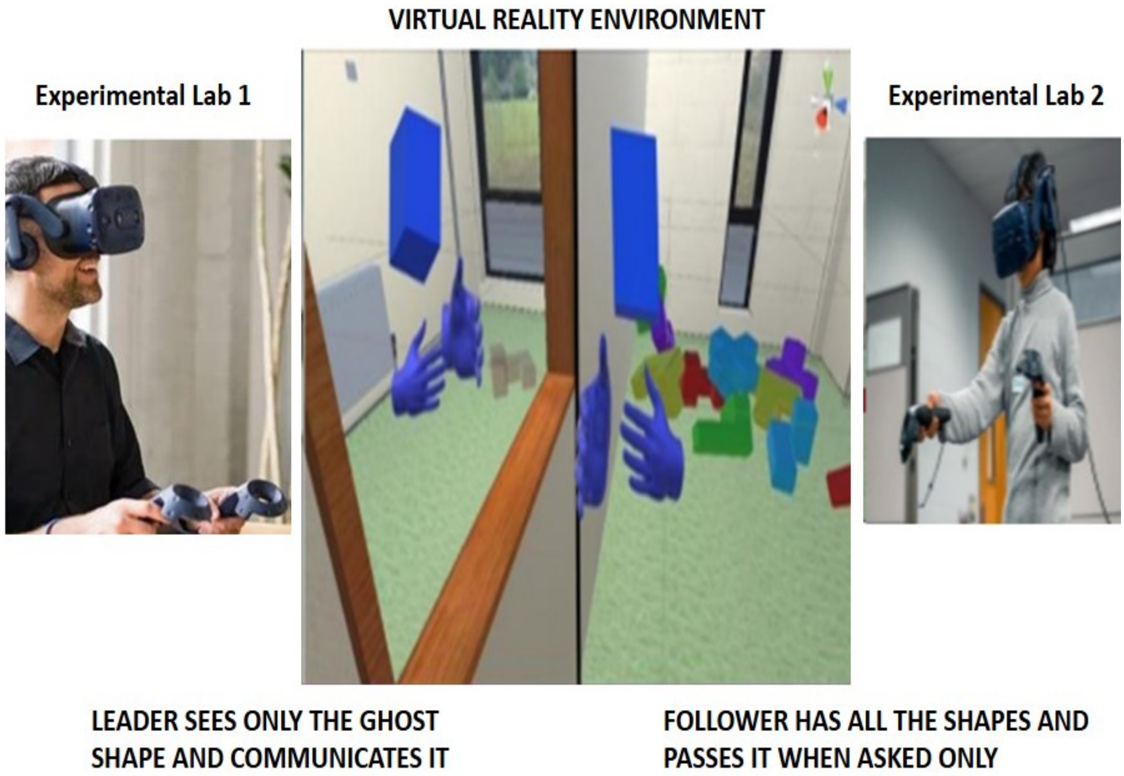


Figure 5-4: *Two participants (Leader in the left virtual room and Follower in the right virtual room with an opening for passing objects) join the VR environment wearing HTC Vive Pro Eye and two hand controllers*

5.2.4 *Equipment Specification*

The system used for the experiment consisted of a VR setup with HTC Vive Pro Eye [103] HMD connected to high-performance Windows PCs. The PCs were identical for each user. They had an Intel Core i7- 8700 processor, an NVIDIA GeForce RTX 2080 graphics card, and 32 GB of Memory RAM. The communication between the two workstations controlling the headsets was facilitated through Photon Unity Networking (PUN) [38] and Photon Voice, ensuring real-time communication over the network as shown in Fig. 2. The tracking space for the two users was set up using SteamVR (v020) [114]. It provided a playable region of (4.0×1.5m^2), allowing for sufficient physical movement and interaction in the virtual space as shown in Figure 5-4. The virtual environment was implemented in Unity (version 2019.3.2f1) [112] with a high-resolution display of 1440×1600 pixels per eye, operating at 120 Hz. The virtual environment was designed to resemble real-life settings, with ceiling lights, box-shaped objects, tables, heaters, and doors. When required, 3D models for objects were created using Blender [124].

5.2.5 *Phases of the Experiment*

The data collection structure of these phases was informed by ITU-T P.913, P.919 [147], [148] and research in QoE of immersive experiences [127], [128]. The experiment was structured into six distinct phases to ensure a systematic and consistent approach:

- **Phase 1: Introduction and Consent**
  Participants were briefed on the study's objectives, procedures, and potential risks. They provided informed consent and completed a demographic questionnaire, including their experience with VR and collaborative tasks.
- **Phase 2: Vision Screening**
  Participants underwent a vision screening using the Ishihara Test to confirm normal colour vision, which was necessary for performing the task accurately.
- **Phase 3: VR Familiarization**
  Participants were introduced to the VR equipment, including the HTC Vive Pro Eye HMDs and hand controllers, followed by a video. They engaged in a brief individual training session to practice basic interactions such as grabbing, moving, and placing virtual objects and navigating the VR environment using the teleportation feature.
- **Phase 4: Baseline Physiological Measurement**
  Participants were fitted with the E4 wristband for EDA, HR, IBI, BVP, and Skin Temperature data collection. They were requested to remain still for five minutes to establish baseline physiological readings.

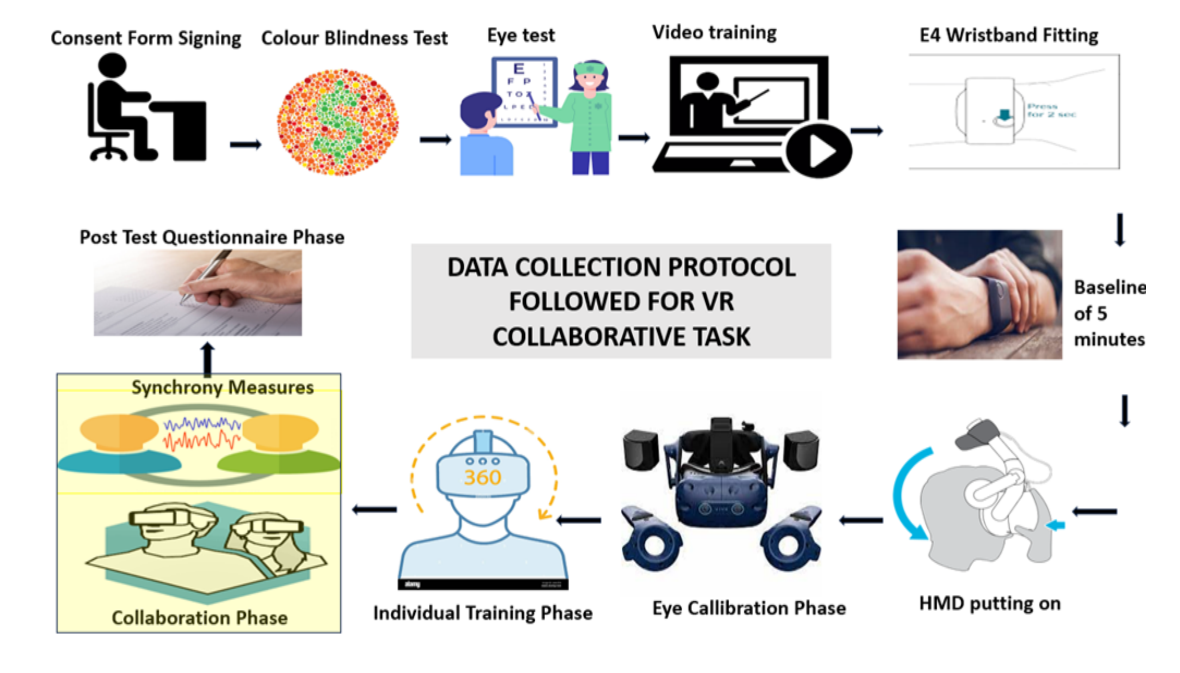


***Figure 5-5**: Experimental Phases of Study 1*

- **Phase 5 Collaborative Task Execution:**
  Participants performed the collaborative 'pick and place' task in their designated roles as Leader and Follower. The Follower selected and passed blocks through the virtual window based on the Leader’s instructions. The Leader then used these blocks to build the cuboid structure in their virtual room. Task performance, physiological responses, and verbal communication were recorded throughout this phase. The average task duration was six to seven minutes.
- **Phase 6: Post-Task Assessment**:
  Immediately after completing the task, participants filled out customized questionnaires and the NASA-TLX to assess their perceived workload, satisfaction, and overall experience.

Figure 5-5 shows the entire process of experimental study 1, starting from consent signing to the collaboration phase. Based on structured experimental procedures and task durations, the total time for the complete experiment per participant pair was approximately 53 to 59 minutes, including briefing, setup, task execution, and post-task assessments.

5.3 Multimodal Data Set

This section presents the Multimodal Data Set collected in Experimental Study 1 to evaluate participants' QoE, cognitive workload, emotional engagement, and task performance during a collaborative VR task. The data was gathered through individual subjective assessments, individual physiological measurements, pair-wise commonality factors, pair-wise recorded audio conversations for sentiment analysis, and pair-wise physiological synchrony computed from each participant’s physiological signals. Additionally, task performance metrics were collected. This approach enabled a comprehensive evaluation of collaborative dynamics, emphasising the impact of role-specific differences, physiological alignment, communication patterns, and task performance on collaboration quality.

5.3.1 *Individual Subjective Assessments*

Individual subjective assessments were conducted immediately after the collaborative VR task to capture participants' perceptions of their experiences, cognitive workload, emotional engagement, and task satisfaction. Two validated tools were used: a Customised Questionnaire and the NASA-TLX [149].

- The NASA-TLX is a widely recognised and standardised instrument to assess perceived cognitive and physical workload. This tool evaluates participants' workload and their overall task experience through six distinct subscales:
    - **Mental Demand**: This subscale measures the mental and perceptual activities required to perform the task.
    - **Physical Demand**: This aspect assesses the physical effort needed to complete the task.
    - **Temporal Demand**: This dimension evaluates the time pressure experienced by participants while engaged in the task.
    - **Performance**: This subscale captures participants' self-perceived success in achieving the task objectives.
    - **Effort**: This component assesses the total mental and physical exertion required to perform the task.
    - **Frustration**: This dimension measures participants' feelings of annoyance, irritation, or stress encountered during the task.

Each subscale is rated on a scale ranging from 0 to 20, enabling a comprehensive measure of the overall workload experienced by participants. The NASA-TLX was selected for this study due to its robust capacity to capture cognitive and physical dimensions of task load, making it particularly suitable for evaluating collaborative VR experiences. For a complete overview of the questionnaire, the full NASA-TLX scale utilised in this study is provided in **Appendix E**

- A customised questionnaire was developed to evaluate the Quality of Experience (QoE) in Collaborative Virtual Environments (CVEs), focusing on five key dimensions: Immersion, Interaction, Collaboration, System Effects, and Post-Usage Acceptability. Participants independently rated their experience using a 5-point Likert scale ranging from 1 (Not at all) to 5 (Very much), enabling detailed analysis of individual perceptions regarding the VR environment, usability, and collaborative quality. The complete version of the questionnaire is included in **Appendix D**, with Table IV summarising the individual items.

The questionnaire design is grounded in validated instruments commonly used in immersive VR research. Items related to immersion (Q1–Q3) were adapted from the Presence Questionnaire, which is widely used to assess the realism, consistency, and sense of being present in virtual environments [150]. Interaction-related items (Q4–Q6), which emphasise usability and control, draw from established usability frameworks like the System Usability Scale (SUS) and prior VR interaction studies [151]. Collaboration-specific questions (Q7–Q8)

Table IV: Customised Questionnaire

QNo	Questions followed by Maximum → Minimum Ratings	Category
Q1	How natural did your interactions with the environment seem? Very natural 5 -------------→ 1 Not at all	Immersion
Q2	How much did your experiences in the virtual environment seem consistent with your real-world experiences? Very consistent 5 -------------→1 Not Consistent	
Q3	How involved were you in the virtual environment? Very Involved 5 -------------→1 Not involved	
Q4	How much did the control devices assisted with the performance of assigned tasks or other activities? Very much 5 -------------→ 1 Not at all	Interaction
Q5	How well could you move or manipulate objects in the virtual environment? Very well 5 -------------→ 1 Not at all	
Q6	How closely were you able to examine objects? Very close 5 -------------→ 1 Not at all	
Q7	How well could you identify and hear sounds of another user? Very clear 5 -------------→ 1 Not at all	Collaboration
Q8	How well could you see the other person's hand manipulating objects in the virtual environment? Very well 5 -------------→ 1 Not at all	
Q9	How much delay did you experience between your actions and expected outcome? Not at all 5 -------------→ 1 Very Much	
Q10	The collaborating task was difficult to complete wearing a Head mounted display Not at all 5 -------------→ 1 Very Difficult	System Effects
Q11	I was restricted in my movements using the system, and therefore could not participate in the task Not at all 5 -------------→ 1 Very restricted	
Q12	Did you feel any motion sickness or dizziness at the end of the experiment? Not at all 5 -------------→ 1 Very much	
Q13	I felt qualified enough in interacting with the virtual environment at the end of the experience Very well 5 -------------→ 1 Not at all	Post-usage
Q14	How proficient in moving and interacting with the virtual environment did you feel at the end of the experience. Very Proficient 5 -------------→ 1 Not at all	
Q15	Did you become so involved in the task that it is as if you were inside the game rather than moving a controller or watching a screen? Very much involved 5 -------------→ 1 Not at all	

are based on research exploring shared task awareness, role clarity, and communication in multi-user VR environments [152]. Items focused on system-related issues such as latency, movement constraints, and motion sickness (Q9–Q12) are influenced by the Simulator Sickness Questionnaire (SSQ), a standardised measure of physical discomfort in VR environments [153]. Post-usage acceptability questions (Q13–Q14), which evaluate users' confidence and perceived task proficiency, are rooted in research on user perception and acceptance in immersive learning and collaboration environments [154]. The final item (Q15), which captures engagement and psychological presence, was inspired by flow theory applied to virtual collaboration contexts [155].

5.3.2 *Individual Physiological Measurements*

Various physiological metrics were recorded using wearable sensors to complement the subjective assessments and provide objective data on participants' cognitive and emotional engagement during the collaborative task. These measurements allowed for an in-depth analysis of roles (Leader vs. Follower) within the VR environment. The raw data for these physiological signals are detailed in Appendices F and G, where they are segmented into three distinct sessions of the experiment: the Baseline Phase, the Individual Training Phase, and the Collaboration Phase.

- PD was recorded using eye-tracking technology embedded in the HTC Vive Pro Eye headsets, capturing individual cognitive load and visual attention during task execution. Detailed raw data on pupil dilation across all three phases can be found in **Appendix F**.
- EDA was measured to assess stress levels and emotional arousal, offering valuable insights into each participant's cognitive engagement and emotional demands**.**
- HR and IBI: HR was monitored to evaluate cardiovascular responses linked to cognitive load and stress.
- ST was measured to provide additional context for physiological arousal and stress. Minimal differences were observed between roles; however, some fluctuations in skin temperature indicated moments of heightened arousal or discomfort, particularly during complex or error-prone task segments.
- BVP was measured using the Empatica E4 wristband, which utilises a PPG sensor to detect changes in blood volume by emitting light into the skin and measuring the reflected light. These fluctuations correspond to cardiovascular activity, enabling accurate HR and HRV measurement. This study used BVP to assess mental and physical workload, emotional arousal, and physiological synchrony between participants during collaborative VR tasks. The E4 wristband recorded BVP at 64 Hz, and the data was processed to extract HR and HRV metrics, providing insights into participants' cognitive load, engagement, and social interaction quality.
- ACC data were collected from both participants to capture physical movements within the VR environment. This metric was beneficial for analysing movements within the task.

Except for PD, all other signals were collected using the Empatica E4 wristband. When analysed alongside the subjective questionnaire data, these physiological measures provided a comprehensive view of participants' experiences during the collaborative VR task. The differences between the Leader and Follower groups' physiological responses will show how task responsibilities influenced cognitive load, stress levels, and physical activity. Details of all the physiological signals (EDA, HR, IBI, ST, BVP, and ACC) segregated by session available can be found in Appendix G.

5.3.3 *Pair-wise Recorded Audio Conversation for Sentiment Analysis*

Pairwise recorded audio conversations were analysed to capture the emotional tone and communication dynamics between the Leader and Follower during the collaborative task. This approach allowed an in-depth exploration of emotional states, communication patterns, and collaborative efficiency. All verbal interactions were recorded and transcribed for each pair. The transcriptions were then analysed using AssemblyAI's LeMUR to classify the spoken content into positive, negative, or neutral sentiments. This classification enabled a detailed examination of emotional dynamics within pairs and provided insights into how sentiment patterns influenced task performance and collaborative efficiency.

5.3.4 *Pair-Wise Commonality Metrics from Pre-filled Consent Form*

Before starting the experiment, participants completed a Pre-Filled Consent Form (Appendix B) that collected demographic and experiential data, including Gender, Age, Educational Background, Social Status, Gamer Status, VR/AR Experience, Video Conferencing Tool Usage, and Place of Origin. These data points were used to map Pairwise Commonality Metrics such as Prior VR Experience, Technical Expertise, Demographics, and Social Relationships. These metrics were essential for analysing pairwise physiological synchrony and collaborative dynamics, particularly in exploring the influence of mutual commonality on task performance and collaboration quality.

5.3.5 *Synchrony Metrics*

Synchrony metrics or Physiological synchrony were computed using EDA, Pupil Dilation, and HR to assess the alignment of physiological responses between the Leader and Follower in each pair. This alignment provided insights into collaborative engagement, cognitive alignment, and emotional synchrony.

- EDA Synchrony: EDA Synchrony was measured using Pearson Correlation Coefficient (PCC) and Dynamic Time Warping (DTW) to capture emotional alignment between the Leader and Follower [149], [156].
- HR Synchrony: HR synchrony was evaluated using DTW to assess temporal variations in cardiovascular responses, reflecting shared cognitive and emotional states during the collaborative task [156].
- Pupil Dilation Synchrony: Pupil Dilation Synchrony was analysed using R-squared (R^2) to measure cognitive load alignment and shared visual attention between the pair participants [157].

These synchrony metrics provided insights into the collaborative dynamics between participants, highlighting how physiological alignment contributes to successful teamwork and task outcomes in VR environments.

5.3.6 *Task Performance Metrics*

To evaluate collaborative efficiency and task complexity, two key task performance metrics were measured:

- **Task Completion Time**: The total time taken by each pair to complete the task, reflecting collaborative efficiency and communication effectiveness.
- **Number of Controller Clicks**: The total hand controller clicks during the task, providing insights into task complexity, interaction effort, and cognitive workload.

These task performance metrics were correlated with subjective assessments, physiological measurements, and voice sentiment analysis to comprehensively understand cognitive workload, emotional engagement, and collaborative efficiency in the VR environment.

5.4 Results

This section presents and discusses the findings from Experimental Study 1. The results are organised into key categories: subjective assessments, physiological measurements, voice sentiment analysis, and synchrony metrics. They provide a holistic view of how the Leader and Follower roles differ in their experiences and performance. Each subsection provides a detailed analysis and interpretation of the data, supported by relevant tables and statistical findings, to offer insights into the effectiveness of VR for facilitating remote collaboration and the unique

Table V: Statistical Analysis of Self-reported Measures with 95% CI

Type	User Group	N	Mean of Scores	Standard Deviation	Mean Rank	Sum of Rank	U-value	W	P value
Immersion	Leader	11	4.51	0.54	12.36	136	51.0	117	0.521
	Follower	11	4.39	0.51	10.64	117			
Interaction	Leader	11	4.54	052	10.91	120	54	120	0.642
	Follower	11	4.66	0.39	12.09	133			
Collaboration	Leader	11	4.48	0.83	11.18	123	57	123	0.804
	Follower	11	4.75	0.26	11.82	130			
System Effects	Leader	11	4.45	0.85	13.05	143	43.5	109.5	0.245
	Follower	11	4.42	0.39	9.95	109			
Post-Usage	Leader	11	4.66	0.36	11.27	124	58	124	0.861
	Follower	11	4.69	0.34	11.73	129			

challenges faced by different roles. The discussion highlights these findings' implications for designing and optimising VR systems for smart manufacturing applications.

5.4.1 *Subjective Questionnaire Results*

The subjective assessment results were gathered using two primary tools: the QoE and the NASA-TLX Questionnaires. This section presents the analyses on experiences and workload during the collaborative VR task. The two questionnaires were given to the participants at the end of the collaboration VR experience.

5.4.1.1 Customised QoE Questionnaire

A Shapiro-Wilk test was performed to assess the normality of the subjective questionnaire data, revealing non-normal distributions for all dimensions ($p < 0.05$). Consequently, the non-parametric Mann-Whitney U test was employed to compare the responses of Leaders and Followers across five dimensions: Immersion, Interaction, Collaboration, System Effects, and Post-Usage Acceptability, with a 95% confidence interval (CI) for the mean differences. **Appendix D** provides the complete customised questionnaire to collect these subjective QoE ratings.

Table V presents the results of the self-reported measures, including Mean Scores, Standard Deviation, Mean Ranks, and P-values for each QoE dimension. The results for each dimension are summarised below.

- **Immersion**: Leaders reported slightly higher immersion scores (Mean = 4.51, SD = 0.54) compared to Followers (Mean = 4.39, SD = 0.51). However, the difference was not statistically significant (U = 51.0, p = 0.521), with a mean difference of 0.12 and a 95% CI of [-0.22, 0.46], indicating substantial overlap in the distribution of scores between the two roles. This suggests that both roles experienced similar levels of immersion in the VR environment, a common finding in studies where participants engage in shared virtual spaces. Research has shown that immersion is often linked to a user's sense of presence in VR, which is typically similar across different roles in collaborative settings [158].
- **Interaction**: Followers (Mean = 4.66, SD = 0.39) reported marginally higher interaction scores than Leaders (Mean = 4.54, SD = 0.52), but the difference was not significant (U = 54, p = 0.642). The mean difference was -0.12 with a 95% CI of [-0.46, 0.22], suggesting that both roles perceived similar interaction experiences in the VR environment. This aligns with findings from prior work suggesting that VR can enhance interaction quality compared to traditional 2D interfaces or desktop-based collaboration tools, due to its immersive nature, which fosters more intuitive engagement and spatial presence among participants [159].
- **Collaboration**: Followers had higher scores (Mean = 4.75, SD = 0.26) compared to Leaders (Mean = 4.48, SD = 0.83), though the difference was not significant (p = 0.804). This indicates a positive perception of collaborative efficiency in both groups. The mean difference of -0.27 with a 95% CI of [-0.87, 0.33] shows considerable overlap in perceived collaboration quality between the roles. This finding reflects previous studies that suggest VR can foster collaborative efficiency in group tasks, with participants often reporting high levels of satisfaction with collaboration features [160].
- **System Effects**: Leaders reported slightly more discomfort or system limitations (Mean = 4.45, SD = 0.85) than Followers (Mean = 4.42, SD = 0.39). However, this difference was not statistically significant (p = 0.245) with a 95% CI of [-0.49, 0.55]. Research shows that users sometimes report minor discomfort in VR environments due to system issues such as latency or visual disturbances, but these effects tend to be minimal and are often not role-dependent [161].

- **Post-Usage Acceptability**: Both groups were satisfied with the VR system's usability, with Leaders scoring a mean of 4.66 (SD = 0.36) and Followers scoring 4.69 (SD = 0.34). The mean difference was -0.03 with a 95% CI of [-0.31, 0.25] and (p=0.0861), indicating similar post-usage acceptability perceptions and a high likelihood of future use [162].

Overall, these customised questionnaire results indicate no statistically significant differences between Leaders and Followers across the five dimensions, as all confidence intervals include zero, reflecting overlapping experiences and perceptions. These findings suggest that the VR environment provided a consistent and equitable collaborative expertise, regardless of the assigned role.

5.4.1.2 NASA-TLX Questionnaire

The NASA-TLX Questionnaire assessed perceived workload across six dimensions: Mental Load, Physical Load, Temporal Load, Performance, Effort, and Frustration. Table VI provides a detailed comparison of the NASA-TLX scores between Leaders and Followers for each workload dimension.

- **Mental Load**: Leaders reported significantly higher mental load (Mean = 53.82, SD = 5.03) compared to Followers (Mean = 41.5.36, SD = 6.27). Although the difference was notable, it was not statistically significant (p = 0.317). This suggests that Leaders experienced more cognitive strain due to giving instructions and manipulating objects. This aligns with a previous study which has found that role-specific task demands in collaborative environments can impact perceived mental load, with followers often reporting higher cognitive demands due to the nature of task execution [163].
- **Physical Load**: Followers also reported higher physical load (Mean = 53.4, SD = 6.58) than Leaders (Mean = 40.91, SD = 8.23). This is expected due to their more physically demanding role. The difference, however, was not statistically significant (p = 0.425). Physical workload variations often stem from differences in task roles and responsibilities, with followers typically experiencing higher physical demands when their tasks involve more hands-on activities [164]
- **Temporal Load**: Temporal load scores were similar between the groups, with Leaders scoring 36.2 (SD = 6.23) and Followers scoring 36.1 (SD = 7.46). The small difference was not significant (p = 0.379), indicating that both roles experienced comparable similar time pressure [165].

- **Performance**: Both groups reported high performance, with Leaders scoring 88.3 (SD = 5.98) and Followers scoring 87.3 (SD = 5.24). The slight difference was not significant (p = 0.681), suggesting that both roles felt effective in their tasks[166]. Performance is the only measure that indicates that a higher score is better in NASA-TLX as opposed to other parameters.
- **Effort**: Followers reported higher effort (Mean = 44.64, SD = 3.71) compared to Leaders (Mean = 33.27, SD = 5.55), reflecting the increased physical and cognitive demands of their role. The difference was not statistically significant (p = 0.154).
- **Frustration**: Frustration levels were similar between Leaders (Mean = 16.55, SD = 8.79) and Followers (Mean = 14.2, SD = 3.08), with no significant difference (p = 0.457). This suggests that both roles experienced similar levels of frustration during the task.

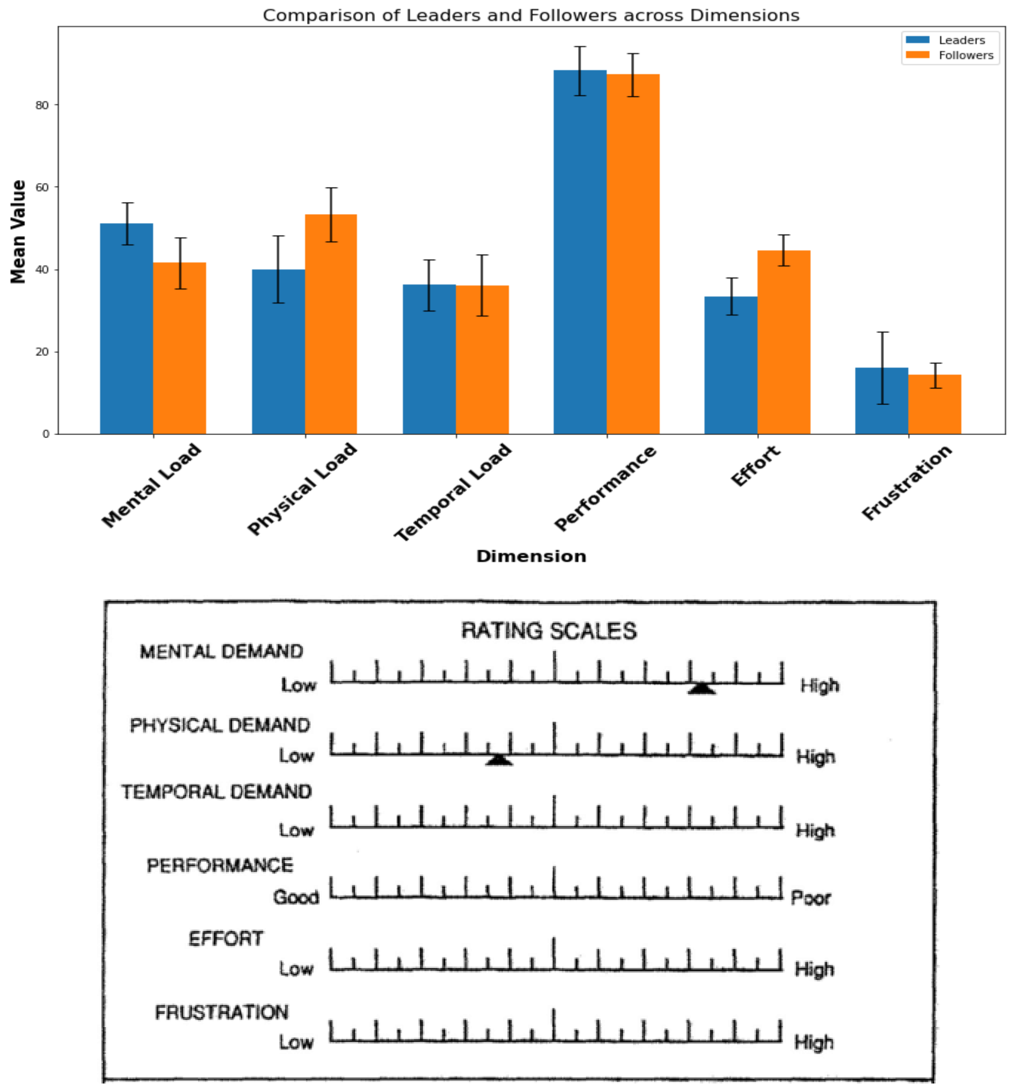


***Figure 5-6**: Group-wise comparison of NASA-TLX compared between Leader and Follower*

Table VI: NASA-TLX Results of 22 participants.

Group	Load type	Mean	Std-Dev
Leader	Mental Load	51.1	5.03
Follower		41.5	6.27
Leader	Physical Load	40	8.23
Follower		53.4	6.58
Leader	Temporal Load	36.2	6.23
Follower		36.1	7.46
Leader	Performance	88.3	5.98
Follower		87.3	5.24
Leader	Effort	33.4	5.55
Follower		44.6	3.71
Leader	Frustration	16	8.79
Follower		14.2	3.08

Overall, these results highlight the differences in perceived workload between the Leader and Follower roles. Followers experienced higher physical demands due to high interaction, while both groups reported high-performance levels and similar frustration levels, indicating practical task completion despite varied workload experiences.

The analysis of the Customised QoE Questionnaire and NASA-TLX indicates that Leaders and Followers experienced high QoE and effective collaboration within the VR environment. However, distinct role-specific differences were observed in workload experiences. Followers reported higher mental and physical load due to their active role in task execution, whereas Leaders experienced increased mental and temporal demands associated with strategic guidance and communication (see Figure 5-6). Despite these variations, both groups maintained high performance and low frustration levels, reflecting effective role distribution and task complexity. These subjective insights suggest mental and emotional demands are role-dependent, warranting further investigation into the underlying physiological mechanisms. The following section explores physiological differences between Leaders and Followers, thereby providing objective validation of the role-specific cognitive and emotional experiences reported in the questionnaires.

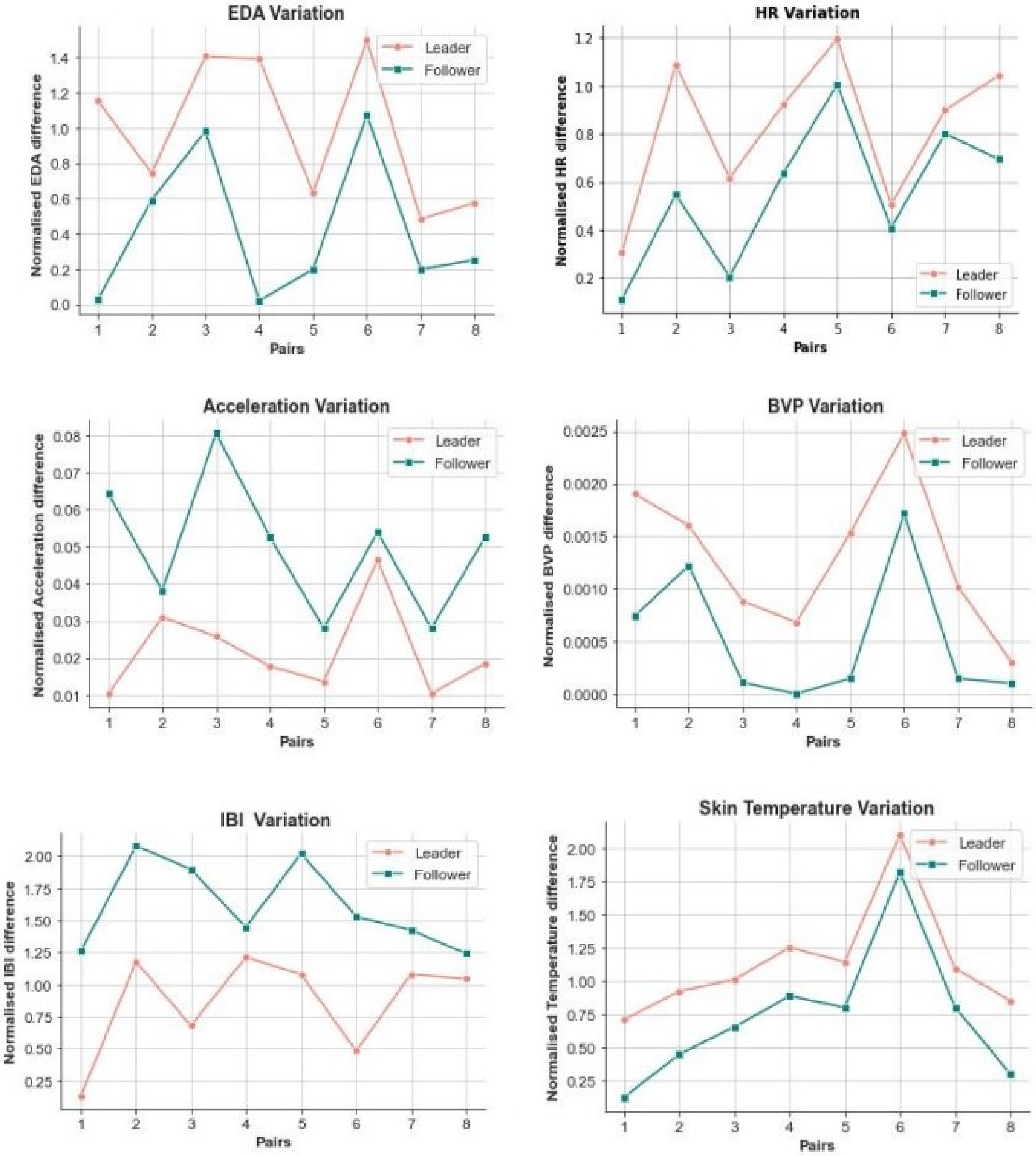


***Figure 5-7**: Variation between different bio signals obtained between Leader and Follower Group*

5.4.2 *Physiological Responses Results*

This section presents the physiological response data collected during the collaborative VR task, comparing the differences between the Leader and Follower roles. The data was standardised using Z-scores, and variations in EDA, HR, IBI, ACC, BVP, ST, and PD were analysed for each role. The raw data for each signal and the statistical descriptives of each signal is provided in **Appendix F** and **Appendix G**. The mean values for each metric, based on the graph data and their statistical significance, are summarised in Table VII.

- **EDA**: EDA is a reliable indicator of cognitive engagement and emotional arousal. Higher EDA values are associated with heightened mental load and stress responses [167], [168]. In this study, Leaders exhibited significantly higher EDA levels (Mean = 0.90) than Followers (Mean = 0.60), with a p-value of 0.020. This physiological finding aligns with the NASA-TLX results, where Leaders reported higher mental and temporal loads. The customised questionnaire further supported this by indicating that Leaders felt more involved and engaged in the VR task, reflecting the elevated cognitive demand experienced in their role.
- **HR**: Heart rate is commonly used to gauge cognitive load, with higher values reflecting increased cognitive demand [169]. Although Leaders consistently showed higher HR values (Mean = 0.85) compared to Followers (Mean = 0.55), the difference was not statistically significant (p = 0.130). This aligns with the customised questionnaire results, which show no significant difference in reported interaction and collaboration quality between Leaders and Followers, indicating comparable cognitive loads.
- **IBI**: IBI, derived from HRV, is an essential indicator of mental load and stress [152]. A lower IBI indicates increased cognitive load and stress. Leaders had significantly lower IBI values (Mean = 1.00) compared to Followers (Mean = 1.75), with a highly significant p-value of 0.000. This physiological measure is consistent with the NASA-TLX results, where Leaders reported higher frustration and effort levels. The IBI findings indicate that the Leaders experienced more significant mental and emotional strain, as corroborated by both the customised questionnaire and NASA-TLX results.
- **STEMP**: Skin temperature (STEMP) is influenced by the sympathetic nervous system and can indicate stress levels [170]. Higher skin temperatures are associated with increased emotional or cognitive stress. In this study, Leaders had significantly higher skin temperature variations (Mean = 1.75) compared to Followers (Mean = 1.50), with a p-value of 0.028. This finding aligns with both the NASA-TLX results, where Leaders reported higher physical and mental effort, and the customised questionnaire, where they indicated higher levels of involvement and stress during the task.

- **BVP**: BVP measures changes in blood volume within small blood vessels, reflecting physiological responses to stress and cognitive load [153]. Leaders exhibited higher BVP values (Mean = 0.001) compared to Followers (Mean = 0.0008), with a significant p-value of 0.037. This suggests that Leaders experienced more pronounced physiological stress, which is consistent with their higher mental and temporal load scores in the NASA-TLX and increased engagement reported in the customised questionnaire. Figure 5-7: Variation between different bio signals obtained between Leader and Follower Group shows the variation between Leader vs Follower for individual normalised EDA, HR, Temp, IBI, ACC, and BVP.
- **ACC**: Acceleration measures the velocity changes in three dimensions (x, y, and z) and provides insights into the physical movement and effort exerted during VR tasks [171]. Higher acceleration values indicate more significant physical movement. Followers showed higher ACC values (Mean = 0.0035) compared to Leaders (Mean = 0.0025), with a significant p-value of 0.002. This difference reflects the increased physical demands placed on Followers, which aligns with the customised questionnaire results where Followers reported higher perceived physical load.

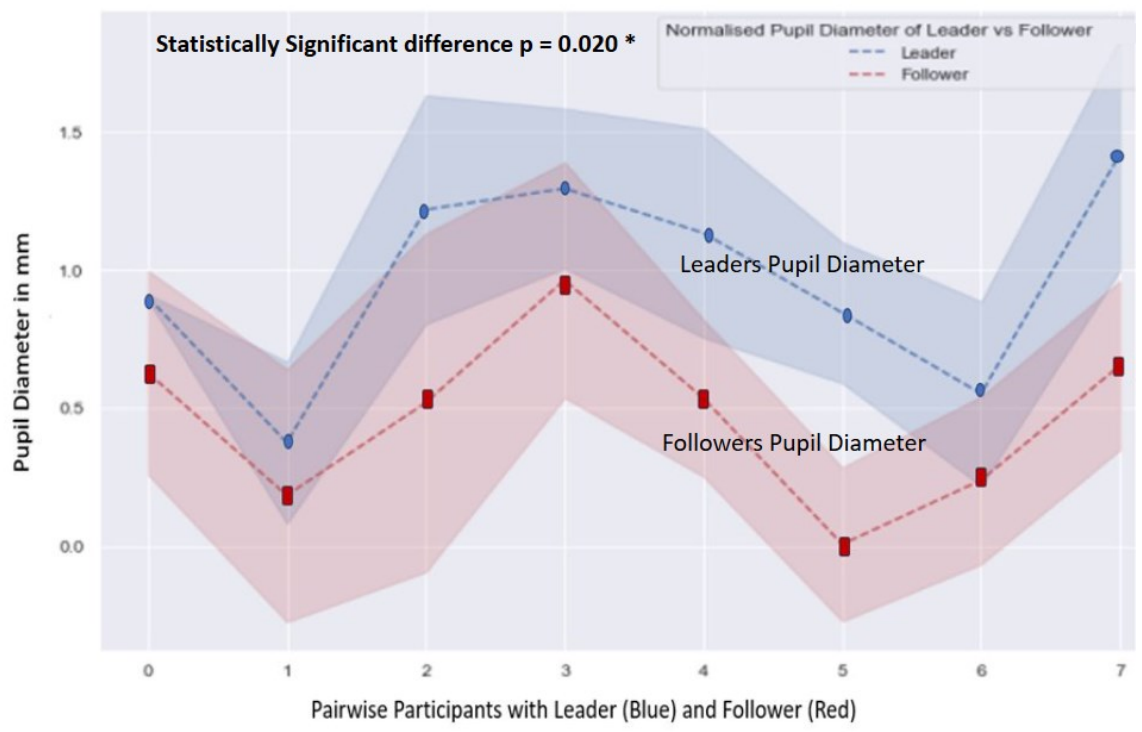


Figure 5-8*: Comparison of normalised pupil diameter between Leaders and Followers across participant pairs. The solid blue line represents Leaders, while the red dashed line represents Followers. Shaded regions indicate the variability (e.g., standard deviation) around the mean pupil diameter for each group, illustrating the range of physiological responses within each role.*

- **PD**: Pupil diameter (PD) is a widely recognised physiological marker of mental effort, visual attention, and overall cognitive load [172]. As illustrated in Figure 5-8 Leaders (blue line) consistently demonstrated greater normalised pupil diameters than Followers (red line) across all participant pairs. On average, Leaders exhibited a mean PD of 1.3 mm, whereas Followers averaged 0.5 mm, with this difference reaching statistical significance ($p = 0.020$). This pronounced difference suggests that Leaders were subject to higher cognitive demands during the VR task, likely due to the increased decision-making, situational awareness, and sustained attentional control required in their role. The variation across pairs shows that while some leader-follower differences were minor, others were substantial, with peak PD differences observed around pairs 3 and 7.

The results of this study highlight significant physiological differences between the Leader and Follower roles, which align with the subjective assessments from both the NASA-TLX and the customised questionnaire. Leaders demonstrated greater cognitive and emotional engagement, as evidenced by higher EDA, IBI, BVP, and PD values, while Followers exhibited more physical activity, indicated by higher ACC values (see Figure 5-7: Variation between different bio signals obtained between Leader and Follower Group). These findings emphasise the importance of role-specific design considerations in VR systems to optimise user experience and performance in collaborative virtual environments. The correlations between different physiological signals and subjective metrics provide novel insights into cognitive workload and user engagement in VR settings, confirming the robustness of combining physiological measures with subjective assessments for comprehensive QoE evaluation. Appendix G provides the details of the raw data captured from 16 participants (starting from 1001 to 1016), with odd numbers representing Leaders and even numbers representing Followers. All the raw data were processed after segregating them into the three phases and standardisation of signals were done using z-score normalisation as mentioned in **Section 4.4.** Following this, each physiological signal was systematically compared between the Leader and Follower groups, enabling the identification of role-based differences in physiological patterns. The results of this study highlight significant physiological differences between the Leader and Follower roles, which align with the subjective assessments from both the NASA-TLX and the customised questionnaire. Leaders demonstrated greater cognitive and emotional engagement, as evidenced by higher EDA, IBI, BVP, and PD values, while Followers exhibited and cognitive states. This level of analysis provides valuable insights into how task roles influence both experience and engagement in collaborative VR.

Table VII: Variation of Physiological Sensors between Leader and Follower

Wearables	Sensors	Group	Mean	U-Value	P-Value
E4 Wrist band	HR	Leader	0.85	47.0	0.130
		Follower	0.55		
	EDA	Leader	0.90	54.0	0.02066*
		Follower	0.60		
	BVP	Leader	0.001	52.0	0.0379*
		Follower	0.0008		
	ST	Leader	1.75	53.0	0.0281*
		Follower	1.50		
	IBI	Leader	1.00	0.0	0.00015***
		Follower	1.75		
	ACC	Leader	0.0025	59.0	0.0029**
		Follower	0.0035		
HTC VIVE	PD	Leader	1.3	54.0	0.020*
		Follower	0.5		

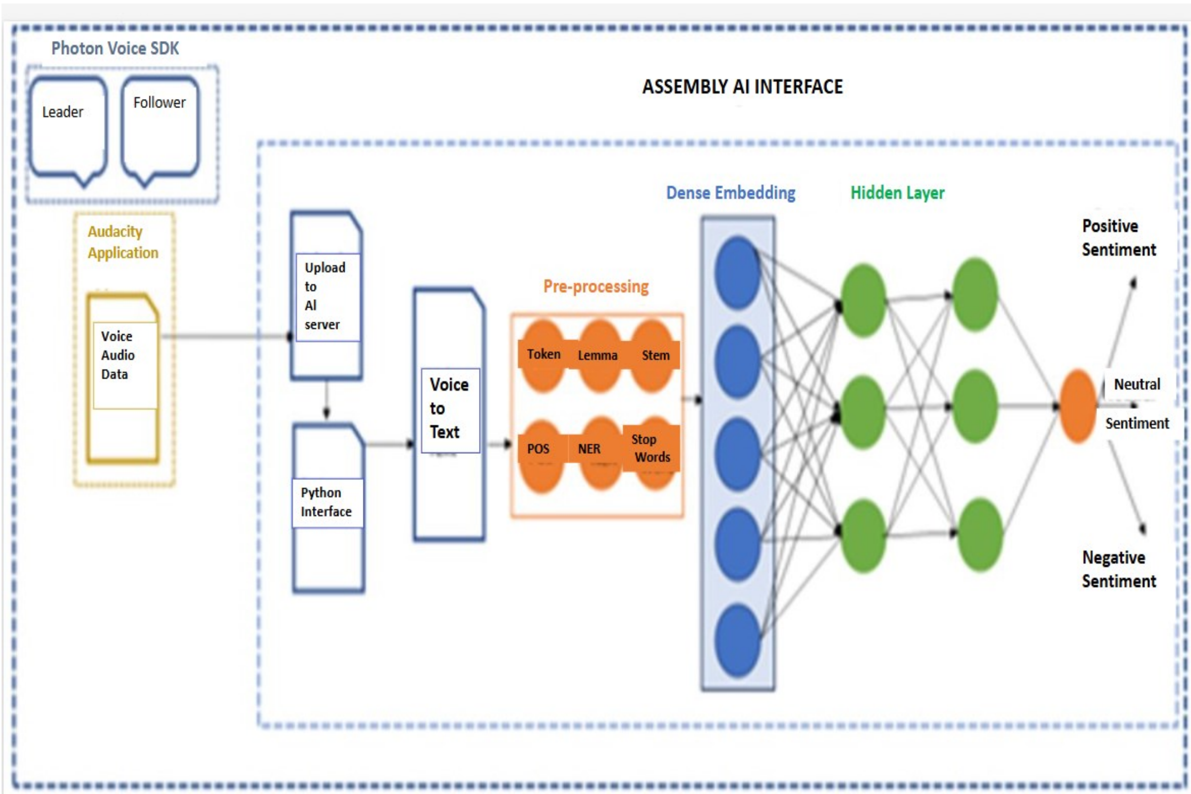
***Significance levels *p < 0.05, **p<0.01, ***p<0.001**

5.4.3 *Voice Sentiment Analysis and Task Performance Metrics Results*

This section investigates the relationship between voice sentiment during collaborative VR tasks and task performance metrics, such as controller clicks and task completion time. Voice This section explores how participants' vocal sentiment during collaborative VR tasks relates to key performance metrics such as controller clicks and task completion time. Voice data from both Leaders and Followers was captured using the *Photon Voice SDK* and recorded via the Audacity application. The recorded voice audio data was then uploaded to the AssemblyAI Interface through a custom Python interface, where it was automatically transcribed (*Voice-to-Text*) [173].

Following transcription, the text underwent a pre-processing pipeline involving tokenization, lemmatization, stemming, part-of-speech (POS) tagging, named entity recognition (NER), and stop-word removal. These processed linguistic features were transformed into dense embeddings, which served as input for a multi-layer neural network. The network's hidden layers classified each spoken segment into positive, neutral, or negative sentiment categories.

The resulting sentiment labels were then statistically correlated with VR performance metrics to assess how emotional tone influenced collaboration quality and task efficiency. For example, positive sentiment was generally associated with smoother coordination and shorter task times, while negative sentiment often coincided with higher controller activity and longer completion durations. This combined pipeline provided a data-driven link between emotional expression



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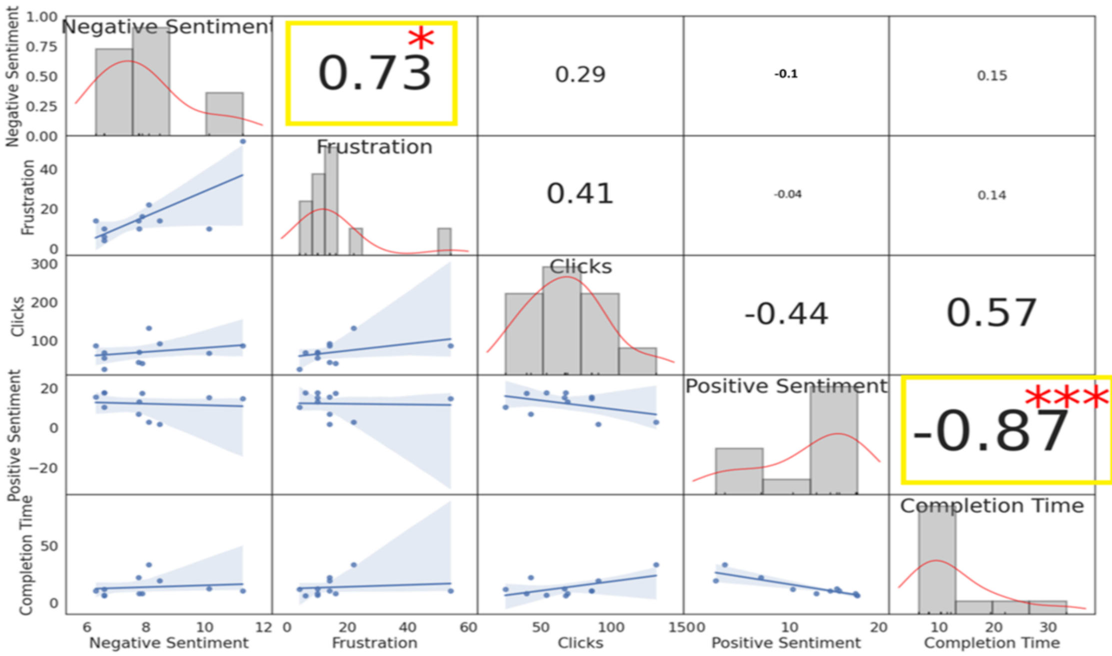

b

***Figure 5-9**: a) Architecture of Voice Sentiment Analysis b) Task Performance Metrics Correlations.*

and collaborative performance, offering valuable insights into the role of vocal sentiment in VR teamwork dynamics, as shown in Figure 5- 9a.

5.4.3.1 Voice Sentiment Analysis

The voice recordings from the VR sessions between the pairs in each team were analysed to identify and classify the emotional tone of the conversations. The recordings were uploaded to the AssemblyAI Interface using Python code, which processed the data and classified the spoken content into positive, negative, and neutral sentiments as shown in Figure 5-9a. The following key findings were derived from this analysis:

- **Negative Sentiment and Frustration**: Higher levels of negative sentiment in the conversations were found to correlate strongly with the perceived frustration levels reported in the NASA-TLX assessments. For example, Team 4 had 11.29% negative sentiment and reported high frustration in the NASA-TLX survey. Similarly, Team 2, which showed 8.1% negative sentiment, also reported high levels of frustration. This suggests that negative communication patterns may be a direct reflection of cognitive overload and emotional strain during complex VR tasks[174].
- **Positive Sentiment and Task Efficiency**: Teams with higher proportions of positive sentiments, such as Team 5 (17.36%) and Team 9 (17.58%), had significantly shorter task completion times and fewer controller clicks. These teams also reported lower mental and physical load in the NASA-TLX assessments, indicating that a positive communication style facilitates smoother collaboration and better task performance [175]. The correlation analysis confirmed that positive sentiment is inversely related to task completion time, emphasising the role of positive emotional engagement in enhancing task efficiency (**Table VIII**).
- **Neutral Sentiment**: Most spoken content was categorised as neutral, with percentages ranging from 74.19% (Team 4) to 89.83% (Team 1). Neutral sentiments often reflected task-focused discussions and procedural instructions, which are crucial for effective collaboration. This baseline of neutral communication ensured that the teams remained focused on the task, contributing to steady task progression and coordination.

5.4.3.2 Task Performance Metrics

Task performance metrics, including the number of controllers clicks and task completion time, were used to evaluate the cognitive and physical demands experienced by participants during the VR task.

The total number of hand controller clicks indicates task complexity and physical interaction. Teams with higher negative sentiment levels, such as Team 2, with 132 clicks, also demonstrated higher click counts, reflecting the cognitive and physical strain experienced during the task. This pattern highlights the impact of physical performance and interaction efficiency on emotional states. The task completion times varied significantly among teams, with shorter times generally associated with higher positive sentiment levels (Figure 5-9b). For example, Team 3, with 12.98% positive sentiment, completed the task in just 08:12. This suggests that effective and supportive communication, marked by positive sentiments, can reduce cognitive load and enhance task performance in collaborative VR environments. Table VIII represents the performance metrics with the voice sentiments for all 22 participants (11 pairs).

***Table VIII**: Voice Sentiment Performance Metrics Table*

	Variation between Leader and Follower Role Groups				
Team Number	***Voice Sentiments*** ***Neutral***	***Positive***	***Negative***	***Completion Time(min: ss)***	***Number of Clicks***
1	89.83%	1.69%	8.47%	19:39	91
2	89.18%	2.7%	8.1%	33:28	132
3	79.2%	12.98%	7.79%	08:12	69
4	74.19%	14.5%	11.29%	10:28	86
5	74.73%	17.36%	7.89%	08.10	40
6	85.43%	6.79%	7.76%	22:11	43
7	74.57%	15.25%	10.16%	12:19	67
8	83.01%	10.37%	6.6%	11:46	25
9	75.82%	17.58%	6.59%	06:24	68
10	75.89%	17.52%	6.59%	06:32	54
11	78.15%	15.53%	6.31%	10:37	86

5.4.3.3 Exploring correlations between Voice Sentiment and Task Performance

The integration of voice sentiment data with task performance metrics provides a holistic view of how emotional states and communication styles affect collaboration quality and efficiency:

- **Negative Sentiment and Frustration**: Teams with higher negative sentiment percentages in their verbal interactions reported higher perceived frustration in the NASA-TLX assessments. This correlation suggests that negative communication patterns may be a reflection of participants' frustration, likely arising from cognitive overload and task complexity, and associated with reduced task efficiency.

- **Positive Sentiment and Task Efficiency**: Teams with higher positive sentiment percentages demonstrated better task performance, characterized by fewer controller clicks and shorter completion times. This suggests that fostering a positive communication environment can mitigate cognitive load, reduce perceived effort, and enhance collaboration quality. The correlation analysis showed a significant inverse relationship between positive sentiment and perceived mental load, as indicated in the NASA-TLX results.
- **Voice Sentiment and Cognitive Load**: The relationship between voice sentiment and task performance metrics corroborates the findings from the subjective assessments. Teams reporting higher negative sentiments in their communication also reported higher mental and temporal loads in the NASA-TLX assessments. Similarly, teams with more positive sentiments in their interactions had lower reported cognitive loads and better task performance, as reflected in both the NASA-TLX and customised questionnaire results.

The results demonstrate a clear relationship between voice sentiment and task performance metrics in collaborative VR tasks (see Figure 5-9b). Analysis of collaborative VR sessions revealed that negative sentiment in speech was strongly linked to higher frustration levels ($r = 0.73$) and tendencies toward less efficient task behaviour, including more controller clicks. Conversely, positive sentiment showed a strong negative correlation with task completion time ($r = -0.87$), indicating faster and smoother collaboration when communication was more positive. These results highlight that emotional tone is a key driver of collaboration quality—positive interactions enhance efficiency, while negative emotions hinder performance. Integrating sentiment analysis with performance metrics offers a robust framework for designing emotionally aware VR systems that support both productivity and user satisfaction.
This study is the **first in VR collaboration research** to deduce participant sentiments directly from **voice audio data** and systematically relate these emotional cues to objective task performance measures. By bridging affective computing with behavioural analytics, it provides a novel methodological approach that moves beyond traditional self-report measures, enabling real-time, data-driven insights into team dynamics within immersive environments. After evaluating the audio recordings on a **pairwise basis** and establishing clear links between sentiment and task efficiency, we extended our analysis to investigate **team-level synchrony metrics**.

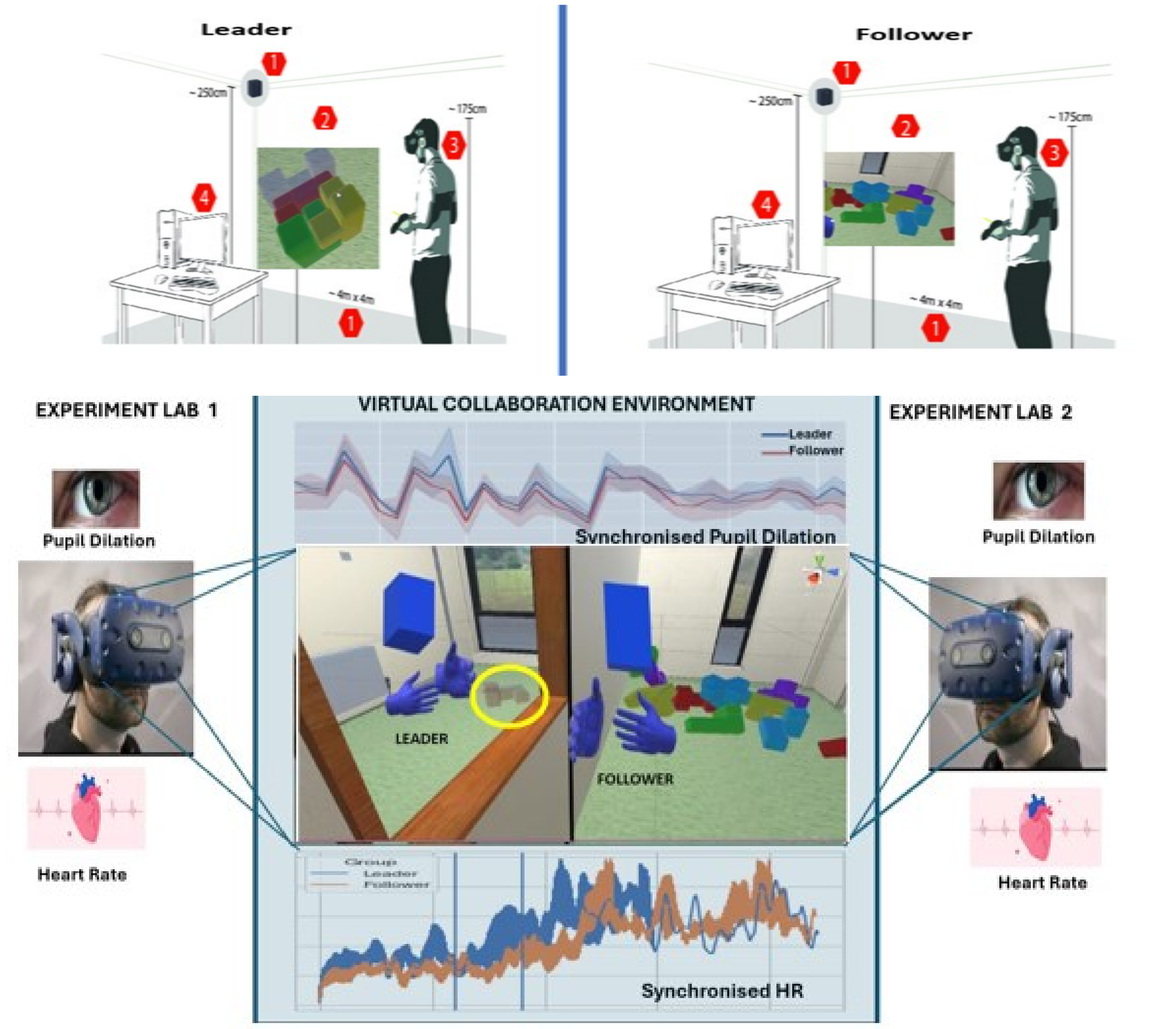

Figure 5-10: Synchrony metric calculation when two users are virtually connected to the same environment although in different physical space

5.4.4 *Collaboration Synchrony Metrics*

Synchrony in physiological signals is a crucial metric for understanding the dynamics of collaborative interactions, especially in VR environments where traditional non-verbal cues are limited. It reflects how participants align their emotional and cognitive states during a task, essential for effective teamwork and communication [176]. In this study, synchrony between Leader and Follower pairs, EDA, HR, and PD were considered (Figure 5-10).

- **EDA synchrony** measures the alignment of arousal levels and emotional states between participants during collaboration tasks. High EDA synchrony has been associated with increased engagement, empathy, and collaborative performance, making it a valuable metric for assessing the quality of social interactions in VR settings [177].

- **HR Synchrony or Cardiac synchrony** refers to the alignment of heart rates between individuals during collaborative tasks, reflecting shared levels of focus, engagement, and physical stress [178]. Research has shown that HR synchrony can be an objective measure of mutual involvement and coordination. High HR synchrony has been linked to improved communication, better coordination, and increased overall satisfaction with the collaborative experience [179].
- **PD synchrony** is another critical physiological response used to gauge cognitive load and emotional states during collaborative tasks. PD reflects changes in the size of the pupils, which can indicate levels of attention, cognitive effort, and emotional arousal [172]. In VR settings, where visual attention and immersion play significant roles, joint pupil dilation synchrony can provide insights into the effectiveness of collaboration [180]. High PD synchrony has been proposed as an objective indicator of teamwork performance, reflecting the engagement and attention participants devote to the collaborative task [157], [181].

5.4.4.1 Biophysiological Data Normalisation:

Due to the inherently individualistic nature of physiological signals, extensive pre-processing is required to enable reliable inter-participant comparisons [182]. To assess physiological synchrony between participants, each of the aforementioned signals undergoes the following three pre-processing steps:

a) Filtering: Removing artefacts and noise to clean the data [183].

b) Decomposition: Extracting additional information from the signals [184].

c) Normalisation: Adjusting signals using Equation (1) to a standard range (0 to 1), enabling fair comparison across participants [185].

These steps are crucial to minimise variability and enhance the quality of the data collected for further analysis. The normalised signals with equation (1) are suggested in [136] , where x' is the normalised signal value, x is the initial signal value, and min () and max () are the minimum and maximum values of signals. By standard practice, the normalised signal value ranges between 0 and 1 [33].

$$x' = (x - Min(x))/(Max(x) - Min\ (x)) \qquad (1)$$

5.4.4.2 Algorithms for Computing Synchrony:

To quantify the synchrony between the physiological signals, seven measures found in the literature were explored: Signal Matching (SM), Instantaneous Derivative Matching (IDM), Directional Agreement (DA), Pearson Correlation Coefficient (PCC) [186], Single Session Index (SSI), Fisher's transform (FZT) [187], DTW [188] and R-squared [189]. Based on their suitability for different types of physiological data, we selected PCC, DTW, and R-squared (R^2) for our analysis of EDA, HR, and PD synchrony, respectively.

- **PCC**: PCC was chosen as a fundamental measure due to its simplicity and effectiveness in capturing the linear relationship between two continuous signals. PCC provides a value between -1 and 1, with values closer to 1 indicating a high positive correlation, reflecting similar arousal patterns in the EDA signals. It has been used to map EDA synchrony with pre-form constructs. PCC provides a straightforward measure of linear relationships [186].
- **DTW**: Used for EDA and HR signals, DTW measures the similarity between two time-based sequences, accommodating differences in speed and timing. Lower DTW distances indicate higher synchrony. It is preferred for time-based physiological signals due to its robustness in handling temporal variations [190].
- **R-squared for Pupil Dilation (PD)**: Measures the degree of correlation between the Leader and Follower's pupil dilation responses. Higher R^2 values indicate greater similarity in pupil dilation, with values ranging from 0 (no correlation) to 1 (exact match).

The formula for R^2:

$$R^2 = (\frac{\sum_{i=1}^{N}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{N}(x_i - \bar{x})^2 \sum_{i=1}^{N}(y_i - \bar{y})^2}} \quad (2)$$

where x_i and y_i are the normalised pupil diameter values of two team members, respectively, and N is the number of samples in the task.

By selecting PCC for EDA, DTW for EDA and HR, and R^2 for pupil dilation, the most appropriate methods for each type of physiological data were selected, providing a comprehensive synchrony assessment in collaborative VR tasks.

5.4.4.3 Additional Metric of Pair- Wise Commonality Metrics from Pre-filled Consent Form

As stated earlier, to measure the physiological synchrony (PS) between the leader and follower, pre-reported experience forms were used derived to reflect participants' mutual responses regarding self-reported experience. To achieve this goal, questions from the pre-filled consent forms were used. As detailed in Table X, these included constructs related to demographics, technical team skills using video conferences, VR experience, and familiarity.

To preprocess the survey data, we began by generating a unified score for each of the four commonality constructs based on the responses from each pair of collaborators. These constructs were: demographics similarity, perceived team player skills, VR experience, and familiarity. For each construct, a binary scoring approach was employed: if both participants in a pair provided matching responses for a given construct, it was assigned a score of 1 (indicating agreement); otherwise, a score of 0 was given (indicating a difference in perception or background). This method allowed for a straightforward assessment of alignment across specific interpersonal characteristics. The total commonality score for each pair was then calculated as the sum of these four binary scores, yielding a value between 0 (no alignment) and 4 (complete alignment).

Mathematically, this can be expressed as

$$\text{Commonality Score} \quad \sum t = \Delta d + \Delta p + \Delta v + \Delta f \tag{3}$$

where: Δd represents the similarity of participants' demographics,

Δp represents perceived team player skills,

Δv represents VR experience, and

Δf represents familiarity.

These aggregated scores served as a comprehensive measure of commonality factors—a concept used to evaluate the extent of shared background or interpersonal resonance within each pair.

In addition to survey data, objective behavioural performance metrics were collected, including the total number of hand controller clicks for each participant during the collaborative VR task. These values were then combined at the pair level to reflect collective physical interaction efforts (see Figure 5-11). Furthermore, task completion time was recorded for each pair, providing a direct indicator of collaboration efficiency.

All of these metrics—commonality score, click count, and task completion time—were analysed in conjunction with physiological synchrony data, specifically EDA synchrony, which was computed using DTW. This analysis aimed to explore the relationship between interpersonal commonalities and physiological synchrony, and how these in turn related to task performance and subjective workload.

Inspired by prior research on dyadic physiological responses [191], this combined approach offers a multidimensional understanding of the human factors that influence collaborative performance in immersive VR environments. The results of this analysis, including the correlation between EDA synchrony, commonality, and performance time, are presented in Table IX.

To the best of the author's knowledge, this is the first study to systematically explore the combined influence of interpersonal commonality factors and physiological synchrony on collaborative task performance within an immersive VR setting. While prior research has examined synchrony in physical or seated environments or considered individual performance metrics in isolation, this study uniquely integrates subjective interpersonal alignment, objective behavioural data, and physiological signal synchrony using DTW as a robust similarity measure. This novel approach provides a more comprehensive understanding of how shared background characteristics and physiological coupling contribute to the efficiency and quality of collaboration in VR. The findings offer valuable insights for the design of human-centred VR systems and open new avenues for research into adaptive, synchrony-aware collaborative environments.

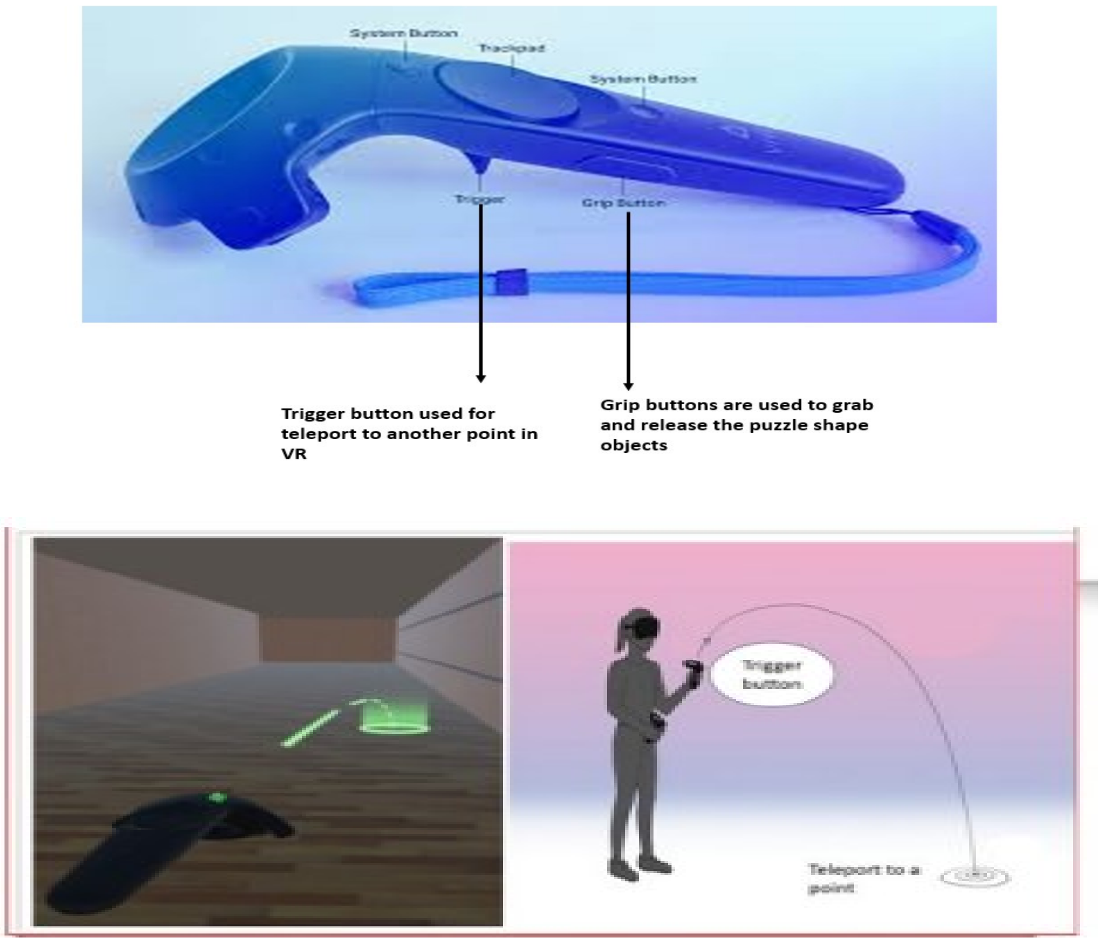

Figure 5-11: *Hand controller clicks used for teleporting and grabbing objects.*

5.5 Results of Synchrony Analysis

Synchrony metrics were computed for each Leader-Follower pair during the collaborative VR task. The results are summarised in Table IX, showing the synchrony metrics along with task performance and total pre-construct values (denoted as t-value) for each pair.

5.5.1.1 Key Findings from EDA Synchrony:

- EDA Synchrony and Pre-Construct Mapping: EDA synchrony was mapped to pre-form constructs using both PCC and DTW. For example, Pair 4, which had a high PCC of 0.86 and a low DTW of 145.0, showed strong synchrony in EDA, aligning well with high pre-construct t-values for mutual focus and verbal coordination. This

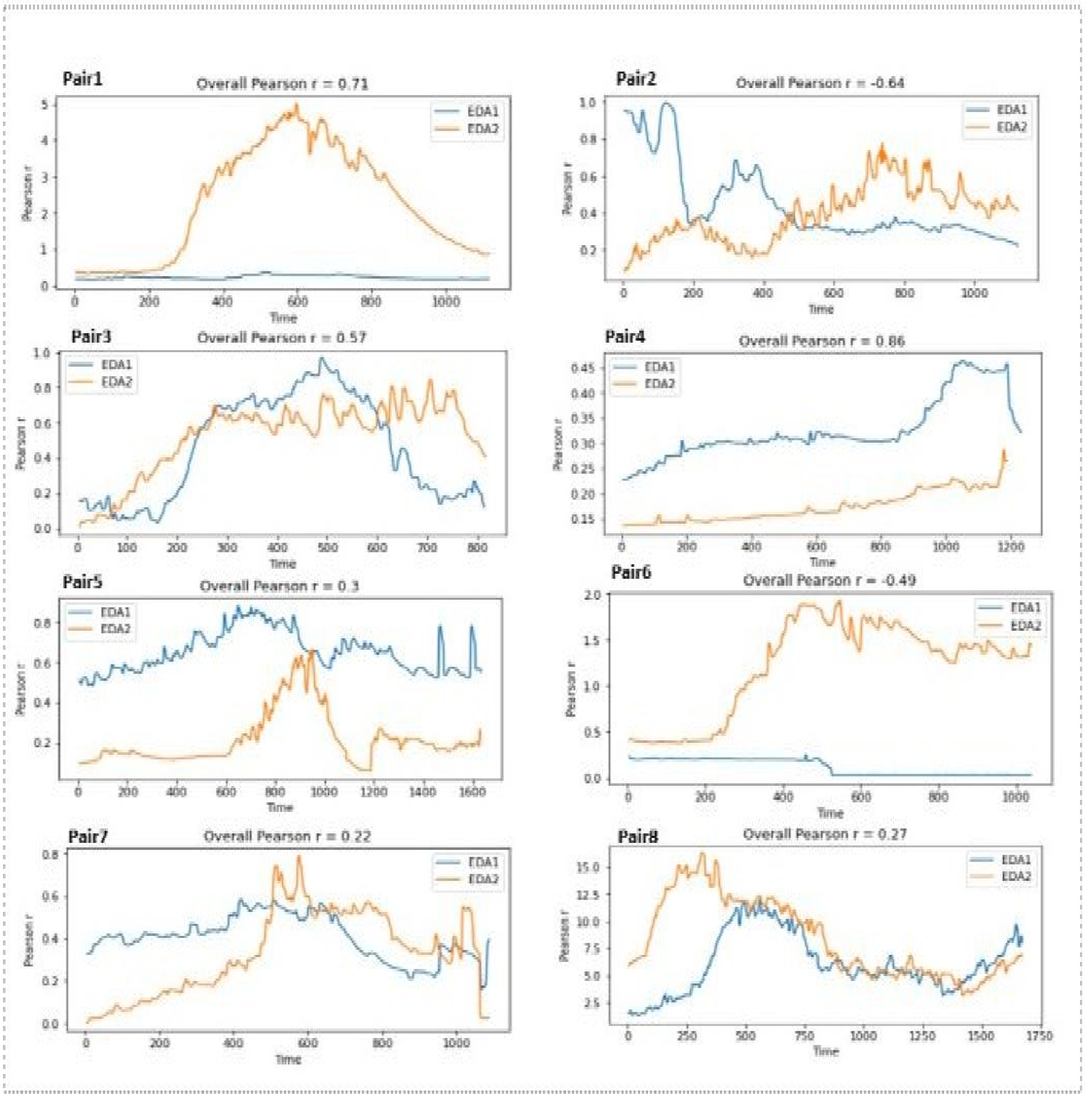

Figure 5-12: *EDA Synchrony plotted with 8 pairs of participants.*

***Table IX**: Synchrony Metrics with EDA, HR and Pupil Dilation Synchrony*

No	Constructs to Map Synchrony form pre-filled-Questionnaire													
	Pre-Form Constructs					*EDA Synchrony*		*HR Synchrony*	*Pupil Dilation*	*Load*		*Task*	*Pair*	*Time*
	Δd	Δp	Δv	Δf	Σt	PCC	DTW	DTW	R^2	Mental Load	Physical Load	Click	Gender	Task
1	0	1	0	0	1	0.70	2309.7	719.26	0.026	58	58	91	F-M	19:39
2	0	1	0	0	1	-0.64	252.2	754.16	0.307	43	66	132	M-M	33:22
3	1	1	1	0	3	0.56	49.5	613.20	0.538	32	36	69	F-F	8:12
4	1	1	1	0	3	0.86	145.0	795.60	0.522	38	46	86	M-F	10:28
5	0	1	0	1	2	0.30	554.6	543.20	0.480	24	32	40	F-M	08:10
6	0	1	0	0	1	-0.48	1125.7	706.82	0.483	52	58	43	F-F	22:11
7	1	1	0	1	3	0.22	120.2	682.68	0.602	39	36	67	F-F	12:19
8	0	1	0	0	1	0.27	1751.8	673.15	0.496	40	28	25	M-M	11:46

suggests that high EDA synchrony is associated with better communication and coordination (see Figure 5-12 and Figure 5-13).

- Pairs with lower EDA DTW values, such as Pair 3 (49.5) and Pair 4 (145.0), also show relatively lower HR DTW values (613.20 and 795.60, respectively). This suggests that better EDA synchrony may be associated with better HR synchrony [192] .
- Higher EDA synchrony (lower DTW values) seems to correspond with higher pupil dilation synchrony (higher R-squared values). For instance, Pair 3 has a low EDA DTW of 49.5 and a high pupil dilation R-squared of 0.538. Similarly, Pair 4 has an

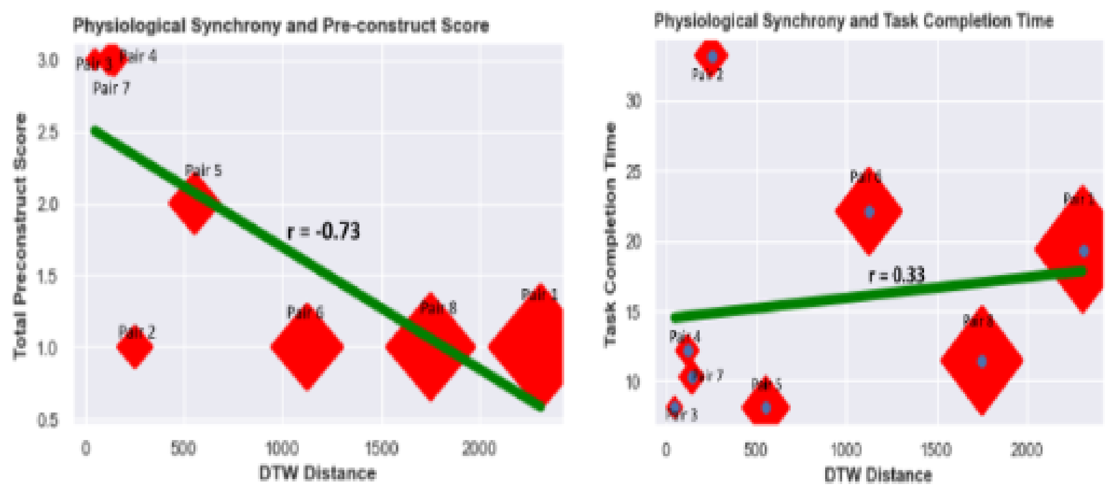

***Figure 5-13**: EDA Synchrony (DTW Distance) computed with Pre-experience Score and Task Completion Time*

EDA DTW of 145.0 and a pupil dilation R-squared of 0.522. This indicates a potential positive relationship between EDA synchrony and pupil dilation synchrony [192].

- Lower EDA DTW values (indicating better synchrony) generally correspond to lower reported mental load, fewer clicks and faster task completion times. For example, Pair 3 has a low EDA DTW (49.5) and a low mental load score (32), while Pair 4, with an EDA DTW of 145.0, also has a moderate mental load score (38) [193].
- There is no clear pattern between EDA synchrony and physical load in the Table IX.

5.5.1.2 Findings from HR Synchrony:

HR synchrony, measured using DTW, provides insights into the physiological alignment of participants during collaborative tasks. Key observations include:

- Positive Correlation with Load: Higher HR synchrony shows a non-significant positive correlation with increased mental ($r = 0.421$, $p = 0.283$) and physical load ($r = 0.482$, $p = 0.204$), indicating potential cognitive strain [194].
- Task Performance: HR synchrony shows no significant relationship with the number of clicks ($r = -0.193$, $p = 0.631$) or task completion time ($r = 0.317$, $p = 0.422$), suggesting it is less predictive of task efficiency.

5.5.1.3 Findings from Pupil Dilation (PD) Synchrony:

Pupil dilation synchrony, measured by R-squared, strongly indicates cognitive engagement and collaboration quality in VR tasks. Key findings include:

- Negative Correlation with Load: Higher pupil dilation synchrony is significantly associated with lower perceived mental ($r = -0.682$, $p = 0.045$) and physical load ($r = -0.712$, $p = 0.037$) [172].
- Task Performance: While not statistically significant, higher synchrony tends to correlate with fewer clicks ($r = -0.521$, $p = 0.165$) and faster task completion times ($r = -0.531$, $p = 0.157$), suggesting more efficient collaboration [195].

Pupil dilation synchrony is a valuable metric for understanding cognitive alignment and reducing perceived effort in collaborative VR environments.

5.5.1.4 Overall Findings

The pre-experience form total value was used to map the synchrony levels of each pair. Higher pre-construct t-values were generally associated with better EDA synchrony, suggesting that pairs with more aligned perceptions of the task and each other had better physiological synchrony and performance. Pair types (e.g., F-M, M-F) demonstrated varying levels of synchrony and task performance.

The results show that high EDA and pupil dilation synchrony are linked to better communication, lower mental load, and improved task efficiency in VR collaborations. HR synchrony, however, is associated with increased cognitive strain. Pre-experience constructs and pair dynamics also impact synchrony and performance. Overall, physiological synchrony, especially in EDA and PD, is a key indicator of effective collaboration in VR tasks as shown in Table X.

***Table X**: Variable Pair with Correlation and p-value*

Variable Pair	Correlation (r)	P-value
Pupil Dilation Synchrony vs. Mental Load	-0.682	0.045
Pupil Dilation Synchrony vs. Physical Load	-0.712	0.037
Pupil Dilation Synchrony vs. Number of Clicks	-0.521	0.165
Pupil Dilation Synchrony vs. Time Completion	-0.531	0.157
HR Synchrony vs. Mental Load	0.421	0.283
HR Synchrony vs. Physical Load	0.482	0.204
HR Synchrony vs. Number of Clicks	-0.193	0.631
HR Synchrony vs. Time Completion	0.317	0.422

Of all three, pupil dilation synchrony appears to be the strongest indicator of task-related experiences, with significant correlations suggesting that higher synchrony correlates with lower perceived load. In contrast, while still relevant, HR synchrony shows a different pattern and weaker correlations, potentially reflecting the physiological cost of collaboration rather

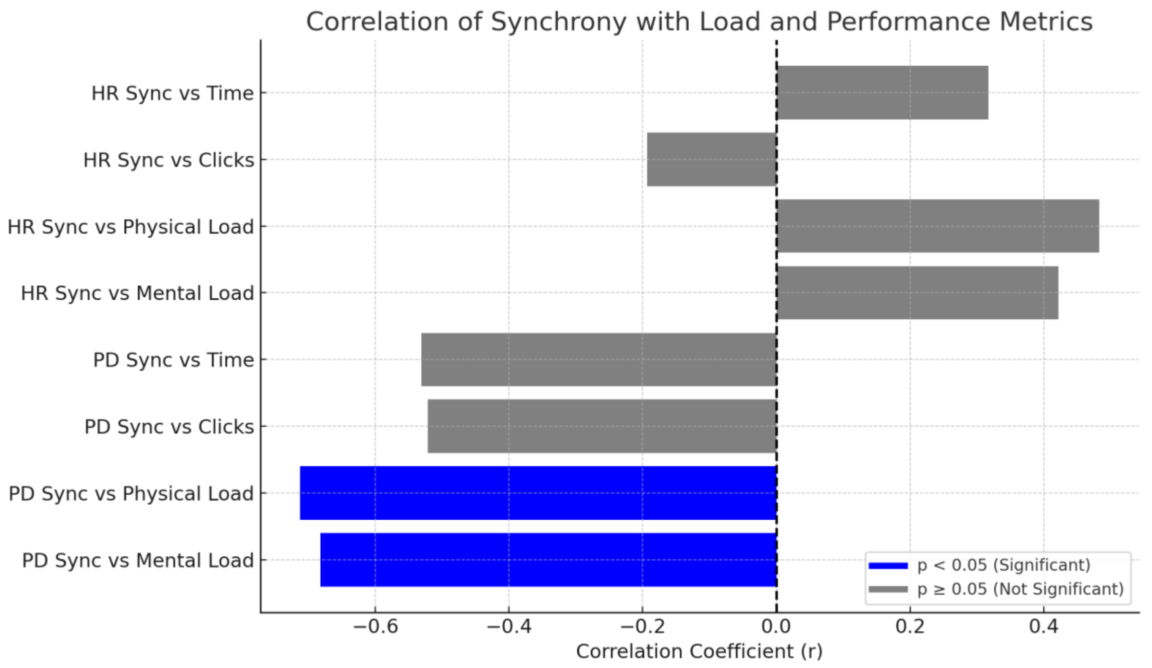

***Figure 5-14**: Significant Correlation between Pupil Dilation Synchrony and Physical and Mental Load.*

than its efficiency. These findings underscore the importance of using multiple physiological measures to capture the complexities of collaborative interactions. The significant negative correlations involving pupil dilation synchrony suggest that this metric might be particularly useful for identifying pairs that work well together with less perceived effort (Figure 5-14).

5.5.1.5 Linear Regressions

Linear regression analysis was conducted to examine the impact of synchrony metrics (pupil dilation and HR synchrony) on perceived mental and physical load, task performance, and efficiency. Unlike correlation analysis, linear regression quantifies the influence of each independent variable. Results show that pupil dilation synchrony had a strong negative effect on mental load (-83.4, p = 0.001) and physical load (-85.28, p = 0.001), indicating that better synchrony is associated with lower perceived demands. While HR synchrony did not show significant correlations earlier, it revealed a positive relationship with both mental and physical load in the regression model, though the effects were weaker and less consistent. Neither synchrony metric significantly influenced the number of clicks or task completion time, indicating minimal impact on task efficiency. All details are summarised in Table XI**,** where pupil dilation synchrony has a significant negative coefficient for both mental and physical load, while HR synchrony shows a smaller positive coefficient. Overall, physiological synchrony, particularly in pupil dilation, appears crucial for reducing perceived cognitive and physical demands during tasks. Future studies should explore these relationships further with larger sample sizes (Figure 5-15).

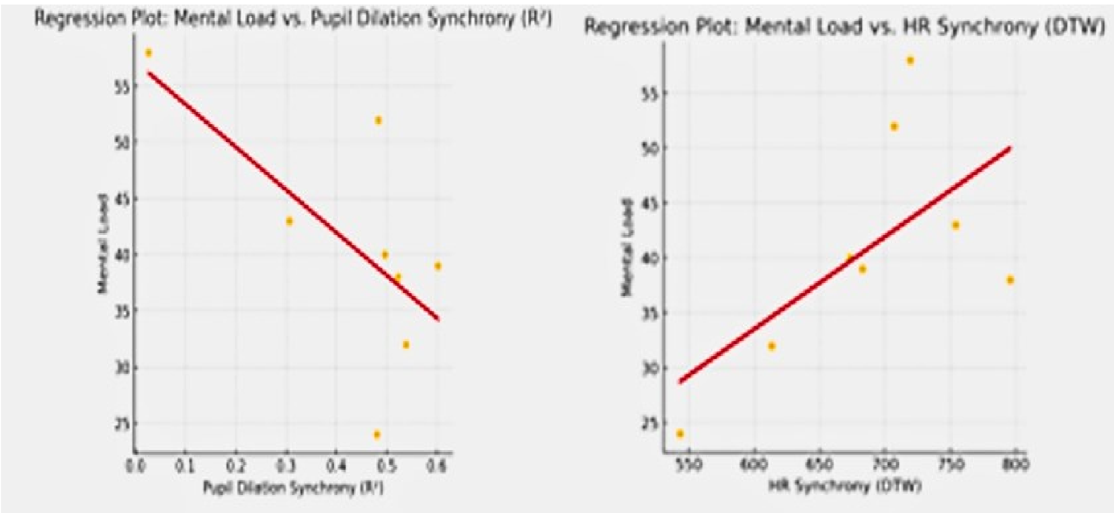

Figure 5-15*: Regression Plot between Mental Load, Pupil Dilation and HR Synchrony*

***Table XI**: Linear Regression of Synchrony Metrics and Physical, Mental Load*

Dependant Variable	Variable	Coeff	Std. Error	t-value	P-value	0.025	0.975
Mental Load	Intercept	96.9	23.98	4.041	0.009	35.44	158.36
Mental Load	Pupil Dilation Synchrony (R2)	-83.4	13.59	-6.134	0.001	-116.94	-49.875
Mental Load	HR Synchrony (DTW)	0.06	0.032	2.031	0.097	-0.015	0.146
Physical Load	Intercept	88.69	25.14	3.527	0.017	24.69	152.69
Physical Load	Pupil Dilation Synchrony (R^2)	-85.28	14.26	-5.978	0.001	-119.46	-51.10
Physical Load	HR Synchrony (DTW)	0.084	0.034	2.516	0.054	-0.002	0.171
Clicks number	Intercept	-7.62	29.09	-0.262	0.804	-82.41	67.169
Clicks number	Pupil Dilation Synchrony (R^2)	-19.68	16.48	-1.194	0.286	-62.048	22.684
Clicks number	HR Synchrony (DTW)	0.046	0.038	1.214	0.279	-0.052	0.145
Time complete	Intercept	25.60	7.23	3.536	0.017	6.351	44.856
Time complete	Pupil Dilation Synchrony (R^2)	-16.45	4.10	-4.011	0.009	-26.737	-6.166
Time complete	HR Synchrony (DTW)	0.0045	0.009	0.514	0.628	-0.015	0.024

For performance metrics, including number of clicks and task completion time, neither synchrony measure showed significant effects. This indicates that while physiological synchrony may influence perceived workload, it may not directly affect task efficiency or physical effort as reflected in these behavioural indicators. These findings suggest that pupil dilation synchrony is a reliable physiological marker for understanding perceived workload in VR-based collaboration. Future research should aim to expand on these insights by including larger participant samples and exploring how synchrony-driven interaction design can enhance Quality of Experience (QoE) in immersive environments.

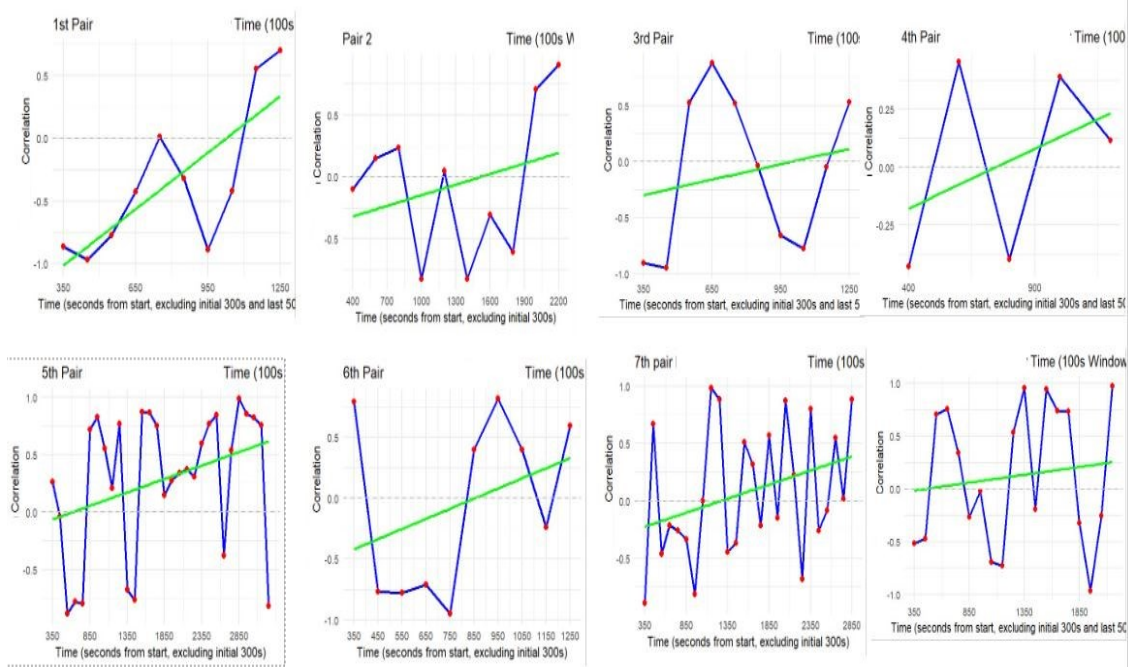

Figure 5-16: *Subplots of each pair showing the trend, how Spearman Rank Correlation of HR synchrony increases with collaboration time plot with a 100-sec rolling window*

5.5.1.6 Temporal Progressions

An analysis of the temporal progression of synchrony, as shown in Figure 5-16, reveals a consistent increase in heart rate synchrony throughout the collaborative VR task. This upward trend suggests that as participants spend more time engaged in the task, their physiological states, particularly heart rate, become more aligned. This adaptation may indicate a learning effect, where pairs better understand the task and begin to mirror each other's responses subconsciously. It could also reflect growing cognitive or emotional rapport, enhancing collaboration and communication over time [12].

5.6 Limitations and Future Work

Several limitations should be considered when interpreting the findings, particularly concerning the synchrony metrics. The small sample size, despite the focus on detecting large effect sizes, may limit the generalizability of the results. The study's power might not have been sufficient to capture smaller effects on physiological synchrony, potentially leading to null findings that do not necessarily reflect the true absence of an effect but rather an inability to detect subtler interactions. Additionally, the majority of participants were experienced VR users, which may not represent the experiences of novice users, children, the elderly, or individuals with disabilities, who may exhibit different responses to VR collaborative tasks and unique synchrony patterns.

The use of the HTC Vive Pro, although providing high fidelity, posed some constraints due to its weight and tethered setup, potentially impacting comfort and natural movement, thereby influencing the physiological synchrony data.

The study employed a customised questionnaire alongside the NASA-TLX to gather subjective metrics, which, while tailored to the research needs, may not have captured all relevant aspects of the user experience and physiological responses in VR. Employing a single, standardized questionnaire that encompasses multiple dimensions of the VR experience could improve the consistency and comparability of findings across different studies. Future research should consider developing such comprehensive tools to better understand VR-induced symptoms and experiences.

Future studies should aim to include a larger and more diverse participant sample to validate and expand these results, exploring the synchrony of additional physiological responses such as HRV, IBI, and acceleration, to achieve a more comprehensive understanding of physiological alignment in collaborative VR settings. Real-time synchrony monitoring could be integrated to provide immediate feedback, potentially enhancing collaboration dynamics.

Streamlining data collection and analysis through the development of simplified questionnaires and automated performance measures could reduce participant burden and yield more precise insights into the relationship between physiological synchrony and collaborative effectiveness in VR environments. Such advancements would benefit content creators and researchers in optimising VR applications for improved user experiences and collaboration quality.

5.7 Conclusion

This study investigated the experience of leaders and followers during VR collaboration using wearable physiological sensors. Data were collected from 22 participants, including 11 leaders and 11 followers. Initial analysis of the customised questionnaire and task load metrics revealed no significant differences between leaders and followers. However, the subsequent analysis incorporating biosensor data and the NASA-TLX questionnaire provided deeper insights. Despite incomplete data from the E4 wristbands, physiological data from 8 pairs were analysed, highlighting IBI as a crucial indicator of QoE. These metrics provided a comprehensive view of human responses to virtual stimuli, along with PD, ACC, BVP, and ST.

The role-dependent QoE measure effectively tailored the VR environment to participants' specific roles, enhancing the understanding of their interactions. Voice sentiment analysis further enriched the assessment, revealing that positive sentiments correlated with shorter task completion times, while frustration levels corresponded to increased negative sentiments. These findings emphasise the value of integrating sentiment analysis with physiological data for a nuanced evaluation of QoE in VR settings.

Physiological synchrony between leaders and followers, particularly in EDA, emerged as a significant factor influencing collaboration dynamics. DTW was found to be more effective than the PCC for analysing synchrony in relation to task performance and pre-construct scores. This suggests that EDA synchrony can serve as an interpretable parameter for evaluating collaboration quality in virtual environments, offering new avenues for developing feedback systems based on synchrony metrics.

The study also revealed a strong association between pupil dilation synchrony and reduced cognitive load, indicating that aligned attentional engagement contributes to a more seamless and satisfying collaborative experience. Conversely, increased heart rate synchrony was linked to higher cognitive demand, highlighting the importance of balanced task design to prevent user overload.

Temporal analysis showed a consistent increase in heart rate synchrony over time, suggesting that prolonged engagement enhances physiological alignment between participants. This trend may reflect a learning effect or the development of cognitive and emotional rapport, ultimately leading to improved collaboration and communication.

The next experiment was motivated to address some of these weaknesses and validate these findings with a more extensive and more diverse sample to strengthen the generalizability of the results. Expanding the study to include various VR scenarios and incorporating real-time synchrony monitoring could further enhance the predictive accuracy and practical application of QoE assessments, ultimately improving user interaction quality in VR collaborations.

Chapter 6: Experimental Study 2

6.1 Research Motivation

Study 1 investigated the impact of Leader-Follower roles on collaboration and physiological synchrony during virtual reality tasks involving 22 participants (11 pairs). The study revealed novel correlations between physiological synchrony, as measured by EDA, HR, and PD, and cognitive load. Cognitive load and user experience were assessed using the NASA-TLX and a customised questionnaire. However, the study was conducted under a single visibility condition, leaving gaps in understanding how varying visibility and extended interaction duration affect collaboration, communication, and synchrony.

To address these gaps, Study 2 expands the investigation by introducing two distinct visibility conditions —complete visibility and no visibility — and increases the sample size to 68 participants (34 pairs). These conditions enable an in-depth examination of how visual access between collaborators influences physiological synchrony, collaboration dynamics, cognitive workload, and QoE across longer task durations. In addition, individual skill levels (assessed via the Interest Questionnaire) are considered to understand their effect on user experience and task performance. The study retains the focus on Leader-Follower roles further to explore their interaction with visibility and task complexity.

Precisely, the research questions explored in this study are aligned with the overarching Sub-Research Questions (SRQs) defined in Chapter 1 and are organised as follows:

Related to SRQ1: How do task roles, physiological responses, and perceived workload influence collaboration in VR?

- How do Leader-Follower task roles influence cognitive workload during collaborative VR tasks?
- How do participants' individual skills (as assessed by the Interest Questionnaire) influence cognitive load and QoE?

Related to SRQ2: How do collaboration factors, specifically visibility, affect team performance and physiological synchrony in collaborative VR?

- How do different visibility conditions (complete visibility vs. no visibility) affect physiological synchrony and collaboration dynamics?

Related to SRQ3: How does physiological synchrony influence collaboration quality and efficiency in VR?

- To what extent does physiological synchrony under different visibility and role conditions correlate with collaboration quality and task efficiency?

By addressing these targeted research questions, this study aims to deepen our understanding of how immersive VR environments can support complex, real-world collaborative manufacturing scenarios. The findings are expected to inform the design of VR-based collaboration tools that enhance productivity, engagement, and team coordination in distributed industrial settings.

6.2 Research Methodology

This section describes the methodology employed for Study 2. It includes an overview of participants' tasks, the instruction modalities, and the experimental protocol. The protocol is largely similar to that of Study 1 as described in Chapter 4; however, the collaboration phase consists of two sub-phases: Closed and Open Scene, and some differences in questionnaires.

6.2.1 *Participants and Teams*

68 subjects, comprising an equal distribution of 34 females and 34 males, were recruited using convenience sampling. The researcher recruited participants using social media, word of mouth, and emails to potential participants in the research department (staff and students). The

Table XII: *Characteristics of participant samples across study 2*

	Participants Info	***Details***
	Number of Participants	68 participants
Age	Age average	33.4 years
	Min Age	19
	Max Age	63
Sex	Female	30
	Male	38
VR experience	Ist Time Experience	30
	Experience	36
Gaming experience	No	46
	Yes	22
Team Experience	Yes	68
	No	0

sample had an average age of 35 ± 11.1 years, with a minimum age of 24 and a maximum age of 63. The details of the Participants are highlighted in Table XII.

6.2.2 *Experimental Protocol*

The experimental protocol was based on a similar structure to that of Study 1 but expanded to include two distinct visibility conditions: complete visibility and no visibility. The protocol was divided into several phases, ensuring consistency in participant preparation and data collection. These phases included participant briefing, data collection setup, task execution, and post-task assessment.

The tasks were performed using HTC Vive Pro Eye HMDs for VR and Empatica E4 wristbands for physiological data collection [103], [104]. Participants were fitted with the wristbands to capture HR, IBI, EDA, BVP, ACC and ST as key physiological measures throughout the experiment. The headsets enabled precise eye-tracking for pupil dilation, which assessed attentional synchrony.

6.2.2.1 Participant Consent Information and Pre-experience Interests

Participants were briefed on the experiment's goals and objectives. They completed a consent form, including a demographics questionnaire and VR experience history. They were also given an Interest Questionnaire (**Appendix H**). Study 1 found that participants with similar demographics, technical experience, VR experience, and pre-existing acquaintanceships exhibited higher EDA synchrony and completed tasks more quickly. For Study 2, we aimed to explore this further. In addition to collecting the details above, we gathered data on practical, social, enterprising, creative, investigative, and organisational skills as examined in the "Examine Your Interests Questionnaire." By identifying more common factors between dyads, we hope to see how these results influence synchrony.

6.2.2.2 Screening Phase

Participants underwent a screening process, including an eye examination using the Snellen chart and the Ishihara test to evaluate visual sharpness and colour vision [196]. After the screening, they watched an instructional video that provided guidance on interacting with the VR environment, including their roles as either leaders or followers.

6.2.2.3 Baseline Physiological Signal Recording

Participants were taken to separate rooms and fitted with Empatica E4 wristbands to collect baseline physiological signals. They rested in a still position for five minutes to establish baseline readings, providing a reference point for comparing physiological responses during the collaborative tasks. After this, participants were equipped with HTC Vive Pro Eye headsets to proceed to the practice phase.

6.2.2.4 Individual Practice Phase

During the practice phase, participants were familiarised with the VR system. They interacted with objects within the virtual environment, learning how to grasp, move, throw, and teleport using the VR controllers. This phase ensured participants were comfortable with the controls and the VR environment before beginning the collaborative tasks. The HTC Vive Pro Eye headsets underwent an automated calibration process to ensure precise eye-tracking data collection.

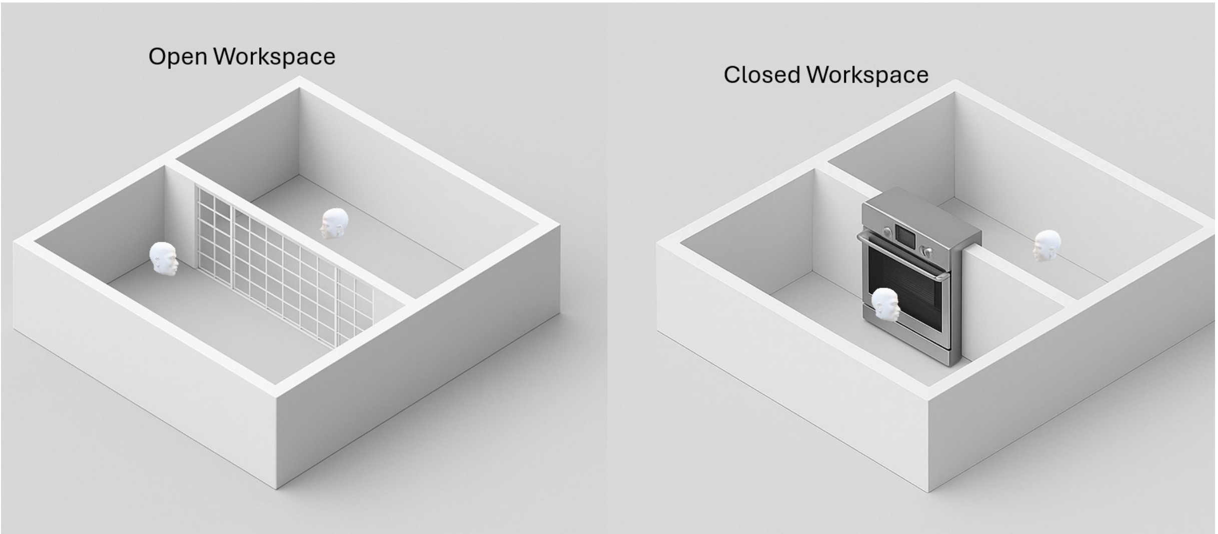

Figure 6-1: *Visual representation of the two experimental conditions used in the collaborative VR study. The Open Workspace (left) allowed participants to maintain visual contact through a semi-transparent mesh, supporting non-verbal cues. The Closed Workspace (right) restricted visibility and physical cues by introducing a solid wall and a virtual object transfer interface (elevator mechanism), simulating remote collaboration without line-of-sight. These conditions were designed to investigate the impact of visibility and spatial layout on physiological synchrony, cognitive load, and collaborative task performance.*

6.2.2.5 Collaboration Phase

Each participant pair was physically located in separate rooms to prevent any unintentional visual or auditory interaction outside the VR environment. All communication and collaboration occurred entirely within the virtual environment.

The collaboration phase consisted of two distinct sub-phases as shown in Figure 6-1.

Open Workspace (Complete Visibility): Participants could see each other through a mesh partition in the virtual room and interact both visually and verbally. The Leader provided instructions on how to manipulate objects, and the Follower followed these directions to complete the task. In this visual contact phase, the avatars consisted of a simplified unisex face, similar to a geometric or abstract model, an upper body, and hands. This limited but functional representation offered essential nonverbal communication cues—such as head orientation, hand gestures, and overall body posture—which were crucial for exploring their impact on physiological synchrony during collaboration.

Closed Scene Phase (No Visibility): Participants were placed in separate virtual rooms where they could communicate verbally but had no visual access to each other. Leaders directed the task, while Followers executed the instructions, sending objects through a virtual elevator for the Leader to place.

The collaborative task involved a structured pick-and-place activity within a custom-designed VR environment. At the start of each scene, the Follower stood in their assigned virtual room with all ten distinct objects laid out randomly on the floor. These objects varied in shape and colour and were fully visible and selectable by the Follower. Meanwhile, the Leader was located in a separate virtual room and could view a ghosted (semi-transparent) representation of the final structure on an assembly platform in front of them. This ghost model served as a spatial reference to guide the placement of incoming objects but was not interactable until the correct object was received.

In both the Open Scene Phase and the Closed Scene Phase, participants were physically located in separate rooms and assigned to non-overlapping virtual spaces. Even during the Open Scene Phase—where the participants could see each other's avatars through a mesh partition—they were not allowed to cross into each other's virtual area or directly access the other's workspace. Only verbal communication and a shared virtual elevator mechanism enabled task coordination and object transfer.

Throughout the activity, the Leader described the desired object based on visual characteristics and instructed the Follower to locate and send it. Once received via the elevator, the Leader placed the object on the corresponding position of the ghost structure. As the task progressed:

- The Follower's side gradually cleared as objects were selected and sent.
- The Leader's assembly area progressively filled, replacing the ghost shapes with solid blocks.

This setup ensured spatial independence while preserving interdependence in communication and task execution, allowing for controlled assessment of collaboration quality, synchrony, and role-based interaction.

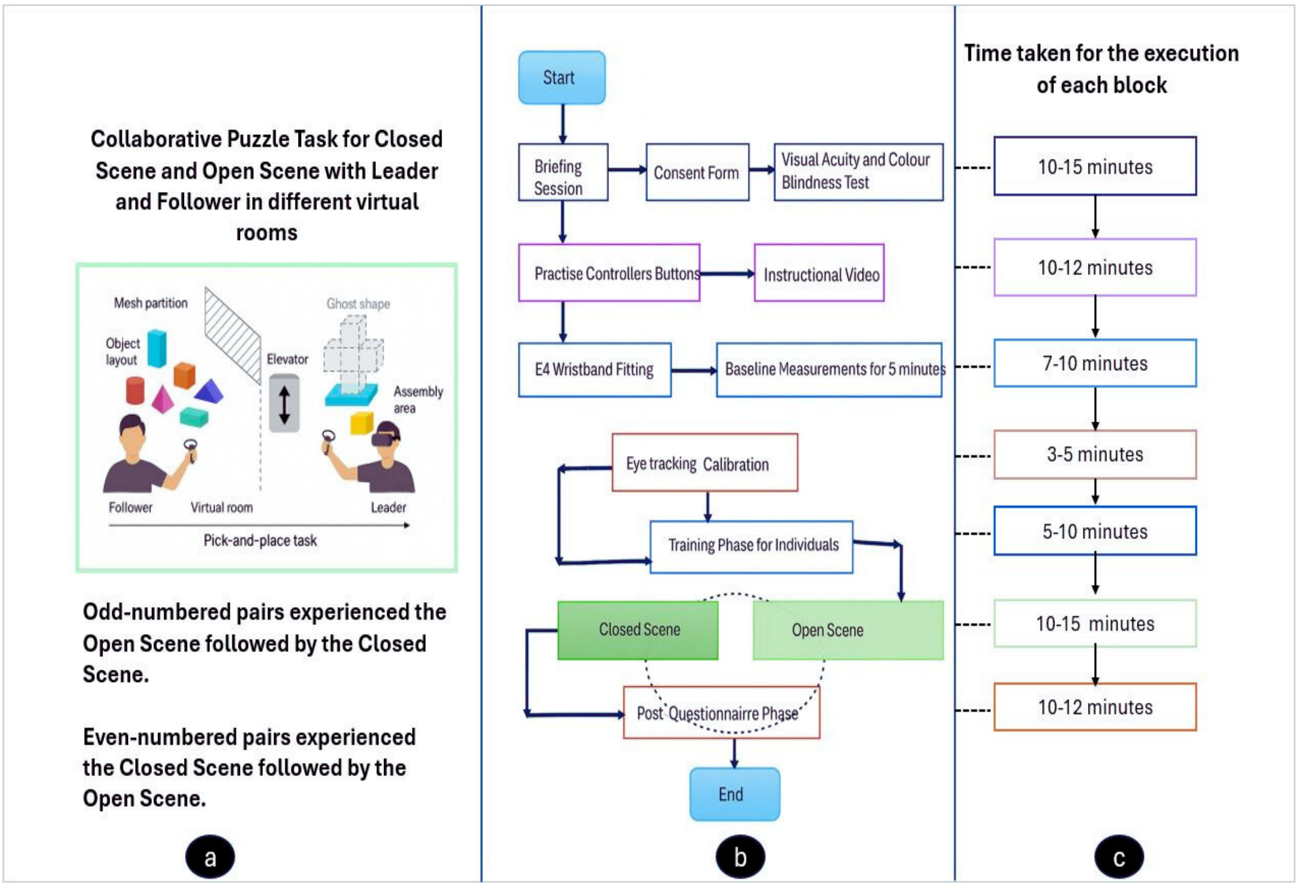

Figure 6-2*: Experimental Phases for Experimental Study 2 with approximate time taken for each phase*

To avoid learning bias, the sequence of sub-phases was counterbalanced: half of the pairs began with the open scene and the other half with the closed scene, ensuring diverse exposure to each collaboration condition.

6.2.2.6 Post-test Assessments: SIM-TLX, VRNQ and Customised Questionnaire Phase

After both tasks were completed, participants were asked to fill out the SIM-TLX and VRNQ as shown in Figure 6-2(b and c). For Study 2, assessment tools were updated to provide more comprehensive and reliable measurements of cognitive load and user experience. SIM-TLX was used instead of NASA-TLX (employed in experiment 1), and the Virtual Reality Neurocognitive Questionnaire (VRNQ) was employed to assess the QoE instead of the customised questionnaire in experiment 1. SIM-TLX was used to determine the cognitive load, capturing factors such as mental demand, effort, frustration, and performance [197]. VRNQ evaluated the overall QoE, including immersion, satisfaction, and perceived effectiveness of collaboration [198]. Additionally, participants completed an Interest Questionnaire to evaluate their skills, allowing for an investigation into how individual skill levels might influence cognitive load and a collaboration outcome. Lastly, participants completed a short, customised collaboration questionnaire with five key questions:

1. Which scene (open or closed) did you prefer for collaboration?
2. How would you rate your performance in the open scene?
3. How would you rate your partner's performance in the open scene?
4. How would you rate your performance in the closed scene?
5. How would you rate your partner's performance in the closed scene?

These assessments provided both quantitative and qualitative insights into participants' experiences across the two visibility conditions. The entire experimental protocol with approximate timings is shown in Figure 6-2 c.

6.2.3 *Equipment Specification*

The experimental setup mirrored that of Study 1, employing two HTC Vive Pro Eye headsets connected to separate PCs equipped with GeForce RTX 3060 graphics cards to generate immersive VR environments [103]. Participants wore Empatica E4 wristbands to collect physiological data throughout the tasks continuously. Each participant was physically located in a separate room during both the open and closed scene phases. In the closed scene, participants could hear each other but were unable to **see** one another in the virtual environment. Object transfers were coordinated through a virtual lift-like system, allowing collaborators to exchange items without visual contact. In contrast, the open scene allowed both visual and auditory interaction via avatars, although participants remained in physically separate rooms.

A key improvement over Study 1 was the suspension of HMD cables from the ceiling, providing participants with greater mobility compared to the earlier setup where cables were floor-based. Experimenters supervised all sessions to ensure participant safety and smooth operation during VR immersion.

6.2.4 *Lighting and Movement in VR*

Lighting control within the VR environment was crucial to maintain consistency in tracking pupil dilation [199]. The lighting in the virtual environment was designed to match the physical room as closely as possible to ensure that participants' pupillary responses were not affected by ambient light changes[200]. Participants navigated the virtual environment using

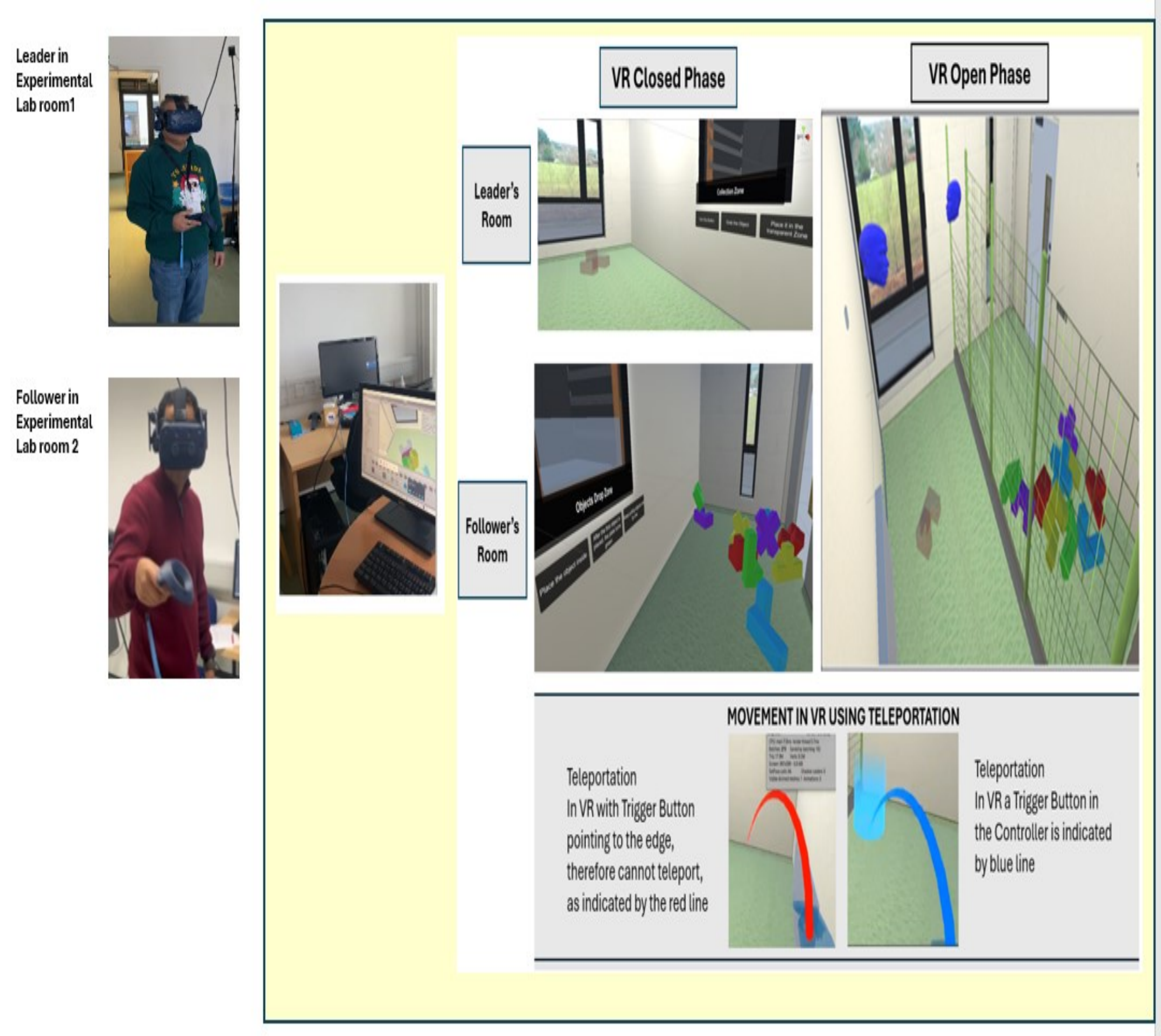

***Figure 6-3**: Collaboration Phase divided into two sub tasks such as Closed Phase and Open Phase*

teleportation, which allowed them to move from one location to another by pointing the controller and triggering the movement. This method minimised the risk of cybersickness and ensured participant comfort during the tasks.

6.2.5 *Task Description*

Study 2's experimental design incorporated two scene conditions—Closed Scene and Open Scene—to evaluate the effect of visual access on collaborative performance and physiological synchrony. Each dyad performed the same pick-and-place task under both conditions, with the sequence of scenes counterbalanced across participant pairs to control for learning effects.

In the Closed Scene, participants were placed in separate virtual rooms. They could communicate verbally but had no visual contact with each other. The Leader viewed a snap zone—a semi-transparent ghost shape representing the target object to be placed—within their workspace. These ghost shapes were already arranged in a pre-defined puzzle layout, meaning the leader did not need to solve the puzzle logic themselves, but had to ensure that the correct object, as described by shape and colour, was placed into the corresponding ghost position. The Follower had access to a scattered pool of 10 distinct objects (comprising 5 shapes × 5 colours) and was responsible for selecting the correct one based on the Leader's verbal instructions. Once selected, the object was dropped into a virtual elevator, a simulated transport mechanism that delivered the object into the Leader's workspace. The Leader then placed it into the matching ghost shape. If correct, the next ghost shape appeared. This continued until all ten objects were placed and the cuboid structure was completed (Figure 6-3 for scene schematic and elevator mechanism).

In the Open Scene, the same task was performed. Still, participants could see each other through a mesh partition (Figure 6-4), providing visual access to nonverbal cues such as hand gestures and avatar posture. Instead of using the elevator, objects were handed directly from the Follower to the Leader through an open slot in the mesh. Visual presence allowed the

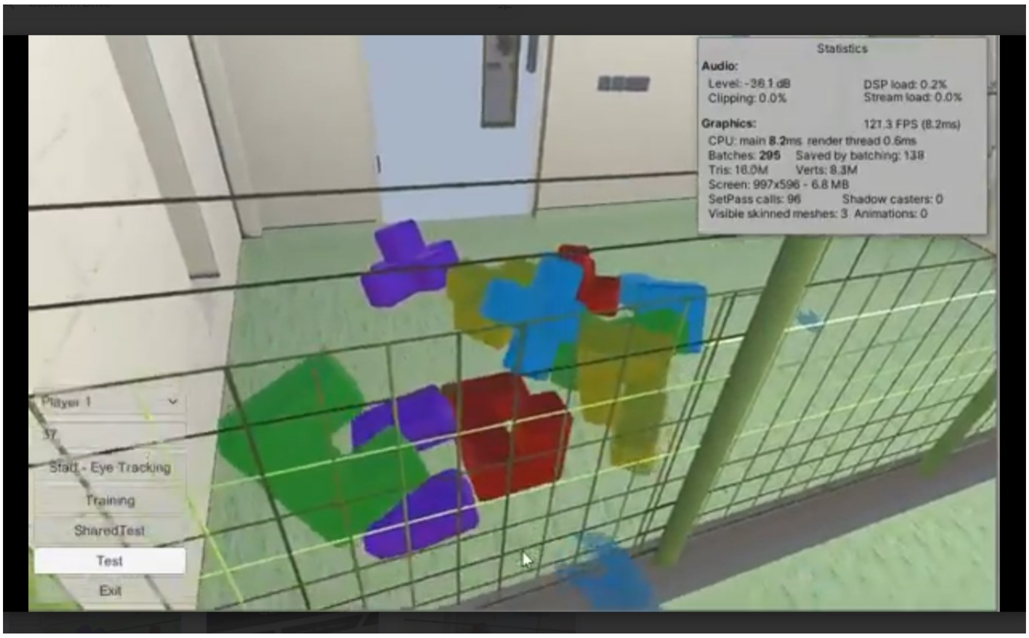

Figure 6-4*: Mesh Grid Wall in the Open Shared Workspace Phase*

Leader to observe the Follower's actions and object selection process, potentially improving coordination.

Both scenes were completed by all participants, with half of the pairs starting with the Open Scene followed by the Closed Scene, and the remaining half in reverse order. To build on findings from Study 1—which indicated that physiological synchrony increased over time—the task duration was extended from approximately 8–10 minutes in Study 1 to 13–15 minutes in Study 2. This allowed for more sustained collaboration and deeper analysis of temporal synchrony patterns.

The vertical lists of objects shown on the Leader's and Follower's sides in the (Figure 6-5) are not visible to participants during the actual task. These are provided for illustration only to depict the types of coloured 3D Soma pieces involved in the pick-and-place activity and to represent key stages in the collaborative task flow.

During the task: On the Follower's side, all ten coloured Soma blocks were visible, scattered randomly in their virtual space. On the Leader's side, only one ghost object (semi-transparent

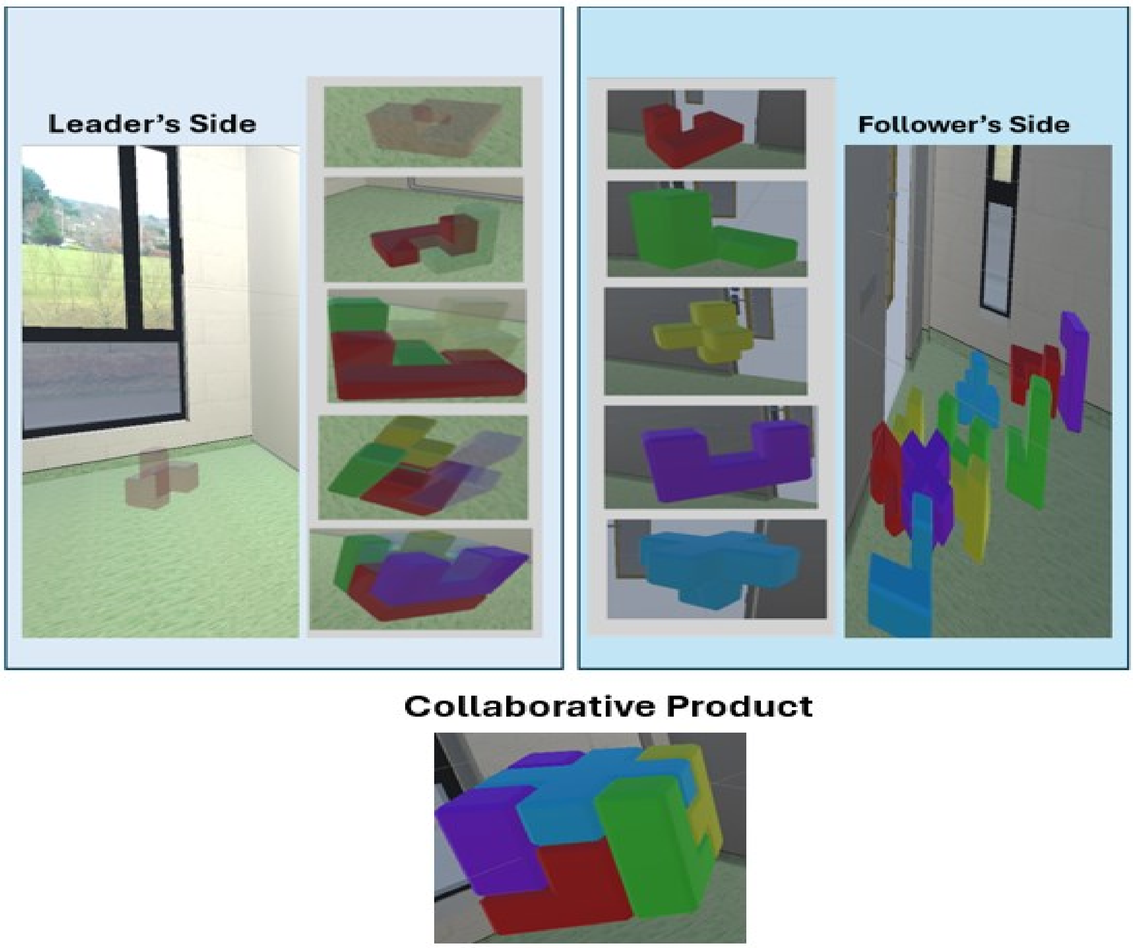

***Figure** 6-5:Pick and Place Collaborative Puzzle Task to form one Soma Cube (Collaborative Product)* red target shape) was shown at a time in the snap zone. The Leader described the object's features (e.g., "blue L-shape") to the Follower, who selected the corresponding piece and passed it via a virtual elevator system.

In both the Open Scene and Closed Scene conditions, the Leader assembled a Soma cube by arranging the received pieces onto the ghost placeholders. Once the first cube was complete, a second cube was stacked on top, forming a cuboid structure made of two Soma cubes. The final image labelled Collaborative Product in (Figure 6-5) shows one completed Soma cube, built collaboratively between the Leader and Follower.

6.2.6 *Physical Laboratory Setup*

Two physical lab rooms were set up, each with a participant and an experimenter. The rooms were separated by a middle room to prevent participants from hearing each other. Each room had a PC connected to an HTC Vive Pro Eye HMD, two hand controllers, an E4 wristband, and pre-and post-questionnaires. The cables for the HTC Vive were suspended from the ceiling to avoid clutter (Figure 6-6).

***Figure 6-6**: Physical room setup with White board and the tethered HTC Vive hanging from the ceiling.*

Experimenters ensured participants were properly fitted with the HMD, collected baseline data from the E4 wristband, and coordinated via WhatsApp to start both participants simultaneously. The virtual rooms mirrored the physical ones for realism. Participants had a table and chair for questionnaires, underwent eye tests, and received video training on the same PC used for the Unity application. Safety was prioritized by preventing collisions with the physical walls during the experiment.

6.2.7 *Communication and Sound Through Network (Photon)*

Participants communicated using Photon Unity Networking and Photon Voice, allowing both participants to connect remotely via the university network via the Photon cloud [108], [110]. This setup enabled them to communicate using the inbuilt microphone in the HMD and the noise-cancelling headphones inside the HMD. Photon effectively reduces noise over the network, ensuring clear and uninterrupted communication between participants. This reliable communication setup was crucial for coordinating tasks and maintaining effective collaboration during the experiment (Figure 6-7).

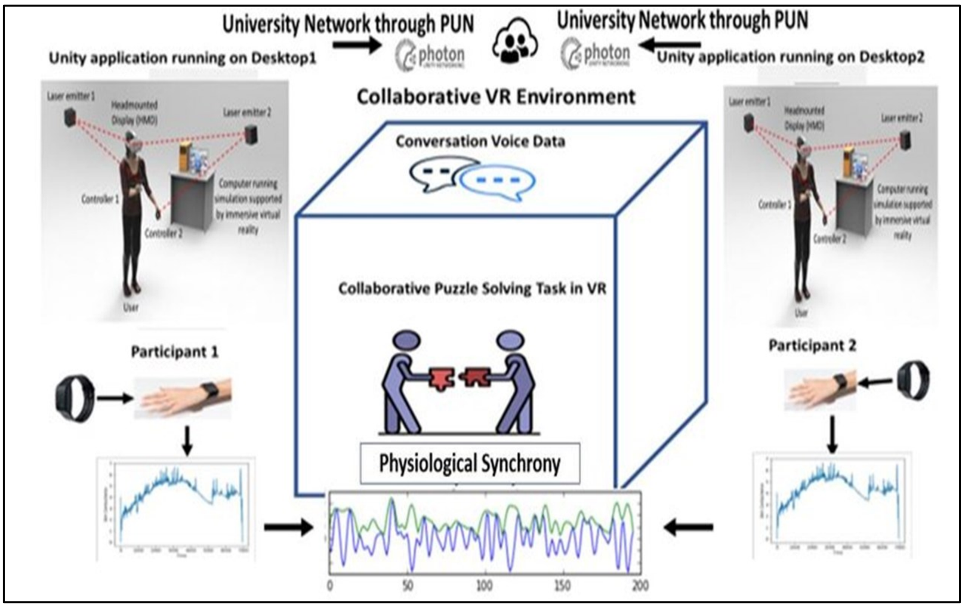

Figure 6-7 *Photon Unity Network through University Network to run multiplayer collaboration using Unity wearing HMD and wristband for multimodal data collection.*

6.3 Multimodal Data Collection for Study 2

A multimodal dataset was collected during the collaborative VR tasks to evaluate collaboration quality, cognitive workload, emotional engagement, and physiological synchrony under different visibility conditions. The data sources and analysis framework mirror those used in

Study 1 (**Chapter** 5), allowing for consistent comparisons across studies. The dataset includes subjective assessments, physiological signals, pair-wise sentiment analysis, synchrony measures, and task performance metrics.

6.3.1 *Individual Subjective Assessments*

Pre- and post-experiment subjective assessments were conducted to evaluate participant characteristics and perceptions of the collaborative VR experience.

6.3.1.1 Pre-Experiment Assessments

1) Experiment Consent and Background Form:

Before commencing the collaborative tasks, participants were asked to complete a Pre-Experiment Consent and Background Form, which collected demographic and experiential data, including:

- Gender
- Age
- Educational background
- Technical expertise
- Prior VR/AR experience
- Familiarity between collaborators (i.e., whether participants knew each other)

2) Interest Questionnaire:

Afterwards, participants completed the Interest Questionnaire to assess their self-perceived skill levels. The Interest Questionnaire provided insights into the participants' preferences and skills across six categories: Practical, Enterprising, Social, Creative, Investigative, and Organisational. Each category had six questions. These data served as baseline indicators for analysing the influence of individual traits and prior experience on collaboration quality, cognitive workload, and synchrony.

6.3.1.2 Post-Experiment Assessments

Immediately after completing the collaborative VR tasks, participants completed three post-task questionnaires:

1) VRNQ (Virtual Reality Neuroscience Questionnaire):

The VRNQ was utilised to assess overall system usability, levels of simulator sickness, and the quality of immersion experienced by participants. The results from the VRNQ were instrumental in evaluating the impact of the VR system on the overall QoE. The complete questionnaire is included in **Appendix I**, and **Appendix J** outlines the VRNQ administration guidelines.

2) Customised Collaboration Questionnaire:

Participants were asked to indicate their preferred collaboration environment by rating their experience in both the open and closed workspace settings. The evaluation focused on their overall preference based on their subjective impressions during the task. Responses were recorded using a 7-point Likert scale ranging from 1 (Low) to 7 (High) for each condition. This allowed for a comparative assessment of scenario favourability under varying visibility constraints. The full set of questions and rating scales used in this evaluation is provided in **Appendix K**.

3) SIM-TLX (Simulated Task Load Index):

The SIM-TLX is a widely used tool for measuring subjective workload across various dimensions of task load experienced by participants. The detailed SIM-TLX questionnaire is provided in **Appendix L**

6.3.2 *Individual Physiological Measurements*

Physiological responses were continuously recorded throughout the collaborative tasks to assess cognitive effort, stress levels, and physical activity. Data was collected using the Empatica E4 wristband, which captured a comprehensive set of biosignals, and the HTC Vive Pro Eye headset, which provided real-time pupil dilation measurements through integrated eye-tracking. PD data were analysed explicitly as indicators of cognitive load and attentional focus, offering insight into individual and shared cognitive states during collaboration. The following physiological signals were collected:

- HR
- EDA.
- PD
- BVP
- IBI
- ST
- ACC

Signal processing followed the methodologies detailed in **Sections 4.4** and **5.3.2.** The signals were then standardised, and the resulting normalised data were used to compare physiological responses between the two participant groups. **Appendix M** presents summary statistics for EDA, including minimum, mean, and maximum values across the baseline, eye calibration, individual training, closed test, shared workspace, and questionnaire phases. Corresponding statistics for HR across the same experimental phases are calculated first in order to compare the normalised score.

6.3.3 *Pair-wise Voice Sentiment Analysis*

All participant conversations were recorded, transcribed, and analysed using AssemblyAI's LeMUR API. Each utterance was categorised into positive, neutral, or negative sentiment, allowing for a detailed evaluation of emotional tone and communication patterns during collaboration. This sentiment analysis approach is consistent with **Section 5.3.3**.

6.3.4 *Synchrony Metrics*

Pair-wise synchrony was computed for EDA, HR, and PD signals to assess physiological alignment between collaborators. These metrics provided insights into cognitive and emotional coordination between Leader and Follower roles. The degree of physiological synchrony between participants was evaluated across two key conditions: Complete Visibility Condition (Open Scene) and No Visibility Condition (Closed Room).

- HR Synchrony
- EDA Synchrony
- Pupil Dilation Synchrony

The computation methods and justification for these synchrony metrics are consistent with those described in **Section 5.4.4.1**.

In Study 2, alongside synchrony metrics, we examined pupil dilation and blink rate over time in both open and closed workspaces to observe how these physiological signals varied throughout task completion. Since blink rate and pupil dilation are established indicators of cognitive load, they were incorporated to provide additional insight into participants' mental effort under different workspace conditions [201], [202], [203].

6.3.5 *Task Performance Metrics*

Two objective performance metrics were used to assess collaboration efficiency:

Task Completion Time: Total time taken to complete the collaborative task.

Number of Controller Clicks: Frequency of interactions via hand controllers, reflecting task complexity and interaction effort.

These performance indicators were analysed in conjunction with physiological and subjective data, using the framework described in **Section** 5.4.3.2 of Chapter 5. Detailed metrics, including task completion time, total clicks, and shared interaction time per pair, are available in **Appendix** O.

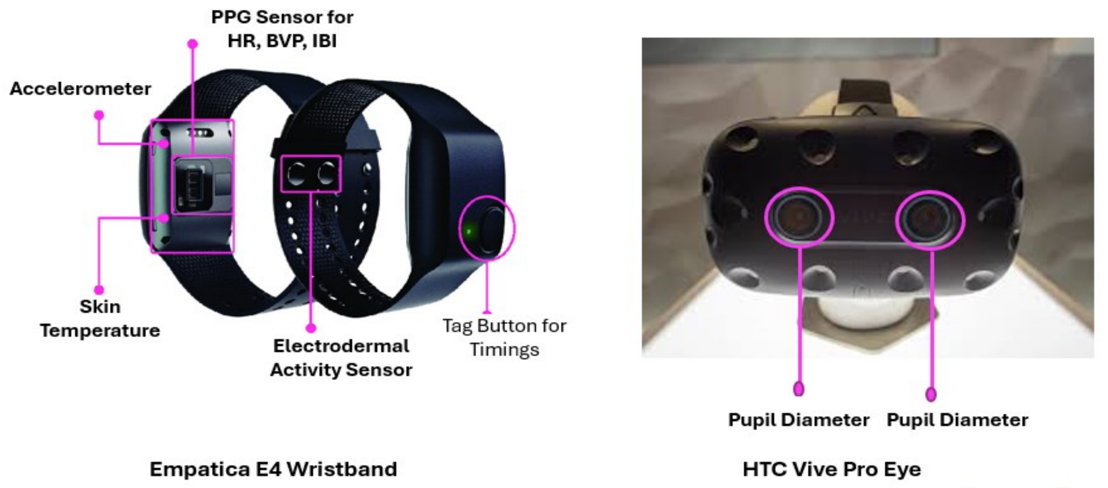

***Figure** 6-8: Metrics collected from Empatica E4 Wristband and HTC Vive Pro Eye*

6.3.6 *Summary of Data Collection Approach*

The data collection strategy for Study 2 followed the same multimodal framework as Study 1, with several key enhancements introduced to enable a more comprehensive evaluation of QoE in collaborative VR under varying visibility conditions. The refinements aimed to strengthen both subjective and objective assessment methods while maintaining methodological consistency to support cross-study comparisons.

Regarding subjective evaluation, the NASA-TLX was replaced with the SIM-TLX, an extended version better suited to simulation-based task environments. The customised QoE questionnaire from Study 1 was replaced with the validated Virtual Reality Neuroscience Questionnaire (VRNQ) to assess system usability, immersion, and simulator sickness. An Interest Questionnaire was also introduced to better map participants' baseline traits and preferences, offering a richer foundation for analysing collaboration dynamics and synchrony. Multimodal data were collected across five major domains:

1. Individual subjective assessments (pre- and post-task),
2. Physiological signals from each collaborator,
3. Voice-based sentiment analysis,
4. Pair-wise synchrony metrics, and

5. Objective task performance indicators.

Physiological data—including HR, EDA, BVP, IBI, ST, ACC, and PD—were captured using the Empatica E4 wristband and the HTC Vive Pro Eye headset. Study 2 also analysed blink rate differences between the Complete Visibility and No Visibility conditions to further explore attentional engagement under different perceptual constraints (Figure 6-8).

All voice conversations were recorded, transcribed, and processed using AssemblyAI's LeMUR API to extract utterance-level sentiment (positive, neutral, negative), enabling detailed analysis of affective tone and communication quality.

Physiological synchrony was computed between each pair of collaborators for EDA, HR, and pupil dilation signals to evaluate interpersonal alignment. These synchrony metrics were analysed across both experimental conditions—Complete Visibility and No Visibility—to understand the impact of visual access on collaboration quality. Additionally, task completion time and controller click frequency were used to assess objective collaboration performance.

Together, these data streams enabled a robust, multi-layered analysis of the collaborative experience.

6.4 Results

The results presented in this section provide comprehensive insights into the user's experience of various aspects of collaborative VR as part of shared tasks. These include participants' preferences and skills (via the Interest Questionnaire), workload and task complexity (from SIM -TLX), VRNQ results, task performance, physiological responses, and synchrony. The data was analysed with a focus on the differences between leaders and followers and gender-based variations. Additionally, the study examined how synchrony considering metrics such as HR, EDA, and pupil dilation align between participants under different conditions (e.g., shared visibility vs. no visibility)

6.4.1 *Interest Questionnaire Results*

The Interest Questionnaire provided insights into the participants' preferences and skills across six categories: Practical, Enterprising, Social, Creative, Investigative, and Organisational. Each category had six questions. The results of the Interest Questionnaire are shown in Table XIII and highlight notable gender-based differences in some skill areas (**Appendix H**)

Table XIII: Interest Questionnaire Examined between Male and Female

Skills examined	Female Mean Value	Male Mean Value	Standard Deviation Female	Standard Deviation Male	t-stat	p-value
Pratical1	2.97	3.62	1.4	1.14	2.11	0.039*
Pratical2	3.72	3.87	1.22	1.15	0.51	0.612
Pratical3	3.72	3.87	1.25	1.22	0.49	0.627
Pratical4	2.55	3.13	0.99	1.1	2.23	0.029*
Pratical5	3.66	3.9	1.37	1.05	0.83	0.411
Pratical6	3.21	3.3	1.32	1.35	0.25	0.805
Social1	3.79	3.87	1.08	1.13	0.29	0.773
Social2	4.07	4.13	0.92	0.89	0.27	0.791
Social3	3.59	4.03	1.12	0.9	1.79	0.078
Social4	3.1	3.31	1.01	1.13	0.77	0.443
Social5	3.81	3.81	1.11	0.88	-0.02	0.987
Social6	2.85	3.14	1.29	1.16	0.92	0.361
Investigative1	4	4.21	0.99	1.02	1.39	0.168
Investigative2	4.21	4.41	0.98	0.91	0.88	0.38
Investigative3	4.28	4.59	1.03	0.68	1.51	0.135
Investigative4	3.38	3.87	1.15	1.03	1.86	0.068
Investigative5	3.1	3.46	1.23	1.1	1.26	0.211
Investigative6	3.07	3.28	1	1.34	0.72	0.473
Enterprise1	3.48	3.69	1.24	1.23	0.13	0.894
Enterprise2	3	3.56	1.24	1.07	0.1	0.917
Enterprise3	3.66	3.72	1.04	0.89	0.27	0.79
Enterprise4	2.31	2.54	1.26	1.31	0.72	0.473
Enterprise5	3.21	3.77	1.01	0.93	2.37	0.021*
Enterprise6	2.32	2.32	1.25	1.21	-0.02	0.985
Organisational1	4.24	4.33	0.94	0.86	0.47	0.64
Organisational2	4.14	4.31	0.95	0.86	0.77	0.446
Organisational3	3.66	4.03	1.17	0.99	1.41	0.163
Organisational4	3.31	3.15	1.34	1.23	-0.5	0.618
Organisational5	3.52	2.97	1.3	0.93	-2.01	0.049*
Organisational6	2.75	2.74	1.14	1	-0.05	0.961

- Practical Skills: Males scored significantly higher in Practical 1 ($t = 2.11$, $p = 0.039$) and Practical 4 ($t = 2.23$, $p = 0.029$), reflecting a stronger preference for hands-on, mechanical tasks.
- Enterprising Skills: Males also scored higher in Enterprise 5 ($t = 2.37$, $p = 0.021$), indicating a greater interest in leadership and risk-taking activities.
- Organisational Skills: Females excelled in Organisational 5 ($t = -2.01$, $p = 0.049$), showing strengths in data management and system organisation.
- No significant gender differences were observed in Social, Creative, or Investigative skills, suggesting similar tendencies in these areas. This data provided a basis for understanding team dynamics and helped identify matching skills among team members.

6.4.2 *Simulation TLX Results*

The Simulation Task Load Index (SIM-TLX) is a widely used tool for measuring subjective workload across various dimensions of task load experienced by participants. It measures the following factors (Table XIV).

***Table XIV**: SIM-TLX Parameters with Description*

Parameters	Description
Mental Demands	How mentally fatiguing the task was
Physical Demands	How physically fatiguing the task was.
Temporal Demands	How hurried or rushed participants felt during the task.
Frustration	How insecure, discouraged, irritated, stressed, or annoyed participants felt.
Task Complexity	How complex the task was perceived to be.
Situational Stress	How stressed participants felt while performing the task.
Distraction	How distracting the task environment was.
Perceptual Strain	How uncomfortable or irritating the visual and auditory aspects of the task were.
Task Control	How difficult the task was to control or navigate.
Presence	How immersed or present participants felt during the task.

These dimensions provided a comprehensive assessment of how the roles of leader and follower influenced task load, and how gender impacted the perceived difficulty of the tasks.

1. Group Wise Comparisons (Leader vs Follower)

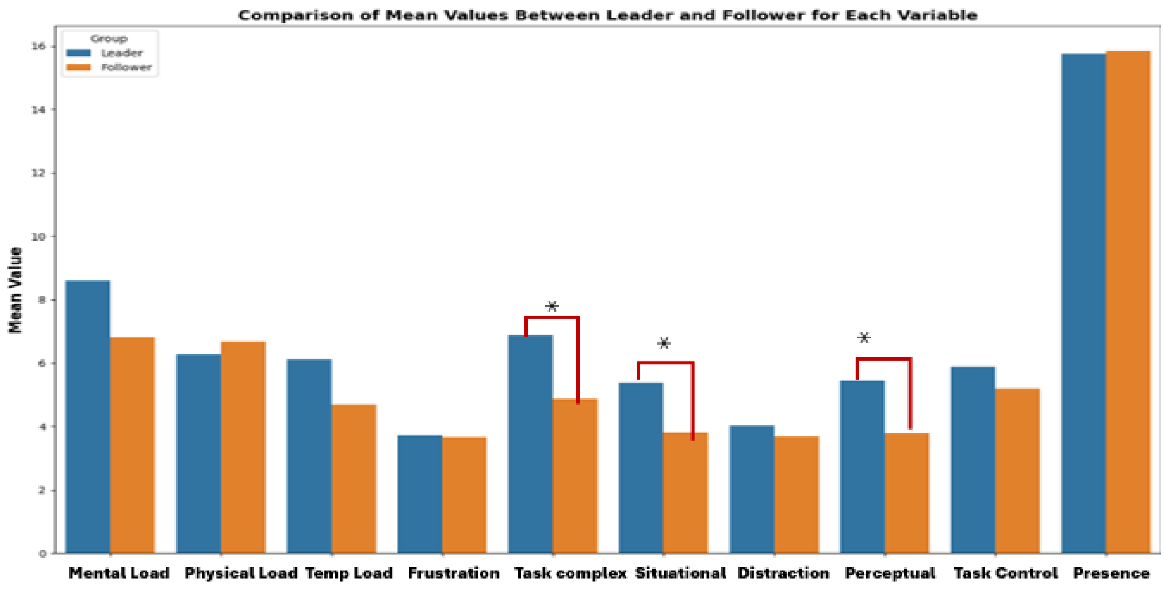

Figure 6-9*: Sim-TLX Results between the Leader vs Follower with significant differences between (Task complexity, Situational stress and Perceptual Strain)*

- Task Complexity: Leaders reported significantly higher task complexity than followers ($p = 0.045$), indicating that leadership roles were perceived as more mentally demanding. This result aligns with the expectation that leaders manage more aspects of the task, leading to increased complexity (see Figure 6-9 and Figure 6-10).
- Perceived Strain: Leaders experienced significantly higher perceived strain ($p = 0.035$), suggesting that they found the tasks more taxing.
- Situational Stress was found to be more for Leaders than Followers ($p= 0.058$)
- Other Metrics: No statistically significant differences were observed between leaders and followers in terms of mental demands, physical demands, temporal demands, frustration, distraction, task control and presence. This indicates that both groups experienced similar levels of fatigue, stress, and environmental distractions.

2. Gender Wise Comparisons (Male vs Female)

- Females vs. Males: Females reported significantly higher task complexity ($p = 0.007$), indicating they either handle more complex tasks or perceive them as more challenging. No significant differences were found in other metrics, suggesting that both genders experience similar mental load, physical demand, frustration, and presence.

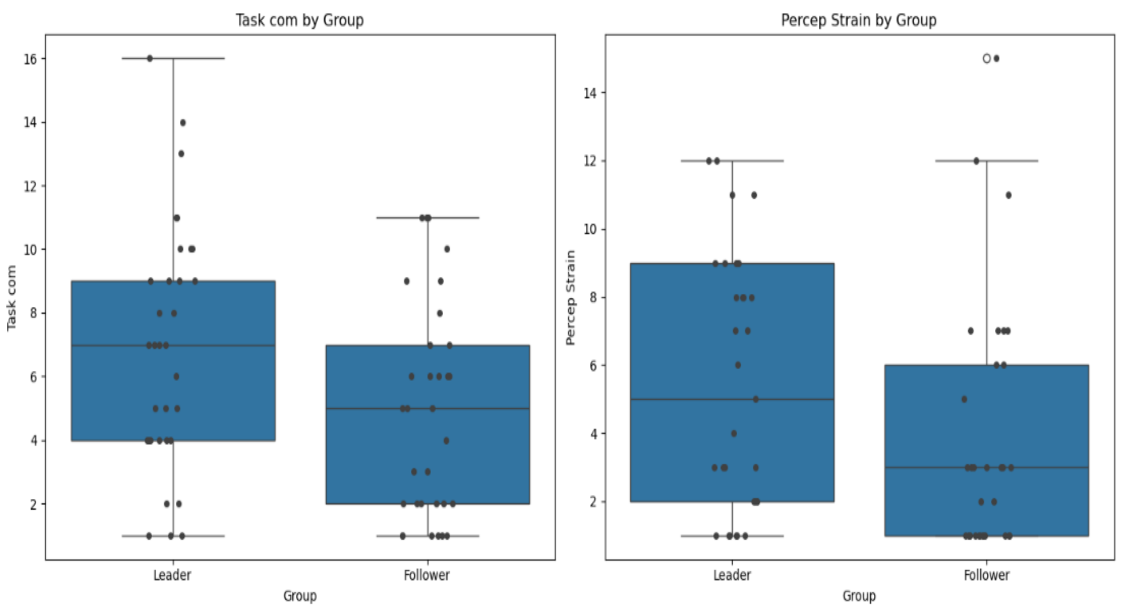

Figure 6-10: Significant Differences between Leader and Follower based on Task complexity and Perceptual Strain

- Male Leaders vs. Female Leaders: Female leaders reported higher mental load ($p = 0.020$) and physical demand ($p = 0.015$), indicating greater perceived difficulty. Male leaders, however, experienced more frustration ($p = 0.032$) and situational stress ($p = 0.048$), highlighting their emotional challenges. Female leaders also reported lower task control ($p = 0.045$) and presence ($p = 0.032$) compared to males.
- Male Followers vs. Female Followers: No significant differences were found between male and female followers, suggesting similar experiences in terms of mental load, physical demand, and other metrics.
- Male Pairs vs. Female Pairs: Male leader-follower pairs showed no significant differences in task metrics. Female pairs reported significantly higher task complexity ($p = 0.007$), indicating they either handle or perceive their tasks as more complex than male pairs.

The analysis reveals several key insights:

- Task Complexity: There is a significant difference between Female Leaders vs Female followers in perceived task complexity.
- Perceptual Strain: Leaders experienced more perceptual strain.
- Mental Load and Physical Demand: Female leaders reported higher levels compared to male leaders.
- Frustration and Situational Stress: Male leaders reported higher levels compared to female leaders.

- Task Control and Presence: Female leaders reported lower levels compared to male leaders.

These findings highlight areas where specific groups might benefit from tailored support or interventions to enhance productivity and well-being. For instance, the higher mental load and physical demand reported by female leaders suggest a need for additional resources or support. Similarly, the higher frustration and situational stress among male leaders indicate areas where stress management and coping strategies could be beneficial. Understanding these differences can help organisations create more effective support systems tailored to the needs of their leaders and followers. The detailed analysis reveals the gendered analysis depicted in Table XV.

***Table XV** Simulation Task Load Assessment Results*

Load Metric	**Leaders vs Followers p-value**	**Males vs Females p-value**	**Male Leaders vs Female Leaders p-value**	**Male Followers vs Female Followers p-value**	**Both Leader and Follower are Male p-value**	**Both Leader and Follower are Female p-value**
Mental Load	0.118	0.146	0.02*	0.082	0.082	0.146
Physical Demand	0.648	0.082	0.015	0.102	0.102	0.082
Temp Demand	0.25	0.578	0.112	0.278	0.278	0.112
Frustration	0.462	0.206	0.032*	0.075	0.112	0.206
Task Complexity	0.045	0.007	0.21	0.075	0.075	0.007*
Situational Stress	0.058	0.382	0.048*	0.098	0.098	0.382
Distraction	0.65	0.954	0.561	0.128	0.128	0.954
Perceptual Strain	0.028	0.621	0.447	0.165	0.165	0.621
Task Control	0.57	0.94	0.045	0.098	0.098	0.94

6.4.3 *Virtual Reality Neuropsychological Questionnaire (VRNQ)*

The Virtual Reality Neuropsychological Questionnaire (VRNQ) is a comprehensive tool designed to assess various factors influencing user experience, usability, and a VR environment's physical and emotional effects. The questionnaire comprises four primary categories: user experience, game mechanics, in-game assistance, and VRISE (virtual reality-induced symptoms and effects). Table XVI shows the various aspects of the primary categories.

- User Experience: The VRNQ ensures that users feel immersed in the environment.
- Game Mechanics: The evaluation of game mechanics ensures that users can navigate, interact with objects, and move in the virtual environment.
- In-Game Assistance: Effective tutorials and in-game instructions ensure that users understand the VR system's features and can complete tasks, enhancing their overall experience.
- VRISE: Measures symptoms like nausea, dizziness, fatigue, and instability[204] .

***Table XVI**: VRNQ Categories*

Domains	Criteria 1	Criteria 2	Criteria 3	Criteria 4	Criteria 5
User Experience	An adequate level of immersion	Pleasant VR experience	High quality graphics	High quality sounds	Suitable hardware (HMD and computer)
Game Mechanics	A suitable navigation system (e.g., teleportation)	Availability of physical movement	Naturalistic picking/placing of items	Naturalistic use of items	Naturalistic 2-handed interaction
In-Game Assistance	Digestible tutorials	Helpful tutorials	Adequate duration of tutorials	Helpful in-game instructions	Helpful in-game prompts
VRISE	Absence or insignificant presence of nausea	Absence or insignificant presence of disorientation	Absence or insignificant presence of dizziness	Absence or insignificant presence of fatigue	Absence or insignificant presence of instability

The results of the four factors are provided as follows:

- User Experience: The comparison between leaders and followers in the User Experience category showed no significant difference, with normalised mean scores of 5.48 for leaders and 5.56 for followers, $t(64) = -0.44$, $p = 0.66$. Gender comparisons within both groups revealed no significant differences. Male leaders had a mean score of 5.39 and female leaders 5.60, with a non-significant $t(64) = -0.81$, $p = 0.42$. Similarly, male followers scored 5.45 and female followers scored 5.74, with no significant difference, $t(64) = -1.16$, $p = 0.26$. Overall, males had a slightly lower mean score of 5.42 compared to females at 5.66, but this difference was also not statistically significant, $t(64) = -1.37$, $p = 0.18$.

- Game Mechanics: In the Game Mechanics category, leaders had a mean score of 5.76, while followers scored 5.67, with no significant difference between the groups, $t(64)$ = -0.44, p = 0.66. The comparison between male and female leaders showed similar scores of 5.79 for males and 5.72 for females, $t(64)$ = 0.27, p = 0.79. Likewise, male and female followers both scored 5.67, showing no significant difference, $t(64)$ = 0.005, p = 0.99. Overall, males scored 5.73 and females scored 5.70, with no significant gender difference, $t(64)$ = 0.15, p = 0.88.
- In-Game Assistance: Leaders had a mean score of 5.65, while followers scored slightly higher at 5.74 in the In-Game Assistance category, though the difference was not significant, $t(64)$ = -0.44, p = 0.66. Male leaders (5.86) scored higher than female leaders (5.40), showing a marginally significant difference, $t(64)$ = 1.96, p = 0.06. No significant differences were observed between male followers (5.75) and female followers (5.72), $t(64)$ = 0.09, p = 0.93. Comparing males (5.80) and females (5.55) overall, the difference approached significance but was not statistically significant, $t(64)$ = 1.35, p = 0.18.
- VRISE (Virtual Reality-Induced Symptoms and Effects): In the VRISE category, leaders had a mean score of 6.31, while followers scored 6.28, with no significant difference, $t(64)$ = -0.44, p = 0.66. Male leaders (6.41) scored higher than female leaders (6.19), but the difference was not significant, $t(64)$ = 0.87, p = 0.39. Likewise, no significant differences were found between male followers (6.39) and female followers (6.12), $t(64)$ = 0.83, p = 0.41. Overall, males scored slightly higher (6.40) than females (6.16) on the VRISE scale, but the difference was not significant, $t(64)$ = 1.20, p = 0.23 Figure 6-11 illustrates the variation between Leader and Follower.

***Table XVII**: Leader vs Follower Comparison for VRNQ*

Comparison	Category	Mean Score (Leaders)	Mean Score (Followers)	t-statistic	p-value
Leader vs Follower	User Experience	5.484848	5.563636	-0.44249	0.659624
Leader vs Follower	Game Mechanics	5.757576	5.674242	0.376	0.708161
Leader vs Follower	In-Game Assistance	5.648485	5.739394	-0.49154	0.624727
Leader vs Follower	VRISE	6.309091	6.284848	0.119955	0.904894

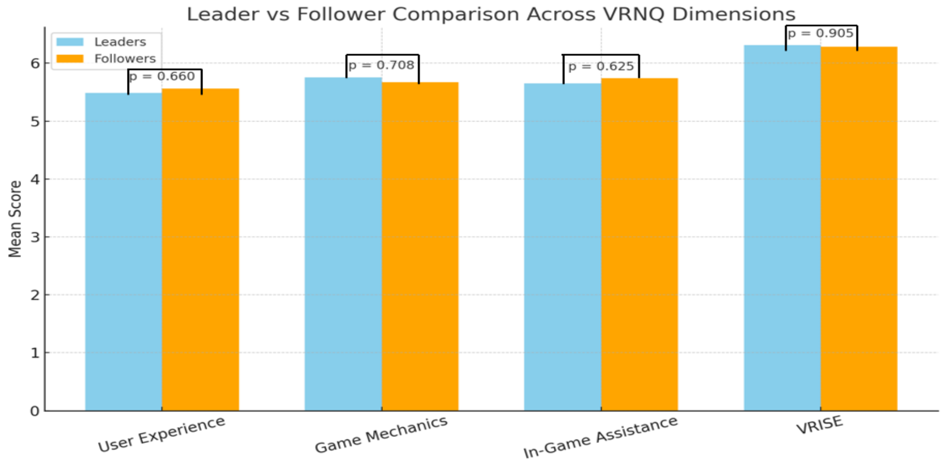

Figure 6-11: *Leader Vs Follower Comparison for VRNQ Questionnaire*

Table XVIII: *VRNQ Results for Male vs Female.*

Comparison	Category	Normalised Mean Score (Males)	Normalised Mean Score (Females)	t-statistic	p-value
Male vs Female	User Experience	5.42	5.66	-1.36779	0.176161
Male vs Female	Game Mechanics	5.73	5.70	0.150759	0.88064
Male vs Female	In-Game Assistance	5.80	5.55	1.352537	0.180963
Male vs Female	VRISE	6.40	6.16	1.200998	0.234178

Overall, no significant differences were found between Leaders and Followers or between Males and Females, as shown Table VII and Table XVIII. The VRNQ scores indicated no statistically significant variation across roles or gender in any of the four evaluated categories: User Experience, Game Mechanics, In-Game Assistance, and VRISE (VR-Induced Symptoms and Effects)

Leader vs Follower: Mean scores were closely aligned across all categories, with *p-values* well above the 0.05 threshold, indicating no meaningful differences in perceived VR quality based on task role.

- **Male vs Female**: Although females reported slightly higher *User Experience* scores and males reported slightly higher scores for *In-Game Assistance* and *VRISE*, none of the differences were statistically significant.

Overall, both role and gender had minimal impact on users' subjective evaluations of the VR system. The term "In-Game Assistance" refers to participants' perceptions of how well the system provided guidance, hints, or support during gameplay—such as clear instructions, tooltips, or feedback mechanisms that aided task execution within the VR environment.

***Table XIX**: Leaders vs Followers Differences*

	Variation	Statistic	p-value
Pupil Dilation	Mann-Whitney U	254.0	0.0215
HR	Mann-Whitney U	0.939	0.350
EDA	Mann-Whitney U	215.5	0.034
BVP	Mann-Whitney U	529.0	0.360
IBI	Mann-Whitney U	489.0	0.044
SKIN TEMP	Mann-Whitney U	441.0	0.021
ACC	Mann-Whitney U	580.0	0.6535

6.4.1 *Leader vs Follower Physiological Metrics Comparison*

Physiological signals were collected through the E4 wristband. Normality testing using the Shapiro Wilk Test was performed to determine whether a dataset is normally distributed [205]. Based on the results of the normality tests, appropriate statistical tests were selected for each metric comparison. All signals fall into the normal distribution for both Leaders and Followers Groups. So, Mann-Whitney U test was selected, calculating z-scores for all signals using the baseline values [206]. The physiological responses measured between the two groups are shown in Error! Reference source not found. , Figure 6-12 and Figure 6-13.

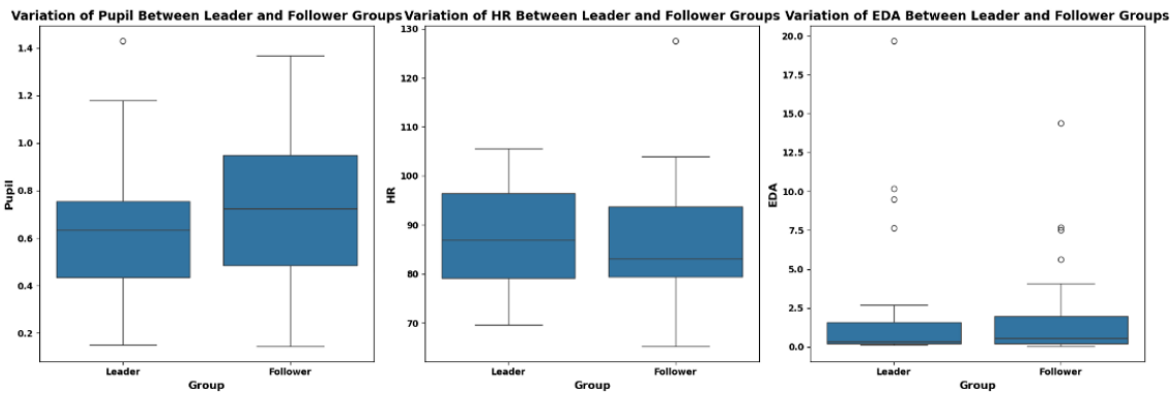

***Figure 6-12**: Variation of Pupil Dilation, HR, and EDA between Leader and Follower Group*

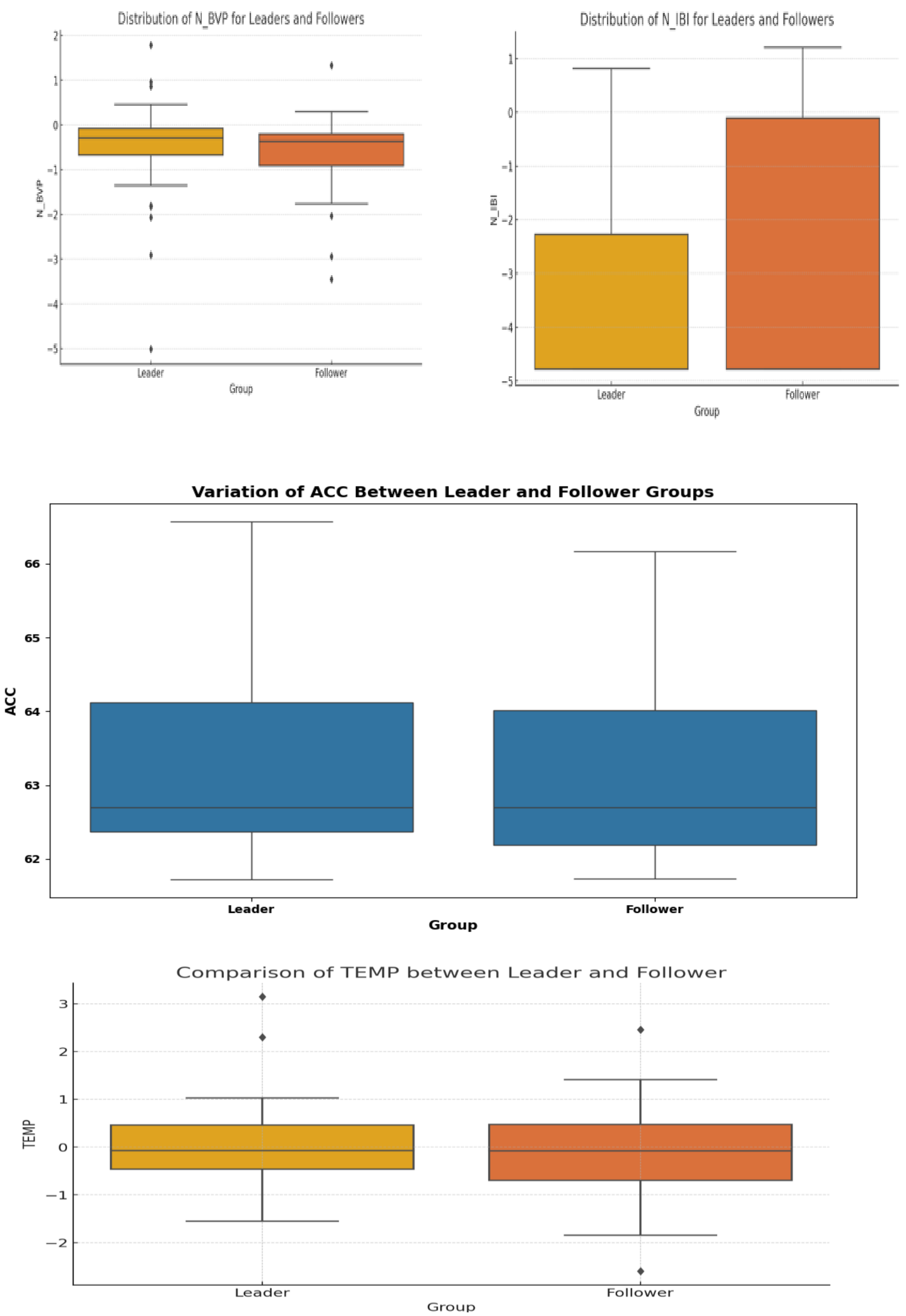

Figure 6-13: *BVP, IBI, ACC, and Temp variation between Leader vs Follower Group*

When comparing the physiological results with the subjective workload measures from the SIM-TLX (Simulation Task Load Index), there are notable correlations and contrasts that provide insight into the cognitive and emotional demands experienced by Leaders and Followers in a VR environment.

1. Pupil Dilation (p = 0.0215) vs. SIM-TLX: The significant difference in pupil dilation between Leaders and Followers suggests higher cognitive load for Leaders. This is consistent with the SIM-TLX dimension of Task Complexity (p = 0.045), indicating that Leaders found the task more mentally demanding. Pupil dilation is a known physiological marker of cognitive effort, typically increasing during tasks that require more attention and decision-making [207]. Although Mental Load in the SIM-TLX (p = 0.118) was not significant, the objective physiological data (larger pupil dilation) indicate that cognitive demands were present, potentially beyond what participants perceived consciously. This highlights the value of incorporating implicit physiological metrics alongside subjective self-reports.
2. HR vs. SIM-TLX parameters: No significant difference in HR was found between the Leader and Follower groups (p = 0.350), which aligns with several non-significant SIM-TLX dimensions, such as Physical Demand (p = 0.648) and Distraction (p = 0.65). This suggests that the VR task did not impose significantly different levels of physical strain or mental distraction between the roles. Previous research indicates that heart rate is not always the best indicator of cognitive load unless the task is particularly demanding or stressful [207] .
3. EDA vs. SIM-TLX parameters: The significant difference in EDA between Leader Follower groups indicates greater physiological arousal in Leaders and suggests higher emotional or cognitive stress (p = 0.034). This aligns with the near-significant Situational Stress (p = 0.058) dimension in the SIM-TLX, which indicates that Leaders may have perceived some stress during the VR task. EDA is a well-known indicator of autonomic nervous system activation, reflecting emotional and attentional engagement [208]. However, the non-significant Frustration (p = 0.462) score indicates that while physiological stress was present, participants may not have consciously perceived themselves as frustrated.
4. BVP vs. SIM-TLX parameters: There was no significant difference in BVP between Leaders and Followers (p = 0.360). This corresponds with the non-significant SIM-TLX scores for Physical Demand (p = 0.648) and Task Control (p = 0.57). BVP is linked to vascular responses, and the lack of difference suggests that both roles experienced similar

levels of physiological activation related to blood flow. This supports the idea that neither role experienced significantly different physical demands or control over task outcomes.

5. IBI vs. SIM-TLX parameters: A significant difference in IBI indicates higher HRV in Leaders than Followers, suggesting more significant autonomic stress or cognitive demand (p=0.044). This is consistent with the Task Complexity (p = 0.045) rating in the SIM-TLX, implying that Leaders found the task more complex, which likely required greater cognitive effort and stress regulation. HRV is a well-established marker of stress, with higher variability often indicating a greater need for cognitive control [209]Interestingly, Mental Load (p = 0.118) was not significant, suggesting that participants may have underestimated the physiological stress response.
6. Skin Temperature (ST) vs. SIM-TLX: The significant difference in ST between Leaders and Followers points to differing thermoregulatory responses, potentially driven by stress (p= 0.021). ST often decreases under stress due to vasoconstriction [210]. While the SIM-TLX Temperature Demand dimension was not significant (p = 0.25), suggesting that participants did not perceive temperature as a significant factor, the physiological response indicates underlying stress-related thermoregulation in Leaders.
7. ACC vs. SIM-TLX: No significant difference in movement ACC was observed between Leaders and Followers (p = 0.6535), which aligns with the non-significant Physical Demand (p = 0.648) and Task Control (p = 0.57) dimensions in the SIM-TLX. Both the physiological and subjective data indicate that neither group experienced a marked difference in physical exertion or the amount of physical control they had over the task, further suggesting that the task demands in VR were cognitively and emotionally oriented rather than physically demanding.

The physiological results, particularly in pupil dilation, EDA, IBI, and ST, highlight significant differences between Leaders and Followers, suggesting that Leaders experienced higher cognitive and emotional demands (Figure 6-9). These findings align with SIM-TLX dimensions such as Task Complexity and Perceptual Strain, which were also significant, supporting the notion that Leaders faced greater cognitive challenges. However, some subjective measures, such as Mental Load and Frustration, did not align with the physiological data, suggesting that participants may not fully perceive the extent of their physiological stress responses. This discrepancy between objective and subjective measures is commonly observed in workload studies, where physiological indicators often capture unconscious stress responses that participants may not consciously report [211]. Overall, the combination of physiological and subjective data provides a comprehensive understanding of the different experiences faced by Leaders and Followers in the VR environment.

6.4.2 *Voice Metrics through Audio Extraction from the VR collaboration*

During the collaborative VR tasks, participants' verbal interactions were recorded using Audacity, capturing the full scope of audio conversations. These recordings were transcribed using AssemblyAI, an advanced speech-to-text API, which produced timestamped text outputs. The resulting transcripts were then processed using the Natural Language Toolkit (NLTK), where standard preprocessing steps were applied—including tokenisation, stop word removal, stemming, part-of-speech tagging, and named entity recognition. This prepared the data for sentiment analysis, enabling the classification of utterances into positive, neutral, or negative categories, thus capturing the emotional tone of participant communication (Figure 6-14).

In contrast to the findings from Study 1, no significant correlation was observed between positive sentiment and total task time, suggesting that overall task duration was not directly associated with affective tone. However, consistent with earlier findings, a weak positive correlation was found between negative sentiment and the total number of controller clicks ($r = 0.22$), indicating that increased interaction effort may be linked to expressions of frustration or difficulty.

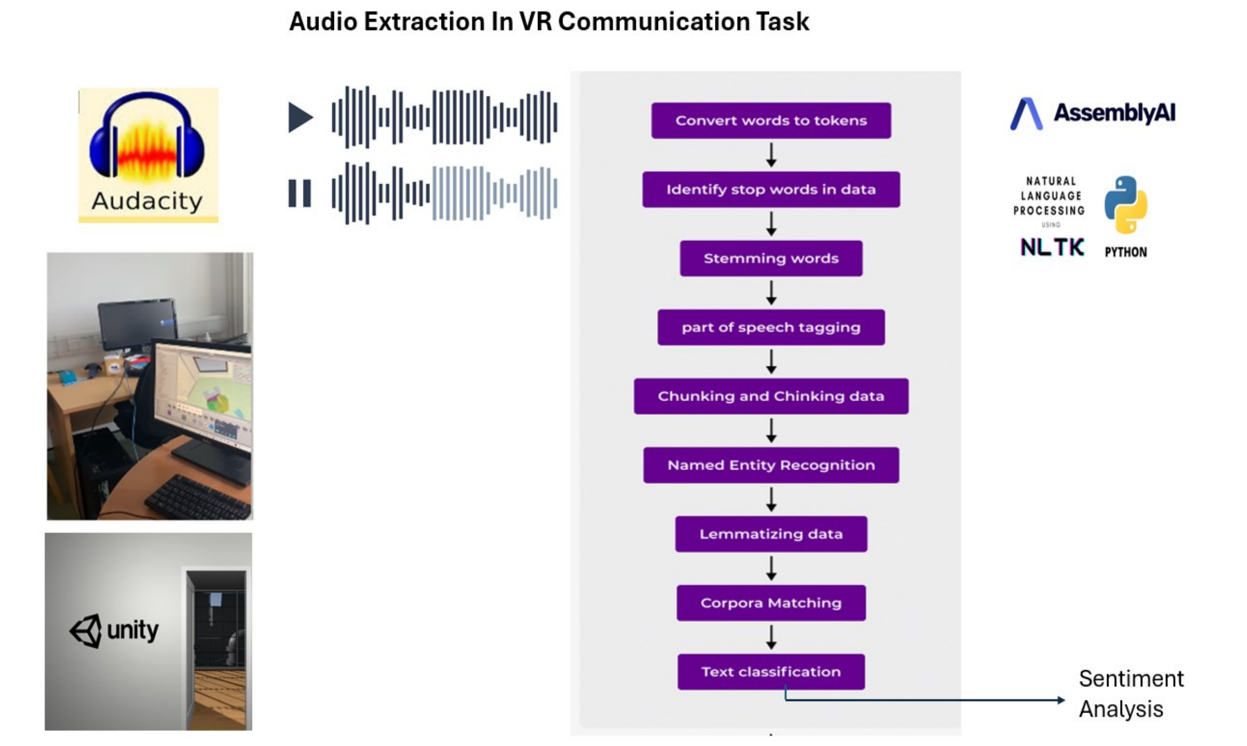

Figure 6-14: *Audio Extraction Using Audacity in Unity Application and then pre-processed using AssemblyAI and NLTK for Sentiment Analysis*

Notably, a moderate negative correlation was observed between positive sentiment and task complexity (r = -0.46), implying that as participants perceived tasks to be more complex, the frequency of positive expressions declined. This suggests that increased cognitive load or decision-making challenges may dampen overt positive emotional responses during collaboration. Conversely, negative sentiment exhibited a weaker positive correlation with task complexity (r = 0.27), suggesting only a slight increase in negative expressions with rising task difficulty.

Additionally, a moderate negative correlation was found between positive sentiment and distraction (r = -0.36), indicating that higher environmental or cognitive distraction levels were associated with fewer positive utterances.

Overall, these findings reveal that task complexity and distraction have a more pronounced effect on reducing positive emotional tone rather than significantly increasing negative

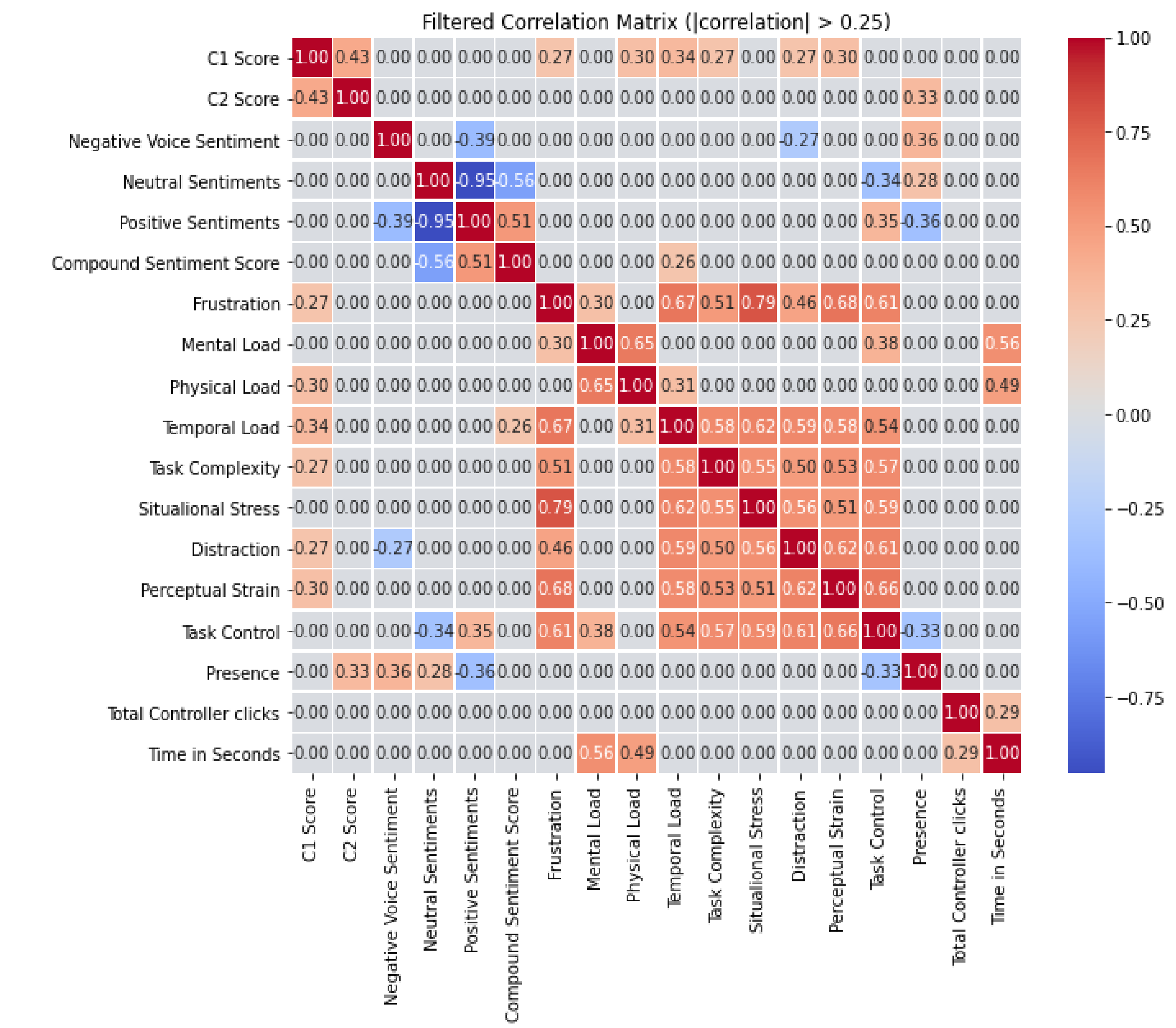

***Figure 6-15**: Correlation Matrix with Voice Sentiments, SIM-TLX factors, Total Controller Clicks, and Total Time in seconds*

sentiment as indicated by the heatmap (Figure 6-15). Notably, total task time did not demonstrate any strong relationship with either sentiment type, suggesting that perceived task demands more influence affective tone during collaboration than by duration.

6.4.3 *Synchrony Metrics for Different Collaboration (Open workspace vs Closed workspace)*

This section presents the results of subjective and physiological synchrony analyses, comparing the Closed (no visual access) and Open (visual access) collaboration conditions in VR. The findings highlight how visual interaction influences perceived collaboration quality and physiological alignment between participants.

6.4.3.1 Subjective Collaboration Experience- Customised Collaboration Questionnaire

To capture participants' subjective impressions of the collaboration experience, a customised questionnaire was administered at the end of both phases. Participants rated six aspects on a 5-point Likert scale, including scene preference, self-contribution, partner contribution, and overall effectiveness.

- **Scene Preference**: A substantial majority—**85.3% (53 out of 68)**—preferred the **Open scene**, indicating strong favour for visual interaction during collaboration (Figure 6-16).
- **Self-Contribution Ratings**:
  - Closed Scene: Mean = 3.6 (SD = 0.7)
  - Open Scene: Mean = 4.4 (SD = 0.5)

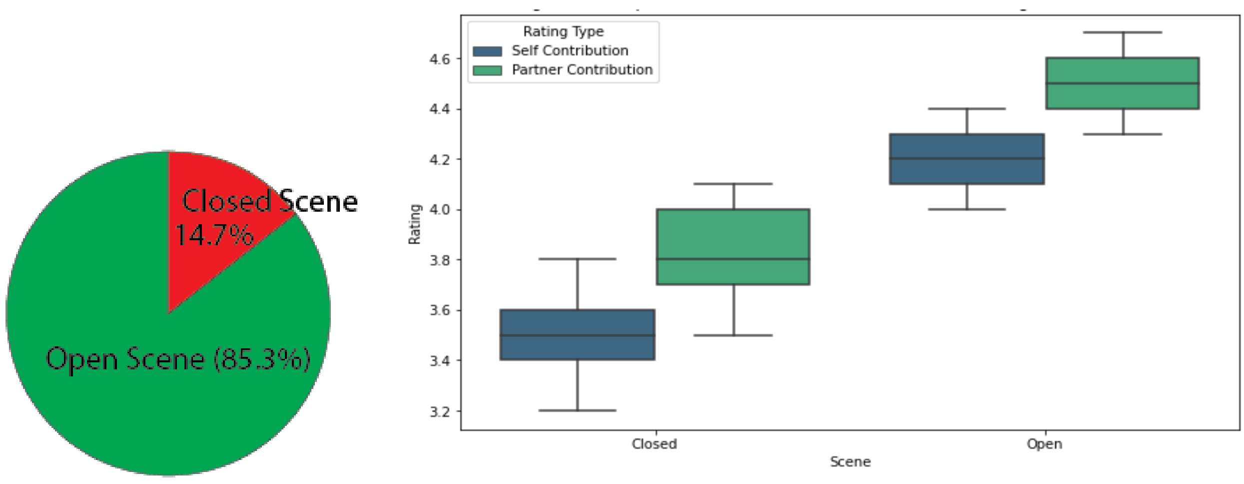

Figure 6-16*: Collaboration Questionnaire Results showing the Preference of Participants of Open Workspace over Closed Workspace*

- **Partner Contribution Ratings**:
  - Closed Scene: Mean = 3.8 (SD = 0.6)
  - Open Scene: Mean = 4.5 (SD = 0.4)

The difference in ratings between the scenes was statistically significant (**p < 0.001**), suggesting that visual contact positively influenced participants' perceptions of both their own and their partner’s effectiveness. These subjective findings align with the physiological and task performance data, reinforcing the importance of visual cues in enhancing collaboration quality.

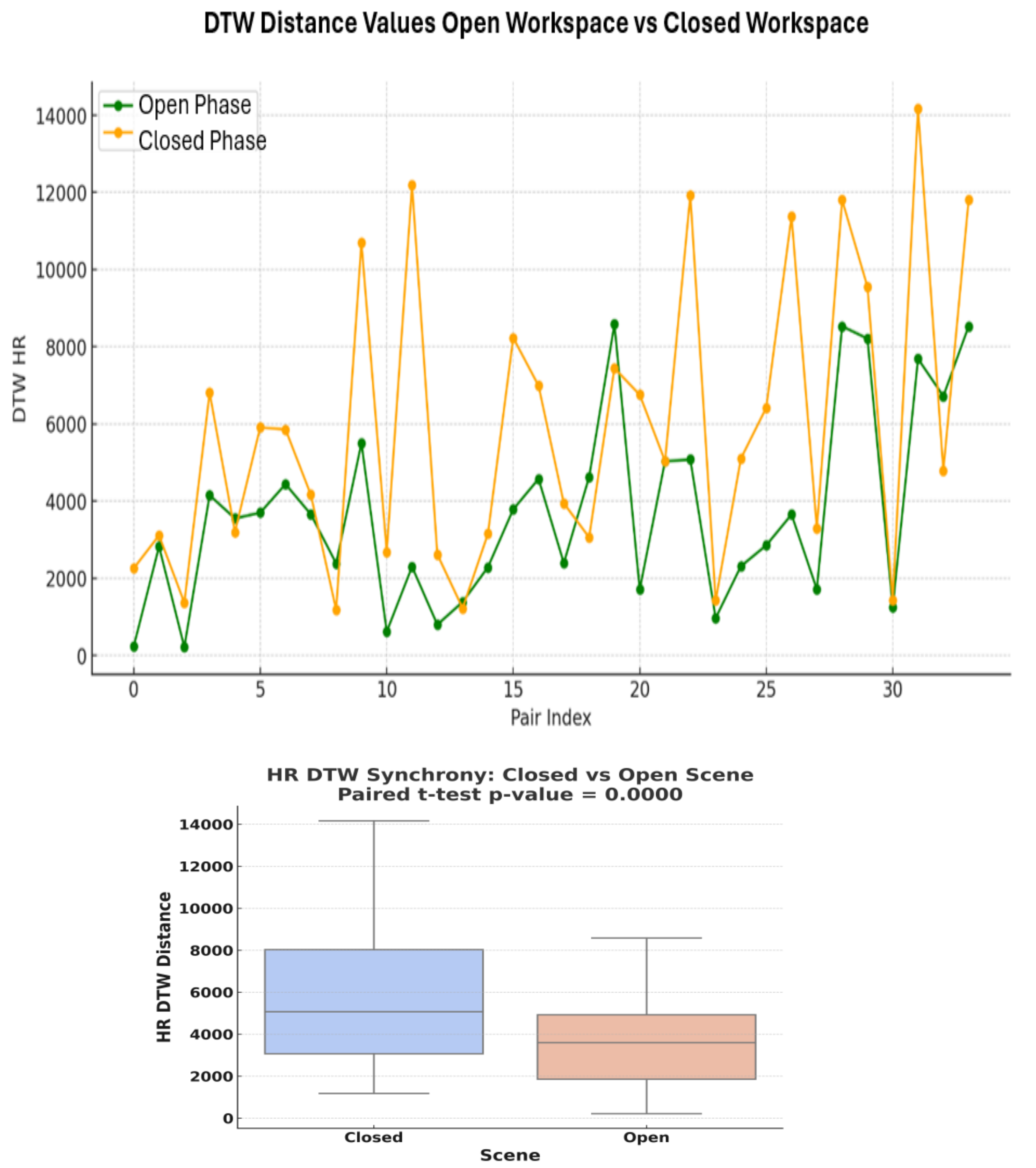

Figure 6-17*: DTW Distance between HR signals in Open Workspace vs Closed Workspace with Box plot*

6.4.3.2 Heart Rate Synchrony

Physiological synchrony was examined through DTW distance comparisons between participant pairs for HR signals in both conditions.

- Closed Phase: The mean DTW distance for HR signals was 5909.42 (SD = 3802.21)
- Open Workspace: The mean DTW distance decreased to 3711.50 (SD = 2491.46)

A paired t-test confirmed that this reduction was statistically significant (t(33) = 5.72, p < 0.001), indicating enhanced HR synchrony when visual contact was available (Figure 6-17). This enhancement aligns with previous findings by [212], who demonstrated increased physiological synchrony with visual interaction in collaborative tasks in real-life scenarios (Figure 6-17). The individual raw data of all participants were given in **Appendix N**.

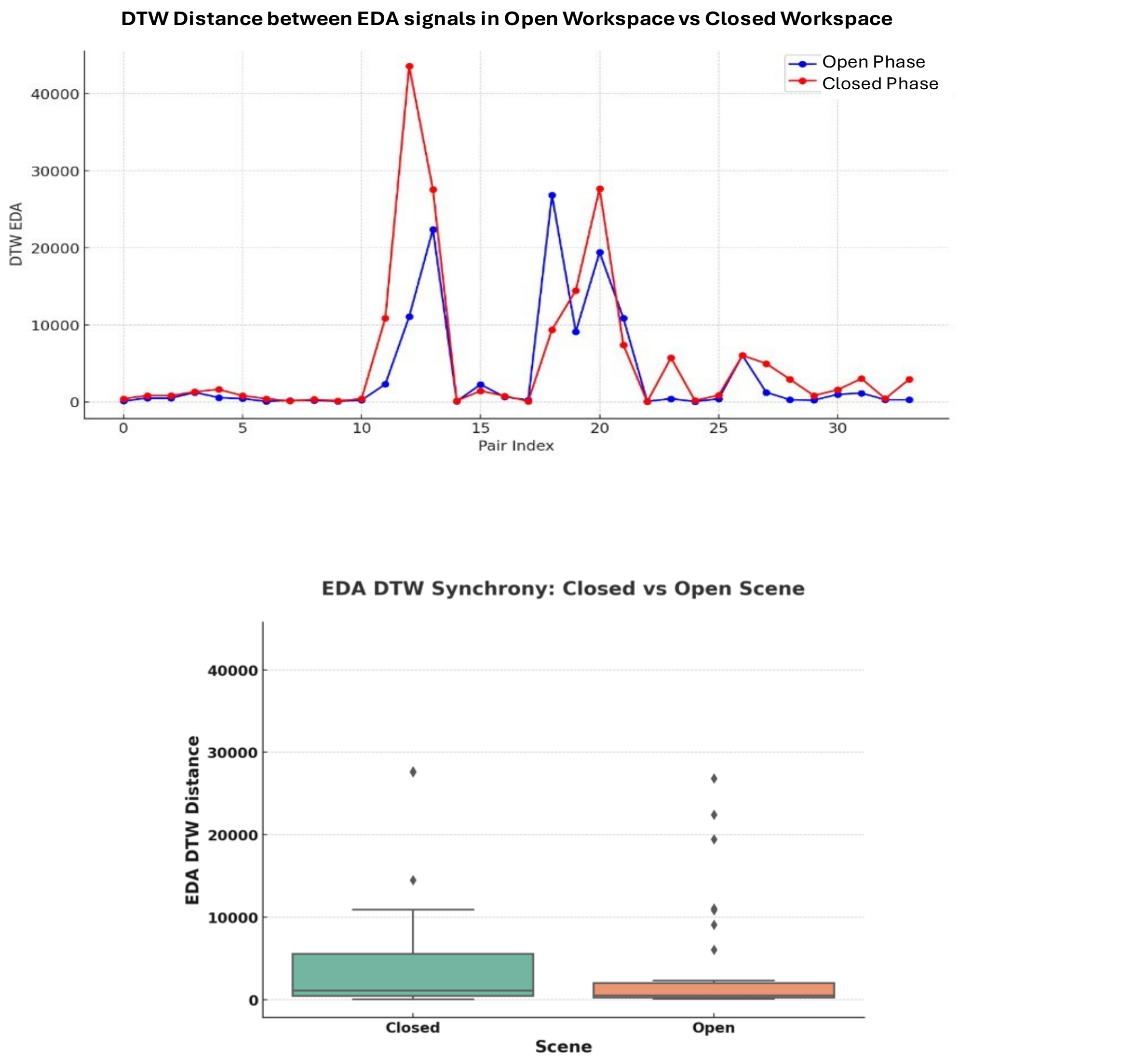

***Figure 6-18**: Figure showing how EDA Synchrony between Leader and Follower Closed Test Scene vs when they are in Open scene.*

6.4.3.3 Electrodermal Activity Synchrony

Similarly, EDA synchrony showed notable improvements:

- Closed Phase: The mean DTW distance for EDA signals was 5313.58 (SD = 9651.77)
- Open: The mean DTW distance reduced to 3567.97 (SD = 6847.10)

The reduction was statistically significant ($t(67) = 6.45$, $p < 0.001$). These results are consistent with the work of [213], who highlighted the role of nonverbal cues, including visual contact, in enhancing physiological synchrony during cooperative activities during real-life team tasks (Figure 6-18). The individual raw data of all participants were given in and pair-wise DTW computations for both EDA and HR are detailed in **Appendix O** .

The significant reduction in DTW distances for both HR and EDA across conditions demonstrates that visual interaction in VR substantially increases physiological synchrony. This enhanced alignment has been linked to improved social cohesion, mutual understanding, and task performance in collaborative settings [144], [179], [214].

These results, when considered alongside the subjective preference data and task efficiency metrics, underscore the value of visual access as a key factor in designing effective and immersive collaborative VR experiences.

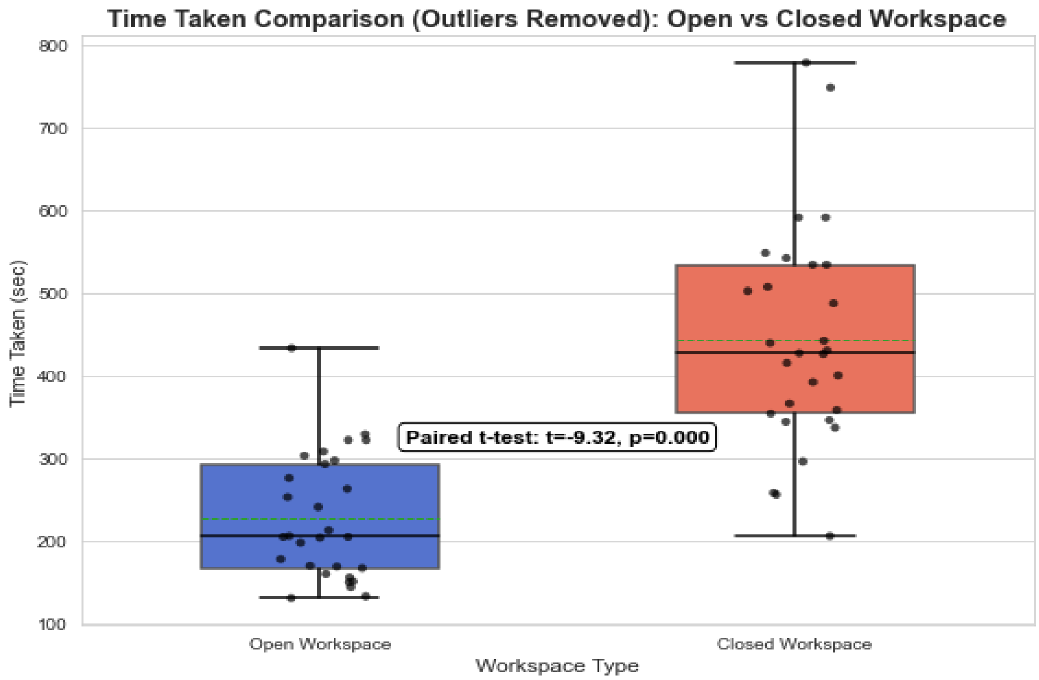

***Figure 6-19**: Task completion Time in Open Workspace vs Closed Workspace*

6.4.3.4 Task Performance

Task performance was evaluated based on task completion time and the number of clicks recorded during each collaboration phase (Figure 6-19).

Task Completion Time

- Closed Phase: The average task completion time was 8.12 minutes (SD = 3.09).
- Open Phase: The average completion time significantly decreased to 4.35 minutes (SD = 1.92)

A paired t-test indicated that this improvement was statistically significant ($t(33) = -8.05$, $p < 0.001$), demonstrating a 21.6% reduction in completion time when visual contact was present. This finding supports prior research by [215] , which emphasised the importance of visual cues in enhancing collaborative task efficiency [216], [217] (Figure 6-15)

Number of Clicks

- Closed Phase: Participants averaged 124 clicks (SD = 25).
- Open Phase: The average number of clicks was reduced to 76 (SD = 20).

The reduction was statistically significant ($t(33) = 4.82$, $p < 0.001$), indicating more efficient and direct interactions in the open scene. This efficiency aligns with findings from [217], highlighting that visual interaction reduces unnecessary actions and streamlines collaboration processes (Figure 6-20).

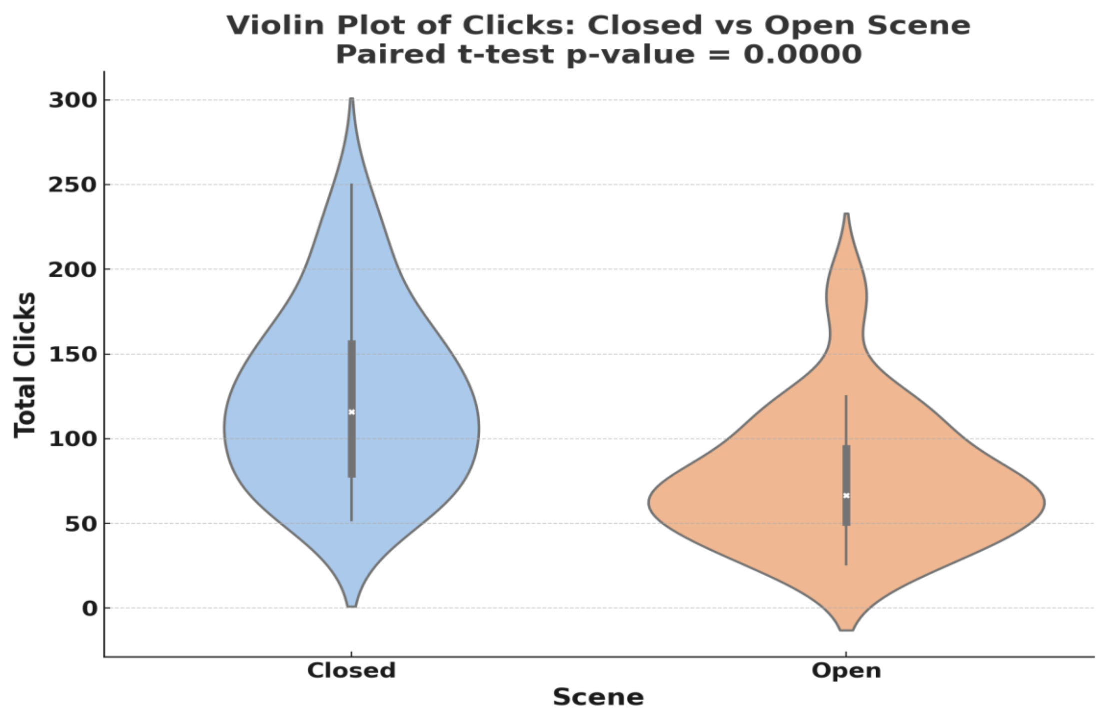

***Figure 6-20**: Significant difference between the number of Clicks in the Closed vs Open Scene significantly higher number of clicks in the closed scenario than open scenario.*

An analysis of physiological synchrony using EDA DTW distance reveals a consistent trend between the open (shared) and closed collaboration scenes. Lower DTW distances in the open scene across participant pairs indicate greater EDA synchrony, reflecting a closer alignment in emotional and arousal responses when visual contact is available. This increased synchrony in the open condition is also associated with shorter task completion times and fewer clicks, suggesting enhanced collaboration efficiency and lower interaction load. Statistical analysis using a paired t-test confirms the significance of this difference in click counts ($p < 0.05$), highlighting that visibility plays a critical role in facilitating mutual understanding and coordinated action. These findings suggest that visual access in collaborative VR tasks enhances physiological alignment and improves task performance (Figure 6-21).

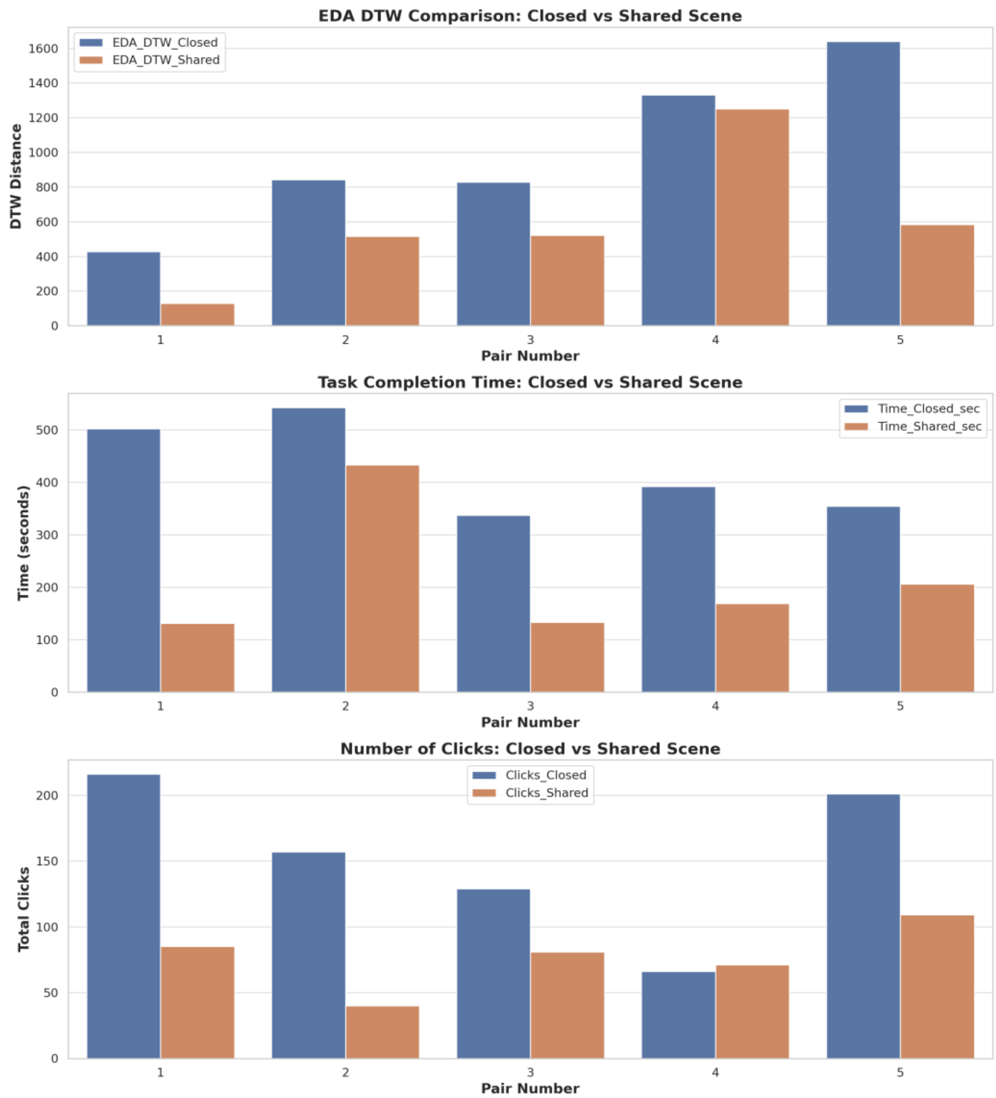

Figure 6-21: *EDA Synchrony with Time Completion and Total Clicks for the first five pairs*

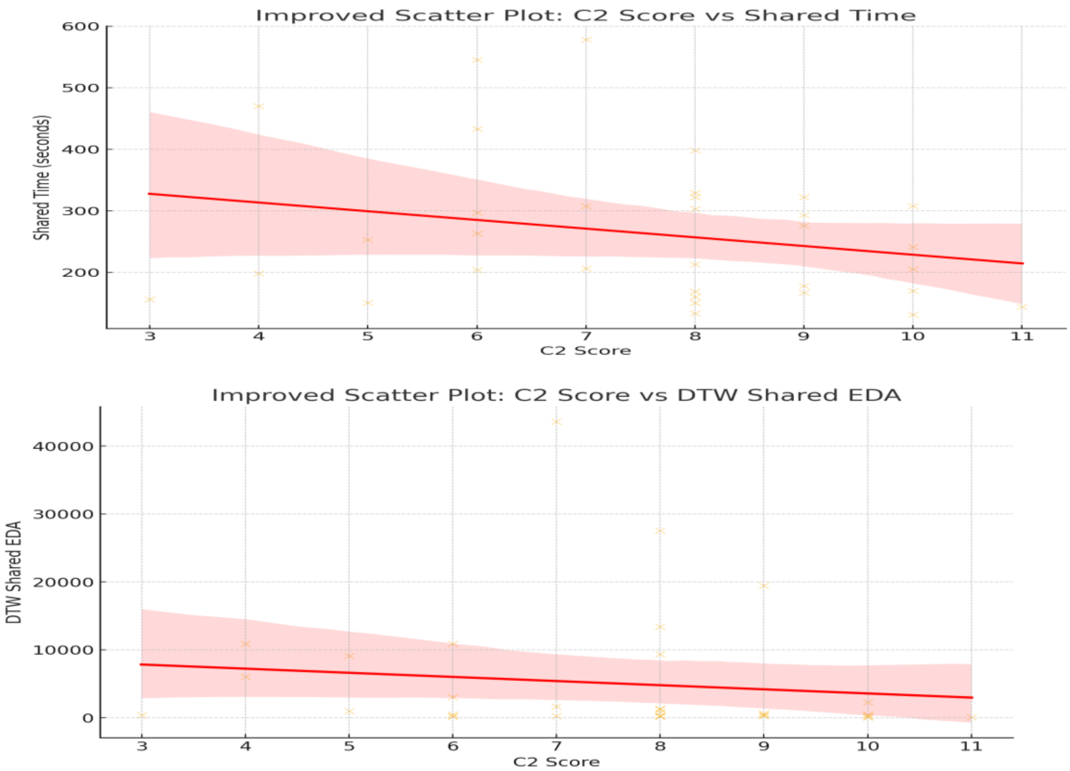

Figure 6-22: *Correlation plot showing negative correlation (0.32) between Commonality Score and Shared Task completion time and the second one is Commonality score vs DTW of Shared EDA*

6.4.3.5 Commonality Score and Synchrony of EDA Signal:

Study 1 and Study 2 both reveal a consistent trend: as the commonality score between collaborators increases—representing shared attributes such as similar skill levels, prior VR experience, demographic similarity, and pre-existing familiarity—the task completion time decreases. In Study 2, this relationship is particularly evident in the open (shared visibility) scenario, where higher commonality scores are associated with faster task completion. Furthermore, increased commonality is also linked to decreased EDA DTW values, indicating greater physiological synchrony between collaborators. These findings suggest that shared experiential and social factors significantly enhance collaboration efficiency and emotional alignment during VR-based tasks, as Figure 6-22 illustrates.

6.4.3.6 Pupil Dilation Synchrony

PD synchrony was calculated using the R-squared method, which measures the correlation between the pupil dilation patterns of two participants. Unlike HR data, which may experience

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2]\,[n\sum y^2 - (\sum y)^2]}}$$

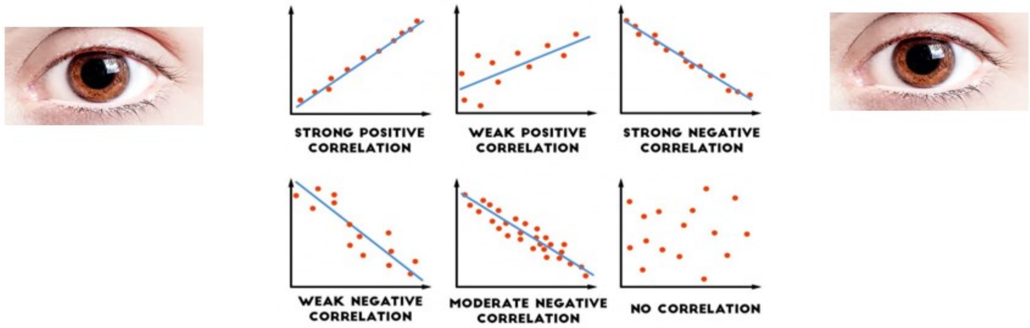

Figure 6-23: *Pupil Dilation Synchrony computed using R-squared*

non-linear variations, pupil dilation is generally more stable and linear, making it suitable for correlation-based methods. An R-squared value closer to 1 indicated higher synchrony, signifying better cognitive alignment during the task [189]. Inspired by a prior study [157], we used R-squared to compute Pupil Dilation Synchrony. Figure 6-23 shows the R-squared synchrony computation visualisation in the overall architecture.

Pupil Dilation (PD) Synchrony: A strong negative correlation was found between PD synchrony and mental load ($r = -0.82$, $p < 0.001$), showing that tasks with higher cognitive demands resulted in lower PD synchrony [218]. PD synchrony was also moderately negatively correlated with physical load ($r = -0.58$, $p < 0.001$), indicating that increased physical effort

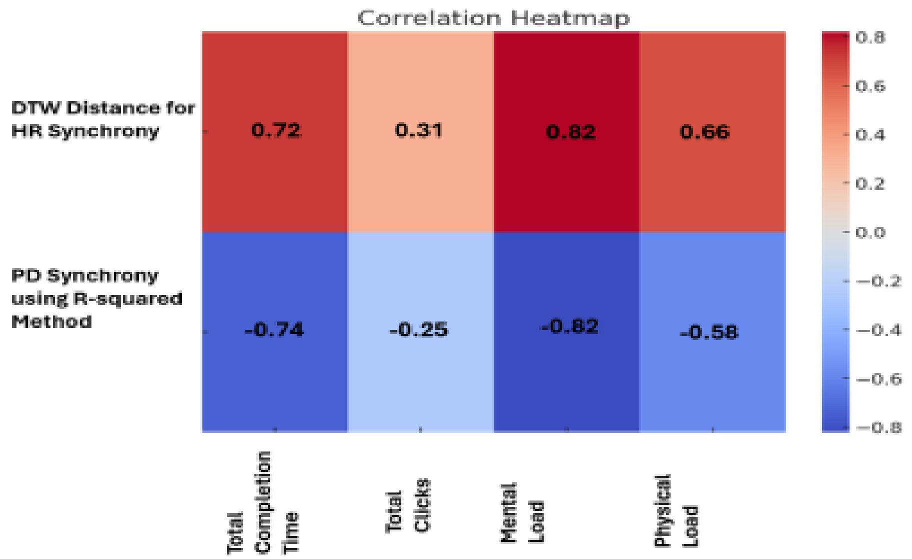

Figure 6-24: *Correlation Heatmap between PD Synchrony, HR Synchrony, Mental Load, Physical Load, Clicks, and Total time for Completion*

decreased pupil dilation synchrony (Figure 6-24). The correlation between PD synchrony and total clicks was weak and non-significant ($r = -0.25$, $p = 0.15$), suggesting little impact of task interaction on PD synchrony [219].

6.4.4 *Pupil Dilation and Blink Rate of Participants in Closed Space Vs Open Workspace.*

In Study 2, pupil dilation and blink rate were continuously recorded using the built-in eye tracker in the VR headset to assess real-time cognitive and emotional responses during collaborative tasks. These physiological signals served as non-invasive indicators of mental workload and stress under two workspace conditions: open (shared visibility) and closed (audio-only). The results revealed that pupil dilation significantly increased in the closed condition, reflecting elevated cognitive effort due to the lack of visual cues and increased reliance on verbal coordination. This effect was particularly pronounced over time, indicating sustained mental demand as participants progressed through the task.

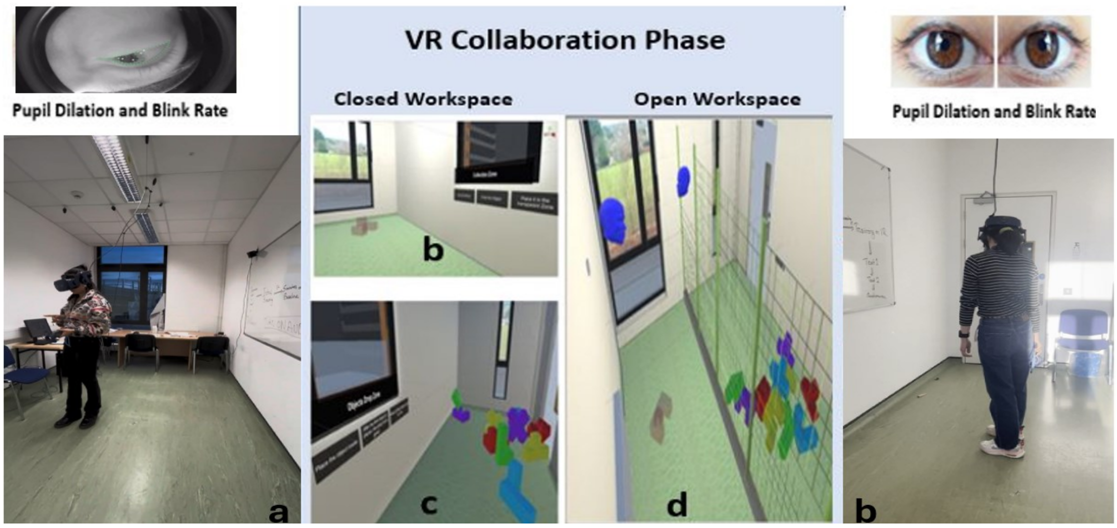

Figure 6-25: *Pupil Dilation and Blink Rate across Closed Workspace vs Open Workspace Collaboration Scenario.*

Figure 6-25 illustrates the setup and conditions of the VR collaboration task used in Study 2, where pupil dilation and blink rate were continuously monitored to assess participants' cognitive and emotional states. Panels a and b show participants in their respective physical spaces wearing VR headsets equipped with built-in eye trackers, enabling the real-time collection of ocular metrics. The central panel highlights the two collaboration conditions: Closed Workspace (b, c) and Open Workspace (d). In the Closed Workspace, users could not see each other or the shared task space, relying solely on audio communication, significantly increasing cognitive load and stress, as evidenced by elevated pupil dilation and blink rate. In contrast, the Open Workspace condition offered shared visual access to the task space and avatar representations of the collaborators, resulting in lower physiological indicators of workload.

Despite their potential, pupil dilation and blink rate have been underutilized in VR research, especially in collaborative scenarios. Although some studies have explored pupil dilation for workload assessment, the implications of blink rate—particularly its increase due to mental

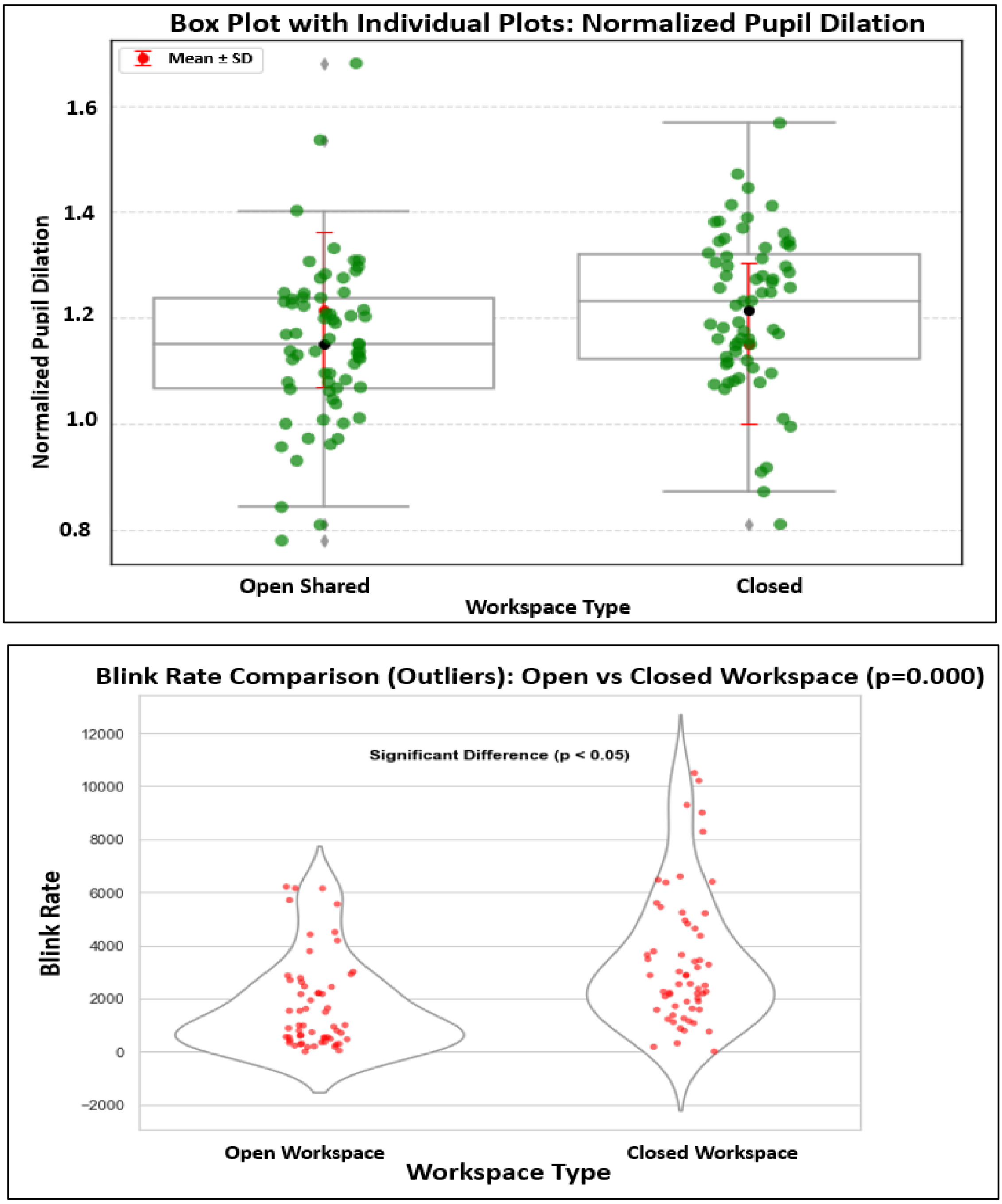

***Figure 6-26**: Pupil Dilation and Blink Rate in Open Shared Workspace vs Closed Workspace*

fatigue over time—remain largely unexplored [220]. Furthermore, no prior research has investigated the combination of these metrics with SIM-TLX to evaluate both real-time and cumulative workload in collaborative VR environments[221], [222].

A paired t-test showed significantly lower normalized pupil dilation in open workspaces compared to closed ones ($t = -5.74$, $p < 0.001$), indicating lower cognitive workload in open environments. Blink rates were also significantly lower in open workspaces ($M = 1735.02$) than in closed ones ($M = 3394.05$), suggesting they may reflect differences in cognitive or environmental demands after removing outliers. Figure 6-26 shows the variation of normalised PD and Blink Rate in Open vs. Closed workspace.

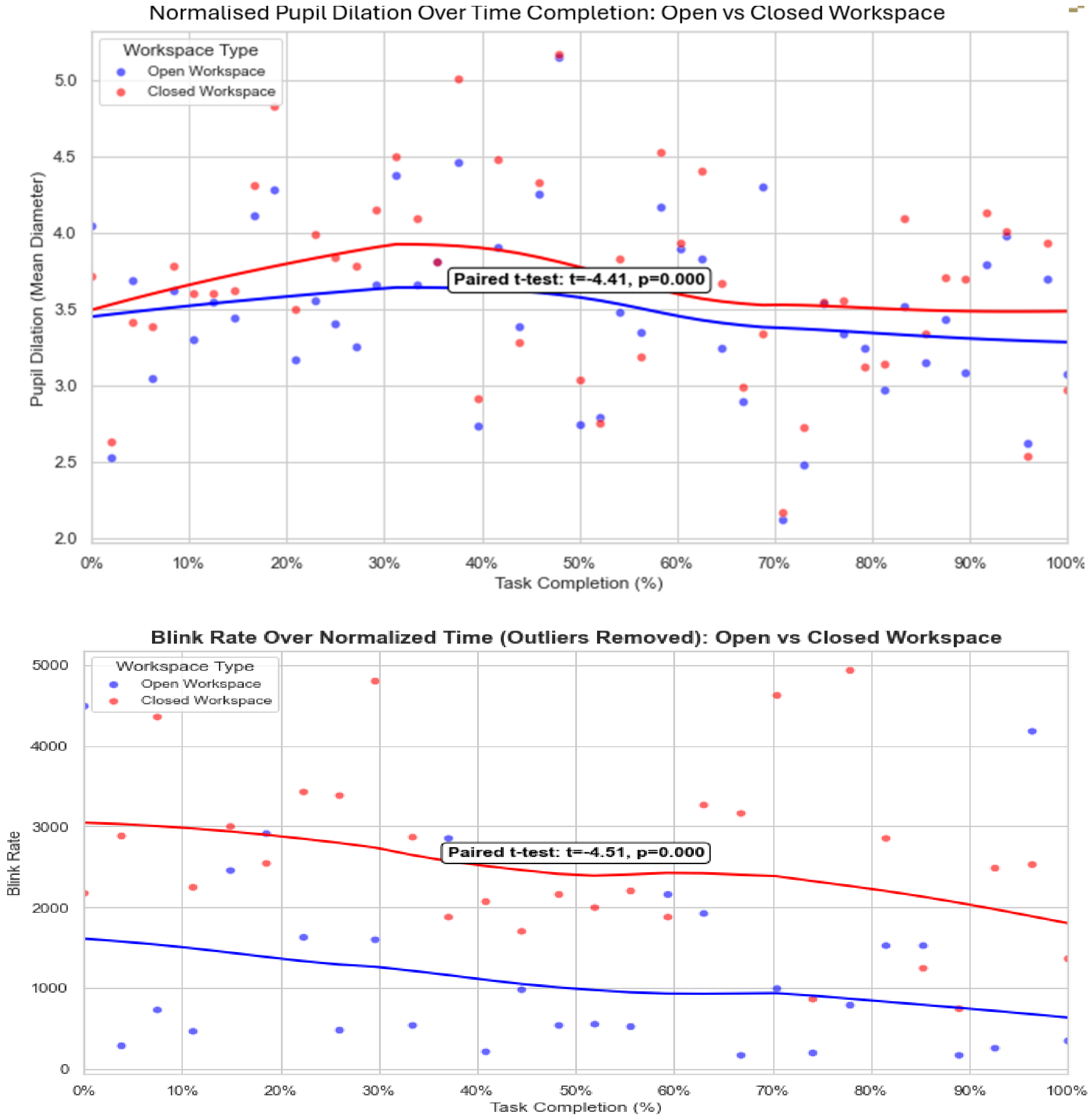

***Figure 6-27**: Normalised Pupil Dilation and Blink Rate over Time (Open vs Closed Workspace)*

6.4.5 *Time Series Analysis of Pupil Dilation and Blink Rate*

Moreover, visual analysis of blink rate and pupil dilation over task completion time supports the earlier findings. As shown in Figure 6-27, participants in closed workspaces consistently demonstrated higher blink rates across all task stages, with a significant difference confirmed by a paired t-test ($t = -4.51$, $p < 0.001$). Likewise, pupil dilation, which is the degree of change or expansion in pupil size from baseline, was significantly greater in the closed workspace condition throughout the task ($t = -4.41$, $p < 0.001$). In this context, it is essential to distinguish between pupil size (the absolute diameter of the pupil at a given time) and pupil dilation (the dynamic change in size in response to cognitive or emotional stimuli). Increased pupil dilation is a well-established indicator of heightened cognitive load, mental effort, or emotional arousal. Therefore, the greater dilation observed in the closed condition suggests that this environment imposed higher cognitive or attentional demands on participants. These physiological trends reinforce the conclusion that limited visual contact may intensify mental workload during collaboration in VR.

6.4.6 *Regression Analysis*

To understand how cognitive and affective factors influence performance, regression analyses were conducted for both the Open and Closed scene conditions. In the Open scene, shorter task completion times were significantly predicted by increased pupil dilation ($p = 0.02$), suggesting heightened attentional engagement. Conversely, higher blink rates ($p = 0.015$) and elevated levels of self-reported mental load and frustration ($p < 0.05$) were

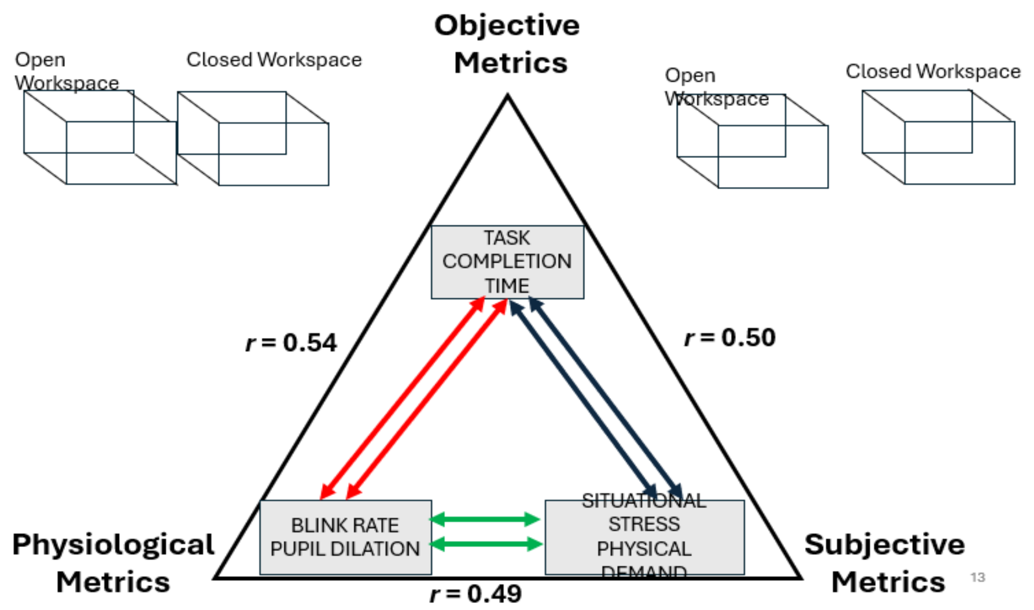

***Figure 6-28**: Relationship between Objective, Physiological and Subjective Metrics*

associated with longer task durations. In the Closed scene, blink rate again significantly predicted longer task time ($p = 0.008$), whereas pupil dilation did not reach statistical significance ($p = 0.09$). Higher physical effort and perceived task complexity also contributed to longer task durations in this condition ($p < 0.05$). Model performance was consistent across scenes, with pupil dilation explaining 48% of the variance in Open scenes and 44% in Closed scenes (Adjusted R^2), while blink rate accounted for 45% and 49% respectively. These findings highlight the utility of physiological signals in predicting collaboration efficiency in immersive VR settings.

6.5 Discussion

The findings from this second experimental study, involving 68 participants in 34 pairs, offer important insights into collaboration in VR environments. Participants performed two collaborative tasks: one where they could see each other as VR avatars (open task), and another where they could only hear each other (closed task). The study evaluated subjective experiences (via custom questionnaires, VRNQ, SIM-TLX, and the Interest Questionnaire), task performance, physiological synchrony (HR, EDA, and PD), and the differences based on leadership roles and gender.

The study revealed that visual interaction significantly enhances physiological synchrony between collaborators aligning with previous studies [121], [223], [224]. Both HR and EDA synchrony, measured using DTW, improved in the open task where participants could see each other. HR synchrony showed a significant improvement during the open task ($t(33) = 5.72$, $p < 0.001$), indicating better coordination and engagement when visual cues were available. This finding aligns with previous research suggesting that visual interaction enhances physiological alignment and collaboration efficiency during team tasks. Similarly, EDA synchrony was significantly higher in the open phase ($t(67) = 6.45$, $p < 0.001$), further highlighting the importance of visual contact in promoting physiological coherence, which has been linked to better social bonding and collaborative performance [225], [226], [227].

PD synchrony, a marker of cognitive load alignment, also showed improved synchrony in the open task, with a strong negative correlation between PD synchrony and both mental load ($r = -0.82$, $p < 0.001$) and physical load ($r = -0.58$, $p < 0.001$). This suggests that tasks with higher cognitive and physical demands resulted in reduced cognitive alignment between participants, further emphasising the role of visual cues in reducing mental and physical strain [228], [229], [230], [231] .

Task performance metrics clearly demonstrated the benefits of visual interaction during collaboration. Task completion time significantly decreased in the open scene ($t(33) = -8.05$, $p < 0.001$), with a reduction of 21.6%, while the number of clicks (a proxy for task interaction efficiency) was significantly lower in the open scene ($t(33) = 4.82$, $p < 0.05$). These results are consistent with existing literature on the role of visual interaction in improving task coordination, efficiency, and overall performance in collaborative settings [232], [233], [234], [235].

Subjective ratings of self- and partner-contribution also showed a clear preference for the open task, with significantly higher ratings in both dimensions ($p < 0.001$). Participants perceived both themselves and their partners as being more effective when visual contact was available. This is in line with previous findings [225] that suggests visual contact in virtual environments fosters a sense of engagement and improves perceptions of collaboration quality.

The SIM-TLX results revealed notable differences in how participants perceived task complexity and workload. Leaders reported significantly higher task complexity ($p = 0.045$), likely due to the cognitive demands of managing and guiding the task. This aligns with previous studies that have highlighted the additional cognitive load associated with leadership roles in collaborative tasks [236]. In contrast, followers experienced significantly higher perceptual strain ($p = 0.035$), possibly because they had to adapt to the leader's decisions, which aligns with findings from research on reactive roles in teamwork [237].

Gender-based comparisons revealed that female participants reported significantly higher task complexity ($p = 0.007$). Female leaders also reported greater mental load ($p = 0.020$) and physical demand ($p = 0.015$) compared to their male counterparts. These findings suggest that female leaders may perceive their role as more demanding in terms of both cognitive and physical effort. Such gender-based differences in task perception have been observed in other studies on leadership in virtual environments, indicating that female leaders may benefit from tailored support.

The VRNQ assessed four key domains: user experience, game mechanics, in-game assistance, and virtual reality-induced symptoms and effects (VRISE). No significant differences were found between leaders and followers across these categories, indicating that both roles had similar experiences in terms of immersion and ease of use in the VR environment. Gender comparisons similarly revealed no significant differences, suggesting that both males and females experienced the VR system in a similar manner regarding usability and potential side effects like nausea or dizziness [238].

However, marginally significant differences were observed between male and female leaders in terms of user experience ($p = 0.06$) and in-game assistance ($p = 0.06$), with male leaders reporting slightly better experiences. Although not statistically significant, these results suggest that male leaders may find the VR system easier to navigate or interact with, a finding consistent with some prior studies on gender differences in technology usage [239].

The Interest Questionnaire, which assessed preferences and skills across six categories (Practical, Enterprising, Social, Creative, Investigative, and Organisational), revealed notable gender-based differences. Males scored higher in practical skills (Practical 1: $t = 2.11$, $p = 0.039$; Practical 4: $t = 2.23$, $p = 0.029$), indicating a stronger preference for hands-on, mechanical tasks. Males also scored higher in enterprising skills (Enterprise 5: $t = 2.37$, $p = 0.021$), reflecting greater interest in leadership and risk-taking activities [240]. Females, on the other hand, scored significantly higher in organisational skills (Organisational 5: $t = -2.01$, $p = 0.049$), indicating strengths in data management and system organisation [241]. No significant gender differences were observed in social, creative, or investigative skills.

In addition, we evaluated pupil dilation and blink rate over time to explore how cognitive load changed throughout task completion. Results showed that pupil dilation and blink rate were consistently higher in the closed workspace, indicating increased cognitive effort without visual cues. These differences were significant ($p < 0.001$), and trends across time revealed that the open workspace gradually reduced cognitive load, as reflected by declining blink rates and pupil dilation throughout the task. This supports previous research showing that both metrics are valid indicators of mental effort [201], [203].

The regression analysis provided valuable insight into the predictive role of physiological signals in collaborative VR performance. Notably, greater pupil dilation was linked to shorter task completion times in the Open scene, while increased blink rates significantly predicted longer durations in both Open and Closed scenes. Additionally, higher subjective ratings of mental load, frustration, and physical effort were associated with extended task times. These patterns affirm the influence of both physiological and perceived user states on collaboration efficiency, with the models explaining up to 48% (pupil dilation) and 49% (blink rate) of the variance in task time.

6.6 Conclusion

This study presents compelling evidence that visual interaction, leadership roles, and gender all significantly influence collaborative performance in virtual reality environments. Building on and confirming the results of Study 1, the findings reinforce that visual cues play a vital role in enhancing collaboration by improving physiological synchrony, reducing perceived workload, and boosting task efficiency.

With a strong sample of 68 participants, this study stands out in the VR research domain as a methodologically robust exploration of team interaction. The results demonstrate that physiological synchrony, as captured through heart rate, electrodermal activity, and pupil dilation, is significantly stronger when visual contact is available, indicating more effective coordination and mutual engagement.

A particularly novel aspect of this work is the use of pupil dilation and blink rate as continuous, time-based indicators of cognitive load in a collaborative remote VR setting. To our knowledge, this is the first study to incorporate these ocular metrics in this context. Both pupil dilation and blink rate were consistently higher in the closed workspace condition, reflecting increased cognitive demand, while a gradual decrease in these signals was observed in the open workspace, suggesting reduced mental strain with visual access.

Further strengthening these findings, correlation analysis revealed that teams with less heart rate synchrony tended to report higher mental and physical load and took longer to complete tasks. Conversely, better synchrony in pupil dilation was associated with lower cognitive strain and more efficient task performance. These results highlight the potential of physiological and ocular synchrony as sensitive markers of collaborative effectiveness.

The implications are far-reaching. By integrating real-time physiological feedback into VR systems, developers can create more adaptive and supportive environments that respond to users' cognitive states. Such systems would be especially beneficial in professional and remote collaboration contexts, where performance and user well-being are equally critical.

While the findings are strong, the study is not without limitations. The use of simplified tasks may not fully reflect real-world complexity and randomly assigned leadership roles may not capture natural team dynamics. Future studies should explore more diverse tasks, allow for self-selection of roles, and address potential influences from technological variables such as headset tethering and ambient lighting, particularly when using pupil-based measures.

Overall, this study offers a rich, multidimensional understanding of collaboration in VR. It advances the field by combining task performance, subjective experience, and both physiological and ocular synchrony, setting the stage for more responsive and inclusive VR environments that enhance productivity and user experience.

Chapter 7: CONCLUSION

This chapter presents the conclusions of this thesis along with the explored RQs, outlines potential future work and applications of the developed technologies, and discusses the main take-home challenges that lay ahead for further research in the field of QoE of users involved in collaborative tasks in VR.

7.1 Revisiting the Research Questions

At the time of writing this thesis, physiological synchrony in VR collaboration remains relatively underexplored. While prior research had examined synchrony in natural team settings and video conferencing, and EEG-based studies had investigated synchrony in some VR contexts, very few studies had explored physiological synchrony using EDA, HR, and PD in immersive VR environments. This research addressed this gap by investigating how task roles, sentiments extracted from leader-follower voice communication, and environmental (time in the experience and visibility of partners) factors shape physiological synchrony and how synchrony, in turn, impacts team performance and QoE.

Through the development of a custom Collaborative VR framework and two controlled experimental studies, this thesis provides empirical evidence that physiological synchrony is not just an outcome of effective teamwork but an active indicator of team cohesion, coordination, and cognitive alignment in VR. The findings are structured around a main research question, further subdivided into three research sub-questions, each addressing a different aspect of synchrony and collaboration in VR.

The first research sub-question focused on how task roles influence physiological responses, perceived workload, and QoE. The results revealed that participants in roles requiring higher cognitive effort exhibited elevated physiological reactions, including increased EDA, HR, BVP, PD, and ACC signals. These physiological differences were also reflected in mental workload measures, as assessed by NASA-TLX in Study 1 and SIM-TLX in Study 2. However, despite differences in physiological responses and workload, QoE scores remained stable across roles, suggesting that cognitive demands did not significantly impact participants' engagement or immersion. Interestingly, prior VR experience and familiarity with collaborators helped regulate workload and physiological responses, indicating that familiarity plays a key role in moderating cognitive strain during collaboration.

The second research sub-question examined how communication and physiological synchrony between collaborators contributes to team performance. While previous research on physiological synchrony focused on natural team settings and video conferencing, its role in VR-based team performance remained underexplored. This study extends prior work by demonstrating that communication directly influences physiological synchrony, enhancing team performance. The results showed that positive vocal sentiments, such as encouragement and supportive feedback, were associated with higher physiological synchrony, which led to greater task efficiency and lower perceived workload. Conversely, negative sentiments, such as frustration and criticism, reduced synchrony and were correlated with increased cognitive strain and inefficient collaboration. This suggests that physiological synchrony is not just a passive reflection of team interactions but an active component of collaborative success. Moreover, synchrony was found to increase over time, as collaborators adapted to each other's working styles, and was further enhanced in teams with prior collaboration experience. The ability of teammates to physiologically align in real-time was a strong predictor of smoother coordination, faster task completion, and fewer unnecessary interactions.

The third and most critical research-sub question explored how physiological synchrony impacts collaboration efficiency and QoE. This question built on the findings from the previous two, providing deeper insights into the role of synchrony as a fundamental driver of collaboration success. The results confirmed that physiological synchrony was the strongest predictor of team efficiency, with highly synchronous teams demonstrating greater coordination, fewer misunderstandings, and reduced frustration levels. Teams with higher synchrony also completed tasks faster, required fewer corrective actions, and exhibited better overall collaboration performance. Beyond efficiency, physiological synchrony was positively correlated with higher QoE. Teams that maintained strong physiological alignment rated their collaboration experience as more immersive and engaging. Additionally, familiarity and shared expertise further strengthened synchrony, leading to higher collaboration effectiveness. These findings establish that VR collaboration is not just about spoken interaction or task execution—physiological synchrony serves as a real-time marker of teamwork efficiency and experience quality.

Differences between Study 1 and Study 2 are shown in the **Table XX**.

Table XX: Comparative Analysis of Study 1 vs Study 2

Aspect	Study 1	Study 2
Physiological Sensors	Used in both studies	Used in both studies
Pre-Experiment Form	VR experience, video conference experience, age, sex, demographics	Same as Study 1, plus Interest Questionnaire
Pre-screening Tools	Ishihara Colour Blindness Test, Eye test	Same as Study 1
Hardware Used	2-HTC Vive Pro Eye 2- Empatica E4 Wristband	Same as Study 1
Setup	Wire over the ground causing disturbance and stamping of the wires while doing the experiment	Wire attached to the roof with a pulley system which made the task easy
Post-Experiment Questionnaires	NASA-TLX, Custom Questionnaire (6 parameters: immersion, presence, interaction, collaboration, side-effects/cybersickness, enjoyment)	SIM-TLX (10 factor), and VRNQ (UX, Game mechanics, In-game assistance, VRISE
Participants	22 (11 Male, 11 Female)	68 (38 Male and 30 Female)
Scenarios	Single scenario	Two different scenarios
Key Metrics	EDA, IBI, PD, ACC, BVP, ST	Same physiological metrics, plus gender analysis
Mental Load	Higher for leaders	Higher for female leaders
Physical Load	Higher for followers	Higher for female leaders, male followers
Emotional Correlations	Positive correlation with task completion times	Similar, with additional gender-specific insights
Physiological Synchrony	Linked to participant backgrounds (EDA and HR) computed using DTW	Increased with commonality factors, scenario-specific differences
Voice Sentiment Analysis	Positive correlation with completion times, negative with frustration	Similar patterns, with gender-specific stress factors
Eye metrics Collected	Pupil Dilations, Synchrony using R-squared value	Same as Study 1, and blink rate.
Task Visibility influence		Introduced in Study 2

7.2 Discussion

This study investigated the role of physiological synchrony in collaborative VR environments, addressing how task roles, participants' communication, and environmental factors shape synchrony and how synchrony, in turn, impacts team performance and QoE. While prior research has explored synchrony in natural team settings and video conferencing, its application to VR collaboration remained largely unexplored. Using EDA, HR, and PD, this study contributes novel insights into how physiological alignment between collaborators influences teamwork efficiency and user experience.

A key finding of this research is that task roles influence physiological responses but do not significantly impact QoE. Participants in cognitively demanding roles exhibited higher physiological activation, as indicated by elevated EDA, HR, and pupil dilation. This aligns with established workload theories that suggest increased cognitive effort is associated with greater physiological arousal. However, despite these workload and physiological strain differences, participants' QoE scores remained stable across roles, suggesting that engagement and immersion were not negatively affected. One possible explanation is that the immersive nature of VR buffers the perceived impact of cognitive strain, allowing users to remain engaged even under high workload conditions. Another factor may be the collaborative nature of the task, as working with a partner could provide a regulatory effect, mitigating perceived stress. These findings suggest that VR teamwork introduces new dynamics in how workload is experienced and managed, reinforcing the need for task designs that optimize cognitive effort while maintaining engagement.

Beyond individual physiological responses, this study found that participants' communication played a crucial role in shaping physiological synchrony and, in turn, team performance. Positive vocal interactions, such as encouragement and supportive feedback, enhanced synchrony, whereas negative sentiments, including frustration and criticism, disrupted it. These findings are consistent with prior research on team dynamics, which suggests that physiological synchrony reflects social bonding and shared cognitive states. However, this study extends those findings by demonstrating that synchrony is not just a passive reflection of team interactions but actively contributes to performance outcomes. Teams that achieved higher synchrony completed tasks more efficiently, experienced lower cognitive strain, and required fewer corrective interactions. This suggests that VR collaboration is not solely dependent on verbal coordination and role distribution but is also shaped by the degree of subconscious physiological alignment between team members.

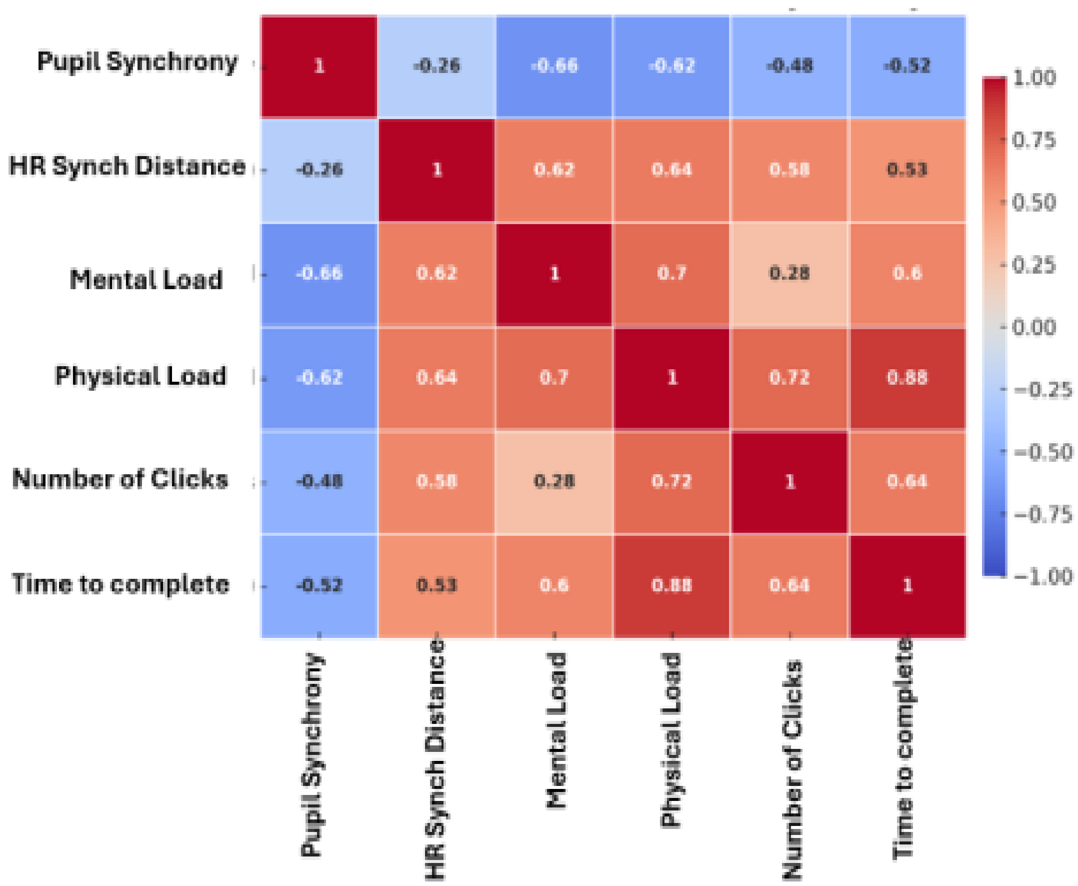

Figure 7-1:*Study 1 Correlation of Pupil Dilation Synchrony, HR Synchrony, Mental Load, Physical Load, Clicks and Time to Complete*

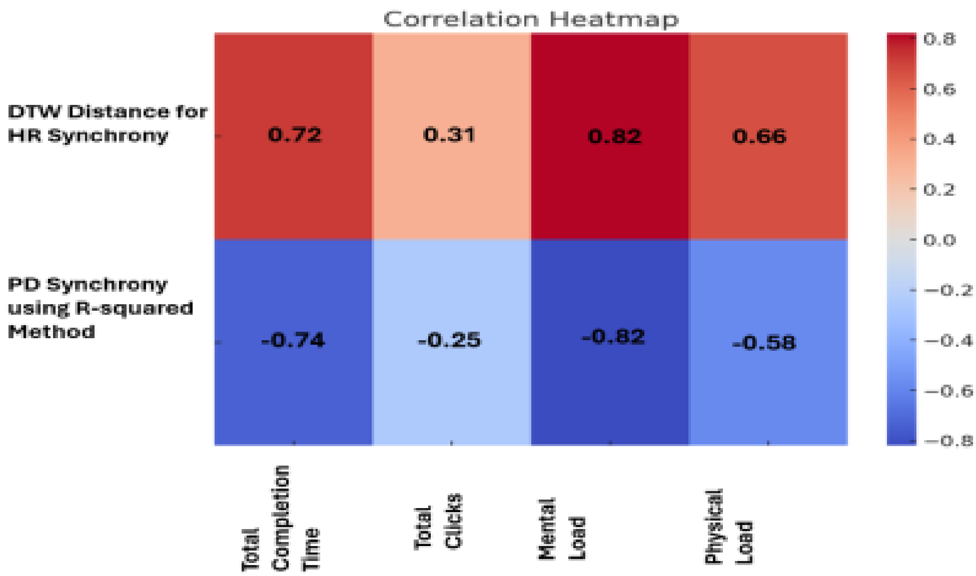

***Figure 7-2**: Study 2 Correlation with Pupil Dilation Synchrony, HR Synchrony, Mental Load, Physical Load, Clicks and Time to Complete*

Another important finding is that physiological synchrony was a strong predictor of collaboration efficiency and QoE. While previous studies have linked synchrony to social and cognitive alignment, this study provides empirical evidence that higher synchrony correlates with task completion indicative of teamwork success in VR environments (Figure 7-1 and Figure 7-2). Participants in highly synchronous teams reported lower perceived cognitive load, higher task efficiency, and greater engagement. This reinforces the idea that effective teamwork in VR is not just about explicit communication and task execution—physiological synchrony itself plays a fundamental role in collaboration success. The ability of teammates to physiologically align in real-time appears to enhance coordination, reduce misunderstandings, and improve overall workflow. Given these findings, future VR collaboration systems could leverage physiological synchrony as a measurable and actionable teamwork metric, opening possibilities for adaptive VR environments that respond to real-time synchrony data.

The two experimental studies in this research provided complementary insights, with notable differences in study design and outcomes. Study 1 primarily focused on task workload and subjective experience using NASA-TLX and a customised QoE questionnaire. Study 2 incorporated additional cognitive load and user experience measures, including SIM-TLX, VRNQ, and an Interest Questionnaire. A significant distinction was the inclusion of avatar visibility in Study 2, which revealed that synchrony was higher when teammates could see each other, confirming the role of visual cues in coordination. Additionally, while both studies demonstrated that positive communication enhanced synchrony, Study 2 provided stronger

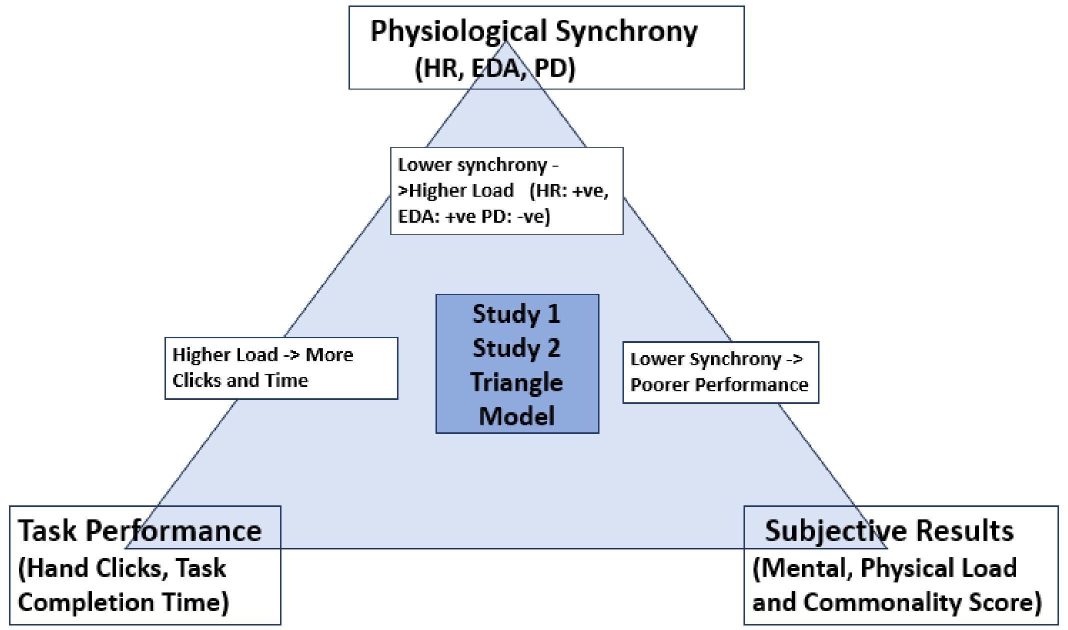

Figure 7-3: *The Triangle Model was developed from the findings of this Research*

evidence that negative communication actively disrupted synchrony, reinforcing the idea that physiological misalignment corresponds to impaired collaboration. These differences suggest that while the core relationship between synchrony and collaboration remained consistent across both studies, additional variables such as task structure, sensory feedback, and measurement techniques influenced the depth of insights gained.

The triangle model (Figure 7-3) illustrates the interrelationships between physiological synchrony (HR and pupil diameter synchrony), task performance metrics (clicks and completion time), and subjective workload (mental, physical load and commonality between collaborators), based on findings from both Study 1 and Study 2. At the top of the triangle, synchrony plays a pivotal role: decreased HR synchrony (indicated by increased DTW distance) and reduced PD synchrony (lower R^2 values) are both associated with increased mental and physical load. This is evidenced by strong correlations such as HR synchrony with mental load ($r = 0.82$) and PD synchrony with mental load ($r = -0.82$). On the lower left, the link between load and task performance is shown, where higher mental and physical demands correspond to greater number of clicks and longer completion times (e.g., physical load with time to complete: $r = 0.88$). This indicates that task difficulty translates into more interaction effort and slower task resolution. Finally, the lower right edge demonstrates that lower synchrony also correlates with poorer performance outcomes, as HR synchrony positively correlates with time ($r = 0.72$) and PD synchrony negatively correlates with time ($r = -0.74$). Together, this triangular framework highlights that physiological alignment between individuals is closely tied to both cognitive demands and efficiency in task execution.

These findings have several important implications for VR collaboration, physiological computing, and human-computer interaction. One of the most promising applications is the use of physiological synchrony as a real-time collaboration metric. By monitoring synchrony between teammates, VR systems could dynamically adapt task difficulty, modify environmental conditions, or provide real-time feedback to optimize team performance. This could be particularly beneficial for high-stakes applications, such as VR-based remote training, emergency response simulations, and collaborative learning environments. Another implication is the potential for biofeedback-driven interventions to improve teamwork. If a team's synchrony drops below a threshold, the system could introduce adaptive features, such as modifying task pacing, prompting relaxation techniques, or providing performance-enhancing cues. Additionally, these findings highlight the role of familiarity and expertise in synchrony development, suggesting that future VR systems could incorporate collaboration history tracking to match users with compatible teammates based on prior synchrony patterns.

While this study focused on EDA, HR, and PD as physiological markers, future research could expand on these findings by integrating EEG, respiration rate, and gaze synchrony to develop a multi-modal physiological synchrony model. Investigating how synchrony patterns vary across social and cultural contexts could further refine our understanding of physiological alignment in teamwork. Additionally, future studies could explore whether synchrony patterns differ based on personality traits, stress tolerance, or task complexity, providing deeper insights into how physiological synchrony operates in diverse collaborative scenarios.

This research demonstrates that physiological synchrony is a core driver of collaboration success. By introducing a novel way to measure and optimise VR collaboration through physiological alignment, this study challenges conventional teamwork models, which often emphasise explicit communication and task-sharing strategies (Figure 7-4). Instead, these findings suggest that VR collaboration should also be understood through the lens of subconscious physiological alignment, reinforcing that successful teamwork in immersive environments is shaped by actions and words and shared cognitive and physiological states. As VR continues to evolve as a platform for remote teamwork, training, and education, leveraging physiological synchrony as a real-time collaboration metric could redefine how we measure and enhance teamwork in virtual spaces.

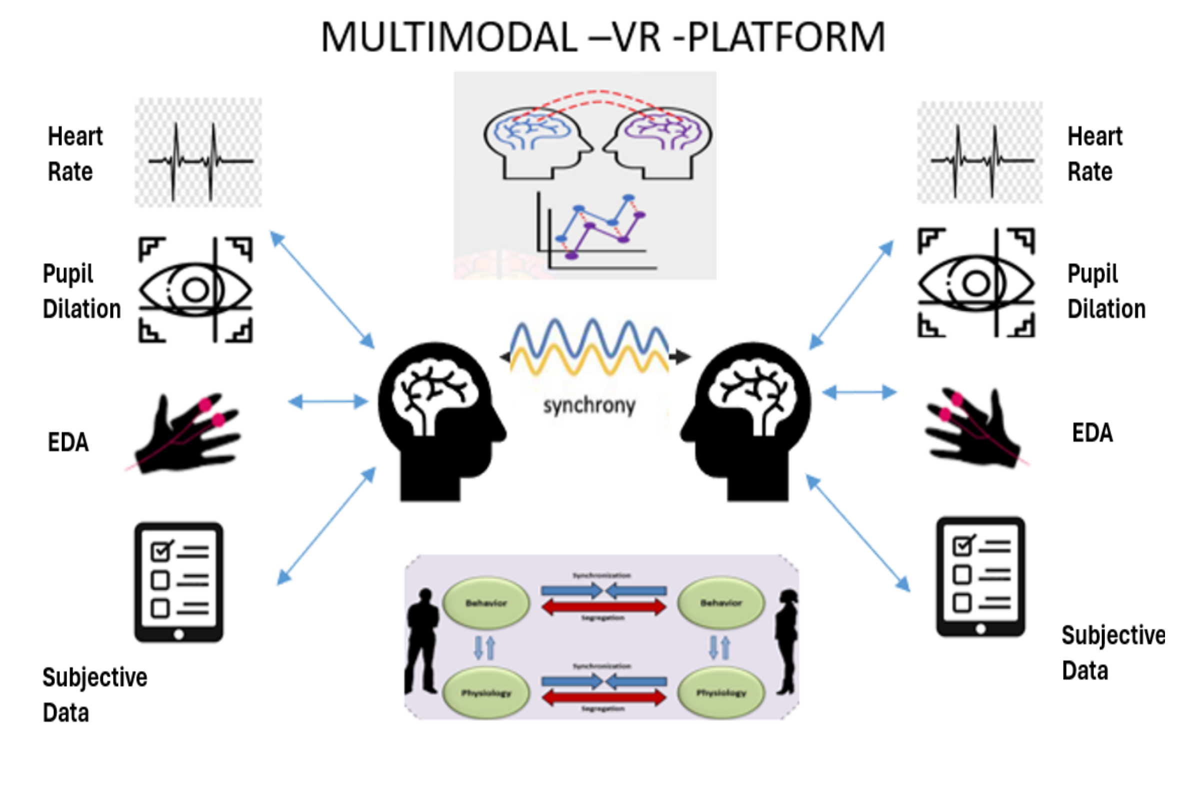

Figure 7-4: *Multimodal VR Platform using Heart Rate, Pupil Dilation and EDA Synchrony with Subjective Metrics*

7.3 Limitations in VR Collaborative Designs

Despite the valuable insights gained from these studies, several limitations should be acknowledged. The sample sizes of 22 participants in Study 1 and 68 in Study 2 may limit the generalizability of the findings. A larger, more diverse sample could provide deeper insights across various demographics and cultural backgrounds.

The VR tasks used in the studies may not fully capture the complexity of real-world activities. Future research could incorporate a broader range of tasks to better understand how different activities influence user experience and performance, while also considering tailored interventions to support gender-specific needs, particularly for female leaders.

While the VRNQ and SIM-TLX questionnaires provided useful data, their reliance on self-reported measures introduces subjectivity. Future work could incorporate more implicit or objective metrics, such as physiological data or performance measures, for a comprehensive assessment of QoE. Additionally, randomly assigning leader and follower roles might have masked role-based preferences and strengths, so allowing participants to choose roles could yield different results.

Another limitation was tethered VR hardware (HTC Vive Pro Eye), which may have affected participant comfort and QoE due to heat and tethering constraints. Standardising VR setups across studies could help minimise equipment-related variability. Moreover, the influence of colour and other visual elements in VR environments may have induced subtle effects on participant behaviour and perception. Future studies should consider these factors to ensure that colour and design choices do not unintentionally affect user experience.

In summary, while these studies offer valuable insights into VR collaboration dynamics and gender differences, addressing these limitations in future research will help build more inclusive and supportive VR applications, enhancing user experiences across diverse populations.

7.4 Future Work and Research Opportunities

The findings from Study 1 and Study 2 provide several avenues for future research to enhance our understanding and application of VR in collaborative tasks and training.

Study 1: Leader and Follower Experiences in VR Collaboration

Future research should aim to expand the sample size beyond 68 participants to improve the generalizability of the findings. Investigating physiological synchrony in a more diverse participant pool could provide deeper insights into how different demographic groups interact in VR environments. Additionally, further analysis is needed to explore the long-term effects of physiological synchrony on collaboration quality. This could involve extended study durations and repeated measurements over time.

Incorporating more advanced wearable sensors and more comprehensive physiological measures may also provide a richer dataset. For instance, combining heart rate variability, galvanic skin response, and advanced facial expression analysis could offer a more nuanced understanding of user experiences. Exploring the impact of different VR hardware setups on user comfort and QoE, mainly focusing on mitigating the heating effects of devices like the HTC Vive Pro Eye, will be crucial.

Given the positive correlations between sentiment analysis and task performance, future studies should integrate more sophisticated sentiment analysis tools to capture real-time emotional states during VR interactions. Investigating the impact of various VR interaction techniques and feedback mechanisms on user engagement and satisfaction can further refine the design of collaborative VR tasks.

Study 2 highlighted significant gender differences in practical skills and VR task performance. Future research should explore these differences in a broader context, examining how various tasks and roles impact male and female users differently. Expanding the range of functions beyond those used in this study will help understand how different activities influence user experience and performance. Deeper analysis of eye gaze data is necessary to investigate how the colours and lighting of the virtual content affect users. Pupillometry results show users are affected by different colour and lighting parameters. In our future studies, we plan to investigate further on these parameters.

The observed lack of correlation between leader and follower based on customised questionnaires and other metrics suggests a need for refining these self-report tools. Future studies should provide more structured, state-of-the-art, standardised questionnaires to participants before and after their experiences to get more information on their experiences.

Future research should investigate collaborative MR experiences. While many studies provide important insights into the dynamics of VR collaboration and gender differences in practical skills and task performance, little work is achieved in collaborative MR.

Lastly given the nature of this research, ensuring privacy in VR is a particularly interesting area for future work. As VR technology continues to evolve and integrate into various sectors, including entertainment, education, and professional training, privacy and security concerns are becoming increasingly prominent. One primary concern is the vast amount of personal data VR systems collect. These systems capture detailed information about users' physical movements, biometric data (such as HR and eye tracking), and behavioural patterns. This data is crucial for enhancing the immersive experience but poses significant privacy risks if mishandled or accessed by unauthorised parties. The potential for misuse of such sensitive information can lead to identity theft, unauthorised surveillance, and other forms of cybercrime, making it imperative to address these concerns proactively. Several directions should be investigated to tackle these privacy and security challenges. These include user encryption and protection, adopting privacy-by-design principles for VR experience creation, and assessment protocols that minimise data collection.

7.5 Implications of this Research

The findings of this research contribute to multiple domains, including VR collaboration design, physiological computing, and team performance assessment. This section discusses the broader implications of using physiological synchrony as a measurable and actionable collaboration metric in immersive virtual environments.

One of the most significant implications of this research is the potential to use physiological synchrony as a real-time indicator of teamwork efficiency. Unlike traditional collaboration metrics that rely on self-reported surveys or post-task evaluations, physiological synchrony provides a continuous, objective measure of interpersonal alignment. By tracking physiological synchrony in real time, VR systems could dynamically adjust task complexity, interaction modalities, or communication channels to enhance collaboration. This could be particularly useful in remote teamwork settings, virtual training programs, and high-stakes simulation environments where efficient coordination is critical.

Another key implication is the role of communication quality in shaping physiological synchrony and overall team performance. The findings showed that positive verbal interactions (e.g., encouragement, affirmation, and cooperative dialogue) enhanced synchrony, whereas negative interactions (e.g., frustration and criticism) reduced synchrony and impaired coordination. This suggests that VR collaboration tools could benefit from sentiment-aware feedback systems that detect speech patterns and provide subtle cues to promote positive communication behaviours. This could be integrated into VR-based teamwork training in professional and educational contexts to help users develop effective interpersonal communication strategies while working in virtual environments.

The study also highlights the importance of familiarity and expertise in synchrony development. Teams with prior experience working together exhibited higher physiological synchrony, reduced cognitive load, and improved task performance. This suggests that VR platforms could leverage collaboration history data to optimise team formation by pairing individuals with compatible working styles. For example, VR-based remote work systems could recommend ideal teammate matches based on previous synchrony patterns, fostering more productive and cohesive virtual teams.

Another implication is the potential for biofeedback-driven interventions to optimise teamwork. Since physiological synchrony was strongly associated with collaboration success, VR systems could integrate real-time biofeedback mechanisms to guide users toward better synchronisation. If a team’s synchrony drops below a certain threshold, the system could adjust environmental stimuli, introduce stress-reducing interventions, or provide visual indicators to encourage improved coordination. Such interventions could be particularly valuable in VR-based therapy, rehabilitation, and crisis response training, where maintaining physiological regulation is essential.

Beyond VR collaboration, this research contributes to the broader field of emphatic computing and adaptive human-computer interaction. The ability to quantify physiological synchrony offers exciting possibilities for emotion-aware AI, intelligent tutoring systems, and real-time adaptive interfaces. Future applications could extend beyond VR teamwork to virtual social interactions, telepresence, and interactive gaming, where real-time physiological alignment could enhance engagement and interpersonal connection.

While this study focused on EDA, HR, and PD, future work could expand upon these findings by incorporating other physiological signals such as EEG, respiration, and gaze synchrony. This could lead to the development of multi-modal synchrony models that provide a richer understanding of cognitive and emotional alignment between users in VR. Additionally, future research could explore cross-cultural variations in physiological synchrony, investigating how social norms and communication styles influence synchrony patterns in global VR collaboration.

Overall, this study demonstrates that physiological synchrony is not just an indicator of collaboration success but a key driver of effective teamwork. By integrating synchrony-based adaptive mechanisms into VR collaboration tools, future systems could offer more intelligent, responsive, and immersive teamwork experiences. As VR continues to shape the future of remote work, education, and training, leveraging physiological synchrony as a core component of human-computer interaction could transform how we measure, evaluate, and enhance collaboration in virtual environments.

7.6 Closing Remarks

This research highlights the pivotal role of physiological synchrony in shaping collaborative experiences within virtual reality. Moving beyond conventional markers of communication and behaviour, the findings demonstrate that effective teamwork in immersive environments arises from the dynamic interplay between cognitive alignment and shared physiological states. This challenges traditional QoE evaluation methods, which are often reliant on self-report measures, and encourages a shift toward a more embodied and data-driven understanding of team interaction in virtual spaces.

The practical implications are considerable. The ability to capture collaborators' physiological signals and derive synchrony metrics from those signals creates new opportunities for evaluating team dynamics. This study's key contribution is identifying an interconnected relationship among three key dimensions: physiological synchrony, subjective workload, and task performance. This positions synchrony as a valid, quantifiable, and meaningful indicator of collaborative quality.

In addition, the potential to enhance synchrony offers promising directions for future research. Investigating how individual differences such as culture, personality, or prior experience affect physiological alignment may lead to the design of more inclusive and personalised strategies for supporting team collaboration across diverse contexts.

As virtual reality technologies become increasingly integrated into professional, educational, and social domains, understanding and leveraging physiological synchrony will be essential for designing collaborative experiences that are not only efficient but also human-centred. The findings presented here contribute to a broader shift in how collaboration is conceived, moving beyond surface-level behaviours to acknowledge the subtle, biological rhythms that shape how people connect, coordinate, and succeed together. In this light, the future of immersive teamwork is not only interactive but deeply attuned to the invisible signals that bring teams into sync.

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APPENDICES

Appendix A

Experimental Study 1 and Study 2 Information Sheet

Principle Investigator: Bhagyabati Moharana Contact: b.moharana@research.ait.ie

Quality of Experience evaluation of Virtual Reality collaborative task using Leader and Follower.

Brief explanation of the title: In this experiment, we aim to evaluate user quality of experience when using head-mounted virtual reality technology. A head-mounted virtual reality display uses technology to project images into a user's field of view. This allows users to view information and interact with virtual data whilst being aware of their surroundings. We aim to capture data while using this device to determine if a user's quality of experience is enriched using a virtual reality device for complex task instruction compared to traditional video instruction.

Introduction

I am inviting you to take part in a research experiment to be carried out in the Technological University of the Shannon, Athlone. This document explains why the research is being carried out and what it will involve. If you are unclear, please do not hesitate to ask questions. Thank you for reading this.

What is the purpose of this investigation?

In this experiment, we aim to evaluate if a higher quality of experience is experienced using Virtual Reality devices between different task roles. This experiment aims to evaluate the quality of experience within a virtual reality experience of solving a 3D puzzle with another participant joining remotely to the same virtual room.

Do I have to take part?

It is entirely up to you to decide whether you wish to participate in this experiment. Refusal to take part is entirely at your discretion. If you choose to take part, you can keep this information sheet and you will be required to sign a consent form.

What does the experiment involve?

This experiment should last for five minutes including questionnaire completion. Participants will be seated in a laboratory in the AIT Engineering Building. The lab will consist of a chair, table, VR headset, PC video display & heart rate smart watch. Participants will attempt to solve a 3D collaborative task by following instructions on either a head mounted virtual reality device or on a PC monitor. The participant will be fitted with a smart watch which will record your heart rate during the test. The participant will be asked to fill out questionnaires before and after the test to give their anticipated and perceived quality of the experience.

What do I have to do?

On the day of the test, participants will undergo a visual screening to ensure they are eligible. The visual screening process involves testing the participant's visual clarity using a Snellen chart and colour perception using the Ishihara test.

If you are pregnant or suspect you may be pregnant, please inform the test administrator.

What are the possible disadvantages and risks of taking part?

Some people may find a testing environment stressful in general. Should a participant feel discomfort at any point, it is essential to communicate this to the PI.

Will my taking part in this project be kept confidential?

Any information collected during this test will be strictly confidential. All data will be stored securely, and recognising you from this experiment will not be possible.

What will happen to the results of the research project?

The results of this experiment will be used to produce a paper for publication as part of my research.

Thank You

Thank you for taking the time to read this information sheet, and I hope you will decide to participate in this investigation. Solving a 3D task is a very fulfilling experience.

Appendix B

Consent Form for Experimental Study 1 and Study 2

Consent Form

Title of Project: *A QoE evaluation of users in collaborative design tasks using Virtual Reality.*

Name of Researcher: *Bhagyabati Moharana*

Please Tick the Box

1. I confirm that I have read the information sheet dated____/___/2023 (Ver:) for the above study and have had the opportunity to ask questions. ☐
2. I am satisfied that I understand the information provided and have had enough time to consider the information. ☐
3. I do not suffer from photosensitive epilepsy or any other form of epilepsy. ☐
4. I'm not pregnant and/or I am not experiencing any symptoms of pregnancy. ☐
5. I understand that my participation is voluntary and that I am free to withdraw at any time, without giving any reason, and without my legal rights being affected. ☐
6. I agree to take part in the above study. ☐
7. Gender Male Female Don't prefer to say.
8. Age
9. Educational Background:
10. Undergraduate Postgraduate Professional
11. Social Status: Single / Married / Others
12. Gamer: Yes / No
13. VR/ AR Experience: Y/N
14. Use of video conferencing tools like Zoom, Google Meet Y/N
15. Place of Origin

______________________________ ______________
Name of Participant Date Signature

______________________________ ______________
Researcher

Appendix C
Experimental Study 1 and Study 2 Protocol

Title of Project: A QoE evaluation of users performing collaborative design tasks in a Virtual Reality application.

Name of Researcher: Bhagyabati Moharana

The QoE evaluation will be achieved by implementing protocols outlined in the ITU-T P910 [1] and ITU-T P913 [2] standards. More specifically, the absolute category rating method will be implemented.

This investigation will use the HTC Vive Reality headset [1] or standard PC video display. Objective data will be captured using a Fitbit Blaze HR activity tracker [4] to record the test subject's heart rate. The following steps are required for successful completion of this experiment:

1. Participant will be seated in the waiting room and provided an information sheet. Any queries will be addressed by the PI at this stage. Once a level of understanding is achieved, participants are requested to sign a consent form.

2. A visual screening process that evaluates visual acuity and colour perception will begin to determine if the assessor is eligible for the experiment. If so:

3. Assessors will be introduced to the non-invasive objective metric device. The Fitbit Blaze HR will be attached to the participant's wrist.

4. The participant will be asked to complete a subjective pre-test questionnaire.

5. A 5-minute waiting period begins, the purpose of this is to allow assessors to gather baseline metrics on heart rate data.

6. Upon completion of the baseline phase, assessors will be brought to the testing room, seated comfortably, and introduced to either the head mounted VR or video device.

7. A training sequence will commence to teach the participant how to do the task in an orientation compatible with the instructions & how to manipulate the cube and move around. This training phase will then

introduce the test subject to the instruction medium & how to use it properly. This involves triggering each instruction on after the other.

8. Upon successful completion of the training sequence, when the assessor is assured that the test subject fully understands the process involved, the assessor will continue to the main test.

9. For the main test, the participant is introduced to the collaborative activity in VR .

10. Next, the test subject is introduced to the instruction medium displaying the first instruction & the test begins. The test subject progresses as far as they can through the test.

11. When the test has completed, i.e. all 21 instructions have been delivered or the participant determines they are finished, the participant is asked to complete the subjective post-test questionnaire.

12. Analyse data.

Appendix D

Experimental Study 1-Questionnaire – Part 1

You will see some statements about experiences. Please indicate, on the scale below if you agree or disagree with the following statements. There are no right or wrong answers, only your opinion counts.

Read through each of the statements carefully, if you have any questions, please ask the principal investigator.

1. **How natural did your interactions with the environment seem?**

Extremely Artificial | Borderline | Completely Natural

2. **How much did your experiences in the virtual environment seem consistent with your real-world experiences?**

Not Consistent | Moderately Consistent | Consistent

3. **How involved were you in the virtual environment?**

Not at all | Somewhat | Completely

4. How much did the control devices assisted with the performance of assigned tasks or with other activities?

Not at all assisted		Assisted Somewhat		Assisted Greatly

5. How well could you move or manipulate objects in the virtual environment?

Not at all		Somewhat		Extensively

6. How closely were you able to examine objects?

Not at all		Pretty Close		Very Closely

7. How well could you identify and hear sounds of another user?

Completely		Moderately		Not at all

8. How well could you see the other person's hand manipulating objects in the environment?

Completely Aware		Somewhat		Not at all

9. The collaborating task was difficult to complete wearing a Head mounted display.

Strongly Disagree	Disagree	Neither Agree or Disagree	Agree	Strongly Agree

10. I felt qualified enough in interacting with the virtual environment at the end of the experience

Strongly Disagree	Disagree	Neither Agree or Disagree	Agree	Strongly Agree

11. I was restricted in my movements using the system and therefore could not participate in the task

Strongly Disagree	Disagree	Neither Agree or Disagree	Agree	Strongly Agree

12. Did you become so involved in a task that it is as if you were inside the game rather than moving a controller and watching the screen?

Completely Involved		Neither Agree or Disagree		Not at all involved

13. How much delay did you experience between your actions and expected outcomes?

No Delay		Moderate Delay		Long Delay

14. How proficient in moving and interacting with the virtual environment did you feel at the end of the experience.

Not proficient		Moderately Proficient		Very Proficient.

15. Did you feel any motion-sickness or dizziness at the end of the experiment?

Not at all		Little bit		Yes somewhat

Appendix E

Experimental Study 1: NASA-TLX Questionnaire

The evaluation you are about to perform is a technique that has been developed by NASA to assess the relative importance of six factors in determining how much workload you experienced while preforming a task. These six factors are defined below on this page.

Read through them to make sure you understand what each factor means. If you have any questions, please ask the principal investigator.

Workload factors	Definition
Mental Demand Level (low/high)	**How much mental and perceptual activity was required** (for example, thinking, deciding, calculating, remembering, looking, searching, etc)? Was the task easy or demanding, simple or complex, forgiving or exacting?
Physical Demand Level (low/high)	**How much physical activity was required** (for example, pushing, pulling, turning, controlling, activating, etc.)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?
Temporal Demand Level (low/high)	**How much time pressure did you feel** due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?
Performance Level(good/poor)	**How successful do you think you were in accomplish the goals of the task set by the experimenter** (or yourself)? How satisfied were you with your performance in accomplish these goals?
Effort Level (low/high)	**How hard did you have to work** (mentally and physically) to accomplish your level of performance?
Frustration Level (low/high)	**How insecure, discouraged, irritated, stressed**, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel during the task?

For each pair, choose the factor that was more important to your experience of the workload in the task that you recently performed:	
☐ Temporal Demand	☐ Mental Demand
☐ Performance	☐ Mental Demand
☐ Mental Demand	☐ Effort
☐ Temporal Demand	☐ Effort
☐ Physical Demand	☐ Performance
☐ Performance	☐ Temporal Demand
☐ Effort	☐ Physical Demand
☐ Mental Demand	☐ Physical Demand
☐ Performance	☐ Frustration
☐ Effort	☐ Performance
☐ Frustration	☐ Effort
☐ Frustration	☐ Mental Demand
☐ Physical Demand	☐ Temporal Demand
☐ Physical Demand	☐ Frustration
☐ Temporal Demand	☐ Frustration

1. Mental Demand (How mentally demanding was the task?/ How much mental and perceptual activity did you spend for this task?)

Very Low Very High

2. Physical Demand (How physically demanding was the task? / How much physical activity did you spend for this task?)

Very Low Very High

3. Temporal Demand (How hurried or rushed was the pace of the task? / How much time pressure did you feel in order to complete this task?

Very Low Very High

4. Performance (How successful were you in accomplishing what you were asked to do? / How successful do you think you were in accomplishing the goals of the task?)

Good Poor

5. Effort (How hard did you have to work to accomplish your level of performance?)

Very Low Very High

6. Frustration (How insecure, discouraged, irritated, stressed, and annoyed were you during this task?)

Very Low Very High

Appendix F

		LE Diameter			RE Diameter				Left Eye Openness			Right Eye Openness		
ID No	Task Group	Calibration Left Diameter	Training Left Diameter	Collab Left Diameter	Calibration Right Diameter	Diff	Training Right Eye Diameter	Collab Right eye Diameter	Callib Left openness	Training Left openness	Collab Left openness	Callib Right openness	Training Right openness	Collab Right openness
1	1	3.96	3.69	3.33	3.96	0.63		3.76	0.92	1.00	**1.00**	0.94	1.00	1.00
2	2	3.88	3.21	2.98	3.88	0.90	3.21	2.98	0.92	0.86	0.89	0.71	0.59	0.86
3	1	2.89	2.59	2.70	2.89	0.18	2.59	2.70	0.79	0.93	**0.93**	0.80	0.95	0.92
4	2	3.35	3.08	2.97	3.35	0.38	3.08	2.97	0.92	0.92	0.92	0.88	0.91	0.85
5	1	3.66	3.16	3.14	3.66	0.52	3.16	3.14	0.97	0.95	**0.94**	0.83	0.92	0.92
6	2	3.83	2.70	2.62	3.83	1.22	2.70	2.62	1.00	0.77	0.75	1.00	0.83	0.79
7	1	4.29	3.92	3.32	4.29	0.96	3.92	3.32	0.95	0.93	**0.95**	0.91	0.91	0.93
8	2	3.83	3.83	2.53	3.83	1.29	3.83	2.53	0.79	0.88	0.75	0.84	0.85	0.89
9	1	3.87	4.09	3.34	3.87	0.54	4.09	3.34	1.00	0.90	**0.70**	1.00	0.88	0.91
10	2	4.52	3.36	3.39	4.52	1.13	3.36	3.39	1.00	0.73	0.86	1.00	0.71	0.86
11	1	3.04	3.11	3.03	3.04	0.01	3.11	3.03	1.00	1.00	**0.96**	1.00	1.00	0.93
12	2	3.91	3.40	3.07	3.91	0.85	3.40	3.07	0.84	0.83	0.86	0.84	0.85	0.83
13	1	2.83	2.78	2.28	2.83	0.55	2.78	2.28	1.00	0.99	**0.88**	1.00	0.99	0.88
14	2	3.64	3.38	3.40	3.64	0.24	3.38	3.40	0.92	0.93	0.88	0.90	0.91	0.73
15	1	4.37	3.86	2.96	4.37	1.42	3.86	2.96	1.00	0.41	0.74	1.00	0.59	0.85
16	2	3.87	3.70	3.21	3.87	0.65	3.70	3.21	0.84	0.80	0.82	0.81	0.76	0.84
17	1	4.06	3.88	3.04	4.06	1.02	3.88	3.04	1.00	0.95	0.84	0.99	0.96	0.89
18	2	3.41	3.23	2.86	3.41	0.55	3.24	2.86	0.89	0.87	0.94	0.90	0.82	0.93
19	1	4.74	3.82	3.42	4.74	1.32	3.82	3.42	0.90	0.94	0.94	0.88	0.93	0.92
20	2	3.13	2.98	2.60	3.13	0.53	2.98	2.60	0.87	0.77	0.94	0.89	0.79	0.93
21	1	3.51	3.48	3.12	3.51	0.39	3.48	3.12	0.79	0.76	0.92	0.77	0.68	0.88
22	2	4.38	3.86	3.03	4.38	1.35	3.86	3.03	0.68	0.58	0.34	0.39	0.45	0.21

Appendix G

Bio-signals Statistical Descriptives (Results from Study 1)

Participant ID	BioSignals	Baseline Data				Training Phase				Collaboration Phase			
		Mean	Standard Deviation	Minimum	Maximum	Mean	Standard Deviation	Minimum	Maximum	Mean	Standard Deviation	Minimum	Maximum
1001	*HRV*	101.63	15.25	52.00	129.12	107.99	20.63	87.60	141.47	108.78	19.21	70.18	180.05
	EDA	0.21	0.01	0.19	0.27	0.22	0.02	0.17	0.29	0.20	0.39	0.00	3.07
	BVP	0.00	0.41	-2.32	3.15	0.00	0.25	-0.99	1.23	0.02	9.59	-141.21	122.03
	TEMP	25.17	0.41	24.25	25.73	26.02	0.15	25.73	26.27	26.28	1.47	24.31	32.27
	IBI	0.01	0.10	0.00	0.86	0.01	0.06	0.00	0.56	0.01	0.06	0.00	1.36
	Beats	0.03	0.18	0.00	2.00	0.01	0.11	0.00	1.00	0.01	0.11	0.00	3.00
	ACC	0.03	0.07	0.00	0.44	0.05	0.06	0.00	0.37	0.05	0.08	0.00	0.78
1002	*HRV*	85.71	5.68	66.00	96.38	88.47	3.31	83.27	94.37	86.42	11.61	60.03	109.58
	EDA	0.43	0.06	0.32	0.62	0.36	0.02	0.32	0.42	2.70	1.35	0.00	5.79
	BVP	-0.15	17.80	-148.33	100.90	-0.13	12.77	-77.35	41.00	-0.12	24.29	-188.13	128.73
	TEMP	30.98	0.23	30.57	31.33	30.67	0.05	30.59	30.75	31.15	0.55	30.57	32.53
	IBI	0.56	0.58	0.00	1.67	0.15	0.37	0.00	1.39	0.04	0.20	0.00	1.41
	Beats	0.79	0.82	0.00	2.00	0.23	0.58	0.00	2.00	0.07	0.33	0.00	2.00
	ACC	0.02	0.04	0.00	0.30	0.06	0.08	0.00	0.36	0.09	0.09	0.00	0.54
1003	*HRV*	77.71	2.92	73.75	87.52	83.29	2.66	79.87	88.62	106.31	34.71	73.78	192.47
	EDA	4.12	0.19	3.68	4.79	8.03	3.59	4.54	19.03	6.93	5.86	0.00	19.95
	BVP	0.01	23.50	-163.16	118.69	-0.02	41.06	-164.42	131.27	0.00	32.27	-306.59	250.95
	TEMP	32.68	0.08	32.49	32.77	32.44	0.10	32.21	32.71	30.03	3.42	23.99	32.99
	IBI	0.70	0.61	0.00	1.83	0.20	0.44	0.00	1.66	0.15	0.38	0.00	1.70
	Beats	0.90	0.78	0.00	2.00	0.28	0.60	0.00	2.00	0.22	0.55	0.00	4.00
	ACC	0.02	0.04	0.00	0.30	0.06	0.07	0.00	0.54	0.06	0.09	0.00	1.14
1004	*HRV*	61.43	1.84	59.35	68.97	76.05	2.96	69.13	81.88	105.29	25.83	63.47	168.63
	EDA	0.31	0.05	0.22	0.40	0.44	0.12	0.10	0.70	1.09	1.37	0.00	5.13
	BVP	-0.05	7.30	-48.90	36.68	0.06	23.74	-86.16	95.51	0.01	14.02	-238.26	125.58
	TEMP	32.83	0.43	32.18	33.41	32.79	0.12	32.59	33.16	28.24	4.14	23.13	34.05
	ACC	0.02	0.06	0.00	0.44	0.08	0.11	0.00	0.73	0.05	0.09	0.00	0.91
	SumIBI	0.80	0.53	0.00	2.16	0.13	0.38	0.00	1.73	0.07	0.27	0.00	1.95
	Beats	0.81	0.55	0.00	2.00	0.16	0.49	0.00	2.00	0.09	0.35	0.00	3.00

Participant ID	BioSignals	Baseline Data Mean	Baseline Data Standard Deviation	Baseline Data Minimum	Baseline Data Maximum	Training Phase Mean	Training Phase Standard Deviation	Training Phase Minimum	Training Phase Maximum	Collaboration Phase Mean	Collaboration Phase Standard Deviation	Collaboration Phase Minimum	Collaboration Phase Maximum
1005	*HRV*	99.67	5.23	72.00	103.95	111.21	6.07	97.88	120.77	100.70	18.37	63.20	132.93
	EDA	4.33	0.56	2.85	5.10	5.26	0.57	4.29	6.69	8.32	3.66	0.00	14.44
	BVP	0.05	7.03	-19.99	38.54	-0.08	24.05	-91.35	89.80	0.05	28.64	-130.82	125.79
	TEMP	31.18	0.03	31.11	31.27	31.59	0.21	31.23	31.99	32.02	1.40	24.21	32.83
	ACC	0.01	0.02	0.00	0.23	0.08	0.09	0.00	0.55	0.09	0.10	0.00	0.63
	IBI	1.06	0.29	0.00	1.66	0.13	0.32	0.00	1.16	0.11	0.34	0.00	1.56
	Beats	1.80	0.49	0.00	3.00	0.24	0.61	0.00	2.00	0.23	0.68	0.00	3.00
1006	*HRV*	70.28	4.50	61.00	83.47	81.36	3.87	75.78	89.65	83.15	3.71	74.43	91.35
	EDA	0.12	0.01	0.11	0.16	0.31	0.16	0.16	0.72	2.72	1.04	0.00	4.77
	BVP	0.01	12.73	-64.98	59.16	0.00	10.67	-75.21	66.05	0.00	16.96	-163.42	135.35
	TEMP	31.63	0.10	31.49	31.89	32.11	0.06	31.87	32.21	32.58	0.96	26.79	33.23
	ACC	0.03	0.07	0.00	0.38	0.04	0.05	0.00	0.31	0.06	0.07	0.00	0.71
	IBI	0.87	0.55	0.00	1.84	0.50	0.59	0.00	1.64	0.47	0.59	0.00	1.69
	Beats	1.02	0.65	0.00	2.00	0.66	0.78	0.00	2.00	0.65	0.80	0.00	2.00
1007	*HRV*	67.96	3.44	61.00	79.25	80.45	4.65	71.88	88.37	77.01	11.45	54.10	108.27
	EDA	0.15	0.01	0.12	0.19	0.20	0.02	0.18	0.25	0.33	0.06	0.00	0.88
	BVP	-0.17	9.03	-40.69	60.30	0.21	13.47	-55.25	104.30	0.03	14.03	-221.80	170.14
	TEMP	31.27	0.16	30.89	31.47	31.53	0.03	31.43	31.57	31.42	0.11	31.19	31.63
	ACC	0.04	0.11	0.00	0.74	0.08	0.08	0.00	0.46	0.09	0.11	0.00	0.83
	IBI	0.91	0.53	0.00	1.94	0.20	0.46	0.00	1.72	0.16	0.41	0.00	1.83
	Beats	1.03	0.61	0.00	2.00	0.25	0.57	0.00	2.00	0.21	0.52	0.00	2.00
1008	*HRV*	76.78	4.33	64.80	83.73	79.88	2.79	74.55	84.52	100.03	35.83	57.73	199.60
	EDA	0.11	0.01	0.08	0.13	0.13	0.00	0.12	0.15	0.10	0.09	0.00	0.43
	BVP	0.01	9.97	-54.07	37.44	-0.01	13.58	-69.10	45.75	0.01	16.44	-195.21	145.56
	TEMP	29.31	0.02	29.25	29.37	29.24	0.04	29.17	29.33	26.52	2.89	20.09	29.19
	ACC	0.02	0.03	0.00	0.23	0.03	0.06	0.00	0.44	0.03	0.06	0.00	0.53
	IBI	0.91	0.54	0.00	1.73	0.70	0.60	0.00	1.83	0.21	0.45	0.00	1.73
	Beats	1.21	0.73	0.00	2.00	0.90	0.78	0.00	2.00	0.28	0.61	0.00	3.00

Participant ID	BioSignals	Baseline Data Mean	Baseline Data Standard Deviation	Baseline Data Minimum	Baseline Data Maximum	Training Phase Mean	Training Phase Standard Deviation	Training Phase Minimum	Training Phase Maximum	Collaboration Phase Mean	Collaboration Phase Standard Deviation	Collaboration Phase Minimum	Collaboration Phase Maximum
1009	*HRV*	73.41	2.70	58.00	94.33	81.32	2.70	75.07	85.97	100.92	21.85	64.88	144.58
	EDA	0.56	0.02	0.52	0.68	0.54	0.03	0.48	0.63	0.32	0.38	0.01	1.34
	BVP	0.01	14.97	-129.08	117.89	0.05	36.63	-171.22	172.62	0.00	17.39	-271.54	188.97
	TEMP	32.78	0.15	32.45	32.99	32.89	0.05	32.79	33.00	27.53	4.20	23.23	33.11
	ACC	0.02	0.03	0.00	0.21	0.07	0.08	0.01	0.41	0.03	0.06	0.00	1.59
	IBI	0.83	0.56	0.00	1.81	0.20	0.43	0.00	1.67	0.08	0.28	0.00	1.88
	Beats	1.03	0.71	0.00	2.00	0.28	0.58	0.00	2.00	0.11	0.38	0.00	2.00
1010	*HRV*	86.61	2.73	76.50	90.78	86.41	3.48	79.17	93.72	102.10	14.41	62.68	132.83
	EDA	0.07	0.01	0.05	0.08	0.09	0.01	0.07	0.11	0.09	0.14	0.00	0.79
	BVP	0.00	0.61	-1.31	1.86	0.00	2.64	-11.07	13.46	0.00	7.26	-110.56	59.84
	TEMP	29.67	0.04	29.59	29.77	29.87	0.07	29.75	30.01	26.97	3.22	22.93	31.27
	ACC	0.00	0.00	0.00	0.02	0.05	0.08	0.00	0.49				
	IBI	0.44	0.54	0.00	1.56	0.29	0.48	0.00	1.48	0.08	0.27	0.00	1.53
	Beats	0.63	0.77	0.00	2.00	0.43	0.71	0.00	2.00	0.12	0.44	0.00	3.00
1011	*HRV*	107.16	21.44	60.50	141.32	109.89	4.82	100.20	120.85	114.76	17.05	91.33	170.03
	EDA	0.21	0.02	0.18	0.26	0.20	0.00	0.19	0.21	0.04	0.08	0.00	0.92
	BVP	0.02	28.26	-107.44	137.33	0.00	30.22	-228.91	125.17	-0.01	14.95	-117.03	157.78
	TEMP	32.45	0.05	32.33	32.55	32.29	0.07	32.18	32.45	23.84	2.01	20.11	32.59
	ACC	0.09	0.12	0.00	0.62	0.07	0.05	0.01	0.32				
	IBI	0.14	0.37	0.00	1.47	0.08	0.26	0.00	1.33	0.03	0.15	0.00	1.30
	Beats	0.21	0.58	0.00	2.00	0.15	0.54	0.00	3.00	0.06	0.29	0.00	3.00
1012	*HRV*	96.74	4.86	77.00	102.17	97.99	3.94	90.22	105.03	89.58	12.73	63.35	115.10
	EDA	0.48	0.03	0.43	0.57	0.45	0.04	0.37	0.55	1.21	0.52	0.00	2.04
	BVP	-0.01	9.67	-108.23	108.21	0.00	17.33	-93.99	98.65	0.01	15.90	-127.05	113.59
	TEMP	27.53	0.39	27.09	28.61	29.24	0.19	28.63	29.41	30.45	0.70	29.29	31.65
	ACC	0.01	0.01	0.00	0.08	0.06	0.07	0.00	0.43	0.05	0.08	0.00	1.14
	IBI	1.03	0.37	0.00	1.47	0.36	0.51	0.00	1.66	0.20	0.42	0.00	1.59
	Beats	1.70	0.61	0.00	2.00	0.65	0.93	0.00	3.00	0.35	0.71	0.00	3.00

Participant ID	BioSignals	Baseline Data Mean	Baseline Data Standard Deviation	Baseline Data Minimum	Baseline Data Maximum	Training Phase Mean	Training Phase Standard Deviation	Training Phase Minimum	Training Phase Maximum	Collaboration Phase Mean	Collaboration Phase Standard Deviation	Collaboration Phase Minimum	Collaboration Phase Maximum
1013	*HRV*	81.33	5.55	48.00	91.27	85.33	3.41	80.68	92.60	107.91	28.11	60.25	170.08
	EDA	0.30	0.01	0.27	0.41	0.28	0.02	0.26	0.33	0.18	0.24	0.00	4.41
	BVP	0.03	6.38	-40.52	41.27	0.01	4.34	-22.59	23.30	0.01	8.24	-131.20	87.22
	TEMP	30.52	0.13	30.33	30.81	30.67	0.07	30.51	30.77	24.77	4.64	20.37	30.65
	ACC	0.01	0.03	0.00	0.18	0.02	0.02	0.00	0.12	0.02	0.07	0.00	0.82
	IBI	0.85	0.55	0.00	1.69	0.44	0.57	0.00	1.59	0.16	0.39	0.00	1.78
	Beats	1.18	0.76	0.00	2.00	0.60	0.78	0.00	2.00	0.24	0.57	0.00	3.00
1014	*HRV*	73.45	3.60	58.67	77.07	81.14	3.94	74.98	88.47	106.84	20.44	64.05	172.35
	EDA	0.30	0.02	0.26	0.34	0.45	0.09	0.29	0.67	0.63	1.21	0.00	6.32
	BVP	0.05	3.83	-29.55	34.93	-0.03	20.40	-94.11	72.30	0.00	14.08	-174.77	167.23
	TEMP	25.24	0.15	24.93	25.51	25.16	0.11	24.95	25.37	22.78	2.63	20.46	28.55
	ACC	0.00	0.00	0.00	0.07	0.06	0.08	0.00	0.40	0.02	0.06	0.00	0.94
	IBI	1.03	0.44	0.00	1.88	0.20	0.42	0.00	1.81	0.06	0.26	0.00	1.73
	Beats	1.29	0.55	0.00	2.00	0.28	0.57	0.00	2.00	0.09	0.37	0.00	2.00
1015	*HRV*	80.31	2.25	71.33	86.79	79.83	3.35	74.33	91.03	119.27	43.20	66.38	200.22
	EDA	1.02	0.06	0.88	1.17	1.48	0.41	0.75	2.51	3.57	4.92	0.01	18.39
	BVP	-0.01	6.21	-34.00	28.94	-0.01	21.48	-91.34	111.99	0.00	16.67	-132.42	125.45
	TEMP	30.83	0.36	30.37	31.53	31.50	0.04	31.43	31.59	25.98	5.30	17.01	32.49
	ACC	0.01	0.01	0.00	0.06	0.09	0.08	0.00	0.40	0.05	0.11	0.00	1.31
	IBI	0.77	0.58	0.00	1.75	0.11	0.35	0.00	1.53	0.06	0.23	0.00	1.61
	Beats	1.01	0.77	0.00	2.00	0.16	0.50	0.00	2.00	0.10	0.41	0.00	3.00
1016	*HRV*	79.51	4.66	51.00	87.87	81.28	3.20	75.75	87.30	89.33	13.14	57.58	127.77
	EDA	4.70	1.21	1.74	6.27	5.79	0.36	4.58	6.94	7.38	4.27	0.00	16.44
	BVP	0.00	24.75	-254.42	215.48	0.03	9.02	-38.29	41.78	0.00	18.89	-168.57	170.77
	TEMP	28.42	0.35	27.97	29.27	28.54	0.18	28.07	28.69	28.04	1.29	23.43	32.59
	ACC	0.01	0.02	0.00	0.21	0.07	0.09	0.00	0.43	0.07	0.09	0.00	0.83
	IBI	0.53	0.60	0.00	1.77	0.40	0.55	0.00	1.58	0.06	0.25	0.00	1.47
	Beats	0.71	0.82	0.00	2.00	0.56	0.77	0.00	2.00	0.09	0.38	0.00	2.00

Appendix H

Examine Your Interests Questionnaire

Completion Instructions

Your interests reflect your preference for doing some activities instead of others and can help you choose an appropriate career. Simply score the following statements from 1 to 5 in the clear box alongside each one. If you strongly disagree with a statement – score 1, and if you strongly agree score 5. Add up your scores for each column and put the overall score boxes overleaf. If you rank your scores in order the top 3 represent your major interest areas.

I like ...	P	E	S	C	I	O
working out how to get things done efficiently						
repairing and fixing machines						
producing designs from my own ideas						
being physically active						
managing a team of people						
working our problems						
working with people						
getting the details right						
to be different						
exploring new ideas for research and purposes						
helping people learn new skills						
making or building things with my hands						
gathering information						
learning new things						
using my imagination in my work						
persuading people to do or to buy something						
organising things, people and events						
providing care for people in some way						
making decisions						
carrying out research projects						
briefing a sales team about a new product						
making lists						
expressing myself in music, painting or writing						
working with community groups						
questioning established theories						
taking calculated risks						
designing or servicing equipment						
analysing statistical data						
working outside in the fresh air						

1

I like ...	P	E	S	C	I	O
listening to people's problems						
analysing a company's annual accounts						
selling something I have created						
writing letters, reports and articles						
using hand/machine tools to make things						
being involved in a community arts project						
giving advice on grants or benefits						
Totals for each column	P	E	S	C	I	O

Appendix I

Experimental Study 2 VRNQ Questionnaire

From 1 to 7, please **circle** the response that closely represents your opinion.

User Experience

What is the level of immersion you experienced?

1	2	3	4	5	6	7
Extremely Low	**Very Low**	**Low**	**Neutral**	**High**	**Very High**	**Extremely High**

What was your level of enjoyment of the VR experience?

1	2	3	4	5	6	7
Extremely Low	**Very Low**	**Low**	**Neutral**	**High**	**Very High**	**Extremely High**

How was the quality of the graphics?

1	2	3	4	5	6	7
Extremely Low	**Very Low**	**Low**	**Neutral**	**High**	**Very High**	**Extremely High**

How was the quality of the sound?

1	2	3	4	5	6	7
Extremely Low	**Very Low**	**Low**	**Neutral**	**High**	**Very High**	**Extremely High**

How was the quality of the VR technology overall (i.e., hardware & peripherals)?

1	2	3	4	5	6	7
Extremely Low	**Very Low**	**Low**	**Neutral**	**High**	**Very High**	**Extremely High**

Game Mechanics

How easy was to use the navigation system (e.g. teleportation) in the virtual environment?

1	2	3	4	5	6	7
Extremely Difficult	Very Difficult	Difficult	Neutral	Easy	Very Easy	Extremely Easy

How easy was to physically move in the virtual environment?

1	2	3	4	5	6	7
Extremely Low	Very Low	Low	Neutral	High	Very High	Extremely High

How easy was to pick up and/or place items in the virtual environment?

1	2	3	4	5	6	7
Extremely Difficult	Very Difficult	Difficult	Neutral	Easy	Very Easy	Extremely Easy

How easy was to do task 2 after performing task 1 in the virtual environment?

1	2	3	4	5	6	7
Extremely Difficult	Very Difficult	Difficult	Neutral	Easy	Very Easy	Extremely Easy

How easy was the 2-handed interaction e.g., grab the tablet with the one hand, and push the button with the other hand?

1	2	3	4	5	6	7
Extremely Difficult	Very Difficult	Difficult	Neutral	Easy	Very Easy	Extremely Easy

How easy was to complete the task?

1	2	3	4	5	6	7
Extremely Difficult	Very Difficult	Difficult	Neutral	Easy	Very Easy	Extremely Easy

How helpful was/were the task?

1 2 3 4 5 6 7

Extremely Unhelpful	Very Unhelpful	Unhelpful	Neutral	Helpful	Very Helpful	Extremely Helpful

How did you feel about the duration of the task?

1	2	3	4	5	6	7
Extremely More Time Needed	Much More Time Needed	More Time Needed	Neutral	Enough Time Available	Much Time Available	Plenty of Time Available

How helpful were the instructions for the task you needed to perform?

1	2	3	4	5	6	7
Extremely Unhelpful	Very Unhelpful	Unhelpful	Neutral	Helpful	Very Helpful	Extremely Helpful

How helpful were the in-game virtual hands in picking objects?

1	2	3	4	5	6	7
Extremely Unhelpful	Very Unhelpful	Unhelpful	Neutral	Helpful	Very Helpful	Extremely Helpful

VRISE

Did you experience nausea?

1	2	3	4	5	6	7
Extremely Intense Feeling	Very Intense Feeling	Intense Feeling	Moderate Feeling	Mild Feeling	Very Mild Feeling	Absent

Did you experience disorientation?

1	2	3	4	5	6	7
Extremely Intense Feeling	Very Intense Feeling	Intense Feeling	Moderate Feeling	Mild Feeling	Very Mild Feeling	Absent

Did you experience dizziness?

1	2	3	4	5	6	7
Extremely Intense Feeling	**Very Intense Feeling**	**Intense Feeling**	**Moderate Feeling**	**Mild Feeling**	**Very Mild Feeling**	**Absent**

<u>Did you experience fatigue?</u>

1	2	3	4	5	6	7
Extremely Intense Feeling	**Very Intense Feeling**	**Intense Feeling**	**Moderate Feeling**	**Mild Feeling**	**Very Mild Feeling**	**Absent**

<u>Did you experience instability?</u>

1	2	3	4	5	6	7
Extremely Intense Feeling	**Very Intense Feeling**	**Intense Feeling**	**Moderate Feeling**	**Mild Feeling**	**Very Mild Feeling**	**Absent**

Appendix J

VIRTUAL REALITY NEUROSCIENCE QUESTIONNAIRE

Guidelines for the Administration, Cut-offs, and Interpretation of the Results.

The Virtual Reality Neuroscience Questionnaire (VRNQ) was developed to enable the appraisal of software qualities such as user experience (UX), game mechanics (GM), and in-game assistance (IGA). Also, the VRNQ appraises the intensity of adverse symptoms pertaining to VR, such as nausea, disorientation, dizziness, instability, and fatigue. These symptoms are clustered under the term "VR induced symptoms and effects" (VRISE), though, there are other terms for this symptomatology such as cybersickness and VR-sickness.The VRNQ was designed to appraise software qualities which are specific to the implementation of immersive VR in clinical and research settings. Hence, VRNQ solely appraises the quality or suitability of VR research/clinical software. We suggest that the researchers or clinicians should clarify to the participant that the VRNQ is specific to the appraisal of research/clinical VR software.

Furthermore, the items of VRNQ under the sections of UX, GM, and IGA are known for increasing the pleasantness of the VR experience and the enablement of a quick familiarisation with the VR systems and controls (e.g., ergonomic and naturalistic interactions with and navigation within the virtual environments). In addition, the items of VRNQ under the sections of UX, GM, and IGA (i.e., of better quality) have been found to assist with alleviating the intensity of VRISE. Finally, the VRISE section includes items corresponding to the predominant adverse symptoms (i.e., nausea, disorientation, dizziness, instability, and fatigue) relevant to VR.

The VRNQ may be used by researchers and/or clinicians to report the quality of the VR software and/or the intensity of VRISE. The researchers or clinicians should consider the **medians** of the population. The medians of the population should also be considered to assess the suitability of the VR software by checking whether the median scores are above the cut-offs of VRNQ (see below the cut-offs of VRNQ). However, the researchers should also consider the **median absolute deviation (MAD)** of the scores. Over-stretched MADs may postulate a significant modulation of the medians by individual differences and/or outliers. Also, extra focus should be given to the VRISE items, since VRISE have been found to compromise neuropsychological, neuroimaging, and physiological data significantly. Thus, individual scores pertaining to VRISE below the cut-off postulate that the data (e.g., neuropsychological or neuroimaging data) of this individual should be excluded from the study. Moreover, the researchers should further scrutinise each symptom (e.g., nausea and dizziness) individually. In case of an intense (or stronger) feeling, we suggest the data of this participant to be excluded from the analysis.

Moreover, the VRNQ may also be used for the development of an immersive VR software. In this case, the participants (e.g., user-testers) should be prompted to provide additional comments and/or suggestions relevant to each question. Under each VRNQ question there are the Likert style responses (i.e., 1-7) which are matched with their respective meaning (e.g., very high). Also, under each VRNQ question there is an available space, where the participant may provide additional comments and/or suggestions relevant to the question. The comments or suggestions of the user-testers may assist with the amelioration of the VR software. Finally, as mentioned above, the medians and MADs of the scores should be considered. We suggest using the parsimonious cut-offs of VRNQ to assess the suitability of the VR software. The developers should build as many versions (e.g., alpha and beta) of the VR software are required until the medians of VRNQ scores are above the cut-offs (e.g., parsimonious).

For a review pertaining to VR software qualities and VRISE, please see:

Kourtesis, P., Collina, S., Doumas, L. A. A., & MacPherson, S. E. (2019). Technological competence is a precondition for effective implementation of virtual reality head mounted displays in human neuroscience: a technological review and meta-analysis. *Frontiers in Human Neuroscience*, 13, 342. https://www.frontiersin.org/articles/10.3389/fnhum.2019.00342/full

When you use the VRNQ please cite:

Kourtesis, P., Collina, S., Doumas, L. A. A., & MacPherson, S. E. (2019). Validation of the Virtual Reality Neuroscience Questionnaire: Maximum Duration of Immersive Virtual Reality Sessions Without the Presence of Pertinent Adverse Symptomatology. *Frontiers in Human Neuroscience*, 13, 417. https://www.frontiersin.org/articles/10.3389/fnhum.2019.00417/full

Virtual Reality Neuroscience Questionnaire (VRNQ) – Scores

Section	Score	Minimum Cut-offs	Parsimonious Cut-offs
User Experience		≥ 25	≥ 30
Game Mechanics		≥ 25	≥ 30
In-Game Assistance		≥ 25	≥ 30
VRISE		≥ 25	≥ 30
Total VRNQ		≥ 100	≥ 120

The median of each sub-score and totals score should meet the suggested cut-offs to support that the evaluated VR software has an adequate quality without any significant VRISE. The utilisation of the parsimonious cut-offs more robustly support the suitability of the VR software.

The VR Neuroscience Questionnaire (VRNQ) was developed by Panagiotis Kourtesis

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Appendix K

Collaboration Questionnaire

What was the level of collaboration between you and your partner during the interaction?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

How pleasant was the interaction between you and your partner during the interaction?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

How much was your partner's contribution central to the interaction?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

How much was your contribution central to the interaction?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

How much rating will you give your collaborating partner?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

Which scenario you would you prefer to work in?

In Open Workspace:

1 2 3 4 5 6 7

Low **High**

In Closed Workspace:

1 2 3 4 5 6 7

Low **High**

Appendix L

Experimental Study 2 – SIM-TLX

TUS

Subject ID:
Group: LG ☐ FG ☐

Questionnaire

The evaluation you are about to perform is a technique that has been developed to assess the relative importance of ten factors in determining how much workload you experienced while preforming a task. These ten factors are defined below on this page.

Read through them to make sure you understand what each factor means. If you have any questions, please ask the principal investigator.

Workload factors	Definition
Mental Demand Level (low/high)	**<u>How much mental and perceptual activity was required</u>** (for example, thinking, deciding, calculating, remembering, looking, searching, etc)? Was the task easy or demanding, simple or complex, forgiving or exacting?
Physical Demand Level (low/high)	**<u>How much physical activity was required</u>** (for example, pushing, pulling, turning, controlling, activating, etc.)? Was the task easy or demanding, slow or brisk, slack or strenuous, restful or laborious?
Temporal Demand Level (low/high)	**<u>How much time pressure did you feel</u>** due to the rate or pace at which the tasks or task elements occurred? Was the pace slow and leisurely or rapid and frantic?
Frustration Level (low/high)	**<u>How insecure, discouraged, irritated, stressed</u>**, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel during the task?
Task Complexity (low/high)	**<u>How complex was the task?</u>**
Situational Stress ((low/high)	**<u>How stressed did you feel while performing the task?</u>**
Distraction (low/high)	**<u>How distracting was the task environment?</u>**
Perceptual Strain (low/high)	**<u>How uncomfortable/ irritating were the visual and auditory aspects of the task?</u>**
Task control (low/high)	**<u>How difficult was the task to control or navigate?</u>**
Presence (low/high)	**<u>How immersed/present did you feel in the task?</u>**

TUS

Subject ID: __
Group: LG ☐ FG ☐

Consider your responses carefully in distinguishing among the different task conditions, and consider each individually.

1. Mental Demand (How mentally demanding was the task?/ How much mental and perceptual activity did you spend for this task?)

   Very Low — Very High

2. Physical Demand (How physically demanding was the task? / How much physical activity did you spend for this task?)

   Very Low — Very High

3. Temporal Demand (How hurried or rushed was the pace of the task? / How much time pressure did you feel in order to complete this task?)

   Very Low — Very High

4. Frustration Level (How insecure, discouraged, irritated, stressed, and annoyed versus secure, gratified, content, relaxed, and complacent did you feel during the task ?)

   High — Low

5. Task complexity (How hard did you have to work to accomplish the task?)

   Very Low — Very High

6. Situational Stress (How stressed did you feel while performing the task?)

   Very Low — Very High

Subject ID: ___
Group: LG ☐ FG ☐

7. Distraction (How distracting was the task environment?)

Very Low Very High

8. Perceptual Strain (How uncomfortable/irritating were the visual and auditory aspects of the task environment?)

Very Low Very High

9. Task Control (How difficult was the task to control or navigate?)

Very Low Very High

10. Presence (How immersed or present did you feel in the task?)

Very Low Very High

Appendix M

ID no	Baseline of EDA Signal			Eye Calibration			Practise Phase Individual			Closed Test Phase			Shared Phase			Questionnaire Phase		
	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
1	0.07	0.16	0.30	0.26	0.26	0.26	0.26	0.26	0.27	0.25	0.29	0.33	0.30	0.34	0.39	0.00	0.22	1.03
2	0.00	0.32	0.70	0.35	0.36	0.36	0.36	0.36	0.37	0.00	0.08	0.58	0.00	0.09	0.59	0.00	0.16	8.07
3	0.22	0.33	0.39	0.38	0.40	0.46	0.39	0.54	0.64	0.46	0.55	0.68	0.49	0.88	1.32	0.00	0.55	3.08
4	0.05	0.12	0.29	0.26	0.27	0.27	0.25	0.27	0.30	0.25	0.31	0.45	0.29	0.37	0.54	0.36	0.45	1.61
5	0.45	0.48	0.59	0.55	0.58	0.60	0.48	0.55	0.75	0.00	0.45	2.59	0.86	0.97	1.26	0.81	0.98	1.42
6	0.05	0.10	0.11	0.10	0.11	0.11	0.11	0.13	0.16	0.00	0.01	0.25	0.01	0.03	0.05	0.00	0.05	0.43
7	0.18	0.26	0.27	0.30	0.30	0.31	0.25	0.28	0.39	0.24	0.27	0.34	0.27	0.29	0.33	0.23	0.30	0.35
8	0.19	0.21	0.28	0.32	0.33	0.35	0.34	0.39	0.73	0.32	1.13	2.45	1.73	2.14	2.78	1.49	2.08	2.83
9	0.84	5.19	13.12	10.86	11.78	12.90	8.34	10.51	12.94	6.77	8.19	10.63	7.76	9.54	12.28	2.35	5.63	9.39
10	0.61	1.48	5.33	5.04	5.33	6.37	4.89	6.47	9.01	6.87	8.43	11.20	5.16	7.01	9.39	8.27	10.56	12.35
11	0.13	0.18	0.36	0.13	0.14	0.14	0.11	0.13	0.15	0.00	1.03	1.98	0.00	0.67	1.87	0.00	0.44	1.37
12	0.18	0.32	0.40	0.39	0.40	0.41	0.39	0.42	0.46	1.60	4.84	10.84	0.43	0.79	1.66	6.42	8.04	10.82
13	0.18	0.21	0.27	0.17	0.18	0.19	0.08	0.47	1.19	1.90	2.57	4.42	0.97	1.73	2.61	1.97	3.18	5.27
14	0.83	1.20	1.29	1.16	1.26	1.32	1.12	1.33	1.52	2.15	2.78	3.40	1.46	1.98	2.37	0.00	2.86	3.73
15	0.09	0.13	0.17	0.13	0.16	0.18	0.18	0.19	0.21	0.21	0.24	0.29	0.18	0.22	0.24	0.01	0.29	0.37
16	0.02	0.03	0.03	0.02	0.03	0.03	0.02	0.03	0.03	0.03	0.13	0.45	0.02	0.03	0.03	0.04	0.13	0.33
17	0.05	0.11	0.13	0.13	0.13	0.13	0.12	0.14	0.15	0.14	0.15	0.20	0.14	0.15	0.19	0.13	0.15	0.16
18	0.03	0.15	0.19	0.19	0.19	0.20	0.19	0.24	0.37	0.33	0.36	0.43	0.28	0.31	0.35	0.00	0.41	2.09
19	0.04	0.10	0.12	0.11	0.11	0.11	0.11	0.12	0.13	0.13	0.15	0.17	0.16	0.17	0.19	0.02	0.11	0.29
20	0.04	0.07	0.09	0.08	0.08	0.08	0.07	0.08	0.09	0.08	0.10	0.12	0.10	0.11	0.12	0.02	0.11	0.25
21	0.09	0.19	0.26	0.32	0.35	0.38	0.34	0.47	2.55	0.44	0.46	0.51	0.43	0.45	0.51	0.00	0.49	0.69
22	0.02	0.03	0.05	0.03	0.03	0.03	0.03	0.05	0.06	0.05	0.06	0.07	0.05	0.06	0.07	0.06	0.07	0.08

	Baseline of EDA Signal			Eye Calibration			Practise Phase Individual			Closed Test Phase			Shared Phase			Questionnaire Phase		
ID no	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
23	0.05	0.28	1.02	0.05	0.05	0.06	0.02	0.06	0.19	0.18	0.77	1.68	1.13	1.44	1.88	0.75	1.32	2.06
24	0.84	3.42	4.05	3.37	3.47	3.65	2.11	2.70	3.44	2.66	4.41	5.80	3.80	4.37	4.98	0.00	4.38	5.77
25	0.62	1.27	2.47	0.59	0.73	0.87	0.86	11.49	18.31	12.63	15.69	18.87	12.16	13.52	15.45	0.00	10.31	13.52
26	0.48	1.22	2.49	0.96	1.12	1.33	1.09	1.88	2.85	0.79	2.67	5.07	3.34	4.53	5.98	0.00	6.06	7.40
27	0.55	0.74	0.87	0.59	0.64	0.71	0.58	0.98	1.37	1.31	1.95	2.55	1.92	2.48	3.34	1.47	2.12	2.78
28	0.84	4.55	5.51	5.27	6.43	7.71	7.58	9.83	12.12	11.47	17.66	21.81	19.97	20.98	21.94	17.70	21.99	24.23
29	0.01	0.02	0.04	0.03	0.03	0.03	0.02	0.04	0.04	0.04	0.11	0.21	0.21	0.32	0.43	0.00	0.35	0.42
30	0.02	0.07	0.12	0.12	0.14	0.25	0.12	0.15	0.24	0.15	0.19	0.25	0.17	0.19	0.22	0.17	0.18	0.22
31	0.04	0.08	0.12	0.10	0.10	0.11	0.10	0.11	0.14	0.11	0.14	0.25	0.15	0.27	0.44	0.00	0.49	0.68
32	0.21	0.60	0.92	0.99	1.00	1.01	1.00	1.12	1.42	1.15	1.47	1.86	1.58	1.77	2.03	0.00	1.17	2.32
33	0.00	0.11	0.83	0.15	0.17	0.20	0.10	0.17	0.24	0.12	0.20	0.30	0.29	0.46	0.55	0.00	0.66	1.46
34	0.00	0.12	0.25	0.06	0.06	0.07	0.00	0.16	0.50	0.00	0.56	1.24	0.00	1.05	1.75	0.00	0.87	2.64
35	0.12	0.24	0.32	0.32	0.33	0.34	0.32	0.35	0.43	0.35	0.38	0.45	0.42	0.43	0.44	0.00	0.43	0.75
36	0.09	0.13	0.20	0.18	0.18	0.20	0.19	0.20	0.23	0.21	0.23	0.30	0.20	0.27	0.30	0.24	0.31	0.48
37	0.30	0.39	0.46	0.39	0.43	0.57	0.37	0.60	1.03	0.71	1.03	1.55	0.52	0.70	1.05	0.95	1.19	1.56
38	0.80	1.20	2.01	1.39	1.99	2.74	2.47	12.74	20.38	7.05	9.66	12.74	8.12	11.71	15.45	6.66	7.49	8.34
39	0.10	0.11	0.15	0.11	0.11	0.11	0.10	0.18	0.32	0.25	0.36	0.69	0.17	0.21	0.30	0.00	0.03	0.48
40	0.84	3.48	8.08	3.06	3.30	3.37	0.00	6.68	11.96	0.34	10.32	12.35	7.40	9.21	10.32	3.44	9.52	13.04
41	0.84	24.68	30.46	28.95	29.90	30.47	15.35	22.89	30.39	14.13	17.19	19.13	14.86	17.08	18.83	6.30	15.94	17.67
42	0.03	0.10	0.27	0.24	0.27	0.35	0.30	0.43	0.75	0.59	0.98	1.46	0.31	0.51	0.84	0.00	0.86	1.52
43	0.07	0.11	0.17	0.12	0.12	0.13	0.11	0.13	0.19	0.15	0.17	0.19	0.13	0.16	0.18	0.00	0.19	1.51
44	0.39	0.58	0.84	0.61	0.61	0.62	0.41	1.38	3.15	4.13	5.38	7.57	2.88	5.15	7.39	4.24	6.10	7.87

	Baseline of EDA Signal			Eye Calibration			Practise Phase Individual			Closed Test Phase			Shared Phase			Questionnaire Phase		
ID no	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max	Min	Mean	Max
45	0.13	0.15	0.21	0.14	0.14	0.14	0.14	0.17	0.20	0.36	0.54	0.97	0.20	0.25	0.38	0.74	1.60	2.38
46	0.15	0.26	0.31	0.31	0.31	0.31	0.31	0.33	0.37	0.48	0.53	0.67	0.35	0.43	0.49	0.62	9.11	15.79
47	0.84	4.52	6.15	7.30	7.34	7.39	7.08	8.47	9.77	8.87	12.16	15.55	0.00	11.75	16.96	NA	NA	NA
48	0.44	0.54	0.78	0.52	0.52	0.53	0.51	0.54	0.60	0.54	0.81	1.69	0.00	1.86	2.42	NA	NA	NA
49	0.05	0.09	0.14	0.13	0.13	0.13	0.13	0.15	0.17	0.16	0.20	0.27	0.23	0.23	0.25	0.00	0.20	0.77
50	0.25	0.31	0.35	0.31	0.31	0.32	0.30	0.32	0.36	0.31	0.32	0.36	0.31	0.32	0.33	0.00	0.30	0.33
51	0.14	0.21	0.25	0.22	0.22	0.22	0.22	0.29	0.61	0.53	1.84	3.89	2.64	3.56	4.40	0.00	3.84	5.22
52	0.08	0.13	0.16	0.14	0.15	0.15	0.14	0.25	0.59	0.42	1.13	2.69	1.55	2.11	3.14	0.00	2.30	3.87
53	0.33	0.95	2.31	0.33	0.34	0.34	0.25	1.61	2.86	0.94	1.76	3.29	0.00	2.09	3.53	0.00	0.00	0.00
54	0.03	0.10	0.16	0.13	0.14	0.14	0.12	0.19	0.23	0.20	0.26	0.35	0.29	0.35	0.39	0.02	0.31	0.38
55	0.05	0.37	1.21	1.46	1.51	1.54	1.51	1.73	2.01	2.33	3.92	6.43	1.44	2.10	2.90	0.00	3.61	6.72
56	0.01	0.16	0.35	0.18	0.19	0.19	0.15	0.19	0.23	0.10	0.29	0.69	0.10	0.14	0.22	0.00	0.48	0.77
57	0.16	0.23	0.28	0.24	0.24	0.26	0.23	0.26	0.28	0.00	0.40	0.54	0.24	0.28	0.40	NA	NA	NA
58	0.31	0.38	0.53	0.35	0.35	0.35	0.31	0.36	0.39	0.95	1.94	3.25	0.33	0.71	1.35	0.02	0.31	0.38
59	0.06	0.08	0.10	0.08	0.08	0.08	0.07	0.12	0.32	0.11	0.44	1.35	0.12	0.21	0.69	0.04	0.20	0.74
60	0.05	0.09	0.11	0.09	0.09	0.09	0.08	0.09	0.10	0.09	0.11	0.13	0.09	0.11	0.12	0.10	0.13	0.19
61	0.84	1.76	2.72	2.24	2.45	3.08	2.18	2.71	3.28	2.36	3.32	4.32	1.97	3.01	3.97	0.00	4.07	6.61
62	0.47	1.18	2.07	1.50	1.59	1.67	0.90	1.49	2.03	1.01	1.68	2.61	1.04	1.37	1.77	0.00	2.68	4.09
63	0.04	0.07	0.09	0.08	0.08	0.08	0.07	0.08	0.09	0.08	0.11	0.20	0.16	0.65	1.66	0.04	0.23	0.82
64	0.83	1.60	2.06	1.44	1.46	1.50	1.20	1.48	2.03	0.99	1.61	2.51	1.52	1.77	2.47	0.47	1.11	2.47
65	0.09	0.12	0.16	0.11	0.12	0.13	0.09	0.12	0.15	0.09	0.15	0.20	0.08	0.12	0.16	0.19	0.23	0.29
66	0.18	0.21	0.25	0.19	0.20	0.21	0.18	0.20	0.23	0.26	0.34	0.45	0.20	0.25	0.27	0.41	0.47	0.56

Appendix N

ID no	Baseline			Eye Callibration HR			Practise Phase HR			Closed Phase HR			Open Phase HR			Post Questionnaire		
	Baseline_mean	Baseline_min	Baseline_max	Eye Tracking Started_mean	Eye Tracking Started _min	Eye Tracking Started _max	Training Launched_mean	Training Launched_min	Training Launched_max	Test Launched_mean	Test Launched_min	Test Launched_max	SharedTest Lauched_mean	Shared Test Lauched_min	Shared Test Lauched_max	Questionnaire Phase_mean	Questionnaire Phase_min	Questionnaire Phase_max
1	71.93	52.00	84.43	86.58	86.58	86.58	86.71	86.63	86.73	74.87	60.30	91.37	82.60	74.53	94.78	97.14	76.83	127.65
2	74.09	59.55	104.78	78.37	78.37	78.37	79.28	78.68	79.78	84.71	72.03	104.42	79.88	74.30	90.95	115.16	85.03	170.73
3	68.65	64.50	75.35	90.76	90.65	90.90	82.59	73.77	91.57	79.24	62.18	99.12	83.13	72.38	94.08	110.78	85.68	144.58
4	70.93	66.29	85.00	104.06	102.78	105.32	84.42	71.48	108.53	79.19	58.50	88.52	79.10	57.92	91.05	76.63	68.20	85.55
5	81.90	68.33	89.07	97.11	83.98	106.67	86.23	58.90	101.68	91.56	73.58	114.00	74.67	66.25	85.95	90.38	84.72	100.07
6	67.11	51.80	99.65	113.03	101.48	118.55	78.64	58.52	123.78	102.54	68.45	131.95	79.71	70.58	91.68	103.92	90.80	114.47
7	79.41	51.00	84.22	82.53	81.98	82.82	80.74	72.15	86.20	77.40	61.45	88.10	83.80	77.12	95.42	88.48	73.42	117.58
8	91.32	67.00	106.63	104.88	103.73	105.97	89.54	64.00	107.00	97.57	79.22	110.75	108.81	94.82	119.00	93.44	71.83	115.18
9	75.89	66.29	86.37	74.96	74.78	75.03	76.96	70.10	85.67	78.60	71.88	90.17	75.14	71.53	79.00	75.06	69.50	83.58
10	82.47	69.00	91.03	86.68	85.82	87.50	88.28	83.43	93.20	90.81	74.73	103.23	93.36	75.12	102.80	91.96	82.97	98.73
11	58.73	55.03	61.59	60.47	60.22	60.98	69.30	60.57	77.92	95.22	74.97	126.30	75.29	63.00	87.70	73.89	62.85	89.20
12	72.55	52.00	92.20	74.16	73.57	74.57	89.38	73.25	106.73	84.36	72.32	90.35	81.43	76.17	87.13	78.13	65.60	93.73
13	107.87	51.00	114.35	114.65	113.67	116.08	109.65	85.87	123.95	96.59	66.57	116.12	112.35	92.23	125.72	101.17	72.45	119.02
14	88.72	76.11	95.37	86.95	85.98	87.48	78.90	67.95	93.10	78.34	56.45	97.38	79.72	72.13	86.92	86.10	75.45	97.23
15	90.45	79.00	98.33	97.03	86.48	101.50	84.89	70.37	95.62	80.94	62.50	98.30	93.35	72.45	112.28	79.22	71.98	89.85
16	88.67	76.11	95.37	85.72	80.62	87.48	79.77	67.95	93.10	84.56	56.45	97.38	76.12	57.68	86.92	90.01	75.45	124.75
17	74.11	56.75	97.02	65.16	64.23	66.58	84.53	63.12	98.83	77.72	63.62	88.27	75.72	56.98	88.90	73.85	62.38	83.03
18	79.11	52.00	86.48	73.92	71.83	75.87	79.02	63.83	96.82	83.85	68.50	112.47	79.15	70.12	88.28	82.62	68.63	94.65
19	72.57	65.35	82.35	77.26	76.88	77.77	85.80	70.27	123.55	81.95	65.47	109.72	82.41	73.55	94.57	83.25	80.22	86.98
20	60.81	49.00	66.30	67.93	67.53	68.10	67.25	60.62	91.62	66.43	56.22	89.38	68.58	64.97	73.88	69.45	66.62	73.65
21	90.95	74.56	103.00	89.49	81.65	95.03	84.16	67.63	109.73	95.03	85.48	110.12	79.68	68.53	94.08	92.82	74.40	107.20
22	94.27	79.00	102.97	82.73	71.28	92.20	88.86	63.52	125.87	92.15	66.37	126.35	77.07	72.45	82.55	85.28	71.82	101.22
23	95.19	80.00	99.97	99.43	98.95	99.75	99.86	96.27	101.73	101.92	65.70	111.33	98.91	77.68	110.07	100.43	85.20	115.62

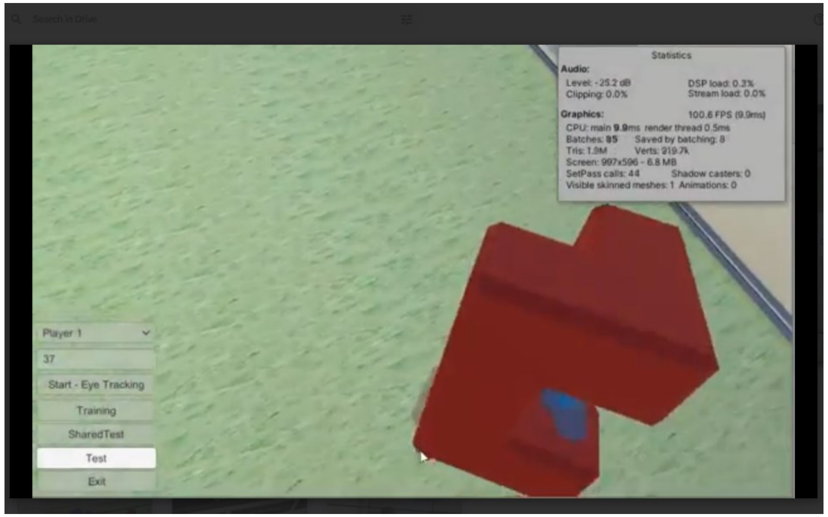
ID no	Baseline			Eye Callibration HR			Practise Phase HR			Closed Phase HR			Open Phase HR			Post Questionnaire		
	Baseline_mean	Baseline_min	Baseline_max	Eye Tracking Started_mean	Eye Tracking Started _min	Eye Tracking Started_max	Training Launched_mean	Training Launched_min	Training Launched_max	Test Launched_mean	Test Launched_min	Test Launched_max	SharedTest Lauched_mean	Shared Test Lauched_min	Shared Test Lauched_max	Questionnaire Phase_mean	Questionnaire Phase_min	Questionnaire Phase_max
24	84.40	52.00	97.71	89.73	88.83	90.22	91.67	72.03	112.73	81.12	64.08	94.92	95.74	82.47	106.87	91.54	84.85	98.28
25	77.10	58.00	83.95	85.52	84.72	86.20	84.73	79.15	90.88	85.66	70.47	104.02	88.05	84.45	90.92	71.87	63.73	86.02
26	75.44	51.00	80.82	80.94	80.18	81.15	81.87	76.07	87.82	84.37	78.92	94.18	83.90	79.00	89.35	87.95	84.90	90.43
27	88.95	76.67	119.00	95.18	86.57	101.05	103.51	85.65	113.50	98.95	82.07	106.08	96.90	81.53	105.85	87.64	60.33	114.88
28	94.97	61.00	122.20	90.98	87.17	97.10	97.14	90.17	104.65	99.07	90.58	104.42	100.20	85.98	108.58	88.25	71.47	106.33
29	72.09	69.82	76.07	77.95	75.28	80.23	83.05	78.73	87.88	86.96	74.25	102.18	78.58	71.63	89.08	83.86	74.78	100.18
30	101.95	49.00	104.97	94.63	89.00	100.38	93.37	77.35	106.40	82.41	61.60	101.53	70.15	62.73	82.48	93.45	67.77	111.12
31	103.34	60.00	130.87	84.07	83.98	84.25	86.22	78.87	97.03	104.62	91.60	109.93	102.64	98.97	105.83	92.06	74.88	99.88
32	80.61	70.58	118.00	72.22	70.95	73.55	107.37	73.83	161.70	95.56	66.83	157.45	86.92	73.47	94.88	94.90	74.52	119.35
33	78.67	70.58	118.00	87.35	85.95	89.15	82.41	73.42	92.57	97.42	66.83	161.70	84.72	73.47	111.85	91.14	74.52	119.35
34	63.45	56.37	80.38	99.47	85.50	110.02	96.56	72.55	116.77	109.95	56.68	143.87	95.89	65.02	134.98	98.22	61.55	123.07
35	74.86	58.07	103.00	84.88	80.85	88.55	88.23	86.43	90.63	84.77	78.10	92.17	87.73	85.75	90.20	84.34	80.87	87.23
36	63.53	54.00	72.97	74.69	73.28	76.30	75.74	70.40	81.15	75.42	65.57	87.43	74.33	68.42	77.92	82.72	66.25	104.08
37	84.70	69.00	94.00	86.31	85.23	88.20	97.61	88.28	108.70	83.58	65.80	94.37	90.62	81.57	96.65	94.62	90.70	98.55
38	81.85	61.00	108.60	81.04	76.73	86.25	76.07	71.87	81.63	77.17	68.28	90.28	76.40	68.12	87.18	80.93	72.17	92.28
39	62.68	59.50	69.67	66.77	66.10	67.63	70.25	55.50	87.05	72.02	68.47	74.75	73.96	69.13	82.93	133.64	68.33	145.15
40	111.26	71.00	118.52	109.94	109.33	111.20	107.11	70.72	127.57	92.71	66.35	118.28	107.96	87.92	120.92	103.06	80.33	118.95
41	82.32	63.00	97.18	71.15	68.88	73.87	86.64	68.85	95.45	87.08	82.30	91.93	92.00	81.82	111.03	86.32	80.43	90.35
42	87.04	63.00	107.40	94.21	89.80	96.38	93.50	78.70	105.88	102.96	91.05	117.93	91.02	77.02	101.32	86.08	67.48	101.93
43	100.06	65.00	138.50	97.23	95.47	98.73	90.71	73.63	105.00	97.95	79.35	109.75	91.21	70.63	107.58	83.43	64.03	98.58
44	79.00	65.83	84.43	81.86	80.80	82.82	81.56	76.78	88.70	83.82	77.68	92.20	82.43	77.38	87.95	82.71	77.68	86.83
45	88.54	73.12	109.00	96.06	95.82	96.18	92.66	74.28	102.42	86.30	77.53	106.82	96.17	74.93	112.27	107.50	87.48	121.17
46	106.77	79.00	120.30	116.19	116.05	116.47	120.38	116.37	128.75	144.09	137.15	150.42	131.54	124.92	138.93	140.07	120.53	151.77

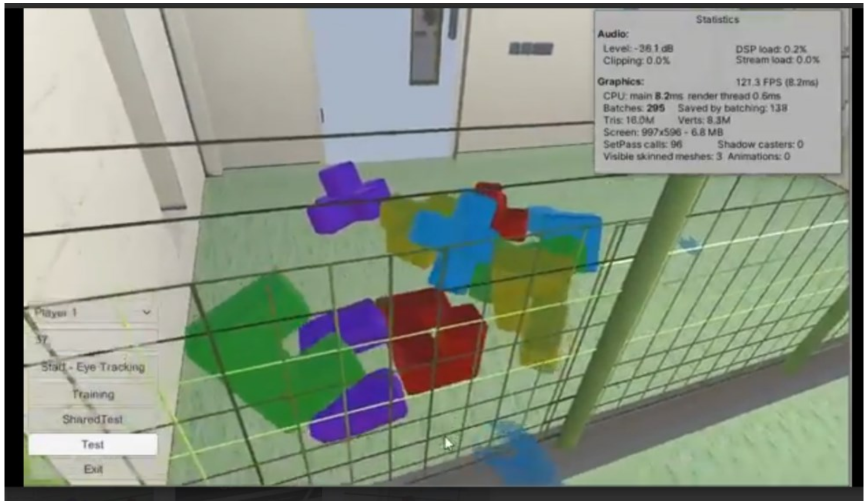
ID no	Baseline			Eye Callibration HR			Practise Phase HR			Closed Phase HR			Open Phase HR			Post Questionnaire		
	Baseline_mean	Baseline_min	Baseline_max	Eye Tracking Started_mean	Eye Tracking Started _min	Eye Tracking Started _max	Training Launched_mean	Training Launched_min	Training Launched_max	Test Launched_mean	Test Launched_min	Test Launched_max	SharedTest Lauched_mean	Shared Test Lauched_min	Shared Test Lauched_max	Questionnaire Phase_mean	Questionnaire Phase_min	Questionnaire Phase_max
47	86.15	68.17	101.28	102.33	102.32	102.33	100.08	92.17	103.93	104.88	100.12	108.10	106.48	101.78	110.38			
48	97.97	88.85	135.33	107.44	107.35	107.53	100.92	90.87	107.22	102.07	86.92	116.42	102.54	83.88	115.18			
49	98.41	53.00	108.45	99.90	99.30	100.47	100.54	85.05	107.45	99.11	75.88	106.02	96.50	76.25	103.38	95.34	83.42	106.88
50	79.40	60.00	85.38	82.57	82.55	82.58	84.58	76.92	98.05	85.20	70.90	94.13	82.71	75.80	89.25	90.17	75.13	101.32
51	83.51	75.33	95.60	86.88	86.43	87.87	84.39	72.22	92.52	89.61	76.18	99.45	89.71	82.75	96.65	87.02	71.95	98.77
52	92.77	66.50	99.75	93.38	92.35	94.92	94.90	81.70	104.97	105.72	92.90	116.55	108.55	94.93	115.32	112.19	92.33	127.42
53	95.33	76.50	102.20	101.55	101.47	101.62	102.49	96.22	107.15	101.01	85.03	108.65	93.92	75.40	109.12	118.17	82.83	156.90
54	77.57	58.50	81.93	75.05	75.03	75.07	77.38	62.08	83.37	83.52	55.32	103.32	84.08	65.95	94.92	82.74	76.10	90.20
55	81.38	60.50	91.85	84.87	84.77	84.97	88.88	85.05	91.00	88.23	83.50	93.47	87.89	87.18	89.67	85.82	77.45	89.63
56	72.32	62.00	82.13	72.96	72.90	73.02	71.17	67.27	73.50	78.40	71.52	84.50	77.13	66.73	86.22	84.18	76.00	90.27
57	93.04	61.00	100.30	94.51	92.37	96.50	100.46	93.78	107.20	99.45	79.52	115.75	112.72	105.95	119.35			
58	66.55	61.12	114.00	74.32	73.65	74.80	68.64	66.18	73.45	74.84	66.63	85.75	71.12	67.47	82.10	65.05	61.75	78.57
59	70.15	64.50	73.90	67.48	67.45	67.50	86.86	65.72	109.30	95.73	81.22	102.47	96.89	86.08	106.78	79.23	71.93	97.00
60	67.97	65.43	72.08	75.59	75.45	75.70	71.62	63.68	78.42	79.13	66.08	89.92	80.53	73.10	89.90	75.59	67.83	86.67
61	92.67	55.00	110.22	108.26	107.98	108.40	106.04	104.72	108.18	96.76	75.15	113.08	106.41	101.73	112.83	83.87	78.50	91.47
62	91.09	76.00	97.43	87.61	85.77	89.42	86.31	78.90	95.32	96.31	75.27	115.10	96.95	93.48	99.42	108.54	101.93	115.13
63	73.26	57.40	88.50	71.45	71.02	71.93	70.18	65.42	73.93	73.38	60.37	83.23	71.92	63.22	82.43	67.95	57.07	81.07
64	94.30	57.86	104.25	100.56	100.37	100.72	100.44	95.90	106.75	101.30	92.10	111.78	101.25	81.23	110.93	94.88	84.05	113.23
65	89.97	58.00	135.03	74.91	74.12	75.13	72.16	69.52	74.45	77.11	68.47	107.83	79.92	72.00	92.35	73.08	64.30	80.82
66	70.49	65.40	88.00	74.66	74.13	75.10	75.95	72.40	81.27	85.70	68.03	96.53	81.09	76.72	86.40	83.16	74.25	91.42

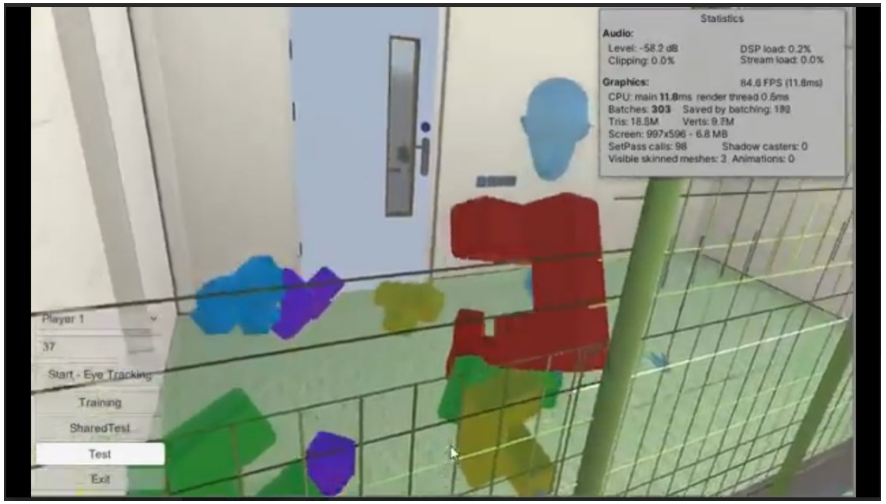
Appendix O

Pair No	Gender- Leader-Follower	DTW Shared EDA	DTW Closed EDA	Closed HR DTW	Open HR DTW	Open Time	Closed Time	Total Open clicks	Total Closed Clicks	Shared Time (seconds)
1	F-M	127.95	427	2256.24	233.24	00:02:11	00:08:22	85	216	131
2	M-M	515.44	841.7596	3108.35	2818.74	00:07:13	00:09:02	40	157	433
3	M-M	520.5652	827.7681	1365.41	215.14	00:02:13	00:05:37	81	129	133
4	F-F	1250.305	1331.403	6804.44	4148.08	00:02:49	00:06:32	71	66	169
5	F-F	582.5197	1640.036	3198.4	3550.41	00:03:26	00:05:54	109	201	206
6	F-M	451.880366	828.003503	5909.64	3697.97	00:05:22	00:09:51	50	56	322
7	M-M	78.24331	434.2768	5855.29	4437.52	00:02:50	00:05:46	50	56	170
8	F-M	223.3737	150.8206	4163.23	3655.41	00:04:57	00:05:58	43	112	297
9	F-M	219.9693	355.719	1179.81	2384.09	00:05:29	00:06:55	176	160	329
10	F-M	92.03351	161.2445	10690.87	5502.84	00:06:38	00:14:15	83	182	398
11	M-M	245.4799	411.3959	2683.45	613.51	00:02:36	00:04:16	67	75	156
12	F-M	2316.333	10908.63	12187.13	2296.93	00:03:18	00:12:28	97	108	198
13	F-M	11064.73	43589.83	2615.75	796.98	00:05:08	00:13:57	119	250	308
14	M-M	22416.54	27578	1221.38	1383.9	00:05:03	00:07:19	74	139	303
15	M-M	102.8209	165.4455	3149.17	2285.29	00:05:08	00:08:07	64	128	308
16	M-M	2280.591	1448.399	8230.03	3784.46	00:04:01	00:07:06	34	182	241
17	M-M	680.5431	795.9969	6988.64	4573.07	00:04:36	00:09:08	53	222	276
18	F-M	250.9392	111.4029	3940.38	2392.21	00:02:58	00:07:10	31	92	178
19	M-M	26854.49	9380.984	3058.73	4622.23	00:03:33	00:12:58	87	75	213
20	F-M	9103.974	14478.68	7441.56	8580.71	00:04:13	00:06:06	74	52	253
21	F-M	19430.14	27686.72	6753.55	1708.65	00:04:53	00:07:07	66	65	293
22	M-M	10877.23	7407.946	5034.25	5030.17	00:09:05	00:05:55	63	80	545
23	M-F	90.50497	38.10188	11929.99	5075.11	00:02:24	00:03:26	65	106	144
24	M-M	432.25529	5746.748961	1422.85	964.15	00:02:30	00:04:56	26	139	150
25	M-F	69.56365	195.113	5102.28	2311.98	00:03:24	00:06:40	26	74	204
26	M-F	414.0202	874.0718	6416.75	2851.73	00:02:47	00:07:22	111	121	167
27	F-M	6049.068	6049.098	11374.05	3650.62	00:07:50	00:16:53	125	118	470
28	F-F	1257.406	4995.183	3297.08	1711.72	00:02:40	00:05:44	59	114	160
29	F-F	302.6047	2944.276	11802.01	8528.72	00:03:25	00:08:54	58	154	205
30	F-F	252.2621	840.0659	9553.86	8210.89	00:09:38	00:10:48	194	104	578
31	M-M	989.5348	1588.291	1428.82	1243.08	00:02:31	00:04:18	28	79	151
32	M-F	1159.9	3048	14163.76	7685.6	00:04:23	00:08:27	120	104	263
33	M-F	305.1941	436.9136	4791.17	6717.13	00:05:22	00:09:51	112	158	322
34	F-F	302.6047	2944.276	11802.01	8528.72	00:03:25	00:08:54	58	154	205

SUPPLEMENTARY FIGURES







Final Notes

"As this research journey comes to an end, I reflect on the valuable knowledge and experiences gained throughout the process. This work has deepened my understanding of collaborative dynamics in immersive environments and highlighted the potential of Collaborative Virtual Reality to enhance human interaction, communication, and performance. I am excited to carry these learnings forward and continue developing innovative solutions that push the boundaries of virtual collaboration."